



OSS XML Gateway User's Guide

For software version DZMS 4.x

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ABOUT THIS GUIDE

DZS Management System (DZMS) is the network management system that provides service-to-subscriber management for DZS™ local loop network elements, including Broadband Access Node (BAN™), Multi-Access Line Concentrator (MALC™), zNID™, Z-Edge 64™, and MXK™, etc. DZMS includes a comprehensive suite of applications that deliver fault, configuration, performance, and security management functions. In addition, DZMS provides for the exposure of its API for automated provisioning by Operations Support Systems (OSS) using web service. The DZS OSS XML Gateway (Optional Software License/Module) provides access to DZMS from service provider OSS systems.

Audience

This guide is written for the system administrator who installs and administers the OSS XML Gateway system and/or the programmer who develops an OSS XML client to provision resources. As a reader of this guide, you should be familiar with basic networking components such as network devices, cards, physical ports, logical interfaces, and permanent virtual circuits (PVCs). To develop an OSS XML client, you need to be familiar with any programming language that supports XML parsing. A basic knowledge of the elements of XSD and web services is also required.

Style and notation conventions

The following conventions are used in this document to alert users to information that is instructional, warns of potential damage to system equipment or data, and warns of potential injury or death. Carefully read and follow the instructions included in this document.



Caution: A caution alerts users to conditions or actions that could damage equipment or data.



Note: A note provides important supplemental or amplified information.



Tip: A tip provides additional information that enables users to more readily complete their tasks.



WARNING! A warning alerts users to conditions or actions that could lead to injury or death.



WARNING! A warning with this icon alerts users to conditions or actions that could lead to injury caused by a laser.

Typographical conventions

The following typographical styles are used in this guide to represent specific types of information.

Bold	Used for names of buttons, dialog boxes, icons, menus, profiles when placed in body text, and property pages (or sheets). Also used for commands, options, parameters in body text, and user input in body text.
Fixed	Used in code examples for computer output, file names, path names, and the contents of online files or directories.
Fixed Bold	Used in code examples for text typed by users.
<i>Fixed Bold Italic</i>	Used in code examples for variable text typed by users.
<i>Italic</i>	Used for book titles, chapter titles, file path names, notes in body text requiring special attention, section titles, emphasized terms, and variables.
PLAIN UPPER CASE	Used for environment variables.
Command Syntax	Brackets [] indicate optional syntax. Vertical bar indicates the OR symbol.

Related documentation

OSS XML Gateway Reference Guide

OSS XML Gateway Release Notes

OSS Gateway User's Guide

OSS Gateway Reference Guide

MXK-F Hardware Installation Guide

MXK-F Configuration Guide

MXK-F Release Notes

DZMS Release Notes

DZMS Administrator's Guide

DZMS Installation Guide

NetHorizon User's Guide

DZMS HA Platform Installation and Administration Guide

Web Client Quick Start Guide

NETCONF User's Guide

Refer to the release notes for software installation information and for changes in features and functionality of the product (if any).

Contacting DZS Quality & Service

All new DZS equipment purchases include one year of HW Warranty and 90 days of Bronze-level Technical Support.

If your product is not within 90 days of the purchase date or you do not have a valid support contract, please contact your local sales representative to get a quote on a support contract.

Customers with a valid support contract or are eligible for 90 days technical support associated with a new product purchase can request technical support by opening a case at:

<https://dzsi.com/support/#TAC>

Customers with a valid support contract have access to technical product documentation, software downloads, knowledge base and consultation on the covered DZS product at the same support portal.

For repair services within the HW Warranty period or under an Extended Warranty support contract, a Return Material Authorization (RMA) must be obtained before sending the equipment for repair. RMA requests can be submitted at:

<https://dzsi.com/support/#RMA>

Technical support

The Technical Assistance Center (TAC) is available with experienced support engineers who can handle questions, assist with service requests, and help troubleshoot systems.

Hours of operation	Monday - Friday, 8 a.m. to 6 p.m, Eastern Time (excluding U.S. holidays)
Telephone (North America)	877-946-6320, prompt #3, #1
Telephone (International)	1-510-777-7133, prompt #3, #1
E-mail	support@dzsi.com
Web - available 24 x 7 to submit and track field issues/ problem reports	https://dzsi.com/support/#TAC Click DZS Problem Reporting System

If you purchased the product from an authorized dealer, distributor, Value Added Reseller (VAR), or third party, contact that supplier for technical assistance and warranty support.

Hardware repair

If the product malfunctions, all repairs must be authorized by DZS with a Return Merchandise Authorization (RMA) and performed by the manufacturer or a DZS-authorized agent. It is the responsibility of users

requiring service to report the need for repair to DZS Quality & Service as follows:

- Complete the RMA Request form (<https://dzsi.com/support/#RMA>) or contact DZS Quality & Service via phone or email:

Hours of operation: Monday - Friday, 8 a.m. to 6 p.m, Eastern Time (excluding U.S. holidays)

E-mail: support@dzsi.com (preferred)

Phone: 877-946-6320 or 510-777-7133, prompt #3, #2

- Provide the part numbers and serial numbers of the products to be repaired.
- All product lines ship with a minimum one year standard warranty (may vary by contract). DZS warrants all repairs for 90 days or the remainder of the standard warranty (whichever is greater).
- DZS will verify the warranty and provide the customer with a repair quote for anything that is not under warranty. DZS requires a purchase order or credit card for out of warranty fees.

1

SYSTEM ADMINISTRATION

DZS Management System (DZMS) is the network management system that provides service-to-subscriber management for DZS local loop network elements, including Broadband Access Node (BAN) and Multi-Access Loop Concentrator (MALC), MXK, MXK-F/Vx Series .

DZMS includes a comprehensive suite of applications that deliver fault, configuration, performance, and security management functions as well as user account management. The DZS OSS XML Gateway (optional software license/module) provides access to DZMS from Operations Support System (OSS). The OSS XML Gateway provides a published web service for bidirectional transfer of management information on the DZS network, thus integrating DZMS network management into service provider back-office systems and third-party applications.

This chapter includes the following sections:

- [Minimum system requirements for OSS XML Gateway server, page 17](#)
- [OSS XML Gateway system component, page 18](#)
- [OSS XML Gateway installation, page 19](#)
- [Administrative tasks, page 23](#)



Note: For instructions on configuring the various components of DZMS and on installing the NetHorizon client, see the *DZMS Administrator's Guide*.

Minimum system requirements for OSS XML Gateway server

This section describes the OSS XML Gateway system requirements. Before you install the OSS XML Gateway server, make sure the DZMS system has been installed either on the local machine or on another machine on the network. Refer to the *DZMS Installation Guide* for DZMS system requirements.



Note: If installing DZMS and OSS XML Gateway on different machines, make sure that the versions of the two systems are compatible.

On Solaris, make sure the default local setting is set to US-english.

Check the following variables in /etc/default/init file:

```
TZ=US/Pacific
CMASK=022
LC_COLLATE=en_US.ISO8859-1
LC_CTYPE=en_US.ISO8859-1
LC_MESSAGES=en
LC_MONETARY=en_US.ISO8859-1
LC_NUMERIC=en_US.ISO8859-1
LC_TIME=en_US.ISO8859-1
```

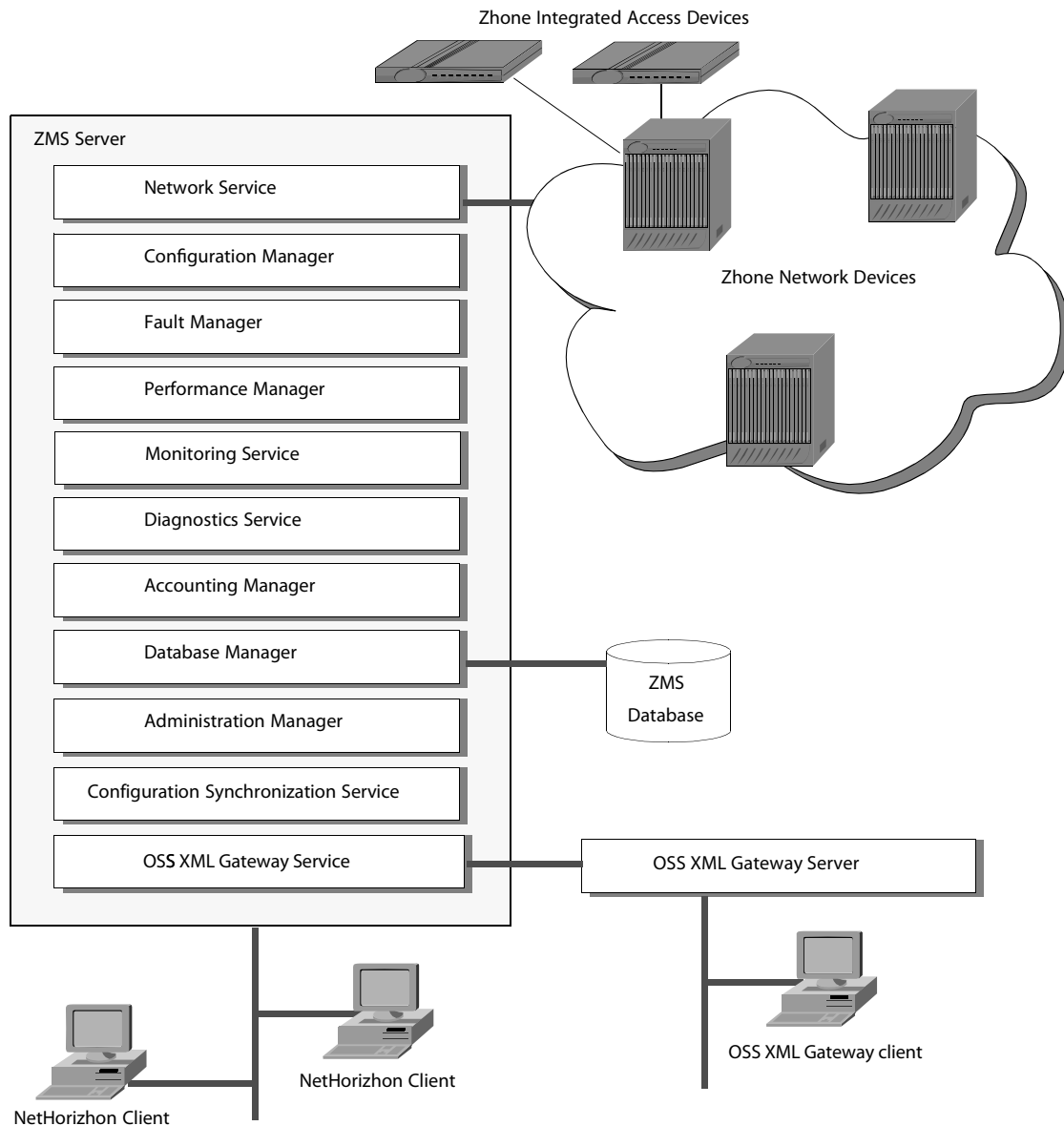
OSS XML Gateway system component

The OSS XML Gateway is based on web services where the client sends its request to the server, in the form of a SOAP message, over HTTP. The components in the OSS XML Gateway environment are:

- **XML Schema Definition (XSD)** is a XML based language for describing the XML that is to be sent to the OSS XML Server.
- **OSS XML Gateway clients** generate requests to manage DZS network elements.
- **TCP/IP client socket communication** is used between OSS XML client and OSS XML server. Thus there is no WSDL required.
- **WebServices** are used between OSS XML server and DZMS server and they are transparent to the OSS XML client implementation.
- **DZMS Component** translates requests and responses between the web service and DZMS domains and interacts with the DZMS network management services.
- **Zhone objects** are network elements and their various components.

[Figure 1](#) illustrates the OSS XML Gateway system environment.

Figure 1: OSS XML Gateway environment



OSS XML Gateway installation

The OSS XML Gateway is distributed over the FTP server containing the following:

1. `install_ossxml.sh`

The installation script for SPARC, x86 Solaris, and x86 Linux.

2. `ossxmlsparcpackageXX_XX_XX_XXX.tar.gz`

The SPARC Solaris OSS XML software package.

3. *ossxmlx86packageXX_XX_XX_XXX.tar.gz*

The x86 Solaris OSS XML software package.

4. *ZHONoss-X.X.X-XXX.x86_64.rpm*

The x86 Oracle linux OSS XML software package.

5. *XX.XX.XX.XXX.md5*

This file containing checksums of files in this directory.

6. *copyright*

(where X stands for the appropriate version of software) and can be accessed by UNIX systems.



Note: Recommended DZMS and OSS XML installation sequence:

1. Install DZMS.
2. Install OSS XML.
3. Start DZMS:

```
/etc/init.d/dzms start
```

Note that starting DZMS will automatically start OSS XML.



Note: To install or uninstall OSS XML on Solaris or Oracle Linux platforms, use the “install_ossxml.sh” script that comes with the OSS XML software distribution.

OSS XML Gateway installation Procedure

- 1 Log in to the server as root.
- 2 Check to see if there is a version of the OSS XML Gateway installed:

```
% pkginfo -l ZHONossxml
```
- 3 Acquire the OSS XML Gateway software package from the FTP server.
- 4 In an Xterm window, change directory to a temp directory:

```
% cd /tmp
```
- 5 Create a temporary directory in */tmp* called **ossxml**:

```
% mkdir ossxml
```
- 6 Change working directory to **ossxml**:

```
% cd ossxml
```
- 7 Copy all six OSS XML Gateway files to */tmp/ossxml* directory:



Note: The upper and lower case lettering of the OSS XML package name must be followed since UNIX is case-sensitive.

- 8** Verify checksum by using the `XX.XX.XX.XXX.md5` file.

- 9** Issue the following command to run the install script:

```
./install_ossxml.sh
```

The script checks for at least 30 MB of free disk space in the `/tmp` directory.

The installation menu appears, listing the installation options:

```
*****
1) Install OSS Gateway Server
2) Remove OSS Gateway Server
q) Quit
*****
Please select the installation option:
```

- 10** If there is a previous version, select option 3) Remove OSS XML Gateway Server from the installation menu. If there is no previous version, skip to [Step 11](#).

- 11** From the installation menu, enter **1** to install the OSS XML Gateway.

The installation utility installs the ZHONossxml package, containing the OSS XML Gateway server core software package.

- 12** When prompted “BY INSTALLING THE ENCLOSED SOFTWARE, YOU AGREE TO BE BOUND BY THE PROVISIONS ABOVE. Do you want to continue (y or n)? <y>”, enter **y** to install the OSS XML Gateway on the machine. The system will display the version of the OSS XML Gateway that is being installed.

- 13** At the confirmation prompt, enter **y** to install the various OSS XML Gateway on the machine. The system checks for disk space.

The installer takes a few minutes to open up all the compressed files.

- 14** The installation utility prompts you to enter the package base directory:

```
Enter base installation directory [/opt]:
```

- 15** Enter the path to the directory where you want the package installed. The server core will be installed into the directory `ossxml` within this base.

For example, if you enter `/opt`, the server core is installed into `/opt/ossxml`.

- 16** At the confirmation prompt, enter **y** to continue.

- 17** If you have installed the DZMS server on a remote machine, when prompted, enter the hostname of the DZMS server. If DZMS is running on the same machine, the name of the local machine will appear as the default value, enter **y** to continue.

18 When prompted, enter the OSS XML Gateway Port Number. Accept default if the port is not used and is available for OSS XML Gateway.

19 At the confirmation prompt, enter **y** to continue.

The installation utility performs various verification functions and displays the message:

```
This package contains scripts which will be executed
with super-user permission during the process of
installing this package.
```

```
Do you want to continue with the installation of
<ZHONEossxml> [y,n,?]
```

20 Enter **y** to continue.

The installation utility completes the installation and performs post installation functions. It creates a symbolic link from `/ossxml` to `/installation_directory/ossxml`.

The install script displays the main menu after successful installation of OSS XML Gateway.

21 Enter **q** to quit the installation process.

22 If desired, change the default port setting (15002) for the OSS XML Gateway using the **ossxmlPort** parameter in the *OSSXMLGateway.properties* file, located in `/opt/ossxml/lib/com/zhone/zms/ossxml/properties`.

23 Remove files from the temporary file:

```
% rm -r /tmp/ossxml
```



Note: The OSS XML Gateway needs to be configured before starting it up. Refer to the *OSS XML Gateway User's Guide* for configuration details.

Removing the OSS XML Gateway server

1 Issue the following command to run the install script:

```
./install_ossxml.sh
```

2 The installation menu appears, listing the installation options:

```
*****
1) Install OSS Gateway Server
2) Remove OSS Gateway Server
q) Quit
*****
Please select the installation option:
```

3 Enter **2** to remove OSS XML Gateway server. The OSS XML Gateway and all its associated components will be removed from the system.

- 4 When prompted “Do you want to remove this package? [y,n,?,q]”, specify **y**.
- 5 When prompted “Do you want to continue with the removal of this package [y,n,?,q]”, specify **y**.
- 6 The “Removal of <ZHONoss> was successful.” message appears.

Changing the JRE in the OSS XML startup script (optional)

- 1 After installation of the OSS XML products, go to `/installed_directory/ossxml/bin`.
- 2 Open file (startOSSXMLgw.sh) file.
- 3 Change the value of the `JAVA_HOME=/zmsjre` to point to the top level of another JRE.
- 4 Save the file and exit.
- 5 Stop and re-start the OSS XML Gateway.

Administrative tasks

This section describes the following administrative tasks:

- [Modifying environment variables](#)
- [Starting/ Restarting the OSS XML server](#)
- [Stopping the OSS XML server](#)
- [Upgrading the OSS XML server](#)
- [Managing operators](#)
- [Adding trap destination, deleting trap destination and changing DZMS IP](#)

Modifying environment variables

The default values and description of the OSS XML Gateway environment variables are listed in [Table 1](#).

Table 1: OSS XML Gateway variables

Variable Name	Default Value	Description
ossxmlPort	15002 (can be reset to any valid port number if there is a port conflict)	The OSS XML Gateway port number.
url	http://<zms server name>:7001/OSSXML/OSSWebservice/OSSWebservice.wsdl	The url of the web service to which the request SOAP xml is transferred.

Table 1: OSS XML Gateway variables

Variable Name	Default Value	Description
serverHost	<dzms server name>	Machine name where DZMS server is installed.
XSDFile	/ossxml/bin/ossxml.xsd	Location of the XSD file.

If you want to change the values to configure the OSS XML server and the DZMS component of the OSS XML Gateway, use a text editor to modify the *OSS XML Gateway.properties* file located in `/<install directory>/ossxml/lib/com/zhone/zms/ossxml/properties`.

Once you make changes to the file, you need to manually restart the OSS XML server for the changes to take effect. For instructions, see [Starting/Restarting the OSS XML server on page 24](#).

Starting/ Restarting the OSS XML server

The DZS OSS XML Gateway server expects an input xml of the format specified in the XSD.

To manually start or restart the OSS XML server, issue the following commands as the root. If the OSS XML is running already, this command will stop the server and then restart the server.

```
/ossxml/bin/ossxml start
```

The OSS XML software displays startup progress to the console.

Stopping the OSS XML server

To manually stop the OSS XML server, issue the following commands as the root.

```
/ossxml/bin/ossxml stop
```

Upgrading the OSS XML server

To upgrade the version of the OSS XML Gateway:

- 1 Un-install the old version. It will automatically stop OSS XML server if they are currently running.
For instructions, see [Removing the OSS XML Gateway server on page 22](#).
- 2 Install the new one.
For instructions, see [OSS XML Gateway installation Procedure on page 20](#).

Managing operators

To use the OSS XML Gateway, the OSS XML Gateway client issues a request to authenticate a DZMS operator in the form of an XML. The OSS XML Gateway identifies the operator to be authenticated, using a valid DZMS operator name and password.

The OSS XML Gateway does not support managing operator accounts. You must use NetHorizon to create, modify, or delete operator accounts. For instructions, see the *DZMS Administrator's Guide*.

For auditing purposes, create unique operator names and passwords for each OSS XML Gateway operator. The DZMS system generates an audit log, listing all operator activity. For details, see the *DZMS Administrator's Guide*.

Adding trap destination, deleting trap destination and changing DZMS IP

Here are the procedures to add trap destination, delete trap destination and change DZMS IP:

- 1** Added OSS XML standalone script to switch all the devices of DZMS from an old DZMS IP to a new DZMS IP. This script will perform the following three operations:
 - a** changes DZMS IP to new DZMS IP,
 - b** removes old DZMS IP trap destination,
 - c** adds a new trap destination for new DZMS IP.
- 2** Successful devices will be logged as part of `/ossxml/log/switch_zms_success.log`.
- 3** Failed devices will be logged as part of `/ossxml/log/switch_zms_fail.log` along with the operations passed and failed. If any one of the 1.a or 1.b or 1.c fails, that device will be treated as failed to switch DZMS IP.
- 4** Each time `switch_zms_server_ip.ksh` runs, logs will be cleaned to log the new run data.

2

THE OSS XML GATEWAY

This chapter describes the basics of how to form requests to the OSS XML Gateway.

This chapter includes the following sections:

- [Overview, page 27](#)
- [Object types, page 29](#)
- [Parent and object handle types, page 30](#)
- [General usage, page 70](#)
- [How to Determine OSS XML Commands, page 86](#)

Overview

The OSS XML Gateway allows you to develop an OSS XML Gateway client. The client provides access to the DZS Management System (DZMS) to perform the following network management functions:

- Configuration requests:
 - **get**, **getTemplate**, **list**, and **listTemplate** requests to retrieve data for objects.
 - **delete** requests to delete objects.
 - **add**, **addUsingTemplate**, and **modify** requests to configure objects.
- Performance requests:

getStatus requests ask for information about an object from the network element. See the *OSS XML Gateway Reference Guide* for more information.

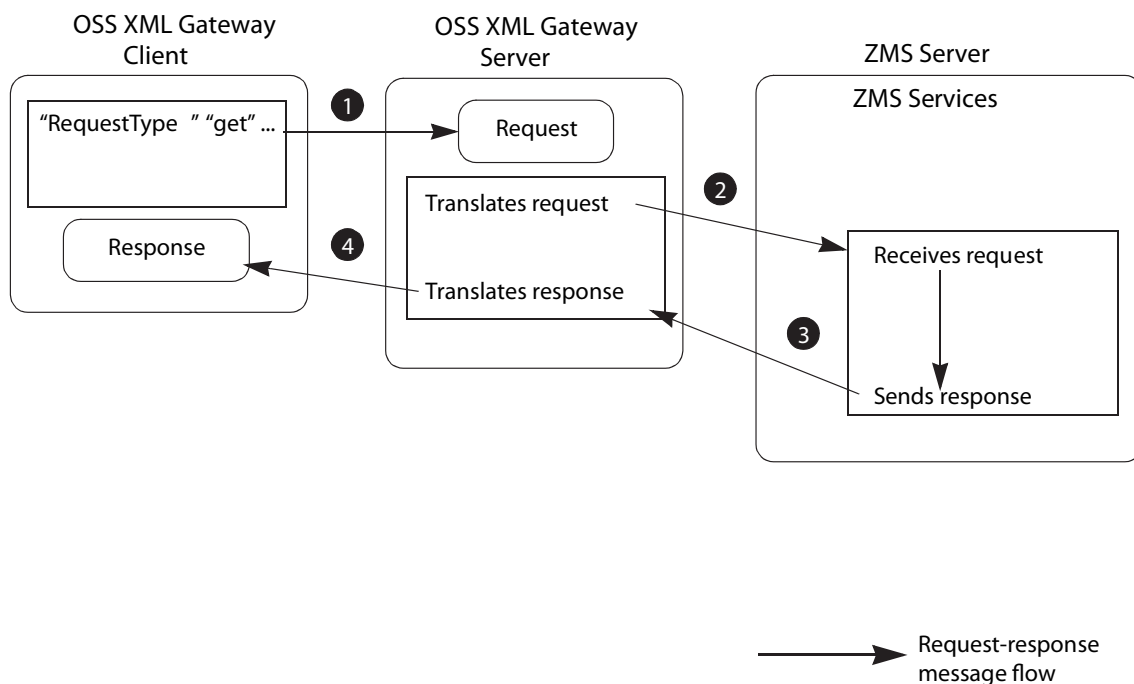
The OSS XML Gateway accepts requests in XML format specified by the XSD.

Synchronous operations

The OSS XML Gateway supports *synchronous* requests, which are requests in which the client waits for a response to the request before continuing with other processing.

Figure 2 shows the flow between the client and the OSS XML Gateway for synchronous requests. The client does not regain control of the program until the response returns from the server.

Figure 2: Synchronous request/response message flow



Steps during a synchronous request:

- 1 The OSS XML Gateway client sends a request XML to the OSS XML Gateway server.
- 2 The OSS XML Gateway server translates the request XML into a SOAP message and then sends it to the web service, from where the request XML is parsed and based on the type of the request, the request is issued to the appropriate DZMS service.
- 3 The DZMS service processes the request and passes the response to the OSS XML Gateway server.
- 4 The OSS XML Gateway server translates the response and returns it to the OSS XML Gateway client.

Object types

An *object* is a DZS network element or its related component, such as a network device, card, physical port, logical interface, or PVC. The OSS XML Gateway represents each object by the object *handle*, which is described by three attributes.

The three handle attributes are sent back to the OSS XML client in the SOAP XML as an OSS XML Response object. This is true for all configuration responses. These three attributes are the first three Name-Value pairs in each of the OSS XML Response object.

To manage an existing object (one that exists in the database), the client specifies the object handle, which is made up of the following attributes:

Handle_Id

An unique identification number of the object.

Handle_type

The type of the object.

Handle_Group

The group_id of an object. Each DZMS object can be put into a group and each group is identified by a unique group identifier. This supports network resource partitioning.

To create objects that do not already exist in the database, the client specifies the parent *handle* (which exists in the database) and the type of object to be created, using the following identifiers:

Parent type

The type of the parent (that exists in the database).

Parent ID

The handle ID for the parent object.

Handle type

The type of the object to be created.

Once the object is created, the database creates a handle ID for it.

Determining Handle IDs

To determine the handle ID of a DZMS object, perform a list operation.

The list response returns all the objects as response elements. Each response element that is returned contains all the data and handle Name-Value pairs of an object including the handle ID Name-Value pair.

Once you have the handle ID, you can perform other requests upon the objects. Start with a list of all the regions. The response will return one OSS XML response element for each region. Each response element will contain

the handle ID Name-Value pair for the region and other Name-Value pairs for other attributes of the region.

The following example shows how the request XML looks like:

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.1.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>Device</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

The OSS XML response XML should be parsed to extract the contents from it. When each of the OSSXMLResponseElement is printed, it will look like the following example:

Name	Value	
Handle_Type	Region	<i>Handle attribute</i>
Handle_Id	21	<i>Handle attribute</i>
Handle_Group	99999	<i>Handle attribute</i>
mapName	world.ivl	<i>Data attribute</i>
globalFocusY	0	<i>Data attribute</i>
globalFocusX	0	<i>Data attribute</i>

Related topic: [How to Determine OSS XML Commands, page 86.](#)

Parent and object handle types

This section describes the parent/object handle type relationships for the BAN, MALC, Sechtor 100A and Z-Edge products and lists the objects by device, card, physical layer, logical layer and other.

Each table lists the parent, the object handle type, which SLMS products support the object, and the commands that can be performed on the object.

Region and device

This section describes the object handle types that have parents of Region or Device.

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
Region	BANDevice	BAN	modify, delete, get, list, add, getStatus
Region	MALC Device	MALC 319 MALC 719 MALC 723	modify, delete, get, list, add, getStatus
Region	MALC23Device	MALC 723	modify, delete, get, list, add, getStatus
Region	MALC19Device	MALC 719	modify, delete, get, list, add, getStatus
Region	MALC319Device	MALC 319	modify, delete, get, list, add, getStatus
Region	Mx19Device	MXK 19	modify, delete, get, list, add, getStatus
Region	Mx23Device	MXK 23	modify, delete, get, list, add, getStatus
Region	Raptor50Device	Raptor 50	modify, delete, get, list, add, getStatus
Region	Raptor100Device	Raptor 100	modify, delete, get, list, add, getStatus
Region	Raptor723aDevice	Raptor 723	modify, delete, get, list, add, getStatus
Region	Raptor719aDevice	Raptor 719	modify, delete, get, list, add, getStatus
Region	Raptor319aDevice	Raptor 319	modify, delete, get, list, add, getStatus
Region	S100AE1Device	Sechtor 100A E1	modify, delete, get, list, add, getStatus
Region	Subscriber	BAN MALC 719 MALC 319 MALC 723 Sechtor 100A E1 Sechtor 100 T1 Z-Edge BH2A Z-Edge H2A	modify, delete, get, list, add, getStatus

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
Region	ZEdgeBH2ADevice	Z-Edge BH2A	modify, delete, get, list, add, getStatus
Region	ZEdgeH2ADevice	Z-Edge H2A	modify, delete, get, list, add, getStatus
BANDevice	AccessList	BAN	modify, delete, get, list, add
BANDevice	AdslAlarmProfile	BAN	modify, get, list
BANDevice	AdslConfigProfile	BAN	modify, get, list
BANDevice	ATMTrafficDesc	BAN	modify, delete, get, list, add
BANDevice	ATMTrafficDescTemplate	BAN	addUsingTemplate, getTemplate, listTemplate
BANDevice	ATMVcCrossConn	BAN	add, modify, delete, list, get, getStatus
BANDevice	AtmVpCrossConn	BAN	add, modify, delete, list, get, getStatus
BANDevice	BridgeCfgGroup	BAN	modify, delete, get, list, add
BANDevice	BridgeLogical	BAN	modify, delete, get, list, add
BANDevice	BridgePath	BAN	modify, delete, get, list, add
BANDevice	BulkStatsCollection	BAN	modify, delete, get, list, add
BANDevice	BulkStatsConfig	BAN	modify, delete, get, list, add
BANDevice	BulkVcCrossConnTemplate	BAN	addUsingTemplate, getTemplate, listTemplate
BANDevice	CardDefault	BAN	None
BANDevice	CommunityName	BAN	modify, delete, get, list, add
BANDevice	DHCPGroup	BAN	modify, delete, get, list, add
BANDevice	DHCPHost	BAN	modify, delete, get, list, add
BANDevice	DHCPServDefault	BAN	get, list
BANDevice	DHCPSubnet	BAN	modify, delete, get, list, add
BANDevice	DslAlarmProfile	BAN	modify, get, list
BANDevice	Hdsl2ConfigProfile	BAN	modify, get, list
BANDevice	IpConfig	BAN	modify, get, list, getStatus
BANDevice	IpDnsResolverConfig	BAN	modify, get, list
BANDevice	IpfilterSpec	BAN	modify, delete, get, list, add, getStatus

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
BANDevice	IpRoutingDomain	BAN	modify, get, list, getStatus
BANDevice	NTPConfig	BAN	modify, delete, get, list, add
BANDevice	SdslConfigProfile	BAN	modify, get, list
BANDevice	ShdslConfigProfile	BAN	modify, get, list, getStatus
BANDevice	Shelf	BAN	None
BANDevice	ShelfFan (use Shelf handle)	BAN	getStatus
BANDevice	ShelfTemperature (use Shelf handle)	BAN	getStatus
BANDevice	SoftwareDownload	BAN	getStatus
BANDevice	Subscriber	BAN	modify, get, list, add
BANDevice	TrapConfig	BAN	modify, delete, get, list, add
BANDevice	UnnumberedIpAlias	BAN	modify, list, get
BANDevice	V52Ig	BAN	modify, delete, get, list, add, getStatus
BANDevice	V52UserPort	BAN	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	AccessList	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	AdslAlarmProfile	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	modify, get, list

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	AdslConfigProfile	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	modify, get, list
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	ATMTrafficDesc	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	ATMTrafficDescTemplate	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	addUsingTemplate, getTemplate, listTemplate
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	ATMVcCrossConn	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	add, modify, delete, list, get, getStatus
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	AtmVpCrossConn	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	add, modify, delete, list, get, getStatus
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	ATMZmcPvc	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	None

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	BridgeCfgGroup	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK19 MXK23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	BridgeLogical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK19 MXK23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	BridgePath	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK19 MXK23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	BulkStatsCollection	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK19 MXK23	modify, delete, get, list, add

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	BulkStatsConfig	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK19 MXK23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	BulkVcCrossConnTemplate	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	addUsingTemplate, getTemplate, listTemplate
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	CardDefault	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	None
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	CardGroup	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	get, list
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	CommunityName	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, delete, get, list, add

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	DHCPGroup	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	DHCPHost	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	DHCPServDefault	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	get, list
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	DHCPSubnet	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, delete, get, list, add

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	DslAlarmProfile	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	modify, get, list
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	Hdsl2ConfigProfile	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	modify, get, list
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	IpConfig	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	modify, get, list getStatus,
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	IpDnsResolverConfig	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK19 MXK23	modify, get, list
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	IpRoutingDomain	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, get, list, getStatus

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
MALC19Device MALC23Device Raptor723Device Raptor719Device	MtacE1Card	MALC 719 MALC 723 Raptor 719 Raptor 723	modify, get, list
MALC19Device MALC23Device Raptor723Device Raptor719Device	MtacT1Card	MALC 719 MALC 723 Raptor 719 Raptor 723	modify, get, list
MALC19Device MALC23Device MALC319Device Mx19Device Mx23Device	McastControlLists	MALC 319 MALC 719 MALC 723 MXK19 MXK23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	MulticastMap	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	NTPConfig	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK19 MXK23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	Ping	MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	modify, get, list

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
MALC19Device MALC23Device MALC319Device Raptor723Device Raptor719Device Raptor319Device	Shelf	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	get, list
MALC19Device MALC23Device Raptor723Device Raptor719Device	ShelfFan (use Shelf handle)	MALC 719 MALC 723 Raptor 719 Raptor 723	getStatus
MALC19Device MALC23Device Raptor723Device Raptor719Device	ShelfTemperature (use Shelf handle)	MALC 719 MALC 723 Raptor 719 Raptor 723	getStatus
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	SoftwareDownload	MALC 719 MALC 319 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	getStatus
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	TrapConfig	MALC 719 MALC 319 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device Mx19Device Mx23Device	UnnumberedIpRef	MALC 719 MALC 319 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, delete, get, list, add

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	VideoCrossConn	MALC 719 MALC 319 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device Raptor100Device Raptor50Device Raptor723Device Raptor719Device Raptor319Device	VideoSystemConfig	MALC 719 MALC 319 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device	V52Ig	MALC 719 MALC 319 MALC 723	modify, delete, get, list, add
MALC19Device MALC23Device MALC319Device	V52UserPort	MALC 719 MALC 319 MALC 723	modify, delete, get, list, add
S100AE1Device S100AT1Device	AccessList	Sechtor 100A E1 Sechtor 100A T1	modify, delete, get, list, add
S100AE1Device S100AT1Device	ATMTrafficDesc	Sechtor 100A E1 Sechtor 100A T1	modify, delete, get, list, add
S100AE1Device S100AT1Device	ATMTrafficDescTemplate	Sechtor 100A E1 Sechtor 100A T1	addUsingTemplate, getTemplate, listTemplate
S100AE1Device S100AT1Device	BridgeLogical	Sechtor 100A E1 Sechtor 100A T1	modify, delete, get, list, add
S100AE1Device S100AT1Device	BridgePath	Sechtor 100A E1 Sechtor 100A T1	modify, delete, get, list, add
S100AE1Device S100AT1Device	CardDefault	Sechtor 100A E1 Sechtor 100A T1	None
S100AE1Device S100AT1Device	CommunityName	Sechtor 100A E1 Sechtor 100A T1	modify, delete, get, list, add, getStatus
S100AE1Device S100AT1Device	IpConfig	Sechtor 100A E1 Sechtor 100A T1	modify, get, list, getStatus

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
S100AE1Device S100AT1Device	IpDnsResolverConfig	Sechtor 100A E1 Sechtor 100A T1	modify, get, list
S100AE1Device S100AT1Device	IpRoutingDomain	Sechtor 100A E1 Sechtor 100A T1	modify, get, list, getStatus
S100AE1Device S100AT1Device	NTPConfig	Sechtor 100A E1 Sechtor 100A T1	modify, get, list, getStatus
S100AE1Device S100AT1Device	Shelf	Sechtor 100A E1 Sechtor 100A T1	get, list
S100AE1Device S100AT1Device	SoftwareDownload	Sechtor 100A E1 Sechtor 100A T1	getStatus
S100AE1Device S100AT1Device	SubscriberPort	Sechtor 100A E1 Sechtor 100A T1	modify, delete, get, list, add
S100AE1Device S100AT1Device	TrapConfig	Sechtor 100A E1 Sechtor 100A T1	modify, delete, get, list, add, getStatus
S100AE1Device	Shelf	Sechtor 100A E1 Sechtor 100A T1	get, list
S100AE1Device	V52Ig	Sechtor 100A E1	modify, delete, get, list, add, getStatus
S100AE1Device	V52UserPort	Sechtor 100A E1	modify, delete, get, list, add
ZEdgeBH2ADevice ZEdgeH2ADevice	AccessList	Z-Edge BH2A Z-Edge H2A	modify, delete, get, list, add
ZEdgeBH2ADevice ZEdgeH2ADevice	ATMTrafficDesc	Z-Edge BH2A Z-Edge H2A	modify, delete, get, list, add
ZEdgeBH2ADevice ZEdgeH2ADevice	ATMTrafficDescTemplate	Z-Edge BH2A Z-Edge H2A	addUsingTemplate, getTemplate, listTemplate
ZEdgeBH2ADevice ZEdgeH2ADevice	BridgeLogical	Z-Edge 64 BH2A Z-Edge 64 H2A	modify, delete, get, list, add
ZEdgeBH2ADevice ZEdgeH2ADevice	BridgePath	Z-Edge 64 BH2A Z-Edge 64 H2A	modify, delete, get, list, add
ZEdgeBH2ADevice ZEdgeH2ADevice	CommunityName	Z-Edge BH2A Z-Edge H2A	modify, delete, get, list, add
ZEdgeBH2ADevice ZEdgeH2ADevice	DHCPGroup	Z-Edge BH2A Z-Edge H2A	modify, delete, get, list, add

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
ZEdgeBH2ADevice ZEdgeH2ADevice	DHCPHost	Z-Edge BH2A Z-Edge H2A	modify, delete, get, list, add
ZEdgeBH2ADevice ZEdgeH2ADevice	DHCPServDefault	Z-Edge BH2A Z-Edge H2A	get, list
ZEdgeBH2ADevice ZEdgeH2ADevice	DHCPSubnet	Z-Edge BH2A Z-Edge H2A	modify, delete, get, list, add
ZEdgeBH2ADevice ZEdgeH2ADevice	DslAlarmProfile	Z-Edge BH2A Z-Edge H2A	modify, get, list
ZEdgeBH2ADevice ZEdgeH2ADevice	Hdsl2ConfigProfile	Z-Edge BH2A Z-Edge H2A	modify, get, list
ZEdgeBH2ADevice ZEdgeH2ADevice	IpConfig	Z-Edge BH2A Z-Edge H2A	modify, get, list, getStatus
ZEdgeBH2ADevice ZEdgeH2ADevice	IpDnsResolverConfig	Z-Edge BH2A Z-Edge H2A	modify, get, list
ZEdgeBH2ADevice ZEdgeH2ADevice	IpRoutingDomain	Z-Edge BH2A Z-Edge H2A	modify, get, list, getStatus
ZEdgeBH2ADevice ZEdgeH2ADevice	NatBindings	Z-Edge BH2A Z-Edge H2A	get, list
ZEdgeBH2ADevice ZEdgeH2ADevice	NatConfig	Z-Edge BH2A Z-Edge H2A	modify, get, list
ZEdgeBH2ADevice ZEdgeH2ADevice	NatStats	Z-Edge BH2A Z-Edge H2A	getStatus
ZEdgeBH2ADevice ZEdgeH2ADevice	PppAuth	Z-Edge BH2A Z-Edge H2A	modify, get, list
ZEdgeBH2ADevice ZEdgeH2ADevice	PppLcp	Z-Edge BH2A Z-Edge H2A	modify, get, list, get Status
ZEdgeBH2ADevice ZEdgeH2ADevice	PppLogical	Z-Edge BH2A Z-Edge H2A	modify, get, list
ZEdgeBH2ADevice ZEdgeH2ADevice	PppNcp	Z-Edge BH2A Z-Edge H2A	modify, get, list, getStatus
ZEdgeBH2ADevice ZEdgeH2ADevice	SdslConfigProfile	Z-Edge BH2A Z-Edge H2A	modify, get, list
ZEdgeBH2ADevice ZEdgeH2ADevice	ShdslConfigProfile	Z-Edge BH2A Z-Edge H2A	modify, get, list

Table 2: Objects contained within region and device

Parent	Object	Supported	Commands
ZEdgeBH2ADevice ZEdgeH2ADevice	SoftwareDownload	Z-Edge BH2A Z-Edge H2A	getStatus
ZEdgeBH2ADevice ZEdgeH2ADevice	TrapConfig	Z-Edge BH2A Z-Edge H2A	modify, delete, get, list, add, getStatus
ZEdgeBH2ADevice ZEdgeH2ADevice	Shelf	Z-Edge BH2A Z-Edge H2A	get, list
ZEdgeBH2ADevice ZEdgeH2ADevice	SubscriberPort	Z-Edge BH2A Z-Edge H2A	modify, delete, get, list, add
ZEdgeBH2ADevice ZEdgeH2ADevice	UnnumberedIpAlias	Z-Edge BH2A Z-Edge H2A	modify, list, get

Card

The following table describes the objects contained by the parent Shelf:

Table 3: Card objects

Parent	Object	Supported	Commands
Shelf	AdslCard	BAN	modify, get, list, getStatus
Shelf	Adsl24PortMalcCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	
Shelf	AdslAC5BMalc24PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	AdslAC6BMalcCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus

Table 3: Card objects (Continued)

Parent	Object	Supported	Commands
Shelf	AdslAC6Malc32PortSplCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	AdslAC6MalcPotsCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	AdslAC6MalcSplCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	AdslPots48PortCard	MALC 319 MALC 719 MALC 723	modify, get, list, getStatus
Shelf	AtmConnAvailable (use Card handle)	BAN (DSL and ATM trunking cards only) MALC 719 MALC 319 MALC 723 Raptor 319 Raptor 719 Raptor 723 (DSL and Uplink cards only)	getStatus
Shelf	DslIma8PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	DS3Card	BAN	modify, get, list, getStatus
Shelf	DS3UnchannelizedCard	BAN	modify, get, list, getStatus
Shelf	DSLCard	BAN	modify, get, list, getStatus
Shelf	E1Atm24PortCard	BAN	modify, get, list, getStatus

Table 3: Card objects (Continued)

Parent	Object	Supported	Commands
Shelf	E1Ima8PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	E1Ima8PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	E1Oc12GigEUpLinkCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	E1Tdm63PortCard	BAN	modify, get, list, getStatus
Shelf	E1UplinkAtm8PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	E1UplinkAtm8PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	E1V52Ima16PortCard	MALC 319 MALC 719 MALC 723	modify, get, list, getStatus
Shelf	E1V52Uni16PortCard	MALC 319 MALC 719 MALC 723	modify, get, list, getStatus
Shelf	E3UplinkAtm4PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus

Table 3: Card objects (Continued)

Parent	Object	Supported	Commands
Shelf	E3UplinkAtm4PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	E3Aal52Atm4PortCard	BAN	modify, get, list, getStatus
Shelf	E3Aal5Atm4PortCard	BAN	modify, get, list, getStatus
Shelf	E3Atm4PortPolyCard	BAN	modify, get, list, getStatus
Shelf	EtherCard	BAN	modify, get, list, getStatus
Shelf	GshdslCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	Gshdsl4Wire24PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	GshdslLifelinePotsCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	InfoServCard	BAN	modify, get, list, getStatus
Shelf	MalcDs1Atm32PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	MALCOltCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus

Table 3: Card objects (Continued)

Parent	Object	Supported	Commands
Shelf	MtacCard	MALC 719 MALC 723 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	MtacE1Card	MALC 719 MALC 723 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	MtacFcE1Card	MALC 319	modify, get, list, getStatus
Shelf	MtacFcT1Card	MALC 319	modify, get, list, getStatus
Shelf	MtacT1Card	MALC 719 MALC 319 MALC 723 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	MxUplink2Tg8gCard	MXK19 MXK23	modify, get, list, getStatus
Shelf	MxUplink8gCard	MXK19 MXK23	modify, get, list, getStatus
Shelf	Mx20ActiveEthCard	MXK19 MXK23	modify, get, list, getStatus
Shelf	Mx4GponCard	MXK19 MXK23	modify, get, list, getStatus
Shelf	Mx4NewGponCard	MXK19 MXK23	modify, get, list, getStatus
Shelf	Mx8GponCard	MXK19 MXK23	modify, get, list, getStatus
Shelf	OC3Card	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	OC3Aal5MultiModeCard	BAN	modify, get, list, getStatus
Shelf	OC3Aal5SingleModeCard	BAN	modify, get, list, getStatus

Table 3: Card objects (Continued)

Parent	Object	Supported	Commands
Shelf	OC3FUplinkAtmCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	OC3MultiModeCard	BAN	modify, get, list, getStatus
Shelf	OC3Aal5MultiModeCard	BAN	modify, get, list, getStatus
Shelf	OC3Aal52MultiModeCard	BAN	modify, get, list, getStatus
Shelf	OC3Aal52SingleModeCard	BAN	modify, get, list, getStatus
Shelf	OC3PolyCard	BAN	modify, get, list, getStatus
Shelf	OC3UplinkAtm2PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	Pots24PortCard	MALC 319 MALC 719 MALC 723	modify, get, list, getStatus
Shelf	Pots24Port900Card	MALC 319 MALC 719 MALC 723	modify, get, list, getStatus
Shelf	S100AE1Card	Sechtor 100A	getStatus
Shelf	T1Atm24PortCard	BAN	modify, get, list, getStatus
Shelf	T1Gr303Ima16PortCard	MALC 319 MALC 719 MALC 723	modify, get, list, getStatus
Shelf	T1Gr303Uni16PortCard	MALC 319 MALC 719 MALC 723	modify, get, list, getStatus
Shelf	T1Ima8PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	T1Tdm64PortCard	BAN	modify, get, list, getStatus

Table 3: Card objects (Continued)

Parent	Object	Supported	Commands
Shelf	T1Oc12GigEUplinkCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	T1FUplinkAtm8PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	T3Aal5Atm4PortCard	BAN	modify, get, list, getStatus
Shelf	T3Atm4PortPolyCard	BAN	modify, get, list, getStatus
Shelf	T3Atm4PortCard	BAN	modify, get, list, getStatus
Shelf	T3AtmPortPolyCard	BAN	modify, get, list, getStatus
Shelf	T3UplinkAtm4PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	T3UplinkAtm4PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	T3UplinkAtm4PortCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	ULCCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	VASPCard	BAN	modify, get, list, getStatus

Table 3: Card objects (Continued)

Parent	Object	Supported	Commands
Shelf	VdslCard	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Shelf	Raptor50Device R50GshdslCard	Raptor 50	modify, get, list, getStatus
Shelf	Raptor100Device Malc100T1ImaCard Malc100GshdslCard	Raptor 100	modify, get, list, getStatus
Shelf	ZEdgeBH2ACard	Z-Edge BH2A	modify, get, list, getStatus
Shelf	ZEdgeH2ACard	Z-Edge H2A	modify, get, list, getStatus

Physical

The following table describes physical layer objects:

Table 4: Physical objects

Parent	Object	Supported	Commands
AdslAC5Malc24PortCard	AdslPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Adsl24PortMalcCard	AdslPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
AdslPots48PortCard	AdslPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
AdslCard	AdslPhysical	BAN	modify, get, list, getStatus

Table 4: Physical objects (Continued)

Parent	Object	Supported	Commands
AdslPhysical	AdslFastPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
AdslPhysical	AdslInterleavedPhysical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
DslIma8PortCard	ImaPhysicalGroup	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
DSLCard	DSLPhysical	BAN	modify, get, list, getStatus
E1Ima8PortCard	MAUPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
E1Ima8PortCard	E1Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
E1Ima8PortCard	EtherPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
E1Ima8PortCard	RS232Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus

Table 4: Physical objects (Continued)

Parent	Object	Supported	Commands
E1Oc12GigEUplinkCard	GigEPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, get, list, getStatus
E1UplinkAtm8PortCard	MAUPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
E1UplinkAtm8PortCard	E1Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
E1UplinkAtm8PortCard	EtherPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
E1UplinkAtm8PortCard	RS232Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
E3Aal5Atm4PortCard	E3UnchannelizedPhysical	BAN	modify, get, list, getStatus
E3Atm4PortPolyCard	E3UnchannelizedPhysical	BAN	modify, get, list, getStatus
E3Uplink4PortCard	E3UnchannelizedPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus

Table 4: Physical objects (Continued)

Parent	Object	Supported	Commands
E3UplinkAtm4PortCard	E3UnchannelizedPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
EtherCard	EtherPhysical	BAN	modify, get, list, getStatus
EtherCard	MAUPhysical	BAN	modify, get, list, getStatus
GponOltPhysical	GponOnuPhysical	MXK 19 MXK 23	modify, get, list, getStatus
GponOltPhysical	GponPortPhysical	MXK 19 MXK 23	modify, get, list, getStatus
GshdslCard	DSLPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
InfoServCard	EtherPhysical	BAN	modify, get, list, getStatus
InfoServCard	MAUPhysical	BAN	modify, get, list, getStatus
InfoServCard	RS232Physical	BAN	modify, get, list, getStatus
MalcOltCard	OltPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
MtacE1Card	E1Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
MtacT1Card	T1Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
MxUplink2Tg8gCard	GigEPhysical	MXK 19 MXK 23	modify, get, list, getStatus

Table 4: Physical objects (Continued)

Parent	Object	Supported	Commands
MxUplink2Tg8gCard	EtherPhysical	MXK 19 MXK 23	modify, get, list, getStatus
MxUplink8gCard	GigEPhysical	MXK 19 MXK 23	modify, get, list, getStatus
MxUplink8gCard	EtherPhysical	MXK 19 MXK 23	modify, get, list, getStatus
Mx20ActiveEthCard	GigEPhysical	MXK 19 MXK 23	modify, get, list, getStatus
Mx20ActiveEthCard	EtherPhysical	MXK 19 MXK 23	modify, get, list, getStatus
Mx4GponCard	GponOltPhysical	MXK 19 MXK 23	modify, get, list, getStatus
Mx4NewGponCard	GponOltPhysical	MXK 19 MXK 23	modify, get, list, getStatus
Mx8GponCard	GponOltPhysical	MXK 19 MXK 23	modify, get, list, getStatus
OC3Aal5MultiModeCard	SonetOC3Physical	BAN	modify, get, list, getStatus
OC3Aal5SingleModeCard	SonetOC3Physical	BAN	modify, get, list, getStatus
OC3MultiModeCard	SonetOC3Physical	BAN	modify, get, list, getStatus
OC3PolyCard	SonetOC3Physical	BAN	modify, get, list, getStatus
OC3UplinkAtm2PortCard	SonetOC3Physical	MALC	modify, get, list, getStatus
OC3FUplinkAtmCard	SonetOC3Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
Pots24PortCard	PotsPhysical	MALC 319 MALC 719 MALC 723	modify, get, list, getStatus
Pots24Port900Card	PotsPhysical	MALC 319 MALC 719 MALC 723	modify, get, list, getStatus
S100AE1Card	E1Physical	Sechtor 100A E1	modify, get, list, getStatus
S100AE1Card	EtherPhysical	Sechtor 100A E1	modify, get, list, getStatus
S100AE1Card	MAUPhysical	Sechtor 100A E1	modify, get, list, getStatus
S100AE1Card	SonetOC3Physical	Sechtor 100A E1	modify, get, list, getStatus

Table 4: Physical objects (Continued)

Parent	Object	Supported	Commands
T1Ima8PortCard	EtherPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
T1Ima8PortCard	MAUPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
T1Ima8PortCard	RS232Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
T1Ima8PortCard	T1Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
T1Oc12GigEUpLinkCard	GigEPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, get, list, getStatus
T1Uplink8PortCard	EtherPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
T1Uplink8PortCard	MAUPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus

Table 4: Physical objects (Continued)

Parent	Object	Supported	Commands
T1Uplink8PortCard	RS232Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
T1Uplink8PortCard	T1Physical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
T3Aal5Atm4PortCard	T3UnchannelizedPhysical	BAN	modify, get, list, getStatus
T3Atm4PortPolyCard	T3UnchannelizedPhysical	BAN	modify, get, list, getStatus
T3UplinkAtm4PortCard	T3UnchannelizedPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
T3UplinkAtm4PortCard	T3UnchannelizedPhysical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
ZEdgeBH2ACard	DslPhysical	Z-Edge BH2A	modify, get, list, getStatus
ZEdgeBH2ACard	EtherPhysical	Z-Edge BH2A	modify, get, list, getStatus
ZEdgeBH2ACard	IpDhcpClientResource	Z-Edge BH2A	modify, get, list, getStatus
ZEdgeBH2ACard	IsdnAlarmProfile	Z-Edge BH2A	modify, get, list
ZEdgeBH2ACard	IsdnBriPhysical	Z-Edge BH2A	modify, get, list, getStatus
ZEdgeBH2ACard	MAUPhysical	Z-Edge BH2A	modify, get, list, getStatus
ZEdgeH2ACard	EtherPhysical	Z-Edge H2A	modify, get, list, getStatus
ZEdgeH2ACard	IpDhcpClientResource	Z-Edge H2A	modify, get, list, getStatus
ZEdgeH2ACard	MAUPhysical	Z-Edge H2A	modify, get, list, getStatus
ZEdgeH2ACard	PotsPhysical	Z-Edge H2A	modify, get, list, getStatus

Logical

[Table 5](#) describes objects that are logical network entities:

Table 5: Logical objects

Parent	Object	Supported	Commands
AdslPhysical	ATMLogical	BAN	modify, get, list, getStatus
DSLPhysical	ATMLogical	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A	modify, get, list, getStatus
E3UnchannelizedPhysical	ATMLogical	BAN	modify, get, list, getStatus
BridgeCfgGroup (Also known as Packet Rule Group)	BridgeCfg (Also known as Packet Rule Member)	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 100 Raptor 50 Raptor 719 Raptor 723 MXK19 MXK23	modify, delete, get, list, add
McastControlLists	McastControlList	MALC 319 MALC 719 MALC 723 MXK19 MXK23	modify, delete, get, list, add

Table 5: Logical objects (Continued)

Parent	Object	Supported	Commands
BANDevice MALC 19Device MALC 23Device MALC 319Device Raptor 319Device Raptor 719Device Raptor 723Device Mx19Device Mx23Device	IpLogical	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, delete, get, list, add, getStatus
EtherPhysical	IpLogical	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add, getStatus
E3UnchannelizedPhysical	ATMLogical	BAN	modify, get, list, getStatus
SonetOC3Physical	ATMLogical	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1	modify, get, list, getStatus
ImaPhysicalGroup	ATMLogical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
T3UnchannelizedPhysical	ATMLogical	BAN	modify, get, list, getStatus

Table 5: Logical objects (Continued)

Parent	Object	Supported	Commands
T1Ima8PortCard	ImaPhysicalGroup	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
ImaPhysicalGroup	ATMLogical	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
EtherPhysical	IpLogical	Sechtor 100A E1	modify, delete, get, list, add, getStatus
DslPhysical	ATMLogical	Z-Edge BH2A	modify, get, list, getStatus
GponOnuPhysical	OmciPortAdmin	MALC 319 MALC 719 MALC 723 MXK 19 MXK 23	modify, get, list
GponOnuPhysical	OmciPortStatus	MALC 319 MALC 719 MALC 723 MXK 19 MXK 23	get
GponOnuPhysical	OmciPortStats	MALC 319 MALC 719 MALC 723 MXK 19 MXK 23	get
GponOnuPhysical	OmciOntReboot	MALC 319 MALC 719 MALC 723 MXK 19 MXK 23	modify, get, list reboot

Other objects

Table 6 describes objects that have parents in tables one through four:

Table 6: Other OSS XML Gateway objects

Parent	Object Handle type	Supported	Commands
AccessList	AccessListEntry	BAN, MALC, Sechtor 100A E1, Z-Edge BH2A MXK 19 MXK 23	modify, delete, get, list, add
AdslCard	IpDhcpClientResource	BAN	modify, get, list, getStatus
Aal2VclAtmEndPoint	ATMCid	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add, getStatus
Aal5VclAtmEndPoint	IpDhcpClientIf	Z-Edge H2A	modify, get, list
ATMLogical	Aal2VclAtmEndPoint	MALC 319 MALC 719 MALC 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add, getStatus
ATMLogical	Aal5VclAtmEndPoint	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add, getStatus

Table 6: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Commands
ATMLogical	AtmVpi	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	add, modify, delete, list, get
ATMLogical	OtherVclAtmEndPoint	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add, getStatus
ATMLogical	VplAtmEndPoint	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	add, modify, delete, list, get, getStatus
CardGroup	CardGroupEntry	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	get, list
EtherPhysical	IpDhcpClientIf	Z-Edge H2A	modify, get, list, getStatus
ImaPhysicalGroup	ImaLink	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus
IpfilterSpec	IpFilterStmnt	BAN	modify, delete, get, list, add

Table 6: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Commands
IpLogical	IpAliasAddr	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A MXK 19 MXK 23	modify, delete, get, list, add
IpLogical	IpDhcpClientDns	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A	getStatus
IpLogical	IpDhcpClientRouter	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A	getStatus
IpLogical	IpNetToMedia	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add

Table 6: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Commands
IpRoutingDomain	DHCPLease	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A MXK 19 MXK 23	get, list
IpRoutingDomain	IpCidrRoute	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge H2A Z-Edge BH2A	getStatus
IpRoutingDomain	IpDnsResolverHost	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge H2A Z-Edge BH2A MXK 19 MXK 23	modify, delete, get, list, add

Table 6: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Commands
IpRoutingDomain	IpRip2Config	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A MXK 19 MXK 23	modify, get, list
IpRoutingDomain	IpRip2Global	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A MXK 19 MXK 23	modify, get, list, getStatus
IpRoutingDomain	IpStaticRoute	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 Z-Edge H2A Z-Edge BH2A MXK 19 MXK 23	modify, delete, get, list, add
MtacE1Card	CardExternalRelay	MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723	modify, get, list, getStatus

Table 6: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Commands
MtacT1Card	CardExternalRelay	MALC 719 MALC 723 Raptor 719 Raptor 723	modify, get, list, getStatus
PotsPhysical	PotsEmConfig	MALC 319 MALC 719 MALC 723	modify, get, list
PotsPhysical	PotsFxoConfig	MALC 319 MALC 719 MALC 723	modify, get, list
PotsPhysical	PotsFxsConfig	MALC 319 MALC 719 MALC 723	modify, get, list
RS232Physical	RS232AsyncConfig	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723 MXK 19 MXK 23	modify, get, list, getStatus
RS232Physical	RS232InSignal	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	getStatus
RS232Physical	RS232OutSignal	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	getStatus

Table 6: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Commands
Subscriber	SubscriberPort	BAN MALC 319 MALC 719 MALC 723 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add
SubscriberPort	SubDataConn	BAN MALC 319 MALC 719 MALC 723 Raptor 50 Raptor 100 Raptor 319 Raptor 719 Raptor 723	Modify, delete, get, list, add, getStatus
SubscriberPort	SubVoiceConn	BAN MALC 319 MALC 719 MALC 723 Raptor 319 Raptor 719 Raptor 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add
SubVoiceConn	SubVoiceAal2EndPoint	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add
SubVoiceConn	SubVoiceElcpAal2EndPoint	Sechtor 100A E1 Sechtor 100A T1	modify, delete, get, list, add
SubVoiceConn	SubVoiceGr303EndPoint	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1, Sechtor 100A T1	modify, delete, get, list, add
SubVoiceConn	SubVoiceIsdnEndPoint	Z-Edge BH2A	modify, delete, get, list, add

Table 6: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Commands
SubVoiceConn	SubVoicePotsEndPoint	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add
SubVoiceConn	SubVoiceV52EndPoint	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1, Sechtor 100A T1	modify, delete, get, list, add
SubVoiceConn	SubVoiceVoipEndPoint	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1, Sechtor 100A T1	modify, delete, get, list, add
SubVoiceConn	Voice	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1 Sechtor 100A T1 Z-Edge H2A Z-Edge BH2A	modify, delete, get, list, add
V52Ig	V52CPath	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1	modify, delete, get, list, add
V52Ig	V52Link	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1	modify, delete, get, list, add, getStatus

Table 6: Other OSS XML Gateway objects (Continued)

Parent	Object Handle type	Supported	Commands
V52Ig	V52UserPort	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1	modify, delete, get, list, add, getStatus
V52Link	V52CChannel	BAN MALC 319 MALC 719 MALC 723 Sechtor 100A E1	modify, delete, get, list, add, getStatus

Handle text

The handle text feature enables ATM VC cross connections and CES cross connections to be stored in both the DZMS database and on the MALC device for MALC devices running software version 1.12.1 and above. For devices running previous versions, the name is stored only in the DZMS database.

The handle text feature enables bridge logical names to be stored in both the DZMS database and on the device for software versions supporting bridging.

The name attribute in the AtmVcCrossConn and BridgeLogical objects contains the value of the handle text. This value can be specified during all OSS operations.

The storing of handle text on the device enables connection names to persist after corresponding devices have been removed and added from the OSS XML or DZMS client.

Table 7: Objects that Support the Handle Text Attribute

Object	Handle Text Attribute
AtmVcCrossConn	name
BridgeLogical	name



Note: The handle ID will continue to be a DZMS-specific field that is stored only in the DZMS database.



Note: The **name** attribute is optional during bridge create for BridgeLogical object.

General usage

This section describes the basic procedures for performing operations with the OSS XML Gateway.

Requests

Each OSS XML Gateway request is made up of a *request* and one or more *request elements*, each of which takes attributes:

Request attributes—Attributes that specify information common to all elements in the request.

Request element attributes—Attributes that specify data relevant to a single operation. For example, a configuration request element identifies an object handle and specifies data attributes to be managed on that object.

The OSS XML Gateway client prepares an XML file with the Request and Request element attributes. The following examples show sample of the input XML files with these attributes:

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <operName>ossuser</operName>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>IPAddress</Name>
      <Value>172.16.90.162</Value>
    </Attribute>
    <Attribute>
      <Name>vpi</Name>
      <Value>0</Value>
    </Attribute>
    <Attribute>
      <Name>netMask</Name>
      <Value>255.255.255.0</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

The attributes required for each request depend on the operation being performed.

The order of elements in an OSS XML request must be maintained to be compliant with OSS XSD schema. The order is:

RequestType, RequestMode, SessionID, operName, Version.

The following examples show sample of a valid list request with the right order:

```
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
```

Responses

Each OSS XML Gateway response XML is made up of a *response* and one or more *response elements*. The response contains overall status and description strings.

Response elements contain information about each corresponding request element and data relevant to the element. The attributes returned depend on the actual request. There can be attributes that represent session-level information, such as operator name and session ID; attributes that represent handle information, indicating what object was operated on; attributes to represent data, such as VPI value, subnet mask, and so on; and, in the case of operation failure, attributes with errors. All attributes are specified as a set of name-value pairs.

While *OverallStatus* represents the response states (success, complete failure or partial failure), *OverAllDescription* provides the reason for failure.

- If the response is successful, *OverallStatus* and *OverallDescription* will be set to Success and CommonSuccess respectively.
- If the response is failed, *OverallStatus* will represent the summary of failure like complete failure or partial failure.

In such cases, *OverallDescription* will provide the reason for failure. [Table 8](#) shows the description for the possible values of *OverallDescription*.

Table 8: Possible Values of OverallDescription

OverallDescription	Description
REQUEST_INIT_FAILED	Request processed failed during initialization.
COMPLETE_FAILURE	Complete failure in request processing.
CREATE_DATA_FAILURE	Failed to create DZMS data object.
DATA_NOT_AVAIL	Indicates that data is not available for the given handle.
ERROR_IN_REQUEST_ELEMENTS	Indicates that some of the request elements contain errors.
UNKNOWN_DESCRIPTION	Generic error and unable to describe the exact reason for failure.

Table 8: Possible Values of OverallDescription

OverallDescription	Description
MISSING_REQ_ATTR	Request is missing one or more mandatory attributes.
INTERNAL_ERROR	Generic internal error..
INVALID_PARENT_TYPE	Invalid parent type was set to create a new object.
PARENT_CHILD_MISMATCH	Invalid parent child combination.
INVALID_HANDLE_TYPE	Invalid parent type was set to create a new object.
INVALID_TEMPLATE_TYPE	Invalid template type provided.
INVALID_TEMPLATE_ID	Invalid template id provided
INVALID_FILTER_TYPE	Filter type provided during list operation is invalid.
OBJECT_DOES_NOT_EXISTS	No real object found for the given Handle_Id.
INVALID_ATTRIBUTE_VALUE	Attribute value out of valid range.

Authentication request

To start an OSS XML Gateway session, the OSS XML Gateway client issues a request to authenticate a DZMS operator.

[Table 9](#) lists the attributes the OSS XML client must specify in an **authentication** request XML.

Table 9: Authentication request attributes

Object Attribute	Description	Valid values (comments)	Scope
RequestType	Request type.	authenticate	Mandatory
Version	DZMS software version.	1.2 and higher	Mandatory

An **authentication** request includes one or more request elements. Each request element identifies an operator to be authenticated. [Table 10](#) lists the request element attributes required to identify an operator.

Table 10: Authentication request element attributes to identify an operator

Object Attribute	Description	Valid values (comments)	Scope
loginName	Operator login name.	<i>Valid DZMS operator name</i>	Mandatory
password	Operator password.	<i>Valid DZMS operator password</i>	Mandatory

Authentication response

[Table 11](#) lists the attributes the OSS XML Gateway returns for an **authentication** request.

Table 11: Authentication response element attributes

Object Attribute	Description
SessionID	The ID that identifies the session with the OSS XML Gateway.
operName	Operator login name.
groupId	Used as Handle_Group in creation of region, customer, and subscriber.

The attributes in an **authentication** response element are required attributes in subsequent requests. Parse the Authentication response XML and get the values of these attributes and use them in each request of the current session with the OSS XML Gateway.

Configuration requests

To manage an object, the OSS XML Gateway client issues a configuration request. The OSS XML Gateway supports the following configuration requests:

- **add**
- **addUsingTemplate**
- **delete**
- **get**
- **getTemplate**
- **list**
- **listTemplate**
- **modify**

The attributes required for each request depend on the specifics of the actual request:

Type of request—In an **add** request, the client manages (creates) an object that does not exist in the database. The other types of requests manage objects that already exist in the database. This difference necessitates **add** requests to specify different information than the other request types.

Type of object—Individual requests specify the data attributes appropriate to the particular object.

Session-level request attribute definitions

Table 12 lists the session-level attributes required to issue a request. Set the value of the **RequestType** attribute to the appropriate request type.

Table 12: Session-level request attributes

Object Attribute	Description	Valid values (comments)	Scope
Overwrite	Overwrite option. If set to true , then OSS XML Client will not receive any warnings during the request processing. If set to false , then OSS XML Client will receive warnings for certain critical operations. (Warnings are usually issued when an operation on one object affects other objects.)	true (<i>Default</i>) false	Optional
RequestType	Request type.	add addUsingTemplate delete get getTemplate list listTemplate modify	Mandatory
Version	DZMS software version.	1.2 and higher	Mandatory
RequestMode	Whether updates are made to the network element and to the object in the database, or to the object in the database only.	online (<i>Default</i>) offline	Optional
operName	Operator login name (obtained from authenticate request).	<i>Valid DZMS operator name</i>	Mandatory
SessionID	ID that identifies the session with the OSS XML Gateway (obtained from authenticate request).	<i>Valid session ID obtained from DZMS for a valid DZMS operator</i>	Mandatory

Database-specific objects

Several attributes which appear when you enter a **list** or a **get** request on various cards are database-specific and should not be modified or database integrity will be compromised.

The following are database-specific objects:

- **active**
- **deviceSubType**
- **lineGroupKey**

- **shelf**
- **Type**

Refer to the *OSS XML Gateway Reference Guide* for more information about these objects.

Add request

An **add** request creates an object on a network element and in the database. The attributes specified by the request are used to initialize the object.

An **add** request requires session-level attributes (see [Table 12 on page 74](#)). Set the value of the **RequestType** attribute to **add**.

Because the object does not exist in the database, no object handle exists for the object. To provide enough information to add an object, an **add** request element must identify the parent handle (handle type and handle ID of the parent object), the type of object to be created, and the data attributes to be configured for that object.

An **add** request includes one or more request elements. For example, you can form multiple request elements to add several ATM traffic descriptors, each of which resides on a separate device. In each request element, identify the parent device, specify the handle type `ATMTrafficDesc`, and specify the data attributes to be configured.

[Table 13](#) lists the request element attributes required to identify an object handle.

Table 13: Request element attributes to identify an object

Object Attribute	Description	Valid values (comments)	Scope
Parent_Group	Valid parent object's group identifier. Every object has an associated group ID. (This attribute must be provided for an operator who does not have view-all privileges.)	<i>Valid group ID.</i>	Optional
Parent_Type	Type of parent object.	<i>Valid parent type for this request.</i> <i>All handle types are listed in the Parent and object handle types on page 30.</i>	Mandatory

Table 13: Request element attributes to identify an object

Object Attribute	Description	Valid values (comments)	Scope
Parent_Id	Unique ID of the parent object. The DZMS database generates an unique key that identifies the object. For information on how to obtain the ID from a get or list response, see Configuration response on page 85 .	<i>Valid parent ID</i>	Mandatory
Handle_Type	Type of object to be added.	<i>Valid handle type for this request.</i> <i>All handle types are listed in Parent and object handle types on page 30.</i>	Mandatory

The data attributes required in an **add** request element depend on the object type.

addUsingTemplate

An **addUsingTemplate** request creates an instance of an object on a network element and in the database. It uses the values from the corresponding template that is specified in the request. Some of the attributes of the template can be overridden and a different set of values can be specified in the **addUsingTemplate** request.

An **addUsingTemplate** request requires session-level attributes (see [Session-level request attribute definitions on page 74](#)). Set the value of the **RequestType** attribute to **addUsingTemplate**.

There is no object handle since the object that is being added using a template does not exist in the database. To provide enough information to add an object using a template, an **addUsingTemplate** request element must identify the parent handle (**Handle_Type**, **Parent_Id** and **Parent_Group** of the parent object), the template handle (**Template_Type**, **Template_Id** and **template** of the template), and the data attributes to be configured for that object.

An **addUsingTemplate** request includes one or more request elements. For example, you can form multiple request elements to add several ATM VC Cross Connects using different **BulkVcCrossConnTemplate** objects. In each request element, identify the parent device, specify the template type **BulkVcCrossConnTemplate**, and specify the data attributes to be configured.

Table 14: Request element attributes to for a **addUsingTemplate** request

Object Attribute	Description	Valid values (comments)	Scope
Parent_Group	Valid parent object's group identifier. Every object has an associated group ID. (This attribute must be provided for an operator who does not have view-all privileges.)	Valid group ID.	Optional
Parent_Type	Type of parent object.	Valid parent type for this request. All handle types are listed in the section parent and object handle types.	Mandatory
Parent_Id	Unique ID of the parent object. The DZMS database generates a unique key that identifies the object. For information on how to obtain the ID from a get or list response, see Configuration response on page 85 .	Valid parent ID.	Mandatory
Template_Group	Valid template object's group identifier. Every template has an associated group ID. (This attribute must be provided for an operator who does not have view-all privileges.)	Valid template group ID.	Optional
Template_Type	Type of the template.	Valid template type for this request. Supported templates are: BulkVcCrossConn ATMTrafficDescTemplate	Mandatory
Template_Id	Unique ID of the template object. The DZMS database generates a unique key that identifies the object. For information on how to obtain the ID from a getTemplate or listTemplate response, see Configuration response on page 85 .	Valid template ID	Mandatory

The data attributes required in an **addUsingTemplate** request element depend on the object type that is being added using a template.

Delete request

A **delete** request deletes an object from the network element and from the database. A **delete** request requires session-level attributes (see [Table 12 on page 74](#)). Set the value of the **RequestType** attribute to **delete**.

A **delete** request includes one or more request elements. Each request element identifies the object handle to be deleted. For example, you can form multiple request elements to delete several objects, each of which resides on a separate device. In each request element, identify the particular object to be deleted. [Table 15](#) lists the request element attributes required to identify an object handle.

Table 15: Request element attributes to identify an object

Object Attribute	Description	Valid values (comments)	Scope
Handle_Group	The group_id of an object. Each DZMS object can be put into a group, and each group is identified by a unique group identifier. This supports network resource partitioning. (This attribute must be provided for an operator who does not have view-all privileges.)	<i>Valid handle group for this request.</i>	Optional
Handle_Type	Type of object.	<i>Valid handle type for this request.</i> <i>All handle types are listed in Parent and object handle types on page 30.</i>	Mandatory
Handle_Id	Unique ID of the object. The DZMS database generates a unique key that identifies the object. For information on how to obtain the ID from a get or list response, see Configuration response on page 85 .	<i>Valid handle ID</i>	Mandatory

A **delete** request element specifies only the object to be deleted. No data attributes are required.

Get request

A **get** request retrieves information about a single object from the database. A **get** request requires session-level attributes (see [Table 12 on page 74](#)). Set the value of the **RequestType** attribute to **get**.

A **get** request includes one request element. Each request element identifies the object handle to be retrieved.

[Table 16](#) lists the request element attributes required to identify an object handle.

Table 16: Request element attributes to identify an object

Object Attribute	Description	Valid values (comments)	Scope
Handle_Group	The group_id of an object. Each DZMS object can be put into a group, and each group is identified by a unique group identifier. This supports network resource partitioning. (This attribute must be provided for an operator who does not have view-all privileges.)	<i>Valid handle group for this request.</i>	Optional
Handle_Type	Type of object.	<i>Valid handle type for this request.</i> <i>All handle types are listed in Parent and object handle types on page 30.</i>	Mandatory
Handle_Id	Unique ID of the object. The DZMS database generates a unique key that identifies the object. For information on how to obtain the ID from a get or list response, see Configuration response on page 85 .	<i>Valid handle ID</i>	Mandatory

By default, a **get** request returns all the data attributes for the specified object. To limit the data attributes to be retrieved, include one or more *filter retrieval* in the **get** request element. For example, to retrieve only VPI and VCI attribute values in the response, specify two filter retrievals, one for each attribute.

[Table 17](#) lists the request element attribute required to specify a filter retrieval.

Table 17: Request element attributes to specify a filter retrieval

Object Attribute	Description	Valid values (comments)	Scope
Filter_Retrieve	Specifies attribute to be retrieved.	<i>Valid attribute</i>	Optional

getTemplate request

A **getTemplate** request retrieves information about a single template from the database. A **getTemplate** request requires session-level attributes, such as what operation to perform, operator name and session ID obtained from the **authentication** request, and so on. In a **getTemplate** request, set the value of the **RequestType** attribute to **getTemplate**.

A **getTemplate** request includes one request element that identifies the template handle to be retrieved. A template handle consists of template ID, template type and template group. Templates are created or added using

DZMS NetHorizon and template handle information can be obtained by performing a list operation on a given template type.

[Table 23](#) lists the request element attributes required to identify an object handle.

Table 18: Request element attributes to identify a template object

Object Attribute	Description	Valid values (comments)	Scope
Template_Group	Valid template object's group identifier. Every template has an associated group ID. (This attribute must be provided for an operator who does not have view-all privileges.)	Valid template group ID.	Optional
Template_Type	Type of the template object.	Valid template type for this request. Supported templates are: BulkVcCrossConn ATMTrafficDescTemplate	Mandatory
Template_Id	Unique ID of the template object. The DZMS database generates an unique key that identifies the template. For information on how to obtain the template ID from a getTemplate or listTemplate response, see Configuration response on page 85 .	Valid template ID	Mandatory

By default, a **getTemplate** request returns all the data attributes for the specified template object. To limit the data attributes to be retrieved, include one or more filter retrieval in the **getTemplate** request element. For example, to retrieve only the **lineVci** and **lineVpi** attribute values of a **BulkVcCrossConnTemplate** in the response, specify two filter retrievals, one for each attribute.

Table 19: Request element attributes to specify filter retrieval list template

Object attribute	Description	Valid values (comments)	Scope
Filter_Retrieve	Specifies attribute to be retrieved.	Valid attribute	Optional

List request

A **list** request retrieves information about objects in the database. The OSS XML Gateway supports the following types of **list** request:

- List all objects on the network that conform to a filter.

- List all objects contained on a specific object that conform to a filter.
- List the parent object of a specific object that conforms to a filter.

A **list** request requires session-level attributes (see [Table 12 on page 74](#)). Set the value of the **RequestType** attribute to **list**.

A **list** request includes one or more request elements:

- To retrieve all objects on the network, the **list** request omits the object handle.
- To retrieve all objects contained on a specific object, the **list** request identifies the object handle of the parent object.
- To retrieve the parent object of a specific object, the **list** request identifies the object handle of the child object.

See [Table 15](#) for a list of the request element attributes required to identify an object handle.

For example, you can form a request element to retrieve:

All ATM logical interfaces on the network—Omit the object handle.

All ATM logical interfaces on a particular card—Specify the object handle of the parent object (the card).

The parent card containing a particular ATM logical interface—Specify the object handle of the child object (the ATM logical interface).

The **list** request requires a *filter*, which is a set of attributes that define the objects and the attributes to be retrieved. Specify a filter *type* to specify the type of objects you want to retrieve. Specify one or more filter *condition(s)* to specify the attribute values that the object must contain. Specify one or more filter *retrieval* to identify what attributes you want returned in the response.

Table 20: Physical filter types

Type	Description
DS1ATMPPhysical	Filter_Type for listing T1/E1 ATM physicals under any Card that has T1/E1 ATM physicals. Refer to the T1/E1 ATM Physical description in the <i>OSS XML Gateway Reference Guide</i> for applicable parent cards.
DS1Physical	Filter_Type for listing T1/E1 physicals. Refer to the T1/E1 Physical description in the <i>OSS XML gateway Reference Guide</i> for applicable parent cards.

Table 20: Physical filter types (Continued)

Type	Description
DS3UnchannelizedPhysical	Filter_Type for listing T3/E3 UnchannelizedPhysicals. Refer to the T3/E3 UnchannelizedPhysicals description in the <i>OSS XML Gateway Reference Guide</i> for applicable parent cards.
DSLPhysical	Filter_Type for listing Adsl/Dsl physicals. Refer to the ADSL/DSL Physicals description in the <i>OSS XML Gateway Reference Guide</i> for applicable parent cards.
IsdnPhysical	Filter_Type for listing IsdnBriPhysical. Usually on ZEdge Bh2a device/card. Refer to the IsdnBriPhysical description in the <i>OSS XML Gateway Reference Guide</i> for applicable parent cards.
Physical	Can be used on any card to list physicals in a generic way. Useful if the OSS client does not know the exact type of the physicals under a given parent card or device object.
SonetPhysical	Filter_Type for listing sonet oc3 physicals. For example: SonetOC3Physicals. Refer to the SONET description in the <i>OSS XML Gateway Reference Guide</i> for applicable parent cards

For example, you can form a request element to retrieve:

All ATM logical interfaces—Set the filter type to the value **ATMLogical**.

All ATM logical interfaces on a particular card—Set the filter type to the value of the child (ATMLogical).

The parent card that contains a particular ATM logical interface—Set the filter type to the value of the parent (**Card**).

Only the ATM logical interfaces with VPI of 0 and VCI of 1—Set the filter type to the value **ATMLogical**. Specify separate filter conditions, one for each attribute name and value
(`<Attribute><Name>Filter_Condition</Name><Value>vpi=0</Value></Attribute>` and `<Attribute><Name>Filter_Condition</Name><Value>vci=1</Value></Attribute>`).

In each case, you can specify which data attributes you want retrieved for the ATM logical interfaces. Specify one or more filter retrieval to identify what attributes you want returned in the response.

[Table 21](#) lists the request element attributes required to specify a filter.

Table 21: Request element attributes to specify a filter

Object Attribute	Description	Valid values (comments)	Scope
Filter_Type	Specifies to retrieve objects of a particular type.	<i>Valid handle type for this request.</i> <i>All handle types are listed in Parent and object handle types on page 30.</i>	Mandatory
Filter_Condition	Specifies to retrieve objects containing a particular attribute value. (A Filter_Condition with attribute Handle_Group and a value equal to the operator's groupID must be provided as one of the request element attributes for an operator who does not have view-all privileges.)	<Attribute><Name>Filter_Condition</Name><Value>Valid Attribute=value</Value></Attribute> <Attribute><Name>Filter_Condition</Name><Value>Handle_Group=operatorId</Value></Attribute>	Optional
Filter_Retrieve	Specifies attribute to be retrieved.	<i>Valid attribute</i>	Optional
Handle_Group	Specifies the handle group of the reference object whose parent or children need to be retrieved in the list request. (This attribute must be provided for an operator who does not have view-all privileges.)	<i>Valid handle group.</i>	Optional
Handle_Type	Specifies the handle type of the reference object whose parent or children need to be retrieved in the list request.	<i>Valid handle type.</i>	Optional
Handle_Id	Specifies the handle ID of the reference object whose parent or children need to be retrieved in the list request.	<i>Valid handle ID.</i>	Optional

listTemplate request

OSS XML Gateway client can perform **listTemplate** operations on templates already created from NetHorizon's default or user-defined templates. The **listTemplate** operation is similar to list operation on any other DZMS objects. A listTemplate operation can be performed to list all the template instances of a given type from the DZMS database. These results can be used to create corresponding object instances through **addUsingTemplate** operations or **getTemplate** requests. A **listTemplate** request attributes requires session-level attributes (see [Session-level request attribute definitions](#) on page 74). Set the

value of the **RequestType** attribute to **listTemplate**. A listTemplate request includes one or more request elements.

Table 22: Request element attributes to for a listTemplate request

Object Attribute	Description	Valid values (comments)	Scope
Filter_Type	Specifies to retrieve templates of a particular type.	<i>Valid template type for this request.</i> Supported templates are: BulkVcCrossConn ATMTrafficDescTemplate	Mandatory
Filter_Condition	Specifies to retrieve templates containing a particular attribute value. (A Filter_Condition with attribute Handle_Group and a value equal to the operator's groupID must be provided as one of the request element attributes for an operator who does not have view-all privileges.)	<Attribute><Name>Filter_Condition</Name><Value> <i>Valid Attribute=value</i> </Value></Attribute> <Attribute><Name>Filter_Condition</Name><Value> <i>Handle_Group=operatorId</i> </Value></Attribute>	Optional
Filter_Retrieve	Specifies attribute to be retrieved	<i>Valid template ID</i>	Optional
Handle_Group	Specifies the handle group of parent object of the template. For templates, parent is always "Root".	<i>Valid handle group.</i> Root handle group is always 99999 .	Optional
Handle_Type	Specifies the handle type of the parent reference object.	<i>Valid handle type.</i> Root handle type is Root .	Optional
Handle_Id	Specifies the handle ID of the parent object of the template.	<i>Valid handle ID.</i> Root handle ID is 0 .	Optional

Modify request

A **modify** request modifies specific attributes of an object on a network element and in the database. By definition, only data attributes that already exist in the database can be modified.

A **modify** request requires session-level attributes, such as what operation to perform, operator name and session ID obtained from the **authentication** request, and so on (see [Table 12 on page 74](#)). In a **modify** request, set the value of the **RequestType** attribute to **modify**.

A **modify** request includes one or more request elements. Each request element identifies the object handle and specifies the data attributes to be modified on that object. For example, you can form multiple request elements

to modify several IP logical interfaces, each of which resides on a separate card. In each request element, identify the particular IP logical interface and specify the data attributes to be modified.

[Table 23](#) lists the request element attributes required to identify an object handle.

Table 23: Request element attributes to identify an object

Object Attribute	Description	Valid values (comments)	Scope
Handle_Group	The group_id of an object. Each DZMS object can be put into a group, and each group is identified by a unique group identifier. This supports network resource partitioning. (This attribute must be provided for an operator who does not have view-all privileges.)	<i>Valid handle group for this request.</i>	Optional
Handle_Type	Type of object.	<i>Valid handle type for this request.</i> <i>All handle types are listed in Parent and object handle types on page 30.</i>	Mandatory
Handle_Id	Unique ID of the object. The DZMS database generates a unique key that identifies the object. For information on how to obtain the ID from a get or list response, see Configuration response on page 85 .	<i>Valid handle ID</i>	Mandatory

The data attributes required in a **modify** request element depend on the object type. Specify only writable data attributes that already exist in the database for that object.

Configuration response

Each OSS XML Gateway response XML is made up of a *response*, one or more *response elements*, overall *status* and a *description*. Configuration response elements contain information about each corresponding request element and data relevant to the element. The attributes returned depend on the actual request.

Each configuration response element contains the **Handle_Type**, **Handle_Id**, and **Handle_Group** of the object. These attributes are the first three name-value pairs of the response element.

Use the unique Handle_Id to identify an object in a subsequent request. For example, when adding an object, use the unique ID as the **Parent_Id** data attribute (the request element attribute required to identify the parent object).

When modifying an object, use the unique ID as the **Handle_Id** data attribute (the request element attribute required to identify the object).

The OSS XML response XML should be parsed using appropriate parsers and the response elements could be processed.

listTemplate response

The following is an sample **listTemplate** response:

```

OverAllStatus          Success
OverAllDescription     COMMON_SUCCESS
-----Response Element-----
Status                Success
Description           COMMON_SUCCESS
Template_Type         BulkVcCrossConnTemplate
Template_Id           5
Template_Group        1
autoSelectVci         false
bufMaxSize            0
templateDescription   Default Template
lineDs0Bundle         0
partialFill           0
cellLossIntegrationPeriod 0
trunkRXTD             2
templateName          Bulk Vc Cross Conn Default
lineVci               32
cas                   0
autoIncrementVpi      false
trunkStartVci         32
lineRXTD              2
templateCategory      0
trunkTXTD             2
adminStatus           1
autoCreation          1
connType              0
lineVpi               0
trunkVpi              0
Timestamp             2000-12-31 21:00:00:0
trunkDs0Bundle        2
lineTXTD              0
service               0
name                  ATMVcCrossConn

```

How to Determine OSS XML Commands

NetHorizon (DZMS Java client) and OSS XML use the same DZMS server API. If you are familiar with NetHorizon, you can figure out the equivalent OSS XML commands.

The DZMS *debug.log* (/zms_logs/<date>_debug.log) shows what commands the DZMS server receives and the responses that it returns. There are four configuration OSS XML commands: add, delete, modify and list. Even

though all the commands follow the same format, there are OSS XML example commands for almost every profile type. Besides the configuration example commands provided under the directory `/ossxml/test_cases/OssXMLCsTest` and the chapter [XML Samples, page 109](#) of this **OSS XML User's Guide**, by using the following method, you can also determine the OSS XML commands by yourself.

In summary, the method to determine the correct OSS XML commands is:

1. Find the request message in the log file,
2. Find the correct example,
3. Update the example with values from the log file.

The following procedure provides an example of how to determine OSS XML command.

Example: Determine OSS XML Command

This procedure shows how to assign an ONU and create a video service on the ONU:

- 1 Find the request message in the request log for assigning the ONU.

The request log for assigning the ONU is:

```
BaseRequest[ type: MODIFY, offline: false, overwrite:
1, SessionData [name=Everett, Id=0.393353706529377]
  0=GponOnu_Physical[
    Handle: Region=1, Mx1419_Device=10000002
    (group=99999), Shelf=1 (group=99999),
    MxKLcNGponS16_Card=1 (group=99999), GponOlt_Physical=1
    (group=99999), GponOnu_Physical=45 (group=99999)
    Modified Attributes: {onuAdded:old=2,new=1,
meProfile:old=,new=zhone-5225,
serialNoVendorSpecificHex:old=0,new=0364C794}]
```

The log file shows that assigning an ONT is a modify `GponOnu_Physical` command. Where `onuAdded` gets set to `true(1)`, `meProfile` is set to the correct ME name and `serialNoVendorSpecificHex` is set to the ONT's serial number.

There are a couple of oddities between the log and OSSXML commands:

- The underscore between "GponOnu" and "Physical" is optional in OSSXML commands. Many times the example commands do not include the underscore. ie., "GponOnu_Physical" = "GponOnuPhysical"
- The `Handle_Id` of an object is referred to as `Instance_Key` in the log file and is only found in the response.

- 2 Lookup the assign ONT example.

To find all `GponOnu_Physical` commands, use the **find** command:

```
cd /ossxml/test_cases/OssXMLCsTest
```

```
find . -name "*.xml" -exec grep "GponOnuPhysical"
{} \; -print
```

The assign ONT example is at */ossxml/test_cases/OssXMLCsTest/Gpon/GponOnu/GponOnuPhysical_AssignOnt.xml*.

3 Update example with values from log file.

Attributes can be added or deleted as needed in order to match the log file.

For each attribute following the log's "Modified Attributes" line, an XML attribute element needs to exist.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Handle_Type</Name>
<Value>GponOnuPhysical</Value>
</Attribute>
<Attribute>
<Name>Handle_Id</Name>
<Value>75783</Value> <!-- Handle_Id of ONU -->
</Attribute>
<Attribute>
<Name>onuAdded</Name>
<Value>1</Value> <!-- 1=true, 2=false-->
</Attribute>
<Attribute>
<Name>serialNoVendorSpecificHex</Name>
<Value>0364C794</Value>
</Attribute>
<!-- If specifying serial numbers in decimal
<Attribute>
<Name>serialNoVendorSpecific</Name>
<Value>12345678</Value>
</Attribute>
-->
<Attribute>
<Name>meProfile</Name>
<Value>zhone-5225</Value>
</Attribute>
</RequestElement>
</OSS>
```


4 Add the video service.

This is the service log:

```
0=obj: DynOmciCpeConnection[
    Handle: Region=1, Mx1419_Device=10000002
    (group=99999), DynOmciCpeConnection=0 (group=99999)
    Attribute List: {vlanId=0,
bridgeIfMcastControlList=, slanXlateId=0,
slanXlateTpId=33024, dscpToCosIndex=0,
vlanXlateId=403, slanTpId=33024, name=,
slanXlateCos=0, slanId=0, tpType=12, tpIndex=4,
groupTemplate=I-5Meg, guidedVlanId=0,
bridgeIfMaxVideoStreams=5, vlanCos=0, ctmIndex=0,
vlanXlateCos=0,
bridgeTemplateName=DownlinkVideoSingleTagged,
egressGroupTemplate=E-50Meg, rgMode=5, groupIndex=0,
egressGroupIndex=0, slanCos=0, mvrVlanId=0,
autoSelectGem=1}]
Gpon Attributes: {phyShelfId=1, phySlotId=1,
phyPortId=1, gponOnuId=45, gponPortId=0,
gtpTemplateName=OTT}
```

5 Find the "add Cpe-Connection" example and substitute the non-default values from above.

/ossxml/test_cases/OssXMLCsTest/USP/services/CpeConnection/
CpeConnection_add_gponAutoGem.xml

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeConnection</Value>
</Attribute>
<Attribute>
```

```

        <!-- values:
physicalPathTerminationPointEthernetUNI(1),
        interworkingVCCTerminationPoint(2),
        mapperServiceProfile(3),
        ipHostConfigData(4),
        gemInterworkingTerminationPoint(5),
multicastGEMInterworkingTerminationPoint(6),
physicalPathTerminationPointxDSLUNIPartOne(7),
        physicalPathTerminationPointVDSLUNI(8),
        ethernetFlowTerminationPoint(9),
        physicalPathTerminationPointUNI(10),
physicalPathTerminationPointRFvideoUNI(11),
physicalPathTerminationPointIPTVUNI(12),
h248VoiceTerminationPoint(13),
        sipVoiceTerminationPoint(14),
        sipPlarVoiceTerminationPoint(15),
        pweTerminationPoint(16),
        virtualEthernetInterfacePoint(17),
        dot11TerminationPoint(18),
        fiberLanTerminationPoint(19),
        tr69TerminationPoint(20),
        mgmtTerminationPoint(21),
        mgcpVoiceTerminationPoint(22),
        ptpTerminationPoint(23),
        dot1xTerminationPoint(24),
        radiusTerminationPoint(25)

-->
        <Name>tpType</Name>
        <Value>12</Value>
</Attribute>
<Attribute>
        <Name>tpIndex</Name>
        <Value>4</Value>
</Attribute>
<Attribute>
        <Name>vlanId</Name>
        <Value>0</Value>
</Attribute>
<Attribute>
        <Name>slanId</Name>
        <Value>0</Value>
</Attribute>
<Attribute>
        <Name>vlanXlateId</Name>
        <Value>403</Value>
</Attribute>
<Attribute>
        <Name>slanXlateId</Name>
        <Value>0</Value>
</Attribute>
<Attribute>
        <Name>phyShelfId</Name>
        <Value>1</Value>
</Attribute>

```

```

<Attribute>
    <Name>phySlotId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>phyPortId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponOnuId</Name>
    <Value>45</Value>
</Attribute>
<!-- Select 1 for autoSelecting GEM -->
<!-- Select 2 for manually selecting GEM. In such
case, request must provide value for gponPortId -->
<Attribute>
    <Name>autoSelectGem</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gtpTemplateName</Name>
    <Value>OTT</Value>
</Attribute>
<Attribute>
    <!--values :
routed(1),brouted(2),rpppoe(3),bpppoe(4),bridged(5),no
tApplicable(6) -->
    <Name>rgMode</Name>
    <Value>5</Value>
</Attribute>
<!-- MeProfile was set during Assign ONT/Modify ONU
request, so not necessary here -->
<!--
<Attribute>
    <Name>internalMe</Name>
    <Value>zhone-2426</Value>
</Attribute>
-->
<Attribute>
    <Name>bridgeTemplateName</Name>
    <Value>DownlinkVideoSingleTagged</Value>
</Attribute>
<Attribute>
    <Name>bridgeIfMaxVideoStreams</Name>
    <Value>5</Value>
</Attribute>
<Attribute>
    <Name>groupTemplate</Name>
    <Value>I-5Meg</Value>
</Attribute>
<Attribute>
    <Name>egressGroupTemplate</Name>
    <Value>E-50Meg</Value>
</Attribute>

```

```
</RequestElement>  
</OSS>
```

3

USING TEMPLATES

This chapter explains how to use templates to configure objects using the OSS XML gateway. It includes:

- [Overview, page 93](#)
- [Listing available templates, page 93](#)
- [Creating ATM traffic descriptors from templates, page 94](#)
- [Creating ATM cross connects from templates, page 100](#)

Overview

Templates enable you to quickly provision devices using configuration templates. You can only perform a list or get operation on templates from the OSS XML gateway. To create, modify, or delete templates you must use NetHorizon. Refer to the NetHorizon documentation for information about creating templates.

The OSS XML gateway supports using templates for configuring the following objects:

- ATM traffic descriptors
- ATM cross connects

Listing available templates

DZMS includes the following default templates:

- BulkVcCrossConnTemplate
- ATMTrafficDescTemplate

The following example shows how to list all the possible templates:

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>listTemplate</RequestType>
```

```

        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>version</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>Root</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Group</Name>
            <Value>99999</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_ID</Name>
            <Value>0</Value>
        </Attribute>
        <Attribute>
            <Name>Filter_Type</Name>
            <Value>BulkVcCrossConnTemplate</Value>
        </Attribute>
        <Attribute>
            <Name>Filter_Retrieve</Name>
            <Value>AttributeName</Value>
        </Attribute>
        <Attribute>
            <Name>Filter_Retrieve</Name>
            <Value>AttributeName</Value>
        </Attribute>
    </RequestElement>
</OSS>

```

Creating ATM traffic descriptors from templates

ATMTrafficDesc objects can also be provisioned using its corresponding ATMTrafficDescTemplate.

The steps involved in provisioning an ATMTrafficDesc using templates are as follows:

Listing the ATM traffic descriptor templates

The following example shows on how to compose a list request for the ATMTrafficDesc template request element in the OSS XML client:

Request attributes and values:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML

```

```

ossxml.xsd">
  <Request>
    <RequestType>listTemplate</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>version</Version>
  </Request>

```

Request element attribute and values:

```

<RequestElement>
  <Attribute>
    <Name>Handle_Type</Name>
    <Value>Root</Value>
  </Attribute>
  <Attribute>
    <Name>Handle_Group</Name>
    <Value>99999</Value>
  </Attribute>
  <Attribute>
    <Name>Handle_ID</Name>
    <Value>0</Value>
  </Attribute>

```

Specify the template type for Filter_Type value.

```

<Attribute>
  <Name>Filter_Type</Name>
  <Value>ATMTrafficDescTemplate</Value>
</Attribute>

```

Specify the attribute names of the given template to retrieve (optional). By default, all attributes are returned:

```

<Attribute>
  <Name>Filter_Retrieve</Name>
  <Value>AttributeName</Value>
</Attribute>
<Attribute>
  <Name>Filter_Retrieve</Name>
  <Value>AttributeName</Value>
</Attribute>
</RequestElement>
</OSS>

```

Compose the OSS XML request with the following 4 steps, and send it to the OSS XML Gateway:

- 1 Initiate the socket connection to the server.

/*

* Initiate the Socket connection

```

*/
socket = new Socket(host,port);
in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
out = new PrintWriter(socket.getOutputStream(),true);

```

2 Read the contents of the file:

```

/*
    * The XML file is read and converted to a string
    */
BufferedReader inputFile=new BufferedReader(new FileReader(inputFile));
StringBuffer tempBuffer=new StringBuffer();
String fileContent="";
while ((fileContent= inputFile.readLine()) != null)
{
    tempBuffer.append(fileContent);
}
requestToServer= tempBuffer.toString();

```

3 Send the contents to the server:

```

/*
* Send the request to the server
*/
out.println(requestToServer);

```

4 Read the Response from the server:

```

/*
    * Read the response from the server
    * The size of the output message that we get from *
    * the OSSXML server can differ
    * This block handles a large output message.
    */
StringBuffer responseFromServer;
int length=in.read();
while (true)
{
    str=in.readLine();
    responseFromServer.append(str);
    if(str == null || str==" " || str.equals(""))
    {
        break;
    }
}

```

5 Close the streams and socket to free resources:

```

in.close()
out.close()
socket.close()

```


Adding a traffic descriptor

After getting the template, the OSS XML client can use the template to create various object instances. In the object create request, the OSS XML client has to specify the following:

- TemplateId
- Type
- TemplateGroup
- The attributes for the traffic descriptor. These include:
 - templateDescription
 - trafficDescrIndex
 - usageControl
 - templateName
 - trafficDescrType
 - trafficDescrParam5
 - trafficDescrParam4
 - trafficDescrParam3
 - trafficDescrParam2
 - trafficDescrParam1
 - templateCategory
 - trafficFrameDiscard
 - cacDivider
 - Timestamp
 - vclRate
 - serviceCategory
 - name

In the request to the OSS XML gateway, the OSS XML client can change some, or all of, the template attributes.

The following example shows how an ATMTrafficDesc object instance can be created using a ATMTrafficDescTemplate instance, which is obtained from the previous list operation:

```
<?xml version="1.0" encoding="UTF-8" ?>

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
```

```

    <RequestType>addUsingTemplate</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.2</Version>
  </Request>
</RequestElement>

```

Specify the template info:

```

    <Attribute>
      <Name>Template_Type</Name>
      <Value>ATMTrafficDescTemplate</Value>
    </Attribute>
    <Attribute>
      <Name>Template_Id</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>Template_Group</Name>
      <Value>10001</Value>
    </Attribute>

```

Specify the parent info under which this object instance needs to be created:

```

    <Attribute>
      <Name>Parent_Type</Name>
      <Value>BANDevice</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>44</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Group</Name>
      <Value>20001</Value>
    </Attribute>

```

Modify attributes (if desired):

```

    <Attribute>
      <Name>name</Name>
      <Value>ossxml_added_ATD</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Compose the request and send it to the OSS XML Gateway with the following 4 steps:

- 1 Initiate the socket connection to the server.

/*

```

        * Initiate the Socket connection
        */
        socket = new Socket(host,port);
        in = new BufferedReader(new
InputStreamReader(socket.getInputStream()));
        out = new
PrintWriter(socket.getOutputStream(),true);

```

2 Read the contents of the file:

```

/*

        * The XML file is read and converted to a string
        */
BufferedReader inputFile=new BufferedReader(new
FileReader(inputFile));
StringBuffer tempBuffer=new StringBuffer();
String fileContent="";
    while ((fileContent= inputFile.readLine()) != null)
    {
        tempBuffer.append(fileContent);
    }
    requestToServer= tempBuffer.toString();

```

3 Send the contents to the server:

```

/*
* Send the request to the server
*/
    out.println(requestToServer);

```

4 Read the Response from the server:

```

/*
        * Read the response from the server
* The size of the output message that we get from *
* the OSSXML server can differ
        * This block handles a large output message.
        */
StringBuffer responseFromServer;
    int length=in.read();
    while (true)
    {
        str=in.readLine();
        responseFromServer.append(str);
        if(str == null || str==" " || str.equals(""))
        {
            break;
        }
    }

```

5 Close the streams and socket to free resources:

```

in.close()
out.close()

```

```
socket.close()
```

Once the OSS XML GW receives the ATM traffic descriptor instance create request, it creates the traffic descriptor using the values from the template and the values specified in the addUsingTemplate request. The request element attribute values specified in the addUsingTemplate request override the default configuration values from the corresponding template that was obtained using the list template request.

Creating ATM cross connects from templates

ATM VC Cross Connects can be provisioned using the BulkVcCrossConnTemplate template. This template can be used to provision one or more cross-connects in the same request.

Before provisioning ATM VC Cross Connects using its template, the client first has to gather the following information regarding device, card and physicals:

1. List the devices and obtain the device Handle_Id.
2. List the trunk card and obtain the trunk card's Handle_Id.
3. List the line card and obtain the line card's Handle_Id.
4. List trunk card's physicals and obtain its LineGroupId. List line card's physicals and obtain its LineGroupId.

A list operation can be performed to list all the user-defined BulkVcCrossConn templates from the DZMS database. Later this list results can be used to create corresponding object instances.

Listing the cross connect templates

The following example shows on how to compose a list request for the BulkVcCrossConn template request element in the OSS client:

Request attributes and values:

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>listTemplate</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>version</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>Root</Value>
```

```

</Attribute>
<Attribute>
    <Name>Handle_Group</Name>
    <Value>99999</Value>
</Attribute>
<Attribute>
    <Name>Handle_ID</Name>
    <Value>0</Value>
</Attribute>

```

Specify the template type:

```

<Attribute>
    <Name>Filter_Type</Name>
    <Value>BulkVcCrossConnTemplate</Value>
</Attribute>

```

Specify the attribute names of the given template to retrieve (optional). By default, all attributes are returned.

```

<Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>AttributeName</Value>
</Attribute>
<Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>AttributeName</Value>
</Attribute>
</RequestElement>
</OSS>

```

Compose the request and send it to the OSS XML Gateway with the following 4 steps:

1 Initiate the socket connection to the server.

```

/*
 * Initiate the Socket connection
 */
socket = new Socket(host,port);
in = new
BufferedReader(newInputStreamReader(socket.getInputStream()));
out = new
PrintWriter(socket.getOutputStream(),true);

```

2 Read the contents of the file:

```

/*
 * The XML file is read and converted to a string
 */

```

```

BufferedReader inputFile=new BufferedReader(new
FileReader(inputFile));
StringBuffer tempBuffer=new StringBuffer();
String fileContent="";
    while ((fileContent= inputFile.readLine()) != null)
    {
        tempBuffer.append(fileContent);
    }
    requestToServer= tempBuffer.toString();

```

3 Send the contents to the server:

```

/*
 * Send the request to the server
 */
    out.println(requestToServer);

```

4 Read the Response from the server:

```

/*
    * Read the response from the server
 * The size of the output message that we get from *
 * the OSSXML server can differ
    * This block handles a large output message.
 */
StringBuffer responseFromServer;
int length=in.read();
while (true)
{
    str=in.readLine();
    responseFromServer.append(str);
if(str == null || str==" " || str.equals(""))
{
    break;
}
}

```

5 Close the streams and socket to free resources:

```

in.close()
out.close()
socket.close()

```

Adding a cross connect

After getting the template, the OSS XML client can use the template to create various object instances. In the object create request, the OSS XML client has to specify the following:

- templateId
- Type
- TemplateGroup

- The attributes for the cross connect. These include:
 - lineLgId or lineCardSlotId (mandatory)
 - trunkLgId (mandatory)
 - name
 - lineRXTD
 - trunkRXTD
 - adminStatus
 - templateDescription
 - autoCreation
 - Timestamp
 - lineTXTD
 - templateCategory
 - trunkTXTD
 - templateName
 - autoSelectVci
 - lineVpi
 - lineVci
 - trunkVpi
 - trunkStartVci

In the request to the OSS XML gateway, the OSS XML client can change some, or all of, the template attributes.

The following example shows how an ATMCrossConnData object instance can be created using a BulkVcCrossConnTemplate instance, which is obtained from the previous list operation:

Compose the OSS XML request and send it to the Gateway:

Addition of cross connection using OSS XML Gateway is done by using following 3 steps.

1 Add the first VCL end point using the following.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
```

```

        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.1.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>ATM_Logical</Value>
        </Attribute>

        <Attribute>
            <Name>Parent_Id</Name>
            <Value>19</Value>
        </Attribute>
        <Attribute>
            <Name>Device_Id</Name>
            <Value>7</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>OtherVclAtmEndPoint</Value>
        </Attribute>
        <Attribute>
            <Name>name</Name>
            <Value>OSSXMLAdded-vcl_1</Value>
        </Attribute>
        <Attribute>
            <Name>vpi</Name>
            <Value>0</Value>
        </Attribute>
        <Attribute>
            <Name>vci</Name>
            <Value>44</Value>
        </Attribute>
        <Attribute>
            <Name>vclAdminStatus</Name>
            <Value>2</Value>
        </Attribute>
        <Attribute>
            <Name>vclCastType</Name>
            <Value>0</Value>
        </Attribute>
        <Attribute>
            <Name>vclRxTrafficDescrIndex</Name>
            <Value>3</Value>
        </Attribute>
        <Attribute>
            <Name>vclTxTrafficDescrIndex</Name>
            <Value>3</Value>
        </Attribute>
        <Attribute>
            <Name>vclAalType</Name>

```



```

        <Value>4</Value>
    </Attribute>
    <Attribute>
        <Name>vclConnKind</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>encapsType</Name>
        <Value>7</Value>
    </Attribute>
    <Attribute>
        <Name>faultDetectionType</Name>
        <Value>1</Value>
    </Attribute>
</RequestElement>
</OSS>

```

- 2** Add the second VCL end point using the above example with different end point information for at least one of the ATM_Logical, vci and vpi.
- 3** Add the cross connection using the previously added end points.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.1.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>Device</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>19</Value>
        </Attribute>
        <Attribute>
            <Name>Device_Id</Name>
            <Value>7</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>OtherVclAtmEndPoint</Value>
        </Attribute>
        <Attribute>
            <Name>name</Name>
            <Value>OSSXMLAdded-vcl_1</Value>
        </Attribute>
    </RequestElement>

```

```

</Attribute>
<Attribute>
  <Name>vpi</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>vci</Name>
  <Value>44</Value>
</Attribute>
<Attribute>
  <Name>vclAdminStatus</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>vclCastType</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>vclRxTrafficDescrIndex</Name>
  <Value>3</Value>
</Attribute>
<Attribute>
  <Name>vclTxTrafficDescrIndex</Name>
  <Value>3</Value>
</Attribute>
<Attribute>
  <Name>vclAalType</Name>
  <Value>4</Value>
</Attribute>
<Attribute>
  <Name>vclConnKind</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>encapsType</Name>
  <Value>7</Value>
</Attribute>
<Attribute>
  <Name>faultDetectionType</Name>
  <Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 XML SCHEMA DEFINITION (XSD)

This chapter presents the XML Schema Definition (XSD) of the expected input XML file.

XSD

```
<?xml version="1.0" encoding="ISO-8859-1" ?>
<xsd:schema xmlns="http://www.zhone.com/OSSXML"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
targetNamespace="http://www.zhone.com/OSSXML"
elementFormDefault="qualified">
  <xsd:element name="OSS">
    <xsd:complexType>
      <xsd:sequence minOccurs="1">
        <xsd:element name="Request"
type="request_type"/>
        <xsd:element name="RequestElement"
type="requelem_type" maxOccurs="unbounded"/>
      </xsd:sequence>
    </xsd:complexType>
  </xsd:element>
  <xsd:complexType name="request_type">
    <xsd:sequence>
      <xsd:element name="RequestType"
type="xsd:string"/>
      <xsd:element name="RequestMode"
type="xsd:string" minOccurs="0" default="online"/>
      <xsd:element name="SessionID"
type="xsd:string" minOccurs="0"/>
      <xsd:element name="operName" type="xsd:string"
minOccurs="0"/>
      <xsd:element name="Version" type="xsd:string"
minOccurs="0"/>
      <xsd:element name="Overwrite"
type="xsd:string" minOccurs="0" default="true"/>
    </xsd:sequence>
  </xsd:complexType>
  <xsd:complexType name="requelem_type">
    <xsd:sequence minOccurs="1">
      <xsd:element name="Attribute"
type="attrib_type" maxOccurs="unbounded"/>
    </xsd:sequence>
  </xsd:complexType>

```

```
        </xsd:complexType>
        <xsd:complexType name="attrib_type">
            <xsd:sequence>
                <xsd:element name="Name" type="xsd:string"
minOccurs="1"/>
                <xsd:element name="Value" type="xsd:string"
minOccurs="1"/>
            </xsd:sequence>
        </xsd:complexType>
    </xsd:schema>
```

5 XML SAMPLES

This appendix presents few listings of the sample request XML files. The OSS XML client interacts with the OSS XML server using sockets.



Note: DZS provides more OSS Client provisioning examples with the software packages. The OSS CORBA examples are located under the `/oss/test_cases` directory. The OSS XML examples are located under the `/ossxml/test_cases` directory.



Note: OSS XML needs to be terminated by a carriage return/line feed for it to work.

This appendix includes the following sections:

- [OSS XML Gateway client interaction with OSS XML Gateway server, page 110](#)
- [Authenticate request, page 111](#)
- [List requests, page 111](#)
- [Get requests, page 126](#)
- [Add requests, page 131](#)
- [Modify requests, page 209](#)
- [Delete requests, page 221](#)
- [Unified Service Provisioning on GPON: Dynamic OMCI service requests, page 227](#)
- [Unified Service Provisioning on GPON: Residential Gateway \(RG\) Feature, page 250](#)
- [Unified Service Provisioning on ActiveE: Residential Gateway \(RG\) Feature, page 292](#)
- [Performance requests, page 331](#)
- [SELT and DELT object retrieval examples, page 340](#)
- [N2N bond group examples, page 343](#)
- [Clear statistics requests, page 347](#)
- [Port deletion example, page 348](#)

- [OSS XML Support on V-Series Devices, page 349](#)

OSS XML Gateway client interaction with OSS XML Gateway server



Note: This example is written using Java socket programming to communicate with the OSS XML server.

Procedure

Sample code that connects the OSS XML client to the OSS XML server in the default port number - 15002 and the host - host name of the OSS XML server. This code shows how the request is sent to the server and the response is read from the server.

- 1 Initiate the socket connection to the server.

```
/*
 * Initiate the Socket connection
 */
socket = new Socket(host,port);
in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
out = new PrintWriter(socket.getOutputStream(),true);
```

- 2 Read the contents of the file:

```
/*
 * The XML file is read and converted to a string
 */
BufferedReader inputFile=new BufferedReader(new FileReader(inputFile));
StringBuffer tempBuffer=new StringBuffer();
String fileContent="";
while ((fileContent= inputFile.readLine()) != null)
{
    tempBuffer.append(fileContent);
}
requestToServer= tempBuffer.toString();
```

- 3 Send the contents to the server:

```
/*
 * Send the request to the server
 */
out.println(requestToServer);
```

- 4 Read the Response from the server:

```
/*
 * Read the response from the server
 * The size of the output message that we get from *
 * the OSSXML server can differ
```

```

        *   This block handles a large output message.
        */
        StringBuffer responseFromServer;
        int length=in.read();
        while (true)
        {
            str=in.readLine();
            responseFromServer.append(str);
            if(str == null || str==" " || str.equals(""))
            {
                break;
            }
        }
    }
}

```

5 Close the streams and socket to free resources:

```

in.close()
out.close()
socket.close()

```

Authenticate request

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
    <Request>
        <RequestType>authenticate</RequestType>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>loginName</Name>
            <Value>userName</Value>
        </Attribute>
        <Attribute>
            <Name>password</Name>
            <Value>password</Value>
        </Attribute>
    </RequestElement>
</OSS>

```

List requests

List request: Get Handle_Id of a device using device IP

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://www.zhone.com
/OSSXML ossxml.xsd">
    <Request>

```

```

<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>Device_Id</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>deviceIP=172.16.80.129</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: Get list of cards of a device using device Handle_Id

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://www.zhone.com
/OSSXML ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>Card</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=628</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: Get lineGroupId and ifIndex of a physical using its device Handle_Id, card slot number and port number

```

<?xml version="1.0" encoding="UTF-8" ?>

```



```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com
/OSSXML ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>Physical</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Retrieve</Name>
      <Value>lineGroupId</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Retrieve</Name>
      <Value>ifIndex</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Device_Id=628</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Card_Id_1=6</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>ifTypePortId=1</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

List request: ATM AAL5 VCL End Point

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com
/OSSXML ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>

```

```

<RequestElement>
  <Attribute>
    <Name>Filter_Type</Name>
    <Value>Aal5VclAtmEndPoint</Value>
  </Attribute>
  <Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=628</Value>
  </Attribute>
  <Attribute>
    <Name>Filter_Condition</Name>
    <Value>vpi=6</Value>
  </Attribute>
  <Attribute>
    <Name>Filter_Condition</Name>
    <Value>vci=34</Value>
  </Attribute>
  <Attribute>
    <Name>Filter_Condition</Name>
    <Value>lineGroupId=340</Value>
  </Attribute>
</RequestElement>
</OSS>

```

List Request: ATM Virtual Circuit Cross Connection

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>ATMvcCrossConn</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>234</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Retrieve</Name>
      <Value>name</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

```

    </Attribute>
  </RequestElement>
</OSS>

```

List Request: User-defined templates

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>listTemplate</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>Root</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Group</Name>
      <Value>99999</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>0</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>BulkVcCrossConnTemplate</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Retrieve</Name>
      <Value>AttributeName</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Retrieve</Name>
      <Value>AttributeName</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

List request: Retrieve static bridge paths



Note: Static bridge paths can be retrieved either at ZMS / Region / Device / parent BridgeLogical level. The response of each static bridge path will have all the attributes filled in.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com
/OSSXML ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>BridgePath</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=628</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>bridgeIfIndex=3846</Value>
</Attribute>
</RequestElement>
</OSS>
```

List request: Retrieve dynamic bridge paths



Note: Dynamic bridge paths are not stored in DZMS database and are directly retrieved from the device. Hence, request parameters required for retrieving dynamic MACs are slightly different from the regular list request of BridgePath. OSS can only retrieve the information for one bridge logical at a time.

And note that the response will not have all bridge path attributes filled in the response data. You will need to change your response parsing logic based on the request. BridgePath attributes **staticBridgeFlags**, **TimeStamp**, **igmpQueryInterval**, **flapControl**, **multicastAging**, **unicastAging** and **flush** will be applicable only for the static bridge paths hence will be included ONLY in static bridge path response but not in dynamic. Other attributes as **name**, **vlanId**, **slanId**, **ipOrMacAddr**, **bridgeIfIndex** and **addrType** will be included in both static and dynamic responses.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://www.zhone.com
/OSSXML ossxml.xsd">
<Request>
<RequestType>list</RequestType>
```

```

<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>BridgeLogical</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>6208</Value>
</Attribute>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>BridgePath</Value>
</Attribute>
</RequestElement>
</OSS>

```

List request: List all variables in an OMCI profile



Note: The OSS XML server does not support filter conditions on **name**, **value** and **description** attributes of **GponOnuGenProfileVariable** and **GponOnuSpecProfileVariable** objects because they are not stored in the DZMS database. If you need to perform this operation, OSS clients should retrieve the entire list of generic/specific variables and scan for the required variable.

Whenever a list operation is run on **ponOnuGenProfileVariable** and **GponOnuSpecProfileVariable** with including **Filter_Retrieve's** on attributes other than **name**, **value** and **description**, OSS will always return **parentMe** and **parentGeneric** attributes as part of the response.

When a list is performed for all variables in an OMCI generic profile or specific profile, all variables defined in the ME profile are returned. If a value has not been assigned, the value will be an empty string. Each variable data contains a name, value, and description attribute. Listing variables from multiple profiles is allowed.

When a variable data value is modified, the appropriate line is updated in the omci profile text. Only "value" may be modified. If "description" is modified, the modification will be ignored. If "name" is modified, the modification will fail. To modify multiple attributes in a single request, multiple request elements need to be added. One request element for each variable to modify. All the modified variables have to be in the same omci profile.

Listing variables in an OMCI Generic profile

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>GponOnuGenProfileVariable</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=405</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>GponOnugenProfile_Id=19</Value>
</Attribute>
</RequestElement>
</OSS>
```

Listing variables in an OMCI specific profile

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>GponOnuSpecProfileVariable</Value>
</Attribute>
<Attribute>
```

```

        <Name>Filter_Condition</Name>
        <Value>Device_Id=405</Value>
    </Attribute>
    <Attribute>
        <Name>Filter_Condition</Name>
        <Value>Card_Id_1=13</Value>
    </Attribute>
    <Attribute>
        <Name>Filter_Condition</Name>
        <Value>ifTypePortId=1</Value>
    </Attribute>
    <Attribute>
        <Name>Filter_Condition</Name>
        <Value>Physical_Id_2=8</Value>
    </Attribute>
</RequestElement>
</OSS>

```

List Request: a sub region one

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com
/OSSXML ossxml.xsd">
    <Request>
        <RequestType>list</RequestType>
        <RequestMode>offline</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.16.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Filter_Type</Name>
            <Value>SubRegionOne</Value>
        </Attribute>
        <Attribute>
            <Name>Filter_Retrieve</Name>
            <Value>name</Value>
        </Attribute>
    </RequestElement>
</OSS>

```

List Request: GponTrafficProfile

```

<?xml version="1.0" encoding="UTF-8" ?>
- <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
- <Request>

```

```

    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.3</Version>
  </Request>
- <RequestElement>
- <Attribute>
  <Name>Filter_Type</Name>
  <Value>GponTrafficProfile</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Device_Id=30</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: GponTrafficProfile_Template

```

<?xml version="1.0" encoding="UTF-8" ?>
- <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
- <Request>
  <RequestType>list</RequestType>
  <RequestMode>offline</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>2.3</Version>
</Request>
- <RequestElement>
- <Attribute>
  <Name>Handle_Type</Name>
  <Value>Root</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Type</Name>
  <Value>GponTrafficProfile_Template</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: MLPPP bond group

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>

```



```

<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.3.2</Version>
</Request>
<RequestElement>
<Attribute>
  <Name>Parent_Type</Name>
  <Value>Card</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Card_Id_1=13</Value>
</Attribute>
<Attribute>
  <Name>Parent_Id</Name>
  <Value>20</Value>
</Attribute>
<Attribute>
  <Name>Filter_Type</Name>
  <Value>MLPPPBond_Physical</Value>
</Attribute>
<Attribute>
  <Name>Filter_Retrieve</Name>
  <Value>name</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: Retrieve next available ONU

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>

</Request>
<RequestElement>
<Attribute>
  <Name>Filter_Type</Name>
  <Value>NextFreeOnu</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Device_Id=2</Value>

```

```

</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Card_Id_1=3</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>ifTypePortId=3</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: Regional Alarms

```

<?xml version="1.0" encoding="UTF-8" ?>
- <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
- <Request>
  <RequestType>listAlarm</RequestType>
  <RequestMode>offline</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>1.1.2</Version>
</Request>
- <RequestElement>
- <Attribute>
  <Name>Filter_Type</Name>
  <Value>AlarmEvent</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Region_Id=1</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: Device Alarms

```

<?xml version="1.0" encoding="UTF-8" ?>
- <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
- <Request>
  <RequestType>listAlarm</RequestType>
  <RequestMode>offline</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>1.1.2</Version>
</Request>

```

```

- <RequestElement>
- <Attribute>
  <Name>Filter_Type</Name>
  <Value>AlarmEvent</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Region_Id=1</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Device_Id=1</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: Card Alarms

```

<?xml version="1.0" encoding="UTF-8" ?>
- <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
- <Request>
  <RequestType>listAlarm</RequestType>
  <RequestMode>offline</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>1.1.2</Version>
</Request>
- <RequestElement>
- <Attribute>
  <Name>Filter_Type</Name>
  <Value>AlarmEvent</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Region_Id=1</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Device_Id=1</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Shelf_Id=1</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Card_Id_1=31</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: OLT Alarms

```

    <?xml version="1.0" encoding="UTF-8" ?>
-   <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
-   <Request>
    <RequestType>listAlarm</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.1.2</Version>
  </Request>
-   <RequestElement>
-   <Attribute>
    <Name>Filter_Type</Name>
    <Value>AlarmEvent</Value>
  </Attribute>
-   <Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>severity</Value>
  </Attribute>
-   <Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>description</Value>
  </Attribute>
-   <Attribute>
    <Name>Filter_Condition</Name>
    <Value>Region_Id=1</Value>
  </Attribute>
-   <Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=1</Value>
  </Attribute>
-   <Attribute>
    <Name>Filter_Condition</Name>
    <Value>Shelf_Id=1</Value>
  </Attribute>
-   <Attribute>
    <Name>Filter_Condition</Name>
    <Value>Card_Id_1=31</Value>
  </Attribute>
-   <Attribute>
    <Name>Filter_Condition</Name>
    <Value>ifTypePortId=1</Value>
  </Attribute>
  </RequestElement>
</OSS>

```

List Request: ONU Alarms

```

<?xml version="1.0" encoding="UTF-8" ?>
- <OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML
ossxml.xsd">
- <Request>
  <RequestType>listAlarm</RequestType>
  <RequestMode>offline</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>1.1.2</Version>
</Request>
- <RequestElement>
- <Attribute>
  <Name>Filter_Type</Name>
  <Value>AlarmEvent</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Retrieve</Name>
  <Value>severity</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Retrieve</Name>
  <Value>description</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Region_Id=1</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Device_Id=1</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Shelf_Id=1</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Card_Id_1=31</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>ifTypePortId=1</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Condition</Name>
  <Value>Physical_Id_2=1</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: CPE manager

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>PseudoDevice</Value>  <!-- this is the
internal name for a cpe-mgr entry -->
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Device_Id=49</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Card_Id=3</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Physical_Id_1=1</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Physical_Id_2=1</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

Get requests

Get request: user-defined templates

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>getTemplate</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
```

```

        <operName>operName</operName>
        <Version>1.6</Version>
</Request>
<RequestElement>
  <Attribute>
    <Name>Template_Type</Name>
    <Value>ATMTrafficDescTemplate</Value>
  </Attribute>
  <Attribute>
    <Name>Template_Group</Name>
    <Value>TemplateGroup</Value>
  </Attribute>
  <Attribute>
    <Name>Template_Id</Name>
    <Value>TemplateId</Value>
  </Attribute>
  <Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>AttributeName</Value>
  </Attribute>
  <Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>AttributeName</Value>
  </Attribute>
</RequestElement>
</OSS>

```

Get request: ATM Traffic Descriptor

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>get</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>ATMTrafficDesc</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>1255</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Retrieve</Name>
      <Value>trafficDescrIndex</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

```

    <Attribute>
      <Name>Filter_Retrieve</Name>
      <Value>serviceCategory</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Get request: GponTrafficProfile

```

<?xml version="1.0" encoding="UTF-8" ?>
- <OSS xmlns="http://www.zhone.com/OSSXML"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.zhone.com/OSSXML
    ossxml.xsd">
- <Request>
  <RequestType>get</RequestType>
  <RequestMode>offline</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>1.1.2</Version>
</Request>
- <RequestElement>
- <Attribute>
  <Name>Handle_Type</Name>
  <Value>GponTrafficProfile</Value>
</Attribute>
- <Attribute>
  <Name>Handle_Id</Name>
  <Value>72</Value>
</Attribute>
- <Attribute>
  <Name>Filter_Retrieve</Name>
  <Value>shared</Value>
</Attribute>
</RequestElement>
</OSS>

```

Get request: onuStatusWord, OLT statusWord, and OLT opticAlarm

The following example shows how to get **onuStatusWord**:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.zhone.com/OSSXML
    ossxml.xsd">
<Request>
<RequestType>getStatus</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.5.1</Version>

```



```

</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>GponOnuPhysical</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>1134</Value>
</Attribute>
<Attribute>
    <Name>Retrieve_Attr</Name>
    <Value>onuStatusWord</Value>
</Attribute>
</RequestElement>
</OSS>

```

The following example shows how to get **statusWord**:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>getStatus</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.5.1</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>GponOltPhysical</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>1125</Value>
</Attribute>
<Attribute>
    <Name>Retrieve_Attr</Name>
    <Value>statusWord</Value>
</Attribute>
</RequestElement>
</OSS>

```

The following example shows how to get **opticAlarms**:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">

```

```

<Request>
  <RequestType>getStatus</RequestType>
  <RequestMode>online</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>2.5.1</Version>
</Request>
<RequestElement>
  <Attribute>
    <Name>Handle_Type</Name>
    <Value>GponOltPhysical</Value>
  </Attribute>
  <Attribute>
    <Name>Handle_Id</Name>
    <Value>1125</Value>
  </Attribute>
  <Attribute>
    <Name>Retrieve_Attr</Name>
    <Value>opticAlarms</Value>
  </Attribute>
</RequestElement>
</OSS>

```

Get request: Alarms

The following example shows how to get DZMS active alarms:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.zhone.com/OSSXML
  ossxml.xsd">
  <Request>
    <RequestType>getAlarm</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.1.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>AlarmEvent</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>28</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Add requests

Add request: ATM AAL5 EndPoint

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>ATMLogical</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>714</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>Aal5VclAtmEndPoint</Value>
    </Attribute>
    <Attribute>
      <Name>name</Name>
      <Value>Aal5Atmendpoint</Value>
    </Attribute>
    <Attribute>
      <Name>vpi</Name>
      <Value>0</Value>
    </Attribute>
    <Attribute>
      <Name>vci</Name>
      <Value>40</Value>
    </Attribute>
    <Attribute>
      <Name>vclAdminStatus</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>vclRxTrafficDescrIndex</Name>
      <Value>12</Value>
    </Attribute>
    <Attribute>
      <Name>vclTxTrafficDescrIndex</Name>
      <Value>12</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

```

    <Attribute>
      <Name>cpcsRxSduSize</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>vclConnKind</Name>
      <Value>1</Value>
    </Attribute>

    <Attribute>
      <Name>vccAalType</Name>
      <Value />
    </Attribute>
    <Attribute>
      <Name>cpcsTxSduSize</Name>
      <Value>2</Value>
    </Attribute>
    <Attribute>
      <Name>vclCastType</Name>
      <Value>1</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Add request: ATM Other EndPoints

These examples show how to form an add request for OtherVclAtmEndPoint.

The following example shows how to form an add request for a low end OtherVclAtmEndPoint:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>ATMLogical</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>564</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>OtherVclAtmEndPoint</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

```

</Attribute>
<Attribute>
  <Name>name</Name>
  <Value>AddOtherVclEndpoint-1</Value>
</Attribute>
<Attribute>
  <Name>vpi</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>vci</Name>
  <Value>56</Value>
</Attribute>
<Attribute>
  <Name>vclAdminStatus</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>vclRxTrafficDescrIndex</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>vclTxTrafficDescrIndex</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>faultDetectionType</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>vclConnKind</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>vccAalType</Name>
  <Value>4</Value>
</Attribute>
<Attribute>
  <Name>vclCastType</Name>
  <Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

The following example shows how to form an add request for a high end OtherVclAtmEndPoint:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
<Request>
  <RequestType>add</RequestType>
  <RequestMode>offline</RequestMode>

```

```

    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
</Request>
<RequestElement>
  <Attribute>
    <Name>Parent_Type</Name>
    <Value>ATMLogical</Value>
  </Attribute>
  <Attribute>
    <Name>Parent_Id</Name>
    <Value>238</Value>
  </Attribute>
  <Attribute>
    <Name>Handle_Type</Name>
    <Value>OtherVclAtmEndPoint</Value>
  </Attribute>
  <Attribute>
    <Name>name</Name>
    <Value>AddOthervclendpoint-2</Value>
  </Attribute>
  <Attribute>
    <Name>vpi</Name>
    <Value>1</Value>
  </Attribute>
  <Attribute>
    <Name>vci</Name>
    <Value>44</Value>
  </Attribute>
  <Attribute>
    <Name>vclAdminStatus</Name>
    <Value>2</Value>
  </Attribute>
  <Attribute>
    <Name>vclRxTrafficDescrIndex</Name>
    <Value>1</Value>
  </Attribute>
  <Attribute>
    <Name>vclTxTrafficDescrIndex</Name>
    <Value>1</Value>
  </Attribute>
  <Attribute>
    <Name>faultDetectionType</Name>
    <Value>1</Value>
  </Attribute>
  <Attribute>
    <Name>vclConnKind</Name>
    <Value>1</Value>
  </Attribute>
  <Attribute>
    <Name>vccAalType</Name>
    <Value>4</Value>
  </Attribute>
  <Attribute>

```

```

        <Name>vclCastType</Name>
        <Value>1</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Add request: ATM Cross Connection

This example shows how to create an add request for ATMVcCrossConn:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
<Request>
    <RequestType>add</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
</Request>
<RequestElement>
    <Attribute>
        <Name>Parent_Type</Name>
        <Value>Device</Value>
    </Attribute>
    <Attribute>
        <Name>Parent_Id</Name>
        <Value>272</Value>
    </Attribute>
    <Attribute>
        <Name>Handle_Type</Name>
        <Value>ATMVcCrossConn</Value>
    </Attribute>
    <Attribute>
        <Name>name</Name>
        <Value>AtmvcCrossConn-1</Value>
    </Attribute>
    <Attribute>
        <Name>vcCrossConnLowIfIndex</Name>
        <Value>154723</Value>
    </Attribute>
    <Attribute>
        <Name>vcCrossConnLowVpi</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>vcCrossConnLowVci</Name>
        <Value>56</Value>
    </Attribute>
    <Attribute>
        <Name>vcCrossConnHighIfIndex</Name>
        <Value>18867522</Value>
    </Attribute>

```

```

    </Attribute>
    <Attribute>
      <Name>vcCrossConnHighVpi</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>vcCrossConnHighVci</Name>
      <Value>44</Value>
    </Attribute>
    <Attribute>
      <Name>vcCrossConnAdminStatus</Name>
      <Value>1</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Add request: ATM traffic descriptor

This example shows how to form an add request for ATMTrafficDesc:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>465</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>ATMTrafficDesc</Value>
    </Attribute>
    <Attribute>
      <Name>name</Name>
      <Value>ATMTraffic-Add</Value>
    </Attribute>
    <Attribute>
      <Name>trafficDescrType</Name>
      <Value>PCR CLP=0+1</Value>
    </Attribute>
    <Attribute>

```



```

        <Name>serviceCategory</Name>
        <Value>6</Value>
    </Attribute>
    <Attribute>
        <Name>trafficFrameDiscard</Name>
        <Value>1</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Add request: Customer

The following example shows how to form an add request for Customer:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
<Request>
    <RequestType>add</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
</Request>
<RequestElement>
    <Attribute>
        <Name>Parent_Type</Name>
        <Value>Root</Value>
    </Attribute>
    <Attribute>
        <Name>Parent_Id</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>Handle_Type</Name>
        <Value>Customer</Value>
    </Attribute>
    <Attribute>
        <Name>Handle_Group</Name>
        <Value>123546</Value>
    </Attribute>
    <Attribute>
        <Name>name</Name>
        <Value>Customer-Add</Value>
    </Attribute>
    <Attribute>
        <Name>addr1</Name>
        <Value>PCR CLP=0+1</Value>
    </Attribute>
    <Attribute>
        <Name>addr2</Name>
        <Value>6</Value>
    </Attribute>

```

```

    </Attribute>
    <Attribute>
        <Name>city</Name>
        <Value>Lowell</Value>
    </Attribute>
    <Attribute>
        <Name>state</Name>
        <Value>MA</Value>
    </Attribute>
    <Attribute>
        <Name>email</Name>
        <Value>ossxml@zhone.com</Value>
    </Attribute>
    <Attribute>
        <Name>phone</Name>
        <Value>51077732</Value>
    </Attribute>
    <Attribute>
        <Name>fax</Name>
        <Value>51098745</Value>
    </Attribute>
    <Attribute>
        <Name>customerType</Name>
        <Value>2</Value>
    </Attribute>
</RequestElement>
</OSS>

```

List available bridge templates

Get the handle_ID and assign it to **BridgeTemplate_Id**.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>listTemplate</RequestType>
        <RequestMode>offline</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.16.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>Root</Value>
        </Attribute>
        <Attribute>
            <Name>Filter_Type</Name>
            <Value>BridgeLogicalTemplate</Value>
        </Attribute>
    </RequestElement>

```

```
</RequestElement>
</OSS>
```

Add bridge logicals using templates

Adding bridge logicals using templates on ATM logical interfaces

1 Given a deviceIP, get the corresponding Device_I

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>deviceIP=1.2.3.4</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

2 Given deviceId, slotIndex, portId, get the lineGroupId of corresponding physical.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
```

```

<Value>Physical</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=100</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Card_Id_1=10</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>ifTypePortId=100</Value>
</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>lineGroupId</Value>
</Attribute>
</RequestElement>
</OSS>

```



Note: There may be multiple interfaces with same Device_Id, Card_Id and Port_Id but only one interface has lineGroupId as non-zero. Interface with non-zero lineGroupId shall be considered for retrieving Handle_Id of ATMLogical object.

3 Given deviceId and lineGroupId, get Handle_Id & lowerIfIndex for ATMLogical object.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>ATMLogical</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=100</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>lineGroupId=1023</Value>

```

```

</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>rfc1483Index</Value>
</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>aal5Index</Value>
</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>ifIndex</Value>
</Attribute>
</RequestElement>
</OSS>

```



Note: Perform the following steps to retrieve the lowerIfIndex for using in the creation of bridge logical on ATM interfaces:

- Check for value of rfc1483Index of ATMLogical object, if it is non-zero, use that as the value to be assigned for lowerIfIndex.
- Else if rfc1483Index is 0, check for the value of aal5Index, if it is non-zero, use that as the value to be assigned for lowerIfIndex.
- Else if aal5Index is 0, use the ifIndex of ATMLogical as lowerIfIndex.

4 Add the corresponding AAL5 type VCL with providing unused VPI and VCI.



Note: Handle_Id of the ATM_Logical retrieved using previous step will be used as Parent_Id for creation of AAL5 VCL.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Parent_Type</Name>
<Value>ATMLogical</Value>
</Attribute>
<Attribute>
<Name>Parent_Id</Name>

```

```

<Value>100</Value>
</Attribute>
<Attribute>
<Name>Handle_Type</Name>
<Value>Aal5VclAtmEndPoint</Value>
</Attribute>
<Attribute>
<Name>vpi</Name>
<Value>0</Value>
</Attribute>
<Attribute>
<Name>vci</Name>
<Value>40</Value>
</Attribute>
<Attribute>
<Name>encapsType</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>adminStatus</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>rxTrafficDescrIndex</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>txTrafficDescrIndex</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>cpcsRxSduSize</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>connKind</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>vccAalType</Name>
<Value>3</Value>
</Attribute>
<Attribute>
<Name>cpcsTxSduSize</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>castType</Name>
<Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

5 Add the bridge logicals using templates on ATM Logical interfaces

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML
ossxml.xsd">
<Request>
  <RequestType>addUsingTemplate</RequestType>
  <RequestMode>online</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>2.3</Version>
</Request>
<RequestElement>
<Attribute>
  <Name>Parent_Type</Name>
  <Value>Device</Value>
</Attribute>
<Attribute>
  <Name>Parent_Id</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>Template_Type</Name>
  <Value>Bridge_LogicalTemplate</Value>
</Attribute>
<Attribute>
  <Name>Template_Id</Name>
  <Value>233</Value>
  <!-- see bridgeTemplate_list.xml for how to get this
value
-->
</Attribute>
<Attribute>
  <Name>lineGroupId</Name>
  <Value>352</Value>
</Attribute>
<Attribute>
  <Name>lowerIfIndex</Name>
  <Value>725</Value>
</Attribute>
<Attribute>
  <Name>vpi</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>vci</Name>
  <Value>40</Value>
</Attribute>
<!--
optional attributes that can be set include :
  vlanId
  stripInsert
  learnIp
  learnUnicast

```

```

maxUnicast
bridgeIfVlanIdCOS
bridgeIfOutgoingCOSOption
bridgeIfOutgoingCOSValue
bridgeIfStagTPID
bridgeIfStagId
bridgeIfStagStripAndInsert
bridgeIfStagOutgoingCOSOption
bridgeIfStagCOS
bridgeIfStagOutgoingCOSValue
bridgeIfMcastControlList
bridgeIfMaxVideoStreams
TableBasedFilter
DhcpLearn
rstpEnable
mstpId
mvrVlanId
vlanXlateId
slanXlateId
incomingCOSOption
stagIncomingCOSOption
# either the profile index or a template name may
be used to specify a egress packet rule group
egressGroupIndex or egressGroupTemplate
# either the profile index or a template name may
be used to specify a ingress packet rule group
groupIndex or groupTemplate
-->
</RequestElement>
</OSS>

```

Adding bridge logicals using templates on Ethernet port physicals

- 1 Given Device_Id, Card-Id and Port_Id, retrieve the lineGroupId/lowerIfIndex of Ethernet port physical.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>Physical</Value>
</Attribute>

```



```

<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=100</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Card_Id_1=1</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>ifTypePortId=1</Value>
</Attribute>

<Attribute>
<Name>Filter_Retrieve</Name>
<Value>lineGroupId</Value>
</Attribute>
</RequestElement>
</OSS>

```

2 Create bridge using bridge logical templates on top of ethernet port physical.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>addUsingTemplate</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.3</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>Template_Type</Name>
      <Value>Bridge_LogicalTemplate</Value>
    </Attribute>
    <Attribute>
      <Name>Template_Id</Name>
      <Value>233</Value>
    </Attribute>
    <!-- see bridgeTemplate_list.xml for how to get this
value
-->

```

```

    </Attribute>
  <Attribute>
    <Name>lineGroupId</Name>
    <Value>16</Value>
  </Attribute>
  <Attribute>
    <Name>lowerIfIndex</Name>
    <Value>16</Value>
  </Attribute>
  <Attribute>
    <Name>vlanId</Name>
    <Value>100</Value>
  </Attribute>
<!--
optional attributes that can be set include :
  vlanId
  stripInsert
  learnIp
  learnUnicast
  maxUnicast
  bridgeIfVlanIdCOS
  bridgeIfOutgoingCOSOption
  bridgeIfOutgoingCOSValue
  bridgeIfStagTPID
  bridgeIfStagId
  bridgeIfStagStripAndInsert
  bridgeIfStagOutgoingCOSOption
  bridgeIfStagCOS
  bridgeIfStagOutgoingCOSValue
  bridgeIfMcastControlList
  bridgeIfMaxVideoStreams
  TableBasedFilter
  DhcpLearn
  rstpEnable
  mstpId
  mvrVlanId
  vlanXlateId
  slanXlateId
  incomingCOSOption
  stagIncomingCOSOption
  # either the profile index or a template name may
be used to specify a egress packet rule group
  egressGroupIndex or egressGroupTemplate
  # either the profile index or a template name may
be used to specify a ingress packet rule group
  groupIndex or groupTemplate
-->
</RequestElement>
</OSS>

```

Adding bridge logicals using templates on GPON port physicals



Note: Since GPON port physicals are created dynamically, lineGroupId and lowerIfIndex will be set to 0. Instead, OSS client will provide the corresponding slot index, OLT and ONU numbers using gponSlotId, gponOltId and gponPortId attributes. If a GPON port already exists for the specified slot, olt and port, DZMS will use the existing GPON port. If does not exists, DZMS will automatically create the GPON port and then continue with the creation of bridge logical.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>addUsingTemplate</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.3</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>Template_Type</Name>
      <Value>Bridge_LogicalTemplate</Value>
    </Attribute>
    <Attribute>
      <Name>Template_Id</Name>
      <Value>233</Value>
    <!-- see bridgeTemplate_list.xml for how to get this
value
-->
    </Attribute>
    <Attribute>
      <Name>gponSlotId</Name>
      <Value>10</Value>
    </Attribute>
    <Attribute>
      <Name>gponOltId</Name>
      <Value>1</Value>
```

```

    </Attribute>
  <Attribute>
    <Name>gponPortId</Name>
    <Value>911</Value>
  </Attribute>
  <Attribute>
    <Name>gtpTemplateName</Name>
    <Value>4M UBR</Value>
  </Attribute>
  <Attribute>
    <Name>groupTemplate</Name>
    <Value>rateLimitPRG</Value>
  </Attribute>
  <Attribute>
    <Name>egressGroupTemplate</Name>
    <Value>rateLimitPRG</Value>
  </Attribute>
<!--
optional attributes that can be set include :
  vlanId
  stripInsert
  learnIp
  learnUnicast
  maxUnicast
  bridgeIfVlanIdCOS
  bridgeIfOutgoingCOSOption
  bridgeIfOutgoingCOSValue
  bridgeIfStagTPID
  bridgeIfStagId
  bridgeIfStagStripAndInsert
  bridgeIfStagOutgoingCOSOption
  bridgeIfStagCOS
  bridgeIfStagOutgoingCOSValue
  bridgeIfMcastControlList
  bridgeIfMaxVideoStreams
  TableBasedFilter
  DhcpLearn
  rstpEnable
  mstpId
  mvrVlanId
  vlanXlateId
  slanXlateId
  incomingCOSOption
  stagIncomingCOSOption
  ctIndex
  # either the profile index or a template name may
be used to specify a gpon-traffic-profile
  gponTrafficIdx or gtpTemplateName
  # either the profile index or a template name may
be used to specify a egress packet rule group
  egressGroupIndex or egressGroupTemplate
  # either the profile index or a template name may
be used to specify a ingress packet rule group
  groupIndex or groupTemplate

```

```
-->
</RequestElement>
</OSS>
```

Add bridge logicals for ATM, GPON, Ethernet, EFM Bondgroup, Linkagg, VDSL

Adding bridge logicals on ATM logical interfaces

The following instruction is good for adding bridge logicals on ATM logical interfaces and gbond group.

1 Given a deviceIP, get the corresponding Device_I

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>Device</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>deviceIP=1.2.3.4</Value>
</Attribute>
</RequestElement>
</OSS>
```

2 Given deviceId, slotIndex, portId, get the lineGroupId of corresponding physical.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
```

```

<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>Physical</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=100</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Card_Id_1=10</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>ifTypePortId=100</Value>
</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>lineGroupId</Value>
</Attribute>
</RequestElement>
</OSS>

```



Note: There may be multiple interfaces with same Device_Id, Card_Id and Port_Id but only one interface has lineGroupId as non-zero. Interface with non-zero lineGroupId shall be considered for retrieving Handle_Id of ATMLogical object.

3 Given deviceId and lineGroupId, get Handle_Id & lowerIfIndex for ATMLogical object.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>ATMLogical</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=100</Value>
</Attribute>

```

```

<Attribute>
<Name>Filter_Condition</Name>
<Value>lineGroupId=1023</Value>
</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>rfc1483Index</Value>
</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>aal5Index</Value>
</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>ifIndex</Value>
</Attribute>
</RequestElement>
</OSS>

```



Note: Perform the following steps to retrieve the lowerIfIndex for using in the creation of bridge logical on ATM interfaces:

- Check for value of rfc1483Index of ATMLogical object, if it is non-zero, use that as the value to be assigned for lowerIfIndex.
- Else if rfc1483Index is 0, check for the value of aal5Index, if it is non-zero, use that as the value to be assigned for lowerIfIndex.
- Else if aal5Index is 0, use the ifIndex of ATMLogical as lowerIfIndex.

4 Add the corresponding AAL5 type VCL with providing unused VPI and VCI.



Note: Handle_Id of the ATM_Logical retrieved using previous step will be used as Parent_Id for creation of AAL5 VCL.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Parent_Type</Name>
<Value>ATMLogical</Value>

```

```

</Attribute>
<Attribute>
<Name>Parent_Id</Name>
<Value>100</Value>
</Attribute>
<Attribute>
<Name>Handle_Type</Name>
<Value>Aal5VclAtmEndPoint</Value>
</Attribute>
<Attribute>
<Name>vpi</Name>
<Value>0</Value>
</Attribute>
<Attribute>
<Name>vci</Name>
<Value>40</Value>
</Attribute>
<Attribute>
<Name>encapsType</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>adminStatus</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>rxTrafficDescrIndex</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>txTrafficDescrIndex</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>cpcsRxSduSize</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>connKind</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>vccAalType</Name>
<Value>3</Value>
</Attribute>
<Attribute>
<Name>cpcsTxSduSize</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>castType</Name>
<Value>1</Value>
</Attribute>
</RequestElement>

```



```
</OSS>
```

5 Add the bridge logicals on ATM Logical interfaces

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.0</Version>
    <Overwrite>false</Overwrite>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>BridgeLogical</Value>
    </Attribute>
  <!--
BRIDGE TYPES
    UP_LINK = 1;
    DOWN_LINK = 2;
    INTRA_LINK = 4;
    TLS = 6;
    RLINK = 8;
    PPPoA = 9;
    WIRE = 10;
    MVR = 11;
    USER = 12;
    DOWNLINK_VIDEO = 13;
    DOWNLINK_DATA = 14;
    DOWNLINK_PPPoE = 15;
    DOWNLINK_P2P = 16;
    DOWNLINK_VOICE = 17;
    DOWNLINK_UPSTREAM_MCAST = 18;
    IPOB_TLS = 19;
    IPOB_UPLINK = 20;
    IPOB_DOWNLINK = 21;
-->
  <Attribute>
    <Name>bridgeType</Name>
    <Value>14</Value>
```

```

    </Attribute>
  <Attribute>
    <Name>lowerIfIndex</Name>
    <Value>725</Value>
    <!-- for adsl = ifIndex of rfc1483 if-translate
record
-->
  </Attribute>
  <Attribute>
    <Name>lineGroupId</Name>
    <Value>352</Value>
    <!-- ifIndex of adsl lineGroup
-->
  </Attribute>
  <Attribute>
    <Name>vlanId</Name>
    <Value>105</Value>
  </Attribute>
  <Attribute>
    <Name>stripInsert</Name>
    <Value>2</Value>
    <!-- 1=true (for untagged), 2=false
-->
  </Attribute>
  <Attribute>
    <Name>bridgeIfStagId</Name>
    <Value>0</Value>
  </Attribute>
  <Attribute>
    <Name>bridgeIfStagStripAndInsert</Name>
    <Value>1</Value>
    <!-- 1=true, 2=false (double tagged)
-->
  </Attribute>
  <Attribute>
    <Name>vpi</Name>
    <Value>0</Value>
  </Attribute>
  <Attribute>
    <Name>vci</Name>
    <Value>40</Value>
  </Attribute>
<!--
optional attributes that can be set include :
  vlanId
  stripInsert
  learnIp
  learnUnicast
  maxUnicast
  bridgeIfVlanIdCOS
  bridgeIfOutgoingCOSOption
  bridgeIfOutgoingCOSValue
  bridgeIfStagTPID
  bridgeIfStagId

```

```

        bridgeIfStagStripAndInsert
        bridgeIfStagOutgoingCOSOption
        bridgeIfStagCOS
        bridgeIfStagOutgoingCOSValue
        bridgeIfMcastControlList
        bridgeIfMaxVideoStreams
        TableBasedFilter
        DhcpLearn
        rstpEnable
        mstpId
        mvrVlanId
        vlanXlateId
        slanXlateId
        incomingCOSOption
        stagIncomingCOSOption
        ctIndex
        # either the profile index or a template name may
        be used to specify a gpon-traffic-profile
        gponTrafficIdx or gtpTemplateName
        # either the profile index or a template name may
        be used to specify a egress packet rule group
        egressGroupIndex or egressGroupTemplate
        # either the profile index or a template name may
        be used to specify a ingress packet rule group
        groupIndex or groupTemplate
    -->
</RequestElement>
</OSS>

```

Adding bridge logicals on Ethernet port physicals, linkagg, EFM bondgroup, and VDSL

The following instruction is good for adding bridge logicals on Ethernet port physicals, linkagg, EFM bondgroup, and VDSL.

- 1 Given Device_Id, Card-Id and Port_Id, retrieve the lineGroupId/lowerIfIndex of Ethernet port physical.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>Physical</Value>

```

```

</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=100</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Card_Id_1=1</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>ifTypePortId=1</Value>
</Attribute>

<Attribute>
<Name>Filter_Retrieve</Name>
<Value>lineGroupId</Value>
</Attribute>
</RequestElement>
</OSS>

```

2 Create bridge on top of ethernet port physical.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.5.1</Version>
    <Overwrite>false</Overwrite>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>BridgeLogical</Value>
    </Attribute>
    <Attribute>
      <Name>bridgeType</Name>
      <Value>6</Value>
    <!-- 6 = TLS
    -->
  </RequestElement>

```

```

    </Attribute>
  <Attribute>
    <Name>lowerIfIndex</Name>
    <Value>10</Value>
    <!-- lowerIfIndex = lineGroupId
    -->
  </Attribute>
  <Attribute>
    <Name>lineGroupId</Name>
    <Value>10</Value>
    <!-- ifIndex of lineGroup
    -->
  </Attribute>
  <Attribute>
    <Name>vlanId</Name>
    <Value>105</Value>
  </Attribute>
  <Attribute>
    <Name>stripInsert</Name>
    <Value>2</Value>
    <!-- 1=true (for untagged), 2=false
    -->
  </Attribute>
  <Attribute>
    <Name>bridgeIfStagId</Name>
    <Value>106</Value>
  </Attribute>
  <Attribute>
    <Name>bridgeIfStagStripAndInsert</Name>
    <Value>2</Value>
    <!-- 1=true, 2=false (for double tagged)
    -->
  </Attribute>
<!--
optional attributes that can be set include :
  vlanId
  stripInsert
  learnIp
  learnUnicast
  maxUnicast
  bridgeIfVlanIdCOS
  bridgeIfOutgoingCOSOption
  bridgeIfOutgoingCOSValue
  bridgeIfStagTPID
  bridgeIfStagId
  bridgeIfStagStripAndInsert
  bridgeIfStagOutgoingCOSOption
  bridgeIfStagCOS
  bridgeIfStagOutgoingCOSValue
  bridgeIfMcastControlList
  bridgeIfMaxVideoStreams
  TableBasedFilter
  DhcpLearn
  rstpEnable

```

```

        mstpId
        mvrVlanId
        vlanXlateId
        slanXlateId
        incomingCOSOption
        stagIncomingCOSOption
        ctIndex
        # either the profile index or a template name may
        be used to specify a gpon-traffic-profile
        gponTrafficIdx or gtpTemplateName
        # either the profile index or a template name may
        be used to specify a egress packet rule group
        egressGroupIndex or egressGroupTemplate
        # either the profile index or a template name may
        be used to specify a ingress packet rule group
        groupIndex or groupTemplate
    -->
</RequestElement>
</OSS>

```

Adding bridge logicals on GPON port physicals



Note: Since GPON port physicals are created dynamically, lineGroupId and lowerIfIndex will be set to 0. Instead, OSS client will provide the corresponding slot index, OLT and ONU numbers using gponSlotId, gponOltId and gponPortId attributes. If a GPON port already exists for the specified slot, olt and port, DZMS will use the existing GPON port. If does not exists, DZMS will automatically create the GPON port and then continue with the creation of bridge logical.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.5.1</Version>
    <Overwrite>false</Overwrite>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>

```

```

    <Value>1</Value>
  </Attribute>
<Attribute>
  <Name>Handle_Type</Name>
  <Value>BridgeLogical</Value>
</Attribute>
<!--
BRIDGE TYPES
    UP_LINK = 1;
    DOWN_LINK = 2;
    INTRA_LINK = 4;
    TLS = 6;
    RLINK = 8;
    PPPoA = 9;
    WIRE = 10;
    MVR = 11;
    USER = 12;
    DOWNLINK_VIDEO = 13;
    DOWNLINK_DATA = 14;
    DOWNLINK_PPPoE = 15;
    DOWNLINK_P2P = 16;
    DOWNLINK_VOICE = 17;
    DOWNLINK_UPSTREAM_MCAST = 18;
    IPOB_TLS = 19;
    IPOB_UPLINK = 20;
    IPOB_DOWNLINK = 21;
-->
<Attribute>
  <Name>bridgeType</Name>
  <Value>14</Value>
</Attribute>
<Attribute>
  <Name>gponSlotId</Name>
  <Value>10</Value>
</Attribute>
<Attribute>
  <Name>gponOltId</Name>
  <Value>4</Value>
</Attribute>
<Attribute>
  <Name>gponPortId</Name>
  <Value>504</Value>
</Attribute>
<Attribute>
  <Name>gtpTemplateName</Name>
  <Value>4M UBR</Value>
</Attribute>
<Attribute>
  <Name>vlanId</Name>
  <Value>106</Value>
</Attribute>
<Attribute>
  <Name>stripInsert</Name>
  <Value>2</Value>

```

```

<!-- 1=true (for untagged), 2=false
-->
</Attribute>
<Attribute>
  <Name>bridgeIfStagId</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>bridgeIfStagStripAndInsert</Name>
  <Value>1</Value>
<!-- 1=true, 2=false (for double tagged)
-->
</Attribute>
<Attribute>
  <Name>groupTemplate</Name>
  <Value>rateLimitPRG</Value>
</Attribute>
<Attribute>
  <Name>egressGroupTemplate</Name>
  <Value>rateLimitPRG</Value>
</Attribute>
<!--
optional attributes that can be set include :
  vlanId
  stripInsert
  learnIp
  learnUnicast
  maxUnicast
  bridgeIfVlanIdCOS
  bridgeIfOutgoingCOSOption
  bridgeIfOutgoingCOSValue
  bridgeIfStagTPID
  bridgeIfStagId
  bridgeIfStagStripAndInsert
  bridgeIfStagOutgoingCOSOption
  bridgeIfStagCOS
  bridgeIfStagOutgoingCOSValue
  bridgeIfMcastControlList
  bridgeIfMaxVideoStreams
  TableBasedFilter
  DhcpLearn
  rstpEnable
  mstpId
  mvrVlanId
  vlanXlateId
  slanXlateId
  incomingCOSOption
  stagIncomingCOSOption
  ctIndex
  # either the profile index or a template name may
be used to specify a gpon-traffic-profile
  gponTrafficIdx or gtpTemplateName
  # either the profile index or a template name may
be used to specify a egress packet rule group

```



```

        egressGroupIndex or egressGroupTemplate
        # either the profile index or a template name may
        be used to specify a ingress packet rule group
        groupIndex or groupTemplate
    -->
</RequestElement>
</OSS>

```

Add request: Subscriber hosts

To add a primary, dynamic, or static subscriber host using OSS XML Gateway, perform the following tasks. Procedure for creation of AAL5 VCL will be same as the procedure described in the bridge logical sections.

Add primary subscriber host on the top of ATM logical interfaces

1 Create AAL5 VCL.

Same as the procedure used in bridge logical sections [Step 4 Add the corresponding AAL5 type VCL with providing unused VPI and VCI. on page 141.](#)

2 Add the IP Logical Object for the floating point IP.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>14</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>IpLogical</Value>
    </Attribute>
    <Attribute>
      <Name>name</Name>
      <Value>oss_float_ip_1</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

```

</Attribute>
<!--Always set the lineGroupId and lineGroupKey to 0
for creation of floating IP -->
<Attribute>
    <Name> lineGroupId</Name>
    <Value>0</Value>
</Attribute>
<Attribute>
    <Name> lineGroupKey</Name>
    <Value>0</Value>
</Attribute>
<Attribute>
    <Name>addr</Name>
    <Value>5.5.5.1</Value>
</Attribute>
<Attribute>
    <Name>netMask</Name>
    <Value>255.255.255.0</Value>
</Attribute>
<Attribute>
    <Name>bcastAddr</Name>
    <Value>5.5.5.255</Value>
</Attribute>
</RequestElement>
</OSS>

```

3 Add DHCP subnet.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>14</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DHCPSubnet</Value>
</Attribute>
<Attribute>

```

```

        <Name>name</Name>
        <Value>OSS_DHCP_Subnet</Value>
</Attribute>
<!--This must be same as IP address (addr) of Float IP
-->
<Attribute>
        <Name>defRouter</Name>
        <Value>5.5.5.1</Value>
</Attribute>
<Attribute>
        <Name>domainName</Name>
        <Value>lowell</Value>
</Attribute>
<Attribute>
        <Name>network</Name>
        <Value>5.5.5.0</Value>
</Attribute>
<Attribute>
        <Name>netMask</Name>
        <Value>255.255.255.0</Value>
</Attribute>
<Attribute>
        <Name>range1Start</Name>
        <Value>5.5.5.1</Value>
</Attribute>
<Attribute>
        <Name>range1End</Name>
        <Value>5.5.5.5</Value>
</Attribute>
<Attribute>
        <Name>range2Start</Name>
        <Value>0.0.0.0</Value>
</Attribute>
<Attribute>
        <Name>range2End</Name>
        <Value>0.0.0.0</Value>
</Attribute>
<Attribute>
        <Name>range3Start</Name>
        <Value>0.0.0.0</Value>
</Attribute>
<Attribute>
        <Name>range3End</Name>
        <Value>0.0.0.0</Value>
</Attribute>
<Attribute>
        <Name>range4Start</Name>
        <Value>0.0.0.0</Value>
</Attribute>
<Attribute>
        <Name>range4End</Name>
        <Value>0.0.0.0</Value>
</Attribute>
<Attribute>

```

```

        <Name>range5Start</Name>
        <Value>0.0.0.0</Value>
    </Attribute>
    <Attribute>
        <Name>range5End</Name>
        <Value>0.0.0.0</Value>
    </Attribute>
</RequestElement>
</OSS>

```

4 Add the IP Logical Object for primary subscriber host (the one that has IP address assigned for addr field).

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.16.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>Device</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>14</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>IpLogical</Value>
        </Attribute>
        <Attribute>
            <Name>name</Name>
            <Value>oss_float_ip_1</Value>
        </Attribute>
        <Attribute>
            <Name> lineGroupId</Name>
            <Value>2565</Value>
        </Attribute>
        <Attribute>
            <Name> lineGroupKey</Name>
            <Value>5116</Value>
        </Attribute>
        <Attribute><Name>addr</Name>
            <Value>5.5.5.1</Value>
        </Attribute>
        <Attribute>

```

```

        <Name>netMask</Name>
        <Value>255.255.255.255</Value>
    </Attribute>
    <Attribute>
        <Name>bcastAddr</Name>
        <Value>0.0.0.0</Value>
    </Attribute>
    <Attribute>
        <Name>farEndAddr</Name>
        <Value>255.255.255.0</Value>
    </Attribute>
    <Attribute>
        <Name>dhcpServerEnable</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>vpi</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>vci</Name>
        <Value>39</Value>
    </Attribute>
    <Attribute>
        <Name>mcastControlList</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>maxVideoStreams</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>natEnabled</Name>
        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>subnetGroup</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>unnumberedIndex</Name>
        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>vlanId</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>stagCOS</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>reasmMaxSize</Name>
        <Value>0</Value>

```

```

</Attribute>
<Attribute>
  <Name>lowerIfIndex</Name>
  <Value>2801</Value>
</Attribute>
<Attribute>
  <Name>ipAddrDynamic</Name>
  <Value>4</Value>
</Attribute>
<Attribute>
  <Name>pointToPoint</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>bcastEnabled</Name>
  <Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 5** Add the IP Logical Object for dynamic subscriber host (the one that has IP address as 0.0.0.0 for addr field and zero IP address for destAddr field and dhcpServerEnable set to 2).

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>14</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>IpLogical</Value>
    </Attribute>
    <Attribute>
      <Name>name</Name>
      <Value>oss_float_ip_1</Value>
    </Attribute>
    <Attribute>

```

```

        <Name> lineGroupId</Name>
        <Value>2565</Value>
    </Attribute>
    <Attribute>
        <Name> lineGroupKey</Name>
        <Value>5116</Value>
    </Attribute>
    <!-- Not specifying this field also assigns the
    0.0.0.0 for addr -->
    <Attribute>
        <Name>addr</Name>
        <Value>0.0.0.0</Value>
    </Attribute>
    <Attribute>
        <Name>netMask</Name>
        <Value>255.255.255.255</Value>
    </Attribute>
    <Attribute>
        <Name>bcastAddr</Name>
        <Value>0.0.0.0</Value>
    </Attribute>
    <Attribute>
        <Name>farEndAddr</Name>
        <Value>255.255.255.0</Value>
    </Attribute>
    <!--Setting dhcpServerEnable to 1 will make the host as
    primary subscriber -->
    <Attribute>
        <Name>dhcpServerEnable</Name>
        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>vpi</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>vci</Name>
        <Value>39</Value>
    </Attribute>
    <Attribute>
        <Name>maxVideoStreams</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>natEnabled</Name>
        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>subnetGroup</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>unnumberedIndex</Name>
        <Value>2</Value>

```

```

</Attribute>
<Attribute>
    <Name>vlanId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>stagCOS</Name>
    <Value>0</Value>
</Attribute>
<Attribute>
    <Name>reasmMaxSize</Name>
    <Value>0</Value>
</Attribute>
<Attribute>
    <Name>lowerIfIndex</Name>
    <Value>2801</Value>
</Attribute>
<Attribute>
    <Name>ipAddrDynamic</Name>
    <Value>4</Value>
</Attribute>
<Attribute>
    <Name>pointToPoint</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>bcastEnabled</Name>
    <Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 6** Add the IP Logical Object for static subscriber host (the one that has IP address as 0.0.0.0 for addr field and non-zero IP address for destAddr field and dhcpServerEnable set to 2).

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>

```



```

        <Name>Parent_Id</Name>
        <Value>14</Value>
    </Attribute>
    <Attribute>
        <Name>Handle_Type</Name>
        <Value>IpLogical</Value>
    </Attribute>
    <Attribute>
        <Name>name</Name>
        <Value>oss_float_ip_1</Value>
    </Attribute>
    <Attribute>
        <Name> lineGroupId</Name>
        <Value>2565</Value>
    </Attribute>
    <Attribute>
        <Name> lineGroupKey</Name>
        <Value>5116</Value>
    </Attribute>
    <!-- Not specifying this field also assigns the
    0.0.0.0 for addr -->
    <Attribute>
        <Name>addr</Name>
        <Value>0.0.0.0</Value>
    </Attribute>
    <Attribute>
        <Name>netMask</Name>
        <Value>255.255.255.255</Value>
    </Attribute>
    <Attribute>
        <Name>bcastAddr</Name>
        <Value>0.0.0.0</Value>
    </Attribute>
    <Attribute>
        <Name>farEndAddr</Name>
        <Value>255.255.255.0</Value>
    </Attribute>
    <!--Setting dhcpServerEnable to 1 will make the host as
    primary subscriber -->
    <Attribute>
        <Name>dhcpServerEnable</Name>
        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>vpi</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>vci</Name>
        <Value>39</Value>
    </Attribute>
    <Attribute>
        <Name>maxVideoStreams</Name>
        <Value>0</Value>

```

```

</Attribute>
<Attribute>
  <Name>natEnabled</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>subnetGroup</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>unnumberedIndex</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>vlanId</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>stagCOS</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>reasmMaxSize</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>lowerIfIndex</Name>
  <Value>2801</Value>
</Attribute>
<Attribute>
  <Name>ipAddrDynamic</Name>
  <Value>4</Value>
</Attribute>
<Attribute>
  <Name>pointToPoint</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>bcastEnabled</Name>
  <Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```

Add subscriber host on the top of Ethernet port physicals



Note: There is no need for creation of AAL5 VCL for creation of subscriber hosts on Ethernet and GPON port physicals.

- 1 Given Device_Id, Card_Id and Physical_ID-1, retrieve the lineGroupId/lowerIfIndex of Ethernet port physical.

Refer to the example in [Step 1 Given Device_Id, Card-Id and Port_Id, retrieve the lineGroupId/lowerIfIndex of Ethernet port physical. on page 144.](#)

2 Create Float IP

Same as the procedure explained in creation of float IP in case of ATM logical interfaces, [Step 2 Add the IP Logical Object for the floating point IP. on page 161.](#)

3 Create DHCP subnet

Same as the procedure explained in creation of DHCP subnet in case of ATM logical interfaces, [Step 3 Add DHCP subnet. on page 162.](#)

4 Create primary subscriber host on top of ethernet port physical

Same as the creation of primary subscriber hosts on ATM logical interfaces, [Step 4 Add the IP Logical Object for primary subscriber host \(the one that has IP address assigned for addr field\). on page 164,](#) except that lineGroupId and lowerIfIndex will refer to same non-zero value in case of ethernet port physicals.

5 Create dynamic subscriber host on top of ethernet port physical

Same as the creation of dynamic subscriber hosts on ATM logical interfaces, [Step 5 Add the IP Logical Object for dynamic subscriber host \(the one that has IP address as 0.0.0.0 for addr field and zero IP address for destAddr field and dhcpServerEnable set to 2\). on page 166,](#) except that lineGroupId and lowerIfIndex will refer to same non-zero value in case of ethernet port physicals.

6 Create static subscriber host on top of ethernet port physical

Same as the creation of static subscriber hosts on ATM logical interfaces, [Step 6 Add the IP Logical Object for static subscriber host \(the one that has IP address as 0.0.0.0 for addr field and non-zero IP address for destAddr field and dhcpServerEnable set to 2\). on page 168,](#) except that lineGroupId and lowerIfIndex will refer to same non-zero value in case of ethernet port physicals.

Add primary subscriber host on the top of GPON port physicals



Note: There is no need for creation of AAL5 VCL for creation of subscriber hosts on Ethernet and GPON port physicals.

1 Create Float IP

Same as the procedure explained in creation of float IP in case of ATM logical interfaces, [Step 2 Add the IP Logical Object for the floating point IP. on page 161.](#)

2 Create DHCP subnet

Same as the procedure explained in creation of DHCP subnet in case of ATM logical interfaces, [Step 3 Add DHCP subnet. on page 162.](#)

3 Create primary subscriber host on top of GPON port physical.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>14</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>IpLogical</Value>
    </Attribute>
    <Attribute>
      <Name>name</Name>
      <Value>oss_float_ip_1</Value>
    </Attribute>
    <Attribute>
      <Name> lineGroupId</Name>
      <Value>0</Value>
    </Attribute>
    <Attribute>
      <Name>addr</Name>
      <Value>5.5.5.1</Value>
    </Attribute>
    <Attribute>
      <Name>netMask</Name>
      <Value>255.255.255.255</Value>
    </Attribute>
    <Attribute>
      <Name>bcastAddr</Name>
      <Value>0.0.0.0</Value>
    </Attribute>
    <Attribute>
      <Name>farEndAddr</Name>
      <Value>255.255.255.0</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

```

<Attribute>
  <Name>dhcpServerEnable</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>mcastControlList</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>maxVideoStreams</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>natEnabled</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>subnetGroup</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>unnumberedIndex</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>vlanId</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>stagCOS</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>reasmMaxSize</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>lowerIfIndex</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>ipAddrDynamic</Name>
  <Value>4</Value>
</Attribute>
<Attribute>
  <Name>pointToPoint</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>bcastEnabled</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>gponSlotId</Name>

```

```

        <Value>13</Value>
    </Attribute>
    <Attribute>
        <Name>gponOltId</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>gponPortId</Name>
        <Value>701</Value>
    </Attribute>
</RequestElement>
</OSS>

```

- 4** Create dynamic subscriber host on top of GPON port physical (the one that has IP address as 0.0.0.0 for *addr* field and zero IP address for *destAddr* field and dhcpServerEnable set to 2).

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.16</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>Device</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>14</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>IpLogical</Value>
        </Attribute>
        <Attribute>
            <Name>name</Name>
            <Value>oss_float_ip_1</Value>
        </Attribute>
        <Attribute>
            <Name> lineGroupId</Name>
            <Value>0</Value>
        </Attribute>
        <Attribute>
            <Name>addr</Name>
            <Value>0.0.0.0</Value>
        </Attribute>
    </RequestElement>

```

```

<Attribute>
  <Name>destAddr</Name>
  <Value>0.0.0.0</Value>
</Attribute>
<Attribute>
  <Name>netMask</Name>
  <Value>255.255.255.255</Value>
</Attribute>
<Attribute>
  <Name>bcastAddr</Name>
  <Value>0.0.0.0</Value>
</Attribute>
<Attribute>
  <Name>farEndAddr</Name>
  <Value>255.255.255.0</Value>
</Attribute>
<Attribute>
  <Name>dhcpServerEnable</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>mcastControlList</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>maxVideoStreams</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>natEnabled</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>subnetGroup</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>unnumberedIndex</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>vlanId</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>stagCOS</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>reasmMaxSize</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>lowerIfIndex</Name>

```

```

        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>ipAddrDynamic</Name>
        <Value>4</Value>
    </Attribute>
    <Attribute>
        <Name>pointToPoint</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>bcastEnabled</Name>
        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>gponSlotId</Name>
        <Value>13</Value>
    </Attribute>
    <Attribute>
        <Name>gponOltId</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>gponPortId</Name>
        <Value>701</Value>
    </Attribute>
</RequestElement>
</OSS>

```

- 5** Create static subscriber host on top of GPON port physical (the one that has IP address as 0.0.0.0 for *addr* field and zero IP address for *destAddr* field and *dhcpServerEnable* set to 2).

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.16</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>Device</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>14</Value>
        </Attribute>
    </RequestElement>
</OSS>

```



```

<Attribute>
  <Name>Handle_Type</Name>
  <Value>IpLogical</Value>
</Attribute>
<Attribute>
  <Name>name</Name>
  <Value>oss_float_ip_1</Value>
</Attribute>
<Attribute>
  <Name> lineGroupId</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>addr</Name>
  <Value>0.0.0.0</Value>
</Attribute>
<Attribute>
  <Name>destAddr</Name>
  <Value>192.168.3.4</Value>
</Attribute>
<Attribute>
  <Name>netMask</Name>
  <Value>255.255.255.255</Value>
</Attribute>
<Attribute>
  <Name>bcastAddr</Name>
  <Value>0.0.0.0</Value>
</Attribute>
<Attribute>
  <Name>farEndAddr</Name>
  <Value>255.255.255.0</Value>
</Attribute>
<Attribute>
  <Name>dhcpServerEnable</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>mcastControlList</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>maxVideoStreams</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>natEnabled</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>subnetGroup</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>unnumberedIndex</Name>

```

```

        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>vlanId</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>stagCOS</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>reasmMaxSize</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>lowerIfIndex</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>ipAddrDynamic</Name>
        <Value>4</Value>
    </Attribute>
    <Attribute>
        <Name>pointToPoint</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>bcastEnabled</Name>
        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>gponSlotId</Name>
        <Value>13</Value>
    </Attribute>
    <Attribute>
        <Name>gponOltId</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>gponPortId</Name>
        <Value>701</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Add request: Bridge filter type – DHCP relay

The bridge filter group (also known as packet rule group) now support multiple bridge filter members (also known as packet rule group members) in OSS XML GW. Those multiple bridge filter members can have same or different filter types. The filter types could be ACL Allow, ACL Deny, Insert Option 82, DHCP Relay, Insert PPPoE Vendor Tag, Forbid Oui, Rate Limit

Discard, Color Aware Rate Limit Discard, Destination MAC Swap Static, Destination MAC Swap Dynamic, Dscp To Cos, Promote First Encapsulation VLAN, Filter First Encapsulation VLAN, Promote Second Encapsulation VLAN, Filter Second Encapsulation VLAN, and Bridge Storm Detection.

Following are relevant fields and their intended usage for each type.

- ACL Allow => cfgType is 104. It requires cfgType, groupIndex and cfgValue. cfgValue represents the allow rule.
- ACL Deny => cfgType is 120. It requires cfgType, groupIndex and cfgValue. cfgValue represents the deny rule.
- Destination MAC Swap Dynamic => cfgType is 38. cfgValue is not applicable for this profile and will be defaulted to 0:0:0:0:0:0 as they will be created dynamically.
- Destination MAC Swap Static => cfgType is 37. cfgValue represents a valid MAC address (Combination of six single or double digit numbers separated with colons).
- DHCP Relay => cfgType is 98. It requires cfgType, groupIndex and cfgValue. cfgValue represents the subnet group index.
- Forbid OUI => cfgType is 114. It requires cfgType, groupIndex and cfgValue. cfgValue represents bit pattern to be matched in a packet, stored as a string, but specified in the pattern 'nn:nn:nn...nn:nn', where 'nn' is a hex number E.g. a packetRuleValue of '00:02:02' specifies the OUI to be dropped. Must be assigned with a minimum of 2 digit string for successful addition of filter.
- Insert Option 82 => cfgType is 97. It requires cfgType, groupIndex. cfgValue is not applicable for this type of filter.
- Insert PPPOE Vendor tag => cfgType is 99. It requires cfgType, groupIndex and cfgValue. cfgValue is a string of up to 48 bytes to be inserted into PPPoE discovery packets (PADI/PADR) which transit this node. String will be inserted as a Vendor-Specific tag.
- Rate Limit Discard => cfgType is 100. It requires cfgType, groupIndex, cfgValue, packetRule2, and packetRule3. cfgValue is same as packetRule1 in this case and represents the maximum rate limit (in kbps). packetRule2 represents Committed Burst Size (CBS) value in bytes. packetRule3 represents Extended Burst Size (EBS) value in bytes.
- Color Aware Rate Limit Discard => cfgType is 103. It requires cfgType, groupIndex, cfgValue, packetRule2, packetRule3, packetRule4, and packetRule5. cfgValue is same as packetRule1 in this case and represents the maximum rate limit in kbps. packetRule2 and packetRule3 represent same values as in the Rate Limit Discard. packetRule4 represents color-aware mode green packets boundary (COS Green). packetRule5 represents color-aware mode yellow packets boundary (COS Yellow).

packetRule2 and packetRule3 are required by both Rate Limit Discard and Color Aware Rate Limit Discard. The default values for the packetRule2 and packetRule3 vary based on the value assigned for cfgValue (i.e. rate) field:

- rate (strictly) less than 1Mbps: CBS=65536, EBS=100000
- rate less (or equal) to 6Mbps: CBS=120000, EBS=130000
- rate less (or equal) to 10Mbps: CBS=200000, EBS=200000
- rate (strictly) larger than 10Mbps: CBS=400000, EBS=400000

The value range for the packetRule4 and packetRule5:

- green packets boundary: $4 \leq \text{cos/tos} \leq 7$ (Default is 4)
- yellow packets boundary: $0 \leq \text{cos/tos} < 4$ (Default is 0)
- Dscp To Cos $\Rightarrow \Rightarrow$ cfgType is 41. It requires cfgType, groupIndex and cfgValue. cfgValue represents the DSCP index.
- Promote First Encapsulation VLAN $\Rightarrow$ cfgType is 42. It requires cfgType, groupIndex, cfgValue, packetRule2, and packetRule3. cfgValue is same as packetRule1 in this case and represents the VLAN id value in range 1 to 4095. packetRule2 represents TP id value in hexadecimal. packetRule3 represents COS value in range 0 to 7.
- Filter First Encapsulation VLAN $\Rightarrow$ cfgType is 43. It requires cfgType, groupIndex, cfgValue and packetRule2. cfgValue represents cfgValue is same as packetRule1 in this case and represents the VLAN id value in range 1 to 4095. packetRule2 represents TP id value in hexadecimal.
- Promote Second Encapsulation VLAN $\Rightarrow$ cfgType is 44. It requires cfgType, groupIndex, cfgValue, packetRule2, and packetRule3. cfgValue is same as packetRule1 in this case and represents the VLAN id value in range 1 to 4095. packetRule2 represents TP id value in hexadecimal. packetRule3 represents COS value in range 0 to 7.
- Filter Second Encapsulation VLAN $\Rightarrow$ cfgType is 45 It requires cfgType, groupIndex, cfgValue and packetRule2. cfgValue represents cfgValue is same as packetRule1 in this case and represents the VLAN id value in range 1 to 4095. packetRule2 represents TP id value in hexadecimal.
- Bridge Storm Detection $\Rightarrow$ cfgType is 46. It requires cfgType, groupIndex, cfgValue, and packetRule2. cfgValue represents the discard type. Discard type can be one of the Discard, Discard and Alarm, and Discard and Alarm and Block. If we select either of Discard and Alarm or Discard and Alarm and Block, it requires packetRule3 value. Here, packetRule2 represents Packets per Seconds and packetRule3 represents Consecutive seconds. The minimum value for Packets per Seconds is 20. The value range for Consecutive seconds is between 5 and 30.



Note: Both Promote Second Encapsulation VLAN and Filter Second Encapsulation VLAN are applicable for only MALC devices.



Note: Even though multiple filters can be assigned to the same filter group, bridge interfaces can only be assigned to use whole filter group. If only some of the filters of the filter group are to be used, a separate group must be created with those filters.



Note: groupIndex represents the groupIndex of Parent BridgeCfgGroup.

Add a packet rule group

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <RequestMode>online</RequestMode>
    <Version>1.12</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>33</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>BridgeCfgGroup</Value>
    </Attribute>
    <Attribute>
      <Name>name</Name>
      <Value>oss_3</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

Add a DHCP relay members to the existing group

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
```

```

<SessionID>sessionID</SessionID>
<operName>operName</operName>
<RequestMode>online</RequestMode>
<Version>1.12</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>BridgeCfgGroup</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>41</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>BridgeCfg</Value>
</Attribute>
<Attribute>
    <Name>cfgType</Name>
    <Value>98</Value>
</Attribute>
<!-- Incidates the index of subgroup -->
<Attribute>
    <Name>cfgValue</Name>
    <Value>3</Value>
</Attribute>
<!-- Indicates the groupIndex of parent bridge cfg
group -->
<Attribute>
    <Name>groupIndex</Name>
    <Value>34</Value>
</Attribute>
</RequestElement>
</OSS>

```

POTS– to– GR-303 Non-ESA and ESA Connections

To create a POTS– to– GR-303 Non-ESA connection and ESA connection, perform the following tasks:

Add Request: Adding POTS– GR303 Non-ESA Voice Connection

- 1 Create the Sub Voice Connection without providing any end point information.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>

```

```

        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.6</Version>
        <RequestMode>online</RequestMode>
    </Request>
</RequestElement>
    <Attribute>
        <Name>Parent_Type</Name>
        <Value>SubscriberPort</Value>
    </Attribute>
    <Attribute>
        <Name>Parent_Id</Name>
        <Value>7</Value>
    </Attribute>
    <Attribute>
        <Name>Handle_Type</Name>
        <Value>SubVoiceConn</Value>
    </Attribute>
    <Attribute>
        <Name>Device_Id</Name>
        <Value>9</Value>
    </Attribute>
    <Attribute>
        <Name>connectionType</Name>
        <Value>6</Value>
    </Attribute>
    <Attribute>
        <Name>connectionDesc</Name>
        <Value>NonESAPotsToGr303</Value>
    </Attribute>
    <Attribute>
        <Name>adminStatus</Name>
        <Value>2</Value>
    </Attribute>
</RequestElement>
</OSS>

```

2 Create the POTS end point, specifying the newly created connection as its parent.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.6</Version>
        <RequestMode>online</RequestMode>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>

```

```

        <Value>SubVoiceConn</Value>
      </Attribute>
      <Attribute>
        <Name>Parent_Id</Name>
        <Value>167</Value>
      </Attribute>

      <Attribute>
        <Name>Handle_Type</Name>
        <Value>SubVoicePots_EndPoint</Value>
      </Attribute>
      <Attribute>
        <Name>Handle_Group</Name>
        <Value>99999</Value>
      </Attribute>
      <Attribute>
        <Name>Device_Id</Name>
        <Value>9</Value>
      </Attribute>
      <Attribute>
        <Name>name</Name>
        <Value>OSS-Added-POTS-Endpoint</Value>
      </Attribute>
      <Attribute>
        <Name>lineGroupId</Name>
        <Value>1609</Value>
      </Attribute>
    </RequestElement>
  </OSS>

```

3 Create the GR-303 CRV.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.3</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Gr303Ig</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>

```



```

        <Value>Gr303CRV</Value>
      </Attribute>
    <Attribute>
      <Name>localAdminState</Name>
      <Value>2</Value>
    </Attribute>
    <Attribute>
      <Name>crvIndex</Name>
      <Value>6</Value>
    </Attribute>
    <Attribute>
      <Name>signalType</Name>
      <Value>2</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

4 Create the GR303 end point by specifying the newly created connection as its parent and igCrv as the crvIndex from step 3.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
    <RequestMode>online</RequestMode>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>SubVoiceConn</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>167</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>SubVoiceGr303EndPoint</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Group</Name>
      <Value>99999</Value>
    </Attribute>
    <Attribute>
      <Name>Device_Id</Name>
      <Value>9</Value>
    </Attribute>
    <Attribute>
      <Name>name</Name>
      <Value>OSSAdded-GR303-EndPoint</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

```

</Attribute>
<Attribute>
  <Name>igCrv</Name>
  <Value>6</Value>
</Attribute>
<Attribute>
  <Name>igName</Name>
  <Value>one</Value>
</Attribute>
</RequestElement>
</OSS>

```

Add Request: Adding POTS– VOIP Non-ESA Voice Connection

- 1 Create the Sub Voice Connection without providing any end point information.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
    <RequestMode>online</RequestMode>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>SubscriberPort</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>7</Value>
    </Attribute>
    <Attribute>
      <Name>Device_Id</Name>
      <Value>9</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>SubVoiceConn</Value>
    </Attribute>
    <Attribute>
      <Name>connectionType</Name>
      <Value>14</Value>
    </Attribute>
    <Attribute>
      <Name>connectionDesc</Name>
      <Value>NonESAPotsToVoip</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

```

        <Attribute>
            <Name>adminStatus</Name>
            <Value>2</Value>
        </Attribute>
    </RequestElement>
</OSS>

```

2 Create the VOIP end point, specifying the newly created connection as its parent.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.6</Version>
    <RequestMode>online</RequestMode>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>SubVoiceConn</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>168</Value>
        </Attribute>
        <Attribute>
            <Name>Device_Id</Name>
            <Value>9</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>SubVoiceVoip_EndPoint</Value>
        </Attribute>
        <Attribute>
            <Name>name</Name>
            <Value>OSSAdded-VOIP-EndPoint</Value>
        </Attribute>
        <Attribute>
            <Name>directoryNumber</Name>
            <Value>2345</Value>
        </Attribute>
        <Attribute>
            <Name>registrationServer</Name>
            <Value>255</Value>
        </Attribute>
        <Attribute>
            <Name>ipIfIndex</Name>
            <Value>69</Value>
        </Attribute>
    </RequestElement>

```

```

    </RequestElement>
</OSS>

```

- 3 Create the POTS end point, specifying the newly created connection as its parent.

Same as the POTS end point creation example that is listed in the [Add Request: Adding POTS– GR303 Non-ESA Voice Connection](#) on page 182 section.

Add Request: Adding POTS– GR-303 ESA Voice Connection

- 1 Create a POTS-GR303 voice connection without providing any end point information by enabling esaMode flag.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
    <RequestMode>online</RequestMode>
  </Request>
</RequestElement>
  <Attribute>
    <Name>Parent_Type</Name>
    <Value>SubscriberPort</Value>
  </Attribute>
  <Attribute>
    <Name>Parent_Id</Name>
    <Value>7</Value>
  </Attribute>
  <Attribute>
    <Name>Device_Id</Name>
    <Value>9</Value>
  </Attribute>
  <Attribute>
    <Name>Handle_Type</Name>
    <Value>SubVoiceConn</Value>
  </Attribute>
  <Attribute>
    <Name>connectionType</Name>
    <Value>6</Value>
  </Attribute>
  <Attribute>
    <Name>connectionDesc</Name>
    <Value>ESAPotsToGr303</Value>
  </Attribute>
  <Attribute>
    <Name>adminStatus</Name>
    <Value>2</Value>

```

```

    </Attribute>
    <Attribute>
      <Name>esaMode</Name>
      <Value>>true</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

- 2** Create the POTS end point, specifying the POTS-GR303 voice connection as its parent.

Refer to the [Add Request: Adding POTS– GR303 Non-ESA Voice Connection on page 182](#) section.

- 3** Create the GR-303 CRV.

Refer to the [Add Request: Adding POTS– GR303 Non-ESA Voice Connection on page 182](#) section.

- 4** Create the GR303 end point, specifying the newly created connection as its parent and igCrv as the crvIndex from step 3.

Refer to the [Add Request: Adding POTS– GR303 Non-ESA Voice Connection on page 182](#) section.

- 5** Create a POTS-VOIP voice connection. Note that for POTS–GR303 voice connections, endPointOneIndex will refer to POTS and endPointTwoIndex will refer to GR303 end points. However, for VOIP–POTS connections, endPointOneIndex will refer to VOIP and endPointTwoIndex will refer to POTS end points.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
    <RequestMode>online</RequestMode>
  </Request>

  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>SubscriberPort</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>7</Value>
    </Attribute>
    <Attribute>
      <Name>Device_Id</Name>
      <Value>9</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

```

</Attribute><Attribute>
  <Name>Handle_Type</Name>
  <Value>SubVoiceConn</Value>
</Attribute><Attribute>
  <Name>connectionType</Name>
  <Value>14</Value>
</Attribute>
<Attribute>
  <Name>connectionDesc</Name>
  <Value>ESAPotsToVoip</Value>
</Attribute><Attribute>
  <Name>adminStatus</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>esaMode</Name>
  <Value>true</Value>
</Attribute><Attribute>
  <Name>endPointTwoIndex</Name>
  <Value>818</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 6 Create the VOIP end point, specifying the POTS-VOIP connection as its parent.

Refer to the [Add Request: Adding POTS– VOIP Non-ESA Voice Connection on page 186](#) section.

- 7 Create the POTS end point, specifying the POTS-VOIP connection as its parent.

Refer to the [Add Request: Adding POTS– GR303 Non-ESA Voice Connection on page 182](#) section.

POTS– to– V5.2 Connections

To create POTS–to–V5.2 connection, perform the following tasks:

Using a Device_Id to find the Handle_Id of SubscriberPort

Using a Device_Id to find the Handle_Id of SubscriberPort.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>

```

```

<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>SubscriberPort</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=100</Value>
</Attribute>
</RequestElement>
</OSS>

```

Creating POTS – to – V5.2 subscriber voice connection

Create a POTS – to – V5.2 subscriber voice connection.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>SubscriberPort</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>SubVoiceConn</Value>
</Attribute>
<!--Always set to id to 1 -->
<Attribute>
    <Name>id</Name>
    <Value>1</Value>
</Attribute>
<!--
// Connection Types ....
// AAL2 to GR303: 1
// AAL2 to V52: 2

```

```

// VOIP to GR303: 3
// POTS to AAL2: 4
// ISDN to AAL2: 5
// POTS to GR303: 6
// POTS to V52: 7
// VOIP to DS1: 8
//SIP_TO_GR303 is not supported as of now on device.
// SIP to GR303: 9
// VOIP to V52: 10
// ELCP to V52: 11
// ISDN to V52: 12
// EBS to GR303: 13
// VOIP to POTS: 14
// ISDNSIG to VOIP: 15
// POTS to DSONE = 16
-->
<Attribute>
    <Name>connectionType</Name>
    <Value>14</Value>
</Attribute>
<Attribute>
    <Name>adminStatus</Name>
    <Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```



Note: Take a note of the Handle_Id of SubVoiceConn as it will be used as Parent_Id in the creation of POTS and V52 subscriber voice connection end points.

Creating a POTS end point

Create the POTS end point, specify the newly created connection as its parent:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>SubVoiceConn</Value>

```



```

</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>231</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>SubVoicePots_EndPoint</Value>
</Attribute>
<Attribute>
    <Name>lineGroupId</Name>
    <Value>457</Value>
</Attribute>
</RequestElement>
</OSS>

```

Creating a V5.2 end point

Create the V52 end point and specify the newly created connection as its parent.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>SubVoiceConn</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>231</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>SubVoiceV52_EndPoint</Value>
</Attribute>
<Attribute>
    <Name>name</Name>
    <Value>oss_v52_1</Value>
</Attribute>
<Attribute>
    <Name>igName</Name>
    <Value>oss_ig_1</Value>
</Attribute>

```

```

<Attribute>
    <Name>userPortId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>userType</Name>
    <Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

VOIP– to– GR-303 voice connections

Before creating VOIP–to–GR-303 voice connection, check the following list to make sure these items are already existed:

- VOIP server,
- VG host with specified external host address that will get the traffic of this card,
- Traffic descriptor with sufficient bandwidth,
- Interface group.

To create VOIP-to-GR-303 voice connection, perform the following tasks:

Using a Device_Id to find the Handle_Id of SubscriberPort

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>SubscriberPort</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=100</Value>
</Attribute>
</RequestElement>
</OSS>

```

Creating VOIP-to-GR303 subscriber voice connection

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Parent_Type</Name>
<Value>SubscriberPort</Value>
</Attribute>
<Attribute>
<Name>Parent_Id</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>Handle_Type</Name>
<Value>SubVoiceConn</Value>
</Attribute>
<!--Always set to id to 1 -->
<Attribute>
<Name>id</Name>
<Value>1</Value>
</Attribute>
<!--
// Connection Types ...
// AAL2 to GR303: 1
// AAL2 to V52: 2
// VOIP to GR303: 3
// POTS to AAL2: 4
// ISDN to AAL2: 5
// POTS to GR303: 6
// POTS to V52: 7
// VOIP to DS1: 8
//SIP_TO_GR303 is not supported as of now on device.
// SIP to GR303: 9
// VOIP to V52: 10
// ELCP to V52: 11
// ISDN to V52: 12
// EBS to GR303: 13
// VOIP to POTS: 14
// ISDNSIG to VOIP: 15
// POTS to DSONE = 16
-->

```

```

<Attribute>
  <Name>connectionType</Name>
  <Value>3</Value>
</Attribute>
<Attribute>
  <Name>adminStatus</Name>
  <Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```

Using a Device_Id to find ipIndex

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>IpLogical</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Device_Id=100</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Creating a VOIP end point

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
</OSS>

```

```

</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>SubVoiceConn</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>231</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>SubVoiceVoip_EndPoint</Value>
</Attribute>
<Attribute>
    <Name>name</Name>
    <Value>oss_voip_1</Value>
</Attribute>
<Attribute>
    <Name>userId</Name>
    <Value>test</Value>
</Attribute>
<Attribute>
    <Name>passWord</Name>
    <Value>zhone</Value>
</Attribute>
<Attribute>
    <Name>directoryNumber</Name>
    <Value>2345</Value>
</Attribute>
<Attribute>
    <Name>registrationServer</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>ipIfIndex</Name>
    <Value>444</Value>
</Attribute>
</RequestElement>
</OSS>

```

Retrieving Interface Group Name

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>

```

```

<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>Gr303Ig</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=100</Value>
</Attribute>
</RequestElement>
</OSS>

```

Creating a GR-303 end point

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>SubVoiceConn</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>231</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>SubVoiceGr303_EndPoint</Value>
</Attribute>
<Attribute>
    <Name>name</Name>
    <Value>oss_gr303_1</Value>
</Attribute>
<Attribute>
    <Name>igName</Name>
    <Value>oss_ig_2</Value>
</Attribute>
<Attribute>
    <Name>igCrv</Name>

```

```

        <Value>1</Value>
    </Attribute>
</RequestElement>
</OSS>

```

VOIP– to– V5.2 Connections

To create VOIP–to–V5.2 connection, perform the following tasks:

Using a Device_Id to find the Handle_Id of SubscriberPort

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>SubscriberPort</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Device_Id=100</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Creating VOIP to V5.2 voice connection

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>

```

```

<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>SubscriberPort</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>SubVoiceConn</Value>
</Attribute>
<!--Always set to id to 1 -->
<Attribute>
    <Name>id</Name>
    <Value>1</Value>
</Attribute>
<!--
// Connection Types ...
// AAL2 to GR303: 1
// AAL2 to V52: 2
// VOIP to GR303: 3
// POTS to AAL2: 4
// ISDN to AAL2: 5
// POTS to GR303: 6
// POTS to V52: 7
// VOIP to DS1: 8
//SIP_TO_GR303 is not supported as of now on device.
// SIP to GR303: 9
// VOIP to V52: 10
// ELCP to V52: 11
// ISDN to V52: 12
// EBS to GR303: 13
// VOIP to POTS: 14
// ISDNSIG to VOIP: 15
// POTS to DSONE = 16
-->
<Attribute>
    <Name>connectionType</Name>
    <Value>10</Value>
</Attribute>
<Attribute>
    <Name>adminStatus</Name>
    <Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```

Using a Device_Id to find ipIflIndex

```
<?xml version="1.0" encoding="UTF-8" ?>
```



```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>IpLogical</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Device_Id=100</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Creating a VOIP end point

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>SubVoiceConn</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>231</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>SubVoiceVoip_EndPoint</Value>
    </Attribute>
    <Attribute>
      <Name>name</Name>

```

```

        <Value>oss_voip_1</Value>
    </Attribute>
    <Attribute>
        <Name>userId</Name>
        <Value>test</Value>
    </Attribute>
    <Attribute>
        <Name>passWord</Name>
        <Value>zhone</Value>
    </Attribute>
    <Attribute>
        <Name>directoryNumber</Name>
        <Value>2345</Value>
    </Attribute>
    <Attribute>
        <Name>registrationServer</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>ipIfIndex</Name>
        <Value>444</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Creating a V5.2 end point

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.16.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>SubVoiceConn</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>231</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>SubVoiceV52_EndPoint</Value>
        </Attribute>
    </RequestElement>

```

```

<Attribute>
    <Name>name</Name>
    <Value>oss_v52_1</Value>
</Attribute>
<Attribute>
    <Name>igName</Name>
    <Value>oss_ig_1</Value>
</Attribute>
<Attribute>
    <Name>userPortId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>userType</Name>
    <Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

Add request: OMCI Profiles



Note: The OSS XML server does not support filter conditions on **name**, **value** and **description** attributes of **GponOnuGenProfileVariable** and **GponOnuSpecProfileVariable** objects because they are not stored in the DZMS database. If you need to perform this operation, OSS clients should retrieve the entire list of generic/specific variables and scan for the required variable.

Whenever a list operation is run on **ponOnuGenProfileVariable** and **GponOnuSpecProfileVariable** with including Filter_Retrieve's on attributes other than **name**, **value** and **description**, OSS will always return **parentMe** and **parentGeneric** attributes as part of the response.

To create OMCI profiles (Me profile, generic profile, specific profile), perform the following tasks:

Creating an OMCI ME profile

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>

```

```

<Version>1.16</Version>
</Request>
<RequestElement>
  <Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
  </Attribute>
  <Attribute>
    <Name>Parent_Id</Name>
    <Value>8</Value>
  </Attribute>
  <Attribute>
    <Name>Handle_Type</Name>
    <Value>GponOnuMeProfile</Value>
  </Attribute>
  <Attribute>
    <Name>name</Name>
    <Value>meosxml1</Value>
  </Attribute>
  <Attribute>
    <Name>fileName</Name>
    <Value>battery.txt</Value>
  </Attribute>
</RequestElement>
</OSS>

```

Creating an OMCI generic profile

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
osxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>8</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>

```

```

<Value>GponOnuGenProfile</Value>
</Attribute>
<Attribute>
<Name>name</Name>
<Value>genoss1</Value>
</Attribute>
<Attribute>
<Name>parentMe</Name>
<Value>meossxml1</Value>
</Attribute>
</RequestElement>
</OSS>

```

Creating an OMCI specific profile

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Parent_Type</Name>
<Value>GponOnuPhysical</Value>
</Attribute>
<Attribute>
<Name>Parent_Id</Name>
<Value>3750</Value>
</Attribute>
<Attribute>
<Name>Handle_Type</Name>
<Value>GponOnuSpecProfile</Value>
</Attribute>
<Attribute>
<Name>parentGeneric</Name>
<Value>specoss1</Value>
</Attribute>
</RequestElement>
</OSS>

```

Add request: GponTrafficProfile

```

<?xml version="1.0" encoding="UTF-8" ?>

```

```

- <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
- <Request>
  <RequestType>add</RequestType>
  <RequestMode>online</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>2.3</Version>
</Request>
- <RequestElement>
- <Attribute>
  <Name>Parent_Type</Name>
  <Value>Device</Value>
</Attribute>
- <Attribute>
  <Name>Parent_Id</Name>
  <Value>30</Value>
</Attribute>
- <Attribute>
  <Name>Handle_Type</Name>
  <Value>GponTrafficProfile</Value>
</Attribute>
- <Attribute>
  <Name>guaranteedUpBw</Name>
  <Value>512</Value>
</Attribute>
- <Attribute>
  <Name>trafficClass</Name>
  <Value>1</Value>
</Attribute>
- <Attribute>
  <Name>name</Name>
  <Value>test1</Value>
</Attribute>
</RequestElement>
</OSS>

```

Add request: MLPPP bond group

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.3.2</Version>
</Request>

```

```

<RequestElement>
<Attribute>
  <Name>Parent_Type</Name>
  <Value>Card</Value>
</Attribute>
<Attribute>
  <Name>Parent_Id</Name>
  <Value>12</Value>
</Attribute>
<Attribute>
  <Name>Handle_Type</Name>
  <Value>MLPPPBond_Physical</Value>
</Attribute>
<Attribute>
  <Name>ifAdminStatus</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>ifType</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>ifTypeExtension</Name>
  <Value>33</Value>
</Attribute>
<Attribute>
  <Name>isActive</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>ifTypePortId</Name>
  <Value>0</Value>
</Attribute>
<Attribute>
  <Name>name</Name>
  <Value>oss_mlppp_bg</Value>
</Attribute>
</RequestElement>
</OSS>

```

Add request: CPE manager

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<OSS xmlns="http://www.zhone.com/OSSXML">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
<RequestElement>
<Attribute>

```

```

<Name>Parent_Type</Name>
<Value>Physical</Value>
</Attribute>
<Attribute>
<Name>Parent_Id</Name>
<Value>43014</Value> <!-- Handle_Id of ONU -->
</Attribute>
<Attribute>
<Name>Handle_Type</Name>
<Value>PseudoDevice</Value> <!-- this is the internal
name for a cpe-mgr entry -->
</Attribute>
<Attribute>
<Name>cpePatPortsMapped</Name>
<!--
    FULL_COMPRESSION = 3;
    SOME_COMPRESSION = 4;
    NO_COMPRESSION   = 9;
-->
<Value>3</Value>
</Attribute>
<Attribute>
<Name>cpePatMappingType</Name>
<!--
    NORMAL = 3;
    SECURE = 4;
-->
<Value>4</Value>
</Attribute>
<Attribute>
<Name>cpeComm</Name>
<Value>ZhonePrivate</Value>
</Attribute>
<Attribute>
<Name>cpeIp</Name>
<Value>0.0.0.0</Value>
</Attribute>
<Attribute>
<Name>cpeGtp</Name>
<Value>1100000001</Value>
</Attribute>
<Attribute>
<Name>cpeMgrCfgMgrUsed</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>cpeGponPort</Name>
<Value>501</Value>
</Attribute>
</RequestElement>
</OSS>

```


Modify requests

Modify request: ATM AAL5 PVC

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>modify</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>Aal5VclAtmEndPoint</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>785</Value>
    </Attribute>
    <Attribute>
      <Name>name</Name>
      <Value>Aal5Vcl_modified</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

Modify request: ATM virtual circuit cross connect

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>modify</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.6</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>ATMVcCrossConn</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
```

```

        <Value>714</Value>
    </Attribute>
    <Attribute>
        <Name>name</Name>
        <Value>Atm-cross-connect</Value>
    </Attribute>
    <Attribute>
        <Name>vcCrossConnAdminStatus</Name>
        <Value>2</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Modify request: OMCI profiles

To modify OMCI generic profile and OMCI specific profile, perform the following procedures:



Note: Modification of ME profile's attribute list is not supported.



Note: The OSS XML server does not support filter conditions on **name**, **value** and **description** attributes of **GponOnuGenProfileVariable** and **GponOnuSpecProfileVariable** objects because they are not stored in the DZMS database. If you need to perform this operation, OSS clients should retrieve the entire list of generic/specific variables and scan for the required variable.

Whenever a list operation is run on **ponOnuGenProfileVariable** and **GponOnuSpecProfileVariable** with including Filter_Retrieve's on attributes other than **name**, **value** and **description**, OSS will always return **parentMe** and **parentGeneric** attributes as part of the response.

Modifying an OMCI generic profile

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>modify</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.16</Version>
    </Request>
</RequestElement>

```

```

<Attribute>
  <Name>Handle_Type</Name>
  <Value>GponOnuGenProfileVariable</Value>
</Attribute>
<Attribute>
  <Name>Handle_Id</Name>
  <Value>3001</Value>
</Attribute>
<Attribute>
  <Name>name</Name>
  <Value>dn1</Value>
</Attribute>
<Attribute>
  <Name>value</Name>
  <Value>1111111</Value>
</Attribute>
</RequestElement>
<RequestElement>
<Attribute>
  <Name>Handle_Type</Name>
  <Value>GponOnuGenProfileVariable</Value>
</Attribute>
<Attribute>
  <Name>Handle_Id</Name>
  <Value>3002</Value>
</Attribute>
<Attribute>
  <Name>name</Name>
  <Value>name1</Value>
</Attribute>
<Attribute>
  <Name>value</Name>
  <Value>nameForPort1</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modifying an OMCI specific profile

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16</Version>
</Request>
<RequestElement>

```

```

<Attribute>
  <Name>Handle_Type</Name>
  <Value>GponOnuSpecProfileVariable</Value>
</Attribute>
<Attribute>
  <Name>Handle_Id</Name>
  <Value>15012</Value>
</Attribute>
<Attribute>
  <Name>name</Name>
  <Value>proxyIp</Value>
</Attribute>
<Attribute>
  <Name>value</Name>
  <Value>4.5.6.7</Value>
</Attribute>
</RequestElement>
<RequestElement>
<Attribute>
  <Name>Handle_Type</Name>
  <Value>GponOnuSpecProfileVariable</Value>
</Attribute>
<Attribute>
  <Name>Handle_Id</Name>
  <Value>12002</Value>
</Attribute>
<Attribute>
  <Name>name</Name>
  <Value>name1</Value>
</Attribute>
<Attribute>
  <Name>value</Name>
  <Value>specNameForPort</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modifying OMCI ports of Specific profile using SNMP



Note: In case of Specific profile modification (non-SNMP) for ports admin up / down, 0 should be used for up and 1 for down. Values applicable for non-SNMP should be used if the Specific profile is being modified.

In case of using SNMP to make the ports of Specific profile as admin up / down, 1 should be used for up and 2 for down. Values are applicable for SNMP should be used for modifying OMCI ports of Specific profile using SNMP.

1 Get request: OMCI port status

```
<?xml version="1.0" encoding="UTF-8" ?>
```

```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>get</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>OmciPortStatus</Value>
    </Attribute>
    <!-- Represents the Handle_Id of parent ONU -->
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>2001</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>OmciMeId=11</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>OmciPortId=1</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Retrieve</Name>
      <Value>statusText</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

2 Modify request: OMCI port admin

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>modify</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>OmciPortAdmin</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

```

</Attribute>
<!-- Represents the Handle_ID of parent ONU -->
<Attribute>
    <Name>Handle_Id</Name>
    <Value>1199</Value>
</Attribute>
<Attribute>
    <Name>OmcId</Name>
    <Value>11</Value>
</Attribute>
<Attribute>
    <Name>OmcPortId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>adminState</Name>
    <Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modify request: Modifying GPON traffic profile of a GEM port using GTP template

GPON traffic profile of GEM port can be modified using OSS XML GW. You cannot modify a GPON traffic profile of a bridge and subscriber hosts at bridge and subscriber hosts level. Instead, you can modify the underneath GEM port. However, please be aware that, modifying GPON traffic profile of GEM port can effect the all the bridge services configured the top of that GEM port.

The attribute **trafficProfileTemplate** in the **GponPortPhysical** object specifies the GTP template name. The attribute **trafficProfileIdx** is still available in the **GponPortPhysical** object and it takes the GTP index directly. If **trafficProfileIdx** is included as part of OSS XML request and it is > 0, **trafficProfileTemplate** value will be ignored. On the other hand, if **trafficProfileIdx** is not specified or specified as 0 as part of OSS XML GEM port modify request, DZMS will check for the existance of matching Gpon Traffic Profile on device and auto-create the GTP if it does not exist already. **trafficProfileTemplate** has access-type as write-only and it is a string of 60 characters.

Following example illustrates the modification of GTP template of a GEM port.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>

```

```

<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>GponPortPhysical</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>641</Value>
</Attribute>
<Attribute>
    <Name>trafficProfileTemplate</Name>
    <Value>test2</Value>
</Attribute>
<Attribute>
    <Name>trafficProfileIdx</Name>
    <Value>0</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modify request: Modifying GPON traffic profile of a GEM port without GTP templates

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>GponPortPhysical</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>641</Value>
</Attribute>
<Attribute>
    <Name>trafficProfileIdx</Name>
    <Value>3</Value>
</Attribute>

```

```
</RequestElement>
</OSS>
```

Modify request: Running ONT apply command

With the addition of zhoneGponCmd table, MxK offers the ONT apply command that will effect the only modified elements of the specific profile. Apply operation on ONT elements can be run using OSS XML by performing modify request on newly introduced OmciOntApply object. OmciOntApply objects are not stored as part of the DZMS database and they take the Handle_Id of the parent ONU as their Handle_Id. In addition to the Handle_Type and Handle_Id attributes, it has only one attribute called apply to perform the ONT Apply operation. apply attribute must be set to 1 for successful completion of ONU apply.

Following example illustrates the procedure required for running ONT apply command using OSS XML client.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Handle_Type</Name>
<Value>OmciOntApply</Value>
</Attribute>
<Attribute>
<Name>Handle_Id</Name>
<Value>1199</Value>
</Attribute>
<Attribute>
<Name>apply</Name>
<Value>1</Value>
</Attribute>
</RequestElement>
</OSS>
```

Modify request: Running ONT resync command

With the addition of zhoneGponCmd table, MxK offers the ONT resync command that will resync the ONT without flipping the onuAdded flag. Resync operation on ONT elements can be run using OSS XML by

performing modify request on OmcOntResync object. OmcOntResync objects are not stored as part of the DZMS database and they take the Handle_Id of the parent ONU as their Handle_Id. In addition to the Handle_Type and Handle_Id attributes, it has only one attribute called resync to perform the ONT Resync operation. resync attribute must be set to 1 for successful completion of ONU resync.

Following example illustrates the procedure required for running ONT resync command using OSS XML client.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Handle_Type</Name>
<Value>OmcOntResync</Value>
</Attribute>
<Attribute>
<Name>Handle_Id</Name>
<Value>1199</Value>
</Attribute>
<Attribute>
<Name>resync</Name>
<Value>1</Value>
</Attribute>
</RequestElement>
</OSS>
```

Modify request: Running ONT reboot command

With the addition of zhoneGponCmd table, MxK offers the ONT reboot command that will reboot the ONT without flipping the onuAdded flag. Reboot operation on ONT elements can be run using OSS XML by performing modify request on OmcOntReboot object. OmcOntReboot objects are not stored as part of the DZMS database and they take the Handle_Id of the parent ONU as their Handle_Id. In addition to the Handle_Type and Handle_Id attributes, it has only one attribute called reboot to perform the ONT Reboot operation. reboot attribute must be set to 1 for successful completion of ONU reboot.

Following example illustrates the procedure required for running ONT reboot command using OSS XML client.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Handle_Type</Name>
<Value>OmciOntResync</Value>
</Attribute>
<Attribute>
<Name>Handle_Id</Name>
<Value>1199</Value>
</Attribute>
<Attribute>
<Name>reboot</Name>
<Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modify request: fullSerialNo and fullSerialNoHex

The serial number attributes fullSerialNo and fullSerialNoHex in the ONU profile will allow a get and set of a serial number in the “ZNTS12345” format.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Handle_Type</Name>
<Value>GponOnuPhysical</Value>
</Attribute>
<Attribute>
<Name>Handle_Id</Name>
<Value>2254</Value>

```

```

</Attribute>
<Attribute>
  <Name>fullSerialNoHex</Name>
  <Value>ZNTS98765</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modify request: the name of sub region two

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com
/OSSXML ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
  <Name>Handle_Type</Name>
  <Value>SubRegionTwo</Value>
</Attribute>
<Attribute>
  <Name>Handle_Id</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>name</Name>
  <Value>sub_2</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modify request: GponTrafficProfile

```

<?xml version="1.0" encoding="UTF-8" ?>
- <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
- <Request>
  <RequestType>modify</RequestType>
  <RequestMode>online</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>1.1.2</Version>
</Request>

```

```

- <RequestElement>
- <Attribute>
  <Name>Handle_Type</Name>
  <Value>GponTrafficProfile</Value>
</Attribute>
- <Attribute>
  <Name>Handle_Id</Name>
  <Value>72</Value>
</Attribute>
- <Attribute>
  <Name>compensated</Name>
  <Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modify request: MLPPP bond group

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">

<Request>

<RequestType>modify</RequestType>

<RequestMode>online</RequestMode>

<SessionID>sessionID</SessionID>

<operName>operName</operName>

<Version>2.3.2</Version>

</Request>
<RequestElement>
<Attribute>
  <Name>Handle_Type</Name>
  <Value>Physical</Value>
</Attribute>
<Attribute>
  <Name>Handle_Id</Name>
  <Value>974</Value>
</Attribute>
<Attribute>
  <Name>name</Name>
  <Value>oss_mlppp_bg6</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modify request: Alarms

```

    <?xml version="1.0" encoding="UTF-8" ?>
-   <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
-   <Request>
    <RequestType>modifyAlarm</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.1.2</Version>
    </Request>
-   <RequestElement>
-   <Attribute>
    <Name>Handle_Type</Name>
    <Value>AlarmEvent</Value>
    </Attribute>
-   <Attribute>
    <Name>Handle_Id</Name>
    <Value>28</Value>
    </Attribute>
-   <Attribute>
    <Name>assignTo</Name>
    <Value>admin</Value>
    </Attribute>
    </RequestElement>
  </OSS>

```

Delete requests

Delete request: ATM AAL5 VCL end point

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://www.zhone.com
/OSSXML ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>Aal5VclAtmEndPoint</Value>
</Attribute>
</Attribute>

```

```

        <Name>Handle_Id</Name>
        <Value>1628</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Delete request: ATM AAL5 PVC

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
    <Request>
        <RequestType>delete</RequestType>
        <RequestMode>offline</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.6</Version>
    </Request>
    <RequestElement>
        <Handle_Type>Aal5VclAtmEndPoint</Handle_Type>
        <Handle_Id>5160</Handle_Id>
    </RequestElement>
</OSS>

```

Delete request: ATM virtual circuit cross connect

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
    <Request>
        <RequestType>delete</RequestType>
        <RequestMode>offline</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.6</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>ATMVcCrossConn</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Id</Name>
            <Value>456</Value>
        </Attribute>
    </RequestElement>
</OSS>

```

Delete request: OMCI profiles



Note: The OSS XML server does not support filter conditions on **name**, **value** and **description** attributes of **GponOnuGenProfileVariable** and **GponOnuSpecProfileVariable** objects because they are not stored in the DZMS database. If you need to perform this operation, OSS clients should retrieve the entire list of generic/specific variables and scan for the required variable.

Whenever a list operation is run on **ponOnuGenProfileVariable** and **GponOnuSpecProfileVariable** with including Filter_Retrieve's on attributes other than **name**, **value** and **description**, OSS will always return **parentMe** and **parentGeneric** attributes as part of the response.



Note: Procedure for deletion of ME and generic profiles is identical to the deletion of specific profile. However, an ME profile can only be deleted if it does not have any descendant generic profiles on it. Similarly, a generic profile can only be deleted if it does not have any descendant specific profiles on it.

Deleting an OMCI specific profile

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>delete</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Handle_Type</Name>
<Value>GponOnuSpecProfile</Value>
</Attribute>
<Attribute>
<Name>Handle_Id</Name>
<Value>241</Value>
</Attribute>
</RequestElement>
</OSS>
```

Delete request: Subscriber Host



Note: Deletion of primary subscriber hosts from DZMS or OSS XML GW will remove underneath VCL and descendant dynamic and static subscriber hosts.



Note: Deletion of static or dynamic subscriber hosts from DZMS or OSS XML GW will not remove VCL.

Use this procedure to delete a primary subscriber host.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>delete</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>Ip_Logical</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>5643</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

Delete request: a sub region three

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com
/OSSXML ossxml.xsd">
  <Request>
    <RequestType>delete</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
```



```

<Attribute>
  <Name>Handle_Type</Name>
  <Value>SubRegionThree</Value>
</Attribute>
<Attribute>
  <Name>Handle_Id</Name>
  <Value>461</Value>
</Attribute>
</RequestElement>
</OSS>

```

Delete request: GponTrafficProfile

```

<?xml version="1.0" encoding="UTF-8" ?>
- <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
- <Request>
  <RequestType>delete</RequestType>
  <RequestMode>online</RequestMode>
  <SessionID>sessionID</SessionID>
  <operName>operName</operName>
  <Version>1.1.2</Version>
</Request>
- <RequestElement>
- <Attribute>
  <Name>Handle_Type</Name>
  <Value>GponTrafficProfile</Value>
</Attribute>
- <Attribute>
  <Name>Handle_Id</Name>
  <Value>71</Value>
</Attribute>
</RequestElement>
</OSS>

```

Delete request: MLPPP bond group

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">

<Request>

<RequestType>delete</RequestType>

<RequestMode>online</RequestMode>

<SessionID>sessionID</SessionID>

```

```

<operName>operName</operName>

<Version>2.3.2</Version>

</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>MLPPPBondPhysical</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>971</Value>
</Attribute>
</RequestElement>
</OSS>

```

Delete request: Alarms

```

<?xml version="1.0" encoding="UTF-8" ?>
- <OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
- <Request>
    <RequestType>deleteAlarm</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.1.2</Version>
</Request>
- <RequestElement>
- <Attribute>
    <Name>Handle_Type</Name>
    <Value>AlarmEvent</Value>
</Attribute>
- <Attribute>
    <Name>Handle_Id</Name>
    <Value>28</Value>
</Attribute>
</RequestElement>
</OSS>

```

Delete request: CPE manager

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>delete</RequestType>

```

```

<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.12</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>PseudoDevice</Value> <!-- this is the
internal name for a cpe-mgr entry -->
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>180</Value>
</Attribute>
</RequestElement>
</OSS>

```

Unified Service Provisioning OSS XML Examples

- [Unified Service Provisioning on GPON: Dynamic OMCI service requests, page 227](#)
- [Unified Service Provisioning on GPON: Residential Gateway \(RG\) Feature, page 250](#)
- [Unified Service Provisioning on ActiveE: Residential Gateway \(RG\) Feature, page 292](#)

Unified Service Provisioning on GPON: Dynamic OMCI service requests

This section describes how to use configuration requests to provision different services with dynamic OMCI.

By setting the **meProfile** of GponOnuPhysical (corresponds to gpon-olt-onu-config profile on device) to an internal ME profile name, the MXK will use the dynamic OMCI for service provisioning. In such scenarios, **genericProfile** attribute will NO longer be used for provisioning.

The internal ME profile is zNID type based. Each DZS supported OMCI zNID has its own internal ME profile, for example, zhone-2520. zhone-default is a universal ME profile, it can be applied to any zNIDs.

The following OSS XML code examples show how to provision different services by using the Dynamic OMCI CPE profiles:

- [Dynamic OMCI CPE profile provisioning for VoIP on page 228](#)
- [Dynamic OMCI CPE profile provisioning for PWE on page 241](#)
- [Dynamic OMCI CPE profile provisioning for IPTV on page 245](#)

- [Dynamic OMCI CPE profile provisioning for Data](#) on page 248

Dynamic OMCI CPE profile provisioning for VoIP

This section includes:

- [Creating the first VoIP service](#), page 228
- [Creating the second VoIP service \(minimum steps\)](#), page 240

Creating the first VoIP service

To create first VoIP service with dynamic OMCI CPE profiles, use the following steps:

- 1 Get the Handle_Id of device (referred as Device_Id in other profiles) by using device IP address.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>deviceIP=1.2.3.4</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

- 2 Get the Handle_Id and ifIndex of ONU by specifying Device_Id, slot ID, OLT ID and ONU ID.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
```

```

<operName>operName</operName>
<Version>1.16</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>GponOnuPhysical</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=304</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Card_Id_1=5</Value>
</Attribute>
<!-- specifies OLT Physical Id -->
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>ifTypePortId=1</Value>
</Attribute>
<!-- specifies ONU Physical Id -->
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Physical_id_2=1</Value>
</Attribute>
<Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>ifIndex</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 3 Set the appropriate internal ME profile name for modifying gpon-olt-onu-config profile.



Note: There is no need to set the genericProfile in case of dynamic OMCI profiles.



Note: For latest list of ME profiles supported on a device, perform a list operation on GponOnuMeProfile with setting filter condition to Device_Id.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>

```

```

<operName>operName</operName>
<Version>2.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>GponOnuPhysical</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>2254</Value>
</Attribute>
<!-- Must be set to one of the following :
zhone-cig
zhone-2648t
zhone-8524
zhone-2517
zhone-2516
zhone-2648p
zhone-2511
zhone-2543
zhone-2510
zhone-2424
zhone-1e
zhone-9480
zhone-2427

zhone-2426
zhone-2425
zhone-5120
zhone-2644a
zhone-2403
zhone-2402
zhone-8224
zhone-2624p
zhone-2301
zhone-9488
zhone-2624a
zhone-2510a
zhone-5114

zhone-2520
zhone-2628t
zhone-2628p
zhone-8424
zhone-2644p
zhone-7310
zhone-2504
zhone-2501
zhone-8324
zhone-2628a
zhone-default
zhone-2608t
zhone-2648a

```

```
-->
<Attribute>
  <Name>meProfile</Name>
  <Value>zhone-2510</Value>
</Attribute>
</RequestElement>
</OSS>
```

4 Create a VoIP bridge.

Create an MXK bridge for VoIP service. Procedure for provisioning bridges on MXK 2.2 devices is same as MXK 2.1.3 devices.

5 Create a CPE connection.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
  ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>6</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeConnection</Value>
    </Attribute>
    <!-- LowerIfIndex of corresponding bridge -->
    <!-- This value is same as the ifIndex of parent GEM
    port -->
    <Attribute>
      <Name>ifIndex</Name>
      <Value>885</Value>
    </Attribute>
    <!-- TpType can be one of the following values based on
    the service.
    VOIP Service :          h248VoiceTerminationPoint(13),

    sipVoiceTerminationPoint(14),

    sipPlarVoiceTerminationPoint(15)
    PWE Service :          pweTerminationPoint(16)
```

```

IPTV Service :
physicalPathTerminationPointIPTVUNI (12)

Ethernet Service :
physicalPathTerminationPointEthernetUNI (1)
-->
<Attribute>
    <Name>tpType</Name>
    <Value>13</Value>
</Attribute>
<Attribute>
    <Name>tpIndex</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>vlanId</Name>
    <Value>0</Value>
</Attribute>
<Attribute>
    <Name>slanId</Name>
    <Value>0</Value>
</Attribute>
<!-- corresponding bridge VLAN ID -->
<Attribute>
    <Name>vlanXlateId</Name>
    <Value>21</Value>
</Attribute>
<!-- corresponding bridge SLAN ID -->
<Attribute>
    <Name>slanXlateId</Name>
    <Value>22</Value>
</Attribute>
<Attribute>
    <Name>gponSlotId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponOltId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponOnuId</Name>
    <Value>8</Value>
</Attribute>
<Attribute>
    <Name>gponPortId</Name>
    <Value>907</Value>
</Attribute>
<Attribute>
    <Name>gponTrafficIdx</Name>
    <Value>3</Value>
</Attribute>
<Attribute>
    <Name>bridgeTemplateName</Name>

```



```

        <Value>DownlinkSingleTagged</Value>
    </Attribute>
</RequestElement>
</OSS>

```

6 Create cpe-voip-server profile.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>Device</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>11</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>DynOmciCpeVoipServer</Value>
        </Attribute>
        <Attribute>
            <Name>serverIndex</Name>
            <Value>2</Value>
        </Attribute>
        <Attribute>
            <Name>primaryServer</Name>
            <Value>172.168.3.1</Value>
        </Attribute>
        <Attribute>
            <Name>secondaryServer</Name>
            <Value>172.168.3.2</Value>
        </Attribute>
        <Attribute>
            <Name>profileName</Name>
            <Value>zhoneServer</Value>
        </Attribute>
        <!-- sip : 1, sipplar : 2, h248 : 3 -->
        <Attribute>
            <Name>sigProtocol</Name>
            <Value>3</Value>
        </Attribute>
    </RequestElement>

```

```
</OSS>
```

7 Create cpe-sip-diaplan profiles for the server (optional)

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>9</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeSipDialPlan</Value>
    </Attribute>
    <Attribute>
      <Name>serverIndex</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>dialPlanIndex</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>dialPlanString</Name>
      <Value>x.T|*x.T</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

8 Create cpe-sip-server profiles (optional)

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
```

```

<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>8</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeIpServer</Value>
</Attribute>
<Attribute>
    <Name>serverIndex</Name>
    <Value>2</Value>
</Attribute>
<Attribute>
    <Name>gateway</Name>
    <Value>172.168.3.254</Value>
</Attribute>
<Attribute>
    <Name>profileName</Name>
    <Value>gatewayServer1</Value>
</Attribute>
<Attribute>
    <Name>primaryDns</Name>
    <Value>172.168.19.1</Value>
</Attribute>
<Attribute>
    <Name>secondaryDns</Name>
    <Value>172.168.19.2</Value>
</Attribute>
<!-- dhcp: 1, static: 2 -->
<Attribute>
    <Name>hostIpOption</Name>
    <Value>2</Value>
</Attribute>
<Attribute>
    <Name>netMask</Name>
    <Value>255.255.255.0</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 9** Create cpe-ip profile for the VoIP service and associate it with a cpe-ip-server profile.

```
<?xml version="1.0" encoding="UTF-8" ?>
```

```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>GponOnuPhysical</Value>
</Attribute>
<!--Handle_Id of parent ONU physical -->
<Attribute>
    <Name>Parent_Id</Name>
    <Value>7792</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeIp</Value>
</Attribute>
<!--IfIndex of parent ONU physical -->
<Attribute>
    <Name>ifIndex</Name>
    <Value>375</Value>
</Attribute>
<!-- VOIP : 1, PWE : 2 -->
<Attribute>
    <Name>serviceType</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>adminState</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>hostIP</Name>
    <Value>172.168.12.3</Value>
</Attribute>
<!-- Index of parent cpe-ip-server profile -->
<!-- If not specified, this will take the index of
default IP server (1) -->
<Attribute>
    <Name>serverIndex</Name>
    <Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

10 Create cpe-voip-features profile (optional).



Note: The cpe-voip-features profile is only applicable for SIP or SIP PLAR VoIP server.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Parent_Type</Name>
<Value>Device</Value>
</Attribute>
<Attribute>
<Name>Parent_Id</Name>
<Value>3</Value>
</Attribute>
<Attribute>
<Name>Handle_Type</Name>
<Value>DynOmciCpeVoipFeatures</Value>
</Attribute>
<Attribute>
<Name>featuresIndex</Name>
<Value>2</Value>
</Attribute>
<Attribute>
<Name>cidFeatures</Name>
<Value>10</Value>
</Attribute>
<Attribute>
<Name>cpotFeatures</Name>
<Value>10</Value>
</Attribute>
<Attribute>
<Name>callWaitingFeatures</Name>
<Value>10</Value>
</Attribute>
<Attribute>
<Name>callPresentationFeatures</Name>
<Value>10</Value>
</Attribute>
<Attribute>
<Name>profileName</Name>
<Value>platinumService</Value>
</Attribute>
</RequestElement>
```

```
</OSS>
```

11 Create cpe-voip-media profile (optional)

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>11</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeVoipMedia</Value>
    </Attribute>
    <Attribute>
      <Name>mediaIndex</Name>
      <Value>2</Value>
    </Attribute>
    <Attribute>
      <Name>profileName</Name>
      <Value>platinumService</Value>
    </Attribute>
    <Attribute>
      <Name>echoCancel</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>faxMode</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>codecFirstOrder</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>periodFirstOrder</Name>
      <Value>10</Value>
    </Attribute>
  </RequestElement>

```

```

        <Name>suppressionFirstOrder</Name>
        <Value>1</Value>
    </Attribute>
</RequestElement>
</OSS>

```

12 Create cpe-voip-subscriber profile, and associate it with VoIP server, VoIP features, and media profiles.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>GponOnuPhysical</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>7792</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>DynOmciCpeVoipSubscriber</Value>
        </Attribute>
        <Attribute>
            <Name>ifIndex</Name>
            <Value>375</Value>
        </Attribute>
        <Attribute>
            <Name>portNumber</Name>
            <Value>2</Value>
        </Attribute>
        <!-- Next 3 attributes are only needed if the VOIP
server is of type SIP or SIP-PLAR. They will be
discarded in case of H.248 VOIP server. -->
        <Attribute>
            <Name>userName</Name>
            <Value>test4</Value>
        </Attribute>
        <Attribute>
            <Name>password</Name>
            <Value>zhone</Value>
        </Attribute>
        <Attribute>

```

```

        <Name>dialNumber</Name>
        <Value>6072375</Value>
    </Attribute>
    <Attribute>
        <Name>adminState</Name>
        <Value>1</Value>
    </Attribute>
    <!-- ohm600(1), ohm900(2), complex1(3), complex2(4),
    complex3(5) -->
    <Attribute>
        <Name>impedance</Name>
        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>rxGain</Name>
        <Value>3</Value>
    </Attribute>
    <Attribute>
        <Name>txGain</Name>
        <Value>3</Value>
    </Attribute>
    <!-- VOIP server profile index -->
    <Attribute>
        <Name>serverProfileIndex</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>featuresProfileIndex</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>mediaProfileIndex</Name>
        <Value>1</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Creating the second VoIP service (minimum steps)

To create the subsequent VoIP services, use the following minimal number of steps (First three steps will only be required if the provisioning is being done on a new ONU.):

- 1** Create a VoIP bridge (Skip this step if the second service is on the same ONU).
Create an MXK bridge for VoIP service.
- 2** Create a CPE connection (Skip this step if the second service is on the same ONU).

Refer to the Creating of first VoIP service section, [Create a CPE connection. on page 231](#). Note that the value of the TpType must be h248VoiceTerminationPoint(13) for H.248, sipVoiceTerminationPoint(14) for SIP, or sipPlarVoiceTerminationPoint(15) for SIP PLAR service. The value of the TpIndex must be 0 for VoIP service.

- 3 Create cpe-ip profile for the VoIP service and associate a cpe-ip-server profile with it. (Skip this step if the second service is on the same ONU).

Refer to the Creating the first VoIP service section, [Create cpe-ip profile for the VoIP service and associate it with a cpe-ip-server profile. on page 235](#).

- 4 Create a cpe-voip-subscriber profile, and associate it with a server, features and media profiles.

Refer to the Creating the first VoIP service section, [Create cpe-voip-subscriber profile, and associate it with VoIP server, VoIP features, and media profiles. on page 239](#).

Dynamic OMCI CPE profile provisioning for PWE

This section includes:

- [Creating the first PWE service, page 241](#)
- [Creating the second PWE service \(minimum steps\), page 244](#)

Creating the first PWE service

To create first PWE service with dynamic OMCI CPE profiles, use the following steps:

- 1 Create a PWE bridge.

Create an MXK bridge for PWE service.

- 2 Create a CPE connection

Refer to the Creating the first VoIP service section, [Create a CPE connection. on page 231](#). Note that the value of the TpType must be pweTerminationPoint(16) for PWE service. The value of the TpIndex must be 0 for PWE service.

- 3 Create cpe-ip-server profile (optional)

Refer to the Creating the first VoIP service section, [Create cpe-sip-server profiles \(optional\) on page 234](#).

- 4 Create cpe-ip profile for the PWE service and associate it with a cpe-ip-server profile. (optional)

Refer to the Creating the first VoIP service section, [Create cpe-ip profile for the VoIP service and associate it with a cpe-ip-server profile. on page 235](#).

Note that the serviceType of the cpe-ip profile must be 2 for PWE service.

5 Create a cpe-pwe profile (optional).

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>6</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpePwe</Value>
    </Attribute>
    <Attribute>
      <Name>profileName</Name>
      <Value>platinumService</Value>
    </Attribute>
    <!-- ds1(1), e1(2), other(3) -->
    <Attribute>
      <Name>lineType</Name>
      <Value>2</Value>
    </Attribute>
    <!-- b8zs(1), ami(2), hdb3(3), b3zs(4) -->
    <Attribute>
      <Name>encoding</Name>
      <Value>3</Value>
    </Attribute>
    <!-- unstructured(1), octetalignedunstruct(2),
    structured(3) -->
    <Attribute>
      <Name>serviceType</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>payloadSize</Name>
      <Value>250</Value>
    </Attribute>
    <Attribute>
```

```

        <Name>jitterBufMax</Name>
        <Value>80</Value>
    </Attribute>
    <Attribute>
        <Name>jitterBufDesired</Name>
        <Value>20</Value>
    </Attribute>
    <Attribute>
        <Name>pweIndex</Name>
        <Value>101</Value>
    </Attribute>
</RequestElement>
</OSS>

```

6 Create a cpe-pwe-subscriber profile, and associate it with a cpe-pwe profile. (optional)

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>GponOnuPhysical</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>7792</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>DynOmciCpePweSubscriber</Value>
        </Attribute>
        <Attribute>
            <Name>ifIndex</Name>
            <Value>375</Value>
        </Attribute>
        <!-- Represents the index port number -->
        <Attribute>
            <Name>portNumber</Name>
            <Value>3</Value>
        </Attribute>
        <!-- Represents the TCP/UDP near end port number -->
        <Attribute>
            <Name>nearEndPortNumber</Name>

```

```

        <Value>5001</Value>
    </Attribute>
    <Attribute>
        <Name>farEndIP</Name>
        <Value>172.168.23.4</Value>
    </Attribute>
    <!-- Represents the TCP/UDP far end port number -->
    <Attribute>
        <Name>farEndPortNumber</Name>
        <Value>6001</Value>
    </Attribute>
    <Attribute>
        <Name>adminState</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>lineLength</Name>
        <Value>0</Value>
    </Attribute>
    <!-- PWE Profile index -->
    <Attribute>
        <Name>profileIndex</Name>
        <Value>101</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Creating the second PWE service (minimum steps)

To create the subsequent PWE services, use the following minimal number of steps (First three steps will only be required if the provisioning is being done on a new ONU.):

- 1** Create a PWE bridge (Skip this step if the second service is on the same ONU).

Create an MXK bridge for PWE service.

- 2** Create a CPE connection (Skip this step if the second service is on the same ONU).

Refer to the Creating the first VoIP service section, [Create a CPE connection. on page 231](#). Note that the value of the TpType must be pweTerminationPoint(16) for PWE service. The value of the TpIndex must be 0 for PWE service.

- 3** Create cpe-ip profile for the PWE service and associate a cpe-ip-server profile with it. (Skip this step if the second service is on the same ONU).

Refer to the Creating the first VoIP service section, [Create cpe-ip profile for the VoIP service and associate it with a cpe-ip-server profile. on page 235](#).

Note that the serviceType of the cpe-ip profile must be 2 for PWE service.

- 4 Create a cpe-pwe-subscriber profile, and associate it with a cpe-pwe profile.

Refer to the Creating the first PWE service section, [Create a cpe-pwe-subscriber profile, and associate it with a cpe-pwe profile. \(optional\)](#) on page 243.

Dynamic OMCI CPE profile provisioning for IPTV

This section includes:

- [Creating the first IPTV service, page 245](#)
- [Creating the second IPTV service \(minimum steps\), page 248](#)

Creating the first IPTV service

To create first IP Video service with dynamic OMCI CPE profiles, use the following steps:

- 1 Create a bridge with video option.

Create an MXK bridge for IPTV service. Procedure for provisioning bridges on MXK 2.2 devices will be same as MXK 2.1.3 devices.

- 2 Create a CPE connection

Refer to the Creating the first VoIP service section, [Create a CPE connection. on page 231](#). Note that the value of the TpType must be physicalPathTerminationPointIPTVUNI(12) for IPTV service. The value of the TpIndex must be the physical ethernet port index for IPTV service.

- 3 Create cpe-video-access-control profile (optional)

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>11</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

```

<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeVideoAcl</Value>
</Attribute>
<Attribute>
    <Name>aclIndex</Name>
    <Value>2</Value>
</Attribute>
<Attribute>
    <Name>aceIndex</Name>
    <Value>3</Value>
</Attribute>
<Attribute>
    <Name>profileName</Name>
    <Value>platinumService</Value>
</Attribute>
<Attribute>
    <Name>srcIP</Name>
    <Value>172.168.35.4</Value>
</Attribute>
<Attribute>
    <Name>destStartIP</Name>
    <Value>172.168.15.21</Value>
</Attribute>
<Attribute>
    <Name>destEndIP</Name>
    <Value>172.168.15.100</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Create a cpe-video profile (optional).

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>6</Value>
</Attribute>

```

```

<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeVideo</Value>
</Attribute>
<Attribute>
    <Name>profileName</Name>
    <Value>platinumService</Value>
</Attribute>
<Attribute>
    <Name>videoIndex</Name>
    <Value>1</Value>
</Attribute>
<!-- As per the MxK 2.2, gemPort and vlan must always
be set to 4095 and 0 respectively -->
<Attribute>
    <Name>gemPort</Name>
    <Value>4095</Value>
</Attribute>
<Attribute>
    <Name>vlan</Name>
    <Value>0</Value>
</Attribute>
</RequestElement>
</OSS>

```

5 Create a cpe-eth-subscriber profile, and associate it with a cpe-video profile (optional).

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>GponOnuPhysical</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>7792</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeEthSubscriber</Value>
</Attribute>
<Attribute>

```

```

        <Name>ifIndex</Name>
        <Value>375</Value>
    </Attribute>
    <Attribute>
        <Name>portNumber</Name>
        <Value>3</Value>
    </Attribute>
    <Attribute>
        <Name>adminState</Name>
        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>videoProfileIndex</Name>
        <Value>1</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Creating the second IPTV service (minimum steps)

To create the subsequent IPTV services, use the following minimal number of steps (Steps will be same for creating the second IPTV service on the same or a new ONU.):

- 1 Create a bridge with video option.

Create an MXK bridge for IPTV service. Procedure for provisioning bridges on MXK 2.2 devices will be same as MXK 2.1.3 devices.

- 2 Create a CPE connection.

Refer to the Creating the first VoIP service section, [Create a CPE connection. on page 231](#). Note that the value of the TpType must be physicalPathTerminationPointIPTVUNI(12) for IPTV service. The value of the TpIndex must be the physical ethernet port index for IPTV service.

- 3 Create cpe-eth-subscriber profile, and associate it with a cpe-video profile. (optional)

Refer to the Creating the first IPTV service section. [Create a cpe-eth-subscriber profile, and associate it with a cpe-video profile \(optional\). on page 247](#)

Note: If changing the default video configurations and adding subscription restrictions are not needed, creation of an IPTV video bridge (step 1) and corresponding CPE connection (step 2) are sufficient for creating an IPTV service.

Dynamic OMCI CPE profile provisioning for Data

This section includes:

- [Creating the first Ethernet data service, page 249](#)
- [Creating the second Ethernet data service \(minimum steps\), page 249](#)

Creating the first Ethernet data service

To create first ethernet data service with dynamic OMCI CPE profiles, use the following steps:

1 Create a data bridge.

Create an MXK bridge for ethernet data service. Procedure for provisioning bridges on MXK 2.2 devices will be same as MXK 2.1.3 devices.

2 Create a CPE connection

Refer to the Creating the first VoIP service section section, [Create a CPE connection. on page 231](#). The value of the TpType must be physicalPathTerminationPointEthernetUNI(1) for ethernet data service. The value of the TpIndex must be the physical ethernet port index for ethernet data service.

3 Create cpe-eth-subscriber profile (optional)

For the creation of cpe-eth-subscriber profile refer to the creation of first IPTV service section, [Create a cpe-eth-subscriber profile, and associate it with a cpe-video profile \(optional\). on page 247](#). For data service, you don't need to associate the cpe-eth-subscriber profile to a cpe-video profile.

Creating the second Ethernet data service (minimum steps)

To create the subsequent Ethernet data services, use the following minimal number of steps (Steps will be same for creating the second ethernet data service on the same or a new ONU.):

1 Create a data bridge.

Create an MXK bridge for providing ethernet data service. Procedure for provisioning bridges on MXK 2.2 devices will be same as MXK 2.1.3 devices.

2 Create a CPE connection.

Refer to the Creating the first VoIP service section section, [Create a CPE connection. on page 231](#). Note that the value of the TpType must be physicalPathTerminationPointEthernetUNI(1) for ethernet data service. The value of the TpIndex must be the physical ethernet port index for ethernet data service.

3 Create cpe-eth-subscriber profile. (optional)

Refer to the Creating the first ethernet data service section. [Create cpe-eth-subscriber profile \(optional\) on page 249](#).

Note: If changing the default Ethernet physical configurations (rate, duplex, etc) are not needed, creation of data bridge (step 1) and CPE connection (step 2) are sufficient for creating a data service.

Unified Service Provisioning on GPON: Residential Gateway (RG) Feature

This section provides information on how to install and provision ActiveE zNIDs with Unified Service Provisioning RG Configuration Mode using OSS XML GW on the MXK.

The following sections provide the RG overview and OSS XML code examples that show how to provision different services by using the Residential Gateway (RG) enabled CPE:

- [*USP on GPON RG: RG Modes on page 251*](#)
- [*USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251*](#)
- [*USP on GPON RG: Residential Gateway enabled CPE provisioning for Video on page 257*](#)
- [*USP on GPON RG: Residential Gateway enabled CPE provisioning for Voice on page 259*](#)
- [*USP on GPON RG: Residential Gateway enabled CPE provisioning for TR-069 on page 262*](#)
- [*USP on GPON RG: Residential Gateway enabled CPE for Wireless Data service on page 265*](#)
- [*USP on GPON RG: Static Configuration on the WAN side interfaces \(without DHCP\) on page 271*](#)
- [*USP on GPON RG: Static configuration on the LAN side interfaces with a new DHCP server on page 275*](#)
- [*USP on GPON RG: Configuration of Static Routes on page 279*](#)
- [*USP on GPON RG: Configuration of Firewall on page 283*](#)
- [*USP on GPON RG: Configuration of Guided VLAN on page 288*](#)



Note: Even though OSS only creates CPE connection in order to create the MxK bridge, MxK and MX 1U 19x devices will automatically create the following profiles if the CPE connection is of type RG bridged or RG brouted.

- WAN side cpe-if-vlan
 - LAN side cpe-if-vlan
-



Note: If the RG mode is of type bridged PPPoE, in addition to the WAN and LAN side cpe-if-vlan profiles, device will also create cpe-pppoe profile.



Note: Users can not create or delete cpe-if-vlan and cpe-pppoe profiles explicitly. They are automatically created or deleted as part of creation and deletion of CPE connection profiles. DZMS receives a partial config sync notification and updates its database automatically to keep in sync with device.

USP on GPON RG: RG Modes

In case of OSS XML, RG mode can be set by setting rgMode to appropriate value while creating CPE connection.

Following rgMode values in OSS XML requests represent the corresponding RG connection types.

Table 24: rgMode values and corresponding RG connection types

rgMode	RG Connection Type
2	Brouted
4	Bridged PPPoE
5	Bridged
6	Not Applicable

Creation of RG enabled GPON CPE connections is very similar to Dynamic OMCI GPON CPE connection creation with the following exceptions:

- rgMode shall be set to brouted or bridged or bridged PPPOE. Not specifying the rgMode will make the CPE connection add request to be treated as dynamic OMCI type connection.
- TR-69 tpType option is only applicable for RG enabled CPE connections.

USP on GPON RG: Residential Gateway enabled CPE provisioning for Data

You can create Data service on RG enabled CPE with rg-brouted mode (preferred mode for Data), rg-bridged mode, or rg-bpppoe mode.

The following example uses rg-bpppoe mode. After specifying PPPoE as RG mode, the users must specify PPPoE user-ID and password too.

Creating an RG enabled data service using RG-bpppoe mode and GPON traffic profile template

Creation of RG enabled data service can be done using the following steps.

- 1 Creating a GPON traffic profile.

GPON traffic profile creation can be automatically taken care by DZMS if users create a Gpon Traffic Profile Template by specifying a template name for gtpTemplateName. Else, users can also explicitly create a GPON traffic profile (GTP) on device using CLI or DZMS and then provide the corresponding GTP index as gponTrafficIdx in OSS XML requests.

2 Specifying internal ME for ONU

Internal ME for an ONU can be automatically set by OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ONU, CPE connection add request will not override the internal ME on ONU. For overriding the internal ME, a separate modify operation can be performed on GponOnuPhysical object to modify meProfile value.

3 Creating an uplink MXK bridge

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>addUsingTemplate</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Template_Type</Name>
      <Value>BridgeLogicalTemplate</Value>
    </Attribute>
    <!-- Tempalte_Id of bridge templates can be retrieved
    by running list operation on BridgeLogicalTemplate
    object -->
    <Attribute>
      <Name>Template_Id</Name>
      <Value>198</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>BridgeLogical</Value>
    </Attribute>
```

```

<Attribute>
  <Name>vlanId</Name>
  <Value>200</Value>
</Attribute>
<!--lineGroupId of parent uplink ethernet physical -->
<Attribute>
  <Name>lineGroupId</Name>
  <Value>19</Value>
</Attribute>
<!--lineGroupId of parent uplink ethernet physical -->
<Attribute>
  <Name>lowerIfIndex</Name>
  <Value>19</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Create a downlink single-tagged bridge and a CPE connection in RG-bpppoe mode.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
  <Name>Parent_Type</Name>
  <Value>Device</Value>
</Attribute>
<Attribute>
  <Name>Parent_Id</Name>
  <Value>34</Value>
</Attribute>
<Attribute>
  <Name>Handle_Type</Name>
  <Value>DynOmciCpeConnection</Value>
</Attribute>
<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
  <Name>tpType</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>tpIndex</Name>

```

```

        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>vlanXlateId</Name>
        <Value>200</Value>
    </Attribute>
        <Name>gponSlotId</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>gponOltId</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>gponOnuId</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>gponPortId</Name>
        <Value>258</Value>
    </Attribute>
    <!--ZMS will check for the existence of matching GTP on
device and will create a new one if there is no
matching GTP and then continue with the CPE connection
creation -->
    <!--This request can also contain gponTrafficIdx
instead of gtpTemplateName. While gponTrafficIdx is
device specific, gtpTempalteName is ZMS specific -->
    <Attribute>
        <Name>gtpTemplateName</Name>
        <Value>Gold</Value>
    </Attribute>
    <!-- If ONU internal ME is not set, it will be set to
the following -->
    <!-- If already set, following set will be ignored -->
    <Attribute>
        <Name>internalMe</Name>
        <Value>zhone-2648a</Value>
    </Attribute>
    <Attribute>
        <Name>bridgeTemplateName</Name>
        <Value>DownlinkSingleTagged</Value>
    </Attribute>
    <!-- Brouted = 2, Bridged PPPOE = 4 and bridged = 5 -->
    <Attribute>
        <Name>rgMode</Name>
        <Value>4</Value>
    </Attribute>
</RequestElement>
</OSS>

```

5 Modify cpe-pppoe profile to contain proper user name and password:

- a** Perform a list operation on GponOnuPhysical to retrieve the ifIndex of ActiveE physical.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>GponOnuPhysical</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=34</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Card_Id_1=1</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>ifTypePortId=1</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Physical_Id_2=1</Value>
</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>ifIndex</Value>
</Attribute>
</RequestElement>
</OSS>
```

- b** Retrieve the Handle_Id of the cpe-pppoe profile given ONU ifIndex and Device_Id

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
```

```

<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>DynOmciCpePppoe</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=34</Value>
</Attribute>

<!-- ifIndex of the parent ONU -->

<Attribute>
    <Name>Filter_Condition</Name>
    <Value>ifIndex=50</Value>
</Attribute>
</RequestElement>
</OSS>

```

c Modify userName and password for the cpe-pppoe profile.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpePppoe</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>9</Value>
</Attribute>
<Attribute>
    <Name>userName</Name>
    <Value>test2</Value>
</Attribute>
<Attribute>
    <Name>password</Name>
    <Value>test2</Value>
</Attribute>

```



```
</RequestElement>
</OSS>
```

USP on GPON RG: Residential Gateway enabled CPE provisioning for Video

The following example creates Video service on RG enabled CPE with rg-bridged mode.

Creating an RG enabled video service using RG-bridged mode

Creation of RG enabled Video service can be done using the following steps.

1 Creating a GPON traffic profile.

GPON traffic profile creation can be automatically taken care by DZMS if users create a Gpon Traffic Profile Template by specifying a template name for gtpTemplateName. Else, users can also explicitly create a GPON traffic profile (GTP) on device using CLI or DZMS and then provide the corresponding GTP index as gponTrafficIdx in OSS XML requests.

2 Specifying internal ME for ONU

Internal ME for an ONU can be automatically set by OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ONU, CPE connection add request will not override the internal ME on ONU. For overriding the internal ME, a separate modify operation can be performed on GponOnuPhysical object to modify meProfile value.

3 Creating an uplink MXK bridge.

Use the example in the [USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251](#), and change the VLAN ID. This example uses VLAN 300.

4 Create a downlink single-tagged bridge and a CPE connection in RG-bridged mode.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
</RequestElement>
```

```

<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeConnection</Value>
</Attribute>
<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
    <Name>tpType</Name>
    <Value>12</Value>
</Attribute>
<Attribute>
    <Name>tpIndex</Name>
    <Value>2</Value>
</Attribute>
<!--vlanXlateId in the downlink bridge matches the
vlanId set in the uplink bridge.--!>
<Attribute>
    <Name>vlanXlateId</Name>
    <Value>300</Value>
</Attribute>
<Attribute>
    <Name>gponSlotId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponOltId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponOnuId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponPortId</Name>
    <Value>258</Value>
</Attribute>
<Attribute>
    <Name>gponTrafficIdx</Name>
    <Value>165</Value>
</Attribute>
<!-- If ONU internal ME is not set, it will be set to
the following -->
<!-- If already set, following set will be ignored -->
<Attribute>
    <Name>internalMe</Name>

```

```

        <Value>zhone-2426</Value>
    </Attribute>
    <Attribute>
        <Name>bridgeTemplateName</Name>
        <Value>DownlinkSingleTagged</Value>
    </Attribute>
    <!-- Brouted = 2, Bridged PPPOE = 4 and bridged = 5 -->
    <Attribute>
        <Name>rgMode</Name>
        <Value>5</Value>
    </Attribute>
    <!--set the Mcast Control List for passing video -->
    <Attribute>
        <Name>bridgeIfMcastControlList</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>bridgeIfMaxVideoStreams</Name>
        <Value>4</Value>
    </Attribute>
</RequestElement>
</OSS>

```

USP on GPON RG: Residential Gateway enabled CPE provisioning for Voice

The following example creates Voice service on RG enabled CPE with rg-bridged mode.

This section also describes how to assign static IP address and IP server on a WAN interface. Note that this step is optional.

Creating an RG enabled voice service using RG-bridged mode

Creation of RG enabled voice service can be done using the following steps.

1 Creating a GPON traffic profile.

GPON traffic profile creation can be automatically taken care by DZMS if users create a Gpon Traffic Profile Template by specifying a template name for gtpTemplateName. Else, users can also explicitly create a GPON traffic profile (GTP) on device using CLI or DZMS and then provide the corresponding GTP index as gponTrafficIdx in OSS XML requests.

2 Specifying internal ME for ONU

Internal ME for an ONU can be automatically set by OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ONU, CPE connection add request will not override the internal ME on ONU. For overriding the internal ME, a separate modify operation can be performed on GponOnuPhysical object to modify meProfile value.

3 Creating an uplink MXK bridge.

Use the example in the [USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251](#), and change the VLAN ID. This example uses VLAN 400.

4 Create a downlink single-tagged bridge and a CPE connection in RG-bridged mode.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeConnection</Value>
    </Attribute>
    <!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
    PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
    17, WLAN / DOT11 = 18, TR69 = 20 -->
    <Attribute>
      <Name>tpType</Name>
      <Value>14</Value>
    </Attribute>
    <!--vlanXlateId here matches the vlanId specified in
    uplink bridge--!>
    <Attribute>
      <Name>vlanXlateId</Name>
      <Value>400</Value>
    </Attribute>
```

```

<!--vlanId specifies the uni-vlan on the CPE
connection.--!>
<Attribute>
    <Name>vlanId</Name>
    <Value>300</Value>
</Attribute>
<Attribute>
    <Name>gponSlotId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponOltId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponOnuId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponPortId</Name>
    <Value>258</Value>
</Attribute>
<Attribute>
    <Name>gponTrafficIdx</Name>
    <Value>165</Value>
</Attribute>
<!-- If ONU internal ME is not set, it will be set to
the following -->
<!-- If already set, following set will be ignored -->
<Attribute>
    <Name>internalMe</Name>
    <Value>zhone-2426</Value>
</Attribute>
<Attribute>
    <Name>bridgeTemplateName</Name>
    <Value>DownlinkSingleTagged</Value>
</Attribute>
<!-- Brouted = 2, Bridged PPPOE = 4 and bridged = 5 -->
<Attribute>
    <Name>rgMode</Name>
    <Value>5</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 5 To perform the other voice related configuration, such as VoIP server profile, VoIP feature profile, VoIP media profile configuration etc., refer to [Dynamic OMCI CPE profile provisioning for VoIP on page 228](#).

USP on GPON RG: Residential Gateway enabled CPE provisioning for TR-069

The following example creates TR-069 service on RG enabled CPE with rg-bridged mode.

Creating an RG enabled TR-069 service using RG-bridged mode

Creation of RG enabled voice service can be done using the following steps.

1 Specifying internal ME for ActiveE port

Internal ME for an ActiveE port can be automatically set by OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ActiveE port, CPE connection add request will not override the internal ME on ActiveE port. For overriding the internal ME, a separate modify operation can be performed on GponOnuPhysical object to modify meProfile value.

2 Creating an uplink MXK bridge.

Use the example in the [USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251](#), and change the VLAN ID. This example uses VLAN 500.

3 Create a downlink single-tagged bridge and a CPE connection in RG-bridged mode.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeConnection</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

```

<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
<Attribute>
    <Name>tpType</Name>
    <Value>20</Value>
</Attribute>
<Attribute>
    <Name>gponSlotId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponOltId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponOnuId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>gponPortId</Name>
    <Value>258</Value>
</Attribute>
<Attribute>
    <Name>gponTrafficIdx</Name>
    <Value>165</Value>
</Attribute>
<!-- If ONU internal ME is not set, it will be set to
the following -->
<!-- If already set, following set will be ignored -->
<Attribute>
    <Name>internalMe</Name>
    <Value>zhone-2426</Value>
</Attribute>
<Attribute>
    <Name>bridgeTemplateName</Name>
    <Value>DownlinkSingleTagged</Value>
</Attribute>
<!--vlanXlateId here matches the vlanId specified in
uplink bridge--!>
<Attribute>
    <Name>vlanXlateId</Name>
    <Value>500</Value>
</Attribute>
<!-- Brouted = 2, Bridged PPPOE = 4 and bridged = 5 -->
<Attribute>
    <Name>rgMode</Name>
    <Value>5</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Create cpe-system-common profile using OSS XML GW.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Parent_Type</Name>
<Value>Device</Value>
</Attribute>
<Attribute>
<Name>Parent_Id</Name>
<Value>34</Value>
</Attribute>
<Attribute>
<Name>Handle_Type</Name>
<Value>DynOmciCpeSystemCommon</Value>
</Attribute>
<Attribute>
<Name>commonIndex</Name>
<Value>12</Value>
</Attribute>
<Attribute>
<Name>profileName</Name>
<Value>sc12</Value>
</Attribute>
<Attribute>
<Name>acsUserName</Name>
<Value>test11</Value>
</Attribute>
<Attribute>
<Name>acsPassword</Name>
<Value>zhone</Value>
</Attribute>
<Attribute>
<Name>acsUrl</Name>
<Value>www.zhone.com</Value>
</Attribute>
<Attribute>
<Name>staticProfileIndex</Name>
<Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 5** Create a cpe-system profile on a specific ONU with setting the cpe-system-common index created earlier.


```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.
zhone.com/OSSXML ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Parent_Type</Name>
<Value>GponOnuPhysical</Value>
</Attribute>
<!-- Handle_Id of parent ONU -->
<Attribute>
<Name>Parent_Id</Name>
<Value>34</Value>
</Attribute>
<Attribute>
<Name>Handle_Type</Name>
<Value>DynOmciCpeSystem</Value>
</Attribute>
<Attribute>
<Name>commonIndex</Name>
<Value>12</Value>
</Attribute>
</RequestElement>
</OSS>

```

USP on GPON RG: Residential Gateway enabled CPE for Wireless Data service

The following example creates wireless Data service on RG enabled CPE with rg-brouted mode.

Creating an RG enabled Wireless Data service using RG-brouted mode

Creation of RG enabled wireless data service can be done using the following steps.

1 Creating a GPON traffic profile.

GPON traffic profile creation can be automatically taken care by DZMS if users provide create a Gpon Traffic Profile Template by specifying a template name for gtpTemplateName. Else, users can also explicitly create a GPON traffic profile (GTP) on device using CLI or DZMS and then provide the corresponding GTP index as gponTrafficIdx in OSS XML requests.

2 Specifying internal ME for ONU

Internal ME for an ONU can be automatically set using OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ONU, CPE connection add request will not override the internal ME on ONU. For overriding the internal ME, a separate modify operation can be performed on GponOnuPhysical object to modify meProfile value.

3 Creating an uplink MXK bridge.

Use the example in the [USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251](#), and change the VLAN ID. This example uses VLAN 600.

4 Create a downlink single-tagged bridge and a CPE connection in RG-brouted mode.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeConnection</Value>
    </Attribute>
    <!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
    PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
    17, WLAN / DOT11 = 18, TR69 = 20 -->
    <Attribute>
      <Name>tpType</Name>
      <Value>18</Value>
    </Attribute>
    <Attribute>
      <Name>gponSlotId</Name>
      <Value>1</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

```

        <Name>gponOltId</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>gponOnuId</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>gponPortId</Name>
        <Value>258</Value>
    </Attribute>
    <Attribute>
        <Name>gponTrafficIdx</Name>
        <Value>165</Value>
    </Attribute>
    <!-- If ONU internal ME is not set, it will be set to
    the following -->
    <!-- If already set, following set will be ignored -->
    <Attribute>
        <Name>internalMe</Name>
        <Value>zhone-2426</Value>
    </Attribute>
    <Attribute>
        <Name>bridgeTemplateName</Name>
        <Value>DownlinkSingleTagged</Value>
    </Attribute>
    <!--vlanXlateId here matches the vlanId specified in
    uplink bridge.--!>
    <Attribute>
        <Name>vlanXlateId</Name>
        <Value>600</Value>
    </Attribute>
    <!-- Brouted = 2, Bridged PPPOE = 4 and bridged = 5 -->
    <Attribute>
        <Name>rgMode</Name>
        <Value>2</Value>
    </Attribute>
</RequestElement>
</OSS>

```

5 Create CPE WLAN subscriber profile and associate it with a CPE WLAN common profile or CPE WLAN common advance profile (Optional).

By default, the admin-state of the CPE Wireless LAN (WLAN) UNI port is up after creation of CPE connection on that CPE WLAN UNI port. Because of that, the Data and Video traffic can run on this WLAN UNI port without further configuration.

If you want to change the default WLAN physical configurations, you can create a CPE WLAN subscriber profile manually. The WLAN UNI port ID specified in this command must match the one assigned when you creating downlink bridge and CPE connection.

a Create WLAN common profile

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Handle_Type</Name>
<Value>DynOmciCpeWlan</Value>
</Attribute>
<Attribute>
<Name>Parent_Type</Name>
<Value>Device</Value>
</Attribute>
<Attribute>
<Name>Parent_Id</Name>
<Value>34</Value>
</Attribute>
<Attribute>
<Name>profileName</Name>
<Value>wlan1</Value>
</Attribute>
<Attribute>
<Name>wlanIndex</Name>
<Value>4</Value>
</Attribute>
<Attribute>
<Name>maxClients</Name>
<Value>4</Value>
</Attribute>
<Attribute>
<Name>radiusServerIp</Name>
<Value>192.168.1.5</Value>
</Attribute>
<Attribute>
<Name>radiusPort</Name>
<Value>1025</Value>
</Attribute>
</RequestElement>
</OSS>

```

b Create WLAN advanced profile

```

<?xml version="1.0" encoding="UTF-8" ?>

```

```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeWlanAdv</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>profileName</Name>
      <Value>wlan1</Value>
    </Attribute>
    <!--wlanIndex here represents WLAN Adv profile
    index.--!>
    <Attribute>
      <Name>wlanIndex</Name>
      <Value>5</Value>
    </Attribute>
    <Attribute>
      <Name>channel</Name>
      <Value>2</Value>
    </Attribute>
    <Attribute>
      <Name>globalMaxClients</Name>
      <Value>10</Value>
    </Attribute>
    <!-- default = 1, all = 2, rateIn2 = 3, stdrates = 4
    -->
    <Attribute>
      <Name>basicRate</Name>
      <Value>2</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

- c Create WLAN subscriber profile and associate it with ONU. Assign the default CPE WLAN common profile and CPE WLAN common advanced profile to WLAN port 1.



Note: The CPE WLAN common advance profile could be used only for WLAN port 1 (i.e. subscriberIndex 1).

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeWlanSubscriber</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>GponOnuPhysical</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>14678</Value>
    </Attribute>
    <Attribute>
      <Name>subscriberIndex</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>ssid</Name>
      <Value>zhone-1</Value>
    </Attribute>
    <Attribute>
      <Name>encryptKey</Name>
      <Value>234567890</Value>
    </Attribute>
    <Attribute>
      <Name>radiusKey</Name>
      <Value>1025</Value>
    </Attribute>
    <Attribute>
      <Name>wlanProfileIndex</Name>
      <Value>4</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

```

        <Name>wlanAdvProfileIndex</Name>
        <Value>5</Value>
    </Attribute>
    <Attribute>
        <Name>internalMe</Name>
        <Value>zhone-2426</Value>
    </Attribute>
</RequestElement>
</OSS>

```

USP on GPON RG: Static Configuration on the WAN side interfaces (without DHCP)

This section describes how to assign a static IP address to the WAN side interfaces on a zNID.

Assigning a Static IP address to the WAN side interface on the zNID

This example is the continuance of the triple-play services that created in the previous sections.

To assign a static IP address to the WAN side interface, use the following procedure:

- 1 Create services on one ONT (using system defaults).

Refer to [USP on GPON RG: Residential Gateway enabled CPE provisioning for Voice on page 259](#).

- 2 Prepare system wide static IP configuration.

If you want to assign static IP address to the zNID, you need to create a static IP common profile on WAN interface and assign the static IP address to the WAN interface.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.4</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>Device</Value>
        </Attribute>
    </RequestElement>

```

```

        <Name>Parent_Id</Name>
        <Value>34</Value>
    </Attribute>
    <Attribute>
        <Name>Handle_Type</Name>
        <Value>DynOmciCpeIpServer</Value>
    </Attribute>
    <Attribute>
        <Name>serverIndex</Name>
        <Value>6</Value>
    </Attribute>
    <Attribute>
        <Name>gateway</Name>
        <Value>192.16.45.254</Value>
    </Attribute>
    <Attribute>
        <Name>profileName</Name>
        <Value>static-ip-config-1</Value>
    </Attribute>
    <Attribute>
        <Name>netMask</Name>
        <Value>255.255.255.0</Value>
    </Attribute>
    <Attribute>
        <Name>primaryDns</Name>
        <Value>192.16.45.1</Value>
    </Attribute>
    <!-- 1: dynamic, 2: static -->
    <Attribute>
        <Name>hostIpOption</Name>
        <Value>2</Value>
    </Attribute>
</RequestElement>
</OSS>

```

3 Perform a list operation on GponOnuPhysical to retrieve the ifIndex of ONU physical.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>list</RequestType>
        <RequestMode>offline</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.4</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Filter_Type</Name>
            <Value>GponOnuPhysical</Value>

```



```

</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Device_Id=34</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Card_Id_1=1</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>ifTypePortId=1</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Physical_Id_2=1</Value>
</Attribute>
<Attribute>
  <Name>Filter_Retrieve</Name>
  <Value>ifIndex</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 4** Perform a list operation on DynOmciCpeIfVlan to retrieve the WAN side cpe-if-vlan profile given ONU ifIndex and bridge vlan.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>DynOmciCpeIfVlan</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Device_Id=34</Value>
    </Attribute>
    <!--ONU ifIndex -->
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>ifIndex=50</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

```

<!-- vlanXlateId here matches the vlanId specified in
the CPE connection for UniVlan-->
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>vlanXlateId=300</Value>
</Attribute>
<!-- Retrieves WAN side cpe-if-vlan -->
<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>tpType=17</Value>
</Attribute>
</RequestElement>
</OSS>

```

5 Perform the modify operation on DynOmciCpeIfVlan to set the static IP address and IP server on WAN interface.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeIfVlan</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>4229</Value>
</Attribute>
<Attribute>
    <Name>vlanIP</Name>
    <Value>192.168.10.200</Value>
</Attribute>
<Attribute>
    <Name>ipServerProfile</Name>
    <Value>6</Value>
</Attribute>
</RequestElement>
</OSS>

```

USP on GPON RG: Static configuration on the LAN side interfaces with a new DHCP server

This example is the continuance of the triple services that created in the previous sections.

You can set LAN side IP and DHCP server if it is in rg-brouted or rg-bpppoe.

This section configures the following settings on the Brouted LAN interfaces those has same VLAN:

- Assigns a static IP address as the local IP address of the LAN interfaces
- Changes the IP address range for the DHCP server
- Changes the firewall access settings on the LAN interfaces

Assigning a Static IP address to the LAN side interface on the ONU

Assign a Static IP address, a DHCP server, and an IP server to the LAN side interface on the zNID.

- 1 Create services on one ONT (using system defaults).

Refer to [USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251](#).

- 2 Create CPE DHCP server.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeDhcpServer</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

```

<Attribute>
    <Name>serverIndex</Name>
    <Value>2</Value>
</Attribute>
<Attribute>
    <Name>startAddress</Name>
    <Value>192.168.10.50</Value>
</Attribute>
<Attribute>
    <Name>profileName</Name>
    <Value>dhcp1</Value>
</Attribute>
<Attribute>
    <Name>endAddress</Name>
    <Value>192.168.10.254</Value>
</Attribute>
</RequestElement>
</OSS>

```

3 Create IP common profile for Data service.

If you want to assign static IP address to the zNID, you need to create a static IP common profile on LAN interface and assign the static IP address to the LAN interface.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeIpServer</Value>
</Attribute>
<Attribute>
    <Name>serverIndex</Name>
    <Value>5</Value>
</Attribute>

```

```

<Attribute>
  <Name>profileName</Name>
  <Value>HSI-LAN</Value>
</Attribute>
<Attribute>
  <Name>netMask</Name>
  <Value>255.255.255.0</Value>
</Attribute>
<!-- 1:dynamic, 2: static -->
<Attribute>
  <Name>host-ip-option</Name>
  <Value>2</Value>
</Attribute>
<!--Bits values for firewallAccess: 0:http, 1:ping,
2:snmp, 3:snmptrap, 4:ssh, 5:telnet. 11111 means
allow all six protocols on this interface--!>
<Attribute>
  <Name>firewallAccess</Name>
  <Value>11111</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Perform a list operation on GponOnuPhysical to retrieve the ifIndex of ONU physical.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
  <Name>Filter_Type</Name>
  <Value>GponOnuPhysical</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Device_Id=34</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Card_Id_1=1</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>ifTypePortId=1</Value>

```

```

</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Physical_Id_2=1</Value>
</Attribute>
<Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>ifIndex</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 5** Perform a list operation on DynOmciCpeIfVlan to retrieve the HandleId of the LAN side cpe-if-vlan profile with the given ONU ifIndex and bridge vlan.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>DynOmciCpeIfVlan</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=34</Value>
</Attribute>
<!--ONU ifIndex -->
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>ifIndex=50</Value>
</Attribute>
<!-- Bridge VLAN -->
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>vlanXlateId=200</Value>
</Attribute>
<!-- Retrieves LAN side cpe-if-vlan -->
<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
<Attribute>
    <Name>Filter_Condition</Name>

```

```

        <Value>tpType=1</Value>
    </Attribute>
</RequestElement>
</OSS>

```

- 6** Perform the modify operation on DynOmciCpeIfVlan to set the DHCP server, static IP address and IP server on the LAN interface.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>modify</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.4</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>DynOmciCpeIfVlan</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Id</Name>
            <Value>4229</Value>
        </Attribute>
        <Attribute>
            <Name>dhcpServerProfile</Name>
            <Value>2</Value>
        </Attribute>
        <Attribute>
            <Name>vlanIP</Name>
            <Value>192.168.10.200</Value>
        </Attribute>
        <Attribute>
            <Name>ipServerProfile</Name>
            <Value>5</Value>
        </Attribute>
    </RequestElement>
</OSS>

```

USP on GPON RG: Configuration of Static Routes

To add static routes to the zNID, user need to create a static route list, and then associate this static route with a CPE system common profile, at last apply this system common profile to the zNID.

Adding a Static Route to the ONU

Adding a static route to the ONU can be done using the following steps.

1 Adding a static route list.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <!-- Handle_Id of parent ONU -->
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeStaticRouteList</Value>
    </Attribute>
    <Attribute>
      <Name>listIndex</Name>
      <Value>3</Value>
    </Attribute>
    <Attribute>
      <Name>profileName</Name>
      <Value>test3</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

2 Adding a static route.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>

```



```

<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<!-- Handle_Id of parent device -->
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeStaticRoute</Value>
</Attribute>
<Attribute>
    <Name>listIndex</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>entryIndex</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>destinationIp</Name>
    <Value>10.2.1.0</Value>
</Attribute>
<Attribute>
    <Name>metric</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>netmask</Name>
    <Value>255.255.255.0</Value>
</Attribute>
<Attribute>
    <Name>gateway</Name>
    <Value>10.1.1.253</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 3** Associate this static route with the CPE system common profile. You can either modify an existing CPE system common profile or create a new CPE system common profile. This example creates a new one.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>

```

```

<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeSystemCommon</Value>
</Attribute>
<Attribute>
    <Name>commonIndex</Name>
    <Value>13</Value>
</Attribute>
<Attribute>
    <Name>profileName</Name>
    <Value>mmr</Value>
</Attribute>
</Attribute>
<!--Index of shared static route list profile index -->
<Attribute>
    <Name>staticProfileIndex</Name>
    <Value>2</Value>
</Attribute>
<!--the vlanRouting is disabled by default. This
example enables the crossing VLAN routing on the
static route. If "enabled" is selected for cross VLAN
routing, routing between VLANs is allowed. Route table
lookups ignore the VLAN ID of the ingress and egress
ports. If there is a match, the packet is routed out
the interface specified in the Route table, regardless
of which VLAN it is a member of.enabled=1,
disabled=2--!>
<Attribute>
    <Name>vlanRouting</Name>
    <Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Apply static route to an ONU.

If this ONU does not have system common profile assigned to it, you can create a cpe-system profile on this ONU with the cpe-system-common index created earlier.

```

<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.

```

```

zhone.com/OSSXML ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
      <Name>Parent_Type</Name>
      <Value>GponOnuPhysical</Value>
</Attribute>
<!-- Handle_Id of parent ONU -->
<Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
</Attribute>
<Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeSystem</Value>
</Attribute>
<Attribute>
      <Name>commonIndex</Name>
      <Value>13</Value>
</Attribute>
</RequestElement>
</OSS>

```

USP on GPON RG: Configuration of Firewall

User can enable or disable firewall on the CPE. Enabling firewall can protect the CPE from unwanted instruction. When firewall is enabled, incoming connections can still be selectively allowed through firewall access and port forwarding settings. The firewall is enabled by default.

Retrieving firewall value

Use the CPE system common profile to retrieve firewall settings. By default, the firewall is enabled.

List example for CPE system common for retrieving firewall value:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>

```

```

<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>DynOmciCpeSystemCommon</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=34</Value>
</Attribute>
<Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>firewall</Value>
</Attribute>
</RequestElement>
</OSS>

```

Configuring firewall access

Firewall access control manages the protocols allowed on the CPE WAN or LAN interfaces. It requires firewall feature to be enabled.

The protocols are listed below:

- HTTP: Web Browser Traffic.
- PING: ICMP Echoes used to test for connectivity.
- SNMP: Simple Network Management Protocol.
- SNMPTRAP: Alarms for Simple Network Management Protocol.
- SSH: Secure Shell.
- TELNET: Remote Terminal support.

By default, those protocol are all enabled on the CPE LAN interfaces. Only PING is enabled on the CPE WAN interfaces.



Note: Modifying a CPE IP Common profile will trigger a partial reconfiguration on each of the CPEs that depend on this profile and may cause service interruptions on these CPEs.

To modify the default firewall access settings on WAN or LAN interfaces, use the following procedure:

- 1 Make sure the CPE has the firewall setting enabled.
- 2 Create an IP common profile with the new firewall access settings:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>

```

```

<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeIpServer</Value>
</Attribute>
<Attribute>
    <Name>serverIndex</Name>
    <Value>4</Value>
</Attribute>
<Attribute>
    <Name>gateway</Name>
    <Value>1.2.3.254</Value>
</Attribute>
<Attribute>
    <Name>profileName</Name>
    <Value>test1</Value>
</Attribute>
<Attribute>
    <Name>netMask</Name>
    <Value>255.255.255.0</Value>
</Attribute>
<!--http(0), ping(1), snmp (2), snmptrap (3), ssh(4),
telnet(5)-->
<Attribute>
    <Name>firewallAccess</Name>
    <Value>110000</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 3** Apply the new firewall access settings to the WAN interface:
 - a** Perform a list operation on GponOnuPhysical to retrieve the ifIndex of ONU physical.
 - b** Perform a list operation on DynOmciCpeIfVlan to retrieve the Handle_Id of the WAN side cpe-if-vlan profile by given ONU ifIndex and bridge VLAN.
 - c** Perform the modify operation on DynOmciCpeIfVlan to modify the IP common profile index to 4.

Similar examples are provided in [Assigning a Static IP address to the WAN side interface on the zNID on page 271](#).

- 4** Apply the new firewall access settings to the LAN interface:
 - a** Perform a list operation on GponOnuPhysical to retrieve the ifIndex of ONU physical.
 - b** Perform a list operation on DynOmciCpeIfVlan to retrieve the Handle_Id of the LAN side cpe-if-vlan profile by given ONU ifIndex and bridge VLAN.
 - c** Perform the modify operation on DynOmciCpeIfVlan to modify the IP common profile index to 4.

Similar examples are provided in [Assigning a Static IP address to the LAN side interface on the ONU on page 275](#).

Configuring port forwarding

The CPE port forwarding list reflects the existing port forwarding rules. The port forwarding list only can be created on a WAN interface that has NAT or NAPT enabled. By default, there is no port-forwarding list associated to the WAN interface.

Port forwarding list requires firewall feature to be enabled.

To configure port forwarding on the WAN interfaces, use the following procedure:

- 1** Make sure the CPE has the firewall setting enabled.
- 2** Create a port forwarding list and add one entry into it:

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <!-- Handle_Id of parent ONU -->
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
  </Attribute>
</Attribute>
```

```

        <Name>Handle_Type</Name>
        <Value>DynOmciCpePortForward</Value>
    </Attribute>
    <Attribute>
        <Name>listIndex</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>entryIndex</Name>
        <Value>1</Value>
    </Attribute>
    <!-- dmz (1), portrange (2), portremap (3) -->
    <Attribute>
        <Name>fwdType</Name>
        <Value>2</Value>
    </Attribute>
    <Attribute>
        <Name>portStart</Name>
        <Value>5500</Value>
    </Attribute>
    <Attribute>
        <Name>portEnd</Name>
        <Value>5500</Value>
    </Attribute>
    <!-- none (1), tcp (2), udp (3), tcpudp (4), icmp (5),
    icmpv4 (6) -->
    <Attribute>
        <Name>protocol</Name>
        <Value>3</Value>
    </Attribute>
    <Attribute>
        <Name>privateIp</Name>
        <Value>192.168.10.2</Value>
    </Attribute>
</RequestElement>
</OSS>

```

3 Apply the port forwarding to the WAN interface:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>modify</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.4</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Handle_Type</Name>

```

```

        <Value>DynOmciCpeIfVlan</Value>
    </Attribute>
    <Attribute>
        <Name>Handle_Id</Name>
        <Value>23</Value>
    </Attribute>
    <!-- Represents the index of shared cpe-port-fwd
    profile -->
    <Attribute>
        <Name>portFwdListProfile</Name>
        <Value>1</Value>
    </Attribute>
</RequestElement>
</OSS>

```

USP on GPON RG: Configuration of Guided VLAN

Guided VLAN is only applicable for RG type CPE connections. There is no guided VLAN support for dynamic OMCI type (i.e. RG Mode = Not Applicable) CPE connections.

A CPE connection is considered to be using guided VLANs if it satisfies the following criteria:

- CPE connection is RG enabled (i.e. RG mode is set to Routed, Bridged PPPoE, or Bridged).
- VLAN ID and SLAN ID of LAN side DynOmciCpeIfVlan (cpe-if-vlan) is 0.
- VLAN ID and Translate VLAN ID of parent CPE connection are different.



Note: Guided SLAN is not supported at this point.

Creating a Data Service Specific CPE connection with setting guided VLAN

Creation of RG enabled CPE connections with enabling guided VLAN is very similar to uni-vlan based RG CPE connections.

1 Creating a GPON traffic profile.

GPON traffic profile creation can be automatically taken care by DZMS if users provide create a Gpon Traffic Profile Template by specifying a template name for gtpTemplateName. Else, users can also explicitly create a GPON traffic profile (GTP) on device using CLI or DZMS and then provide the corresponding GTP index as gponTrafficIdx in OSS XML requests.

2 Specifying internal ME for ONU

Internal ME for an ONU can be automatically set using OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ONU, CPE connection add request will not override the internal ME on ONU. For overriding the internal ME, a separate modify operation can be performed on GponOnuPhysical object to modify meProfile value.

3 Creating an uplink MXK bridge.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>addUsingTemplate</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Template_Type</Name>
      <Value>BridgeLogicalTemplate</Value>
    </Attribute>
    <!-- Tempalte_Id of bridge templates can be retrieved
    by running list operation on BridgeLogicalTemplate
    object -->
    <Attribute>
      <Name>Template_Id</Name>
      <Value>198</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>BridgeLogical</Value>
    </Attribute>
    <Attribute>
      <Name>vlanId</Name>
      <Value>200</Value>
    </Attribute>
    <!--lineGroupId of parent uplink ethernet physical -->
    <Attribute>
      <Name>lineGroupId</Name>
      <Value>19</Value>
    </Attribute>
```

```

<!--lineGroupId of parent uplink ethernet physical -->
<Attribute>
    <Name>lowerIfIndex</Name>
    <Value>19</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Create a downlink single-tagged bridge and a CPE connection in RG-bridged mode.



Note: For creating CPE connection in RG brouted and bridge PPPoE modes, set rgMode to 2 and 4 respectively in the previous example.



Note: Procedure for creating CPE connection for voice and video services with guided VLAN is analogous to creating services with uni-VLAN and uni-SLAN with the exception that guided VLAN will be set instead of uni-VLAN and uni-SLAN.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeConnection</Value>
</Attribute>
<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
    <Name>tpType</Name>
    <Value>1</Value>
</Attribute>

```

```

<Attribute>
  <Name>tpIndex</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>guidedVlan</Name>
  <Value>203</Value>
</Attribute>
<Attribute>
  <Name>vlanXlateId</Name>
  <Value>200</Value>
</Attribute>
<Attribute>
  <Name>gponSlotId</Name>
  <Value>3</Value>
</Attribute>
<Attribute>
  <Name>gponOltId</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>gponOnuId</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>gponPortId</Name>
  <Value>258</Value>
</Attribute>
<Attribute>
  <Name>gponTrafficIdx</Name>
  <Value>165</Value>
</Attribute>
<Attribute>
  <Name>bridgeTemplateName</Name>
  <Value>DownlinkSingleTagged</Value>
</Attribute>
<!-- Brouted is 2, Bridged Pppoe is 4 and bridged is 5
-->
<!-- If ONU internal ME is not set, it will be set to
the following -->
<!-- If already set, following set will be ignored -->
<Attribute>
  <Name>internalMe</Name>
  <Value>zhone-2426</Value>
</Attribute>
<Attribute>
  <Name>rgMode</Name>
  <Value>5</Value>
</Attribute>
</RequestElement>
</OSS>

```

Unified Service Provisioning on ActiveE: Residential Gateway (RG) Feature

This section provides information on how to install and provision ActiveE zNIDs with Unified Service Provisioning RG Configuration Mode using OSS XML GW on the MXK.

The following sections provide the RG overview and OSS XML code examples that show how to provision different services by using the Residential Gateway (RG) enabled CPE:

- [*USP on AE: RG Modes on page 293*](#)
- [*USP on AE: Residential Gateway enabled CPE provisioning for Data on page 293*](#)
- [*USP on AE: Residential Gateway enabled CPE provisioning for Video on page 298*](#)
- [*USP on AE: Residential Gateway enabled CPE provisioning for Voice on page 300*](#)
- [*USP on AE: Residential Gateway enabled CPE provisioning for TR-069 on page 302*](#)
- [*USP on AE: Residential Gateway enabled CPE for Wireless Data service on page 306*](#)
- [*USP on AE: Static Configuration on the WAN side interfaces \(without DHCP\) on page 311*](#)
- [*USP on AE: Static configuration on the LAN side interfaces with a new DHCP server on page 315*](#)
- [*USP on AE: Configuration of Static Routes on page 319*](#)
- [*USP on AE: Configuration of Firewall on page 323*](#)
- [*USP on AE: Configuration of Guided VLAN on page 328*](#)



Note: Even though OSS only creates CPE connection in order to create the MxK bridge, MxK and MX 1U 19x devices will automatically create the following profiles if the CPE connection is of type RG bridged or RG brouted.

- WAN side cpe-if-vlan
 - LAN side cpe-if-vlan
-



Note: If the RG mode is of type bridged PPPoE, in addition to the WAN and LAN side cpe-if-vlan profiles, device will also create cpe-pppoe profile.



Note: Users can not create or delete cpe-if-vlan and cpe-pppoe profiles explicitly. They are automatically created or deleted as part of creation and deletion of CPE connection profiles. DZMS receives a partial config sync notification and updates its database automatically to keep in sync with device.

USP on AE: RG Modes

In case of OSS XML, RG mode can be set by setting rgMode to appropriate value while creating CPE connection.

Following rgMode values in OSS XML requests represent the corresponding RG connection types.

Table 25: rgMode values and corresponding RG connection types

rgMode	RG Connection Type
2	Bridged Routed
4	Bridged PPPoE
5	Bridged
6	Not Applicable

Creation of RG enabled ActiveE CPE connections is very similar to Dynamic OMCI GPON CPE connection creation with the following exceptions:

- rgMode shall be set to brouted or bridged or bridged PPPOE. Not specifying the rgMode will make the CPE connection add request to be treated as dynamic OMCI type connection.
- TR-69 tpType option is only applicable for RG enabled CPE connections.

USP on AE: Residential Gateway enabled CPE provisioning for Data

You can create Data service on RG enabled CPE with rg-brouted mode (preferred mode for Data), rg-bridged mode, or rg-bpppoe mode.

The following example uses rg-bpppoe mode. After specifying PPPoE as RG mode, the users must specify PPPoE user-ID and password too.

Creating an RG enabled data service using RG-bpppoe mode

Creation of RG enabled data service can be done using the following steps.

- 1 Specifying internal ME for ActiveE port

Internal ME for an ActiveE port can be automatically set by OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ActiveE port, CPE connection add request will not override the internal ME on ActiveE port. For overriding the internal ME, a separate modify operation can be performed on EtherPhysical object to modify meProfile value.

2 Creating an uplink MXK bridge

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>addUsingTemplate</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Template_Type</Name>
      <Value>BridgeLogicalTemplate</Value>
    </Attribute>
    <!-- Tempalte_Id of bridge templates can be retrieved
    by running list operation on BridgeLogicalTemplate
    object -->
    <Attribute>
      <Name>Template_Id</Name>
      <Value>198</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>BridgeLogical</Value>
    </Attribute>
    <Attribute>
      <Name>vlanId</Name>
      <Value>200</Value>
    </Attribute>
    <!--lineGroupId of parent uplink ethernet physical -->
    <Attribute>
      <Name>lineGroupId</Name>
      <Value>19</Value>
    </Attribute>
```

```

<!--lineGroupId of parent uplink ethernet physical -->
<Attribute>
  <Name>lowerIfIndex</Name>
  <Value>19</Value>
</Attribute>
</RequestElement>
</OSS>

```

3 Create a downlink single-tagged bridge and a CPE connection in RG-bpppoe mode.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
  <Name>Parent_Type</Name>
  <Value>Device</Value>
</Attribute>
<Attribute>
  <Name>Parent_Id</Name>
  <Value>34</Value>
</Attribute>
<Attribute>
  <Name>Handle_Type</Name>
  <Value>DynOmciCpeConnection</Value>
</Attribute>
<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
  <Name>tpType</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>tpIndex</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>vlanXlateId</Name>
  <Value>200</Value>
<Attribute>
  <Name>phySlotId</Name>
  <Value>1</Value>
</Attribute>

```

```

<Attribute>
    <Name>phyPortId</Name>
    <Value>1</Value>
</Attribute>
<!-- If ActiveE port internal ME is not set, it will be
set to
the following -->
<!-- If already set, following set will be ignored -->
<Attribute>
    <Name>internalMe</Name>
    <Value>zhone-2648a</Value>
</Attribute>
<Attribute>
    <Name>bridgeTemplateName</Name>
    <Value>DownlinkSingleTagged</Value>
</Attribute>
<!-- Brouted = 2, Bridged PPPOE = 4 and bridged = 5 -->
<Attribute>
    <Name>rgMode</Name>
    <Value>4</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Modify cpe-pppoe profile to contain proper user name and password:

- a** Perform a list operation on EtherPhysical to retrieve the ifIndex of ActiveE physical.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>EtherPhysical</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=34</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Card_Id_1=1</Value>
</Attribute>

```



```

<Attribute>
  <Name>Filter_Condition</Name>
  <Value>ifTypePortId=1</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Physical_Id_2=1</Value>
</Attribute>
<Attribute>
  <Name>Filter_Retrieve</Name>
  <Value>ifIndex</Value>
</Attribute>
</RequestElement>
</OSS>

```

b Retrieve the Handle_Id of the cpe-pppoe profile given ActiveE port ifIndex and Device_Id

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
  <Name>Filter_Type</Name>
  <Value>DynOmciCpePppoe</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Device_Id=34</Value>
</Attribute>

<!-- ifIndex of the parent ActiveE port -->

<Attribute>
  <Name>Filter_Condition</Name>
  <Value>ifIndex=50</Value>
</Attribute>
</RequestElement>
</OSS>

```

c Modify userName and password for the cpe-pppoe profile.

```

<?xml version="1.0" encoding="UTF-8" ?>

```

```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>modify</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpePppoe</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>9</Value>
    </Attribute>
    <Attribute>
      <Name>userName</Name>
      <Value>test2</Value>
    </Attribute>
    <Attribute>
      <Name>password</Name>
      <Value>test2</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

USP on AE: Residential Gateway enabled CPE provisioning for Video

The following example creates Video service on RG enabled CPE with rg-bridged mode.

Creating an RG enabled video service using RG-bridged mode

Creation of RG enabled Video service can be done using the following steps.

1 Specifying internal ME for ActiveE port

Internal ME for an ActiveE port can be automatically set by OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ActiveE port, CPE connection add request will not override the internal ME on ActiveE port. For overriding the internal ME, a separate modify operation can be performed on EtherPhysical object to modify meProfile value.

2 Creating an uplink MXK bridge.

Use the example in the [USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251](#), and change the VLAN ID. This example uses VLAN 300.

3 Create a downlink single-tagged bridge and a CPE connection in RG-bridged mode.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeConnection</Value>
    </Attribute>
    <!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
    PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
    17, WLAN / DOT11 = 18, TR69 = 20 -->
    <Attribute>
      <Name>tpType</Name>
      <Value>12</Value>
    </Attribute>
    <Attribute>
      <Name>tpIndex</Name>
      <Value>2</Value>
    </Attribute>
    <!--vlanXlateId in the downlink bridge matches the
    vlanId set in the uplink bridge.--!>
    <Attribute>
      <Name>vlanXlateId</Name>
      <Value>300</Value>
    </Attribute>
    <Attribute>
      <Name>phySlotId</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
```

```

        <Name>phyPortId</Name>
        <Value>1</Value>
    </Attribute>
    <!-- If ActiveE port internal ME is not set, it will be
    set to the following -->
    <!-- If already set, following set will be ignored -->
    <Attribute>
        <Name>internalMe</Name>
        <Value>zhone-2648a</Value>
    </Attribute>
    <Attribute>
        <Name>bridgeTemplateName</Name>
        <Value>DownlinkSingleTagged</Value>
    </Attribute>
    <!-- Brouted = 2, Bridged PPPOE = 4 and bridged = 5 -->
    <Attribute>
        <Name>rgMode</Name>
        <Value>5</Value>
    </Attribute>
    <!--set the Mcast Control List for passing video -->
    <Attribute>
        <Name>bridgeIfMcastControlList</Name>
        <Value>1</Value>
    </Attribute>
    <Attribute>
        <Name>bridgeIfMaxVideoStreams</Name>
        <Value>4</Value>
    </Attribute>
</RequestElement>
</OSS>

```

USP on AE: Residential Gateway enabled CPE provisioning for Voice

The following example creates Voice service on RG enabled CPE with rg-bridged mode.

This section also describes how to assign static IP address and IP server on a WAN interface. Note that this step is optional.

Creating an RG enabled voice service using RG-bridged mode

Creation of RG enabled voice service can be done using the following steps.

1 Specifying internal ME for ActiveE port

Internal ME for an ActiveE port can be automatically set by OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ActiveE port, CPE connection add request will not override the internal ME on ActiveE port. For overriding the internal ME, a separate modify operation can be performed on EtherPhysical object to modify meProfile value.

2 Creating an uplink MXK bridge.

Use the example in the [USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251](#), and change the VLAN ID. This example uses VLAN 400.

3 Create a downlink single-tagged bridge and a CPE connection in RG-bridged mode.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeConnection</Value>
    </Attribute>
    <!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
    PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
    17, WLAN / DOT11 = 18, TR69 = 20 -->
    <Attribute>
      <Name>tpType</Name>
      <Value>14</Value>
    </Attribute>
    <!--vlanXlateId here matches the vlanId specified in
    uplink bridge--!>
    <Attribute>
      <Name>vlanXlateId</Name>
      <Value>400</Value>
    </Attribute>
    <!--vlanId specifies the uni-vlan on the CPE
    connection.--!>
    <Attribute>
      <Name>vlanId</Name>
      <Value>300</Value>
    </Attribute>
    <Attribute>
      <Name>phySlotId</Name>
      <Value>1</Value>
```

```

</Attribute>
<Attribute>
    <Name>phyPortId</Name>
    <Value>1</Value>
</Attribute>
<!-- If ActiveE port internal ME is not set, it will be
set to the following -->
<!-- If already set, following set will be ignored -->
<Attribute>
    <Name>internalMe</Name>
    <Value>zhone-2648a</Value>
</Attribute>
<Attribute>
    <Name>bridgeTemplateName</Name>
    <Value>DownlinkSingleTagged</Value>
</Attribute>
<!-- Brouted = 2, Bridged PPPOE = 4 and bridged = 5 -->
<Attribute>
    <Name>rgMode</Name>
    <Value>5</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 4 To perform the other voice related configuration, such as VoIP server profile, VoIP feature profile, VoIP media profile configuration etc., refer to [Dynamic OMCI CPE profile provisioning for VoIP on page 228](#).

USP on AE: Residential Gateway enabled CPE provisioning for TR-069

The following example creates TR-069 service on RG enabled CPE with rg-bridged mode.

Creating an RG enabled TR-069 service using RG-bridged mode

Creation of RG enabled voice service can be done using the following steps.

1 Specifying internal ME for ActiveE port

Internal ME for an ActiveE port can be automatically set by OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ActiveE port, CPE connection add request will not override the internal ME on ActiveE port. For overriding the internal ME, a separate modify operation can be performed on EtherPhysical object to modify meProfile value.

2 Creating an uplink MXK bridge.

Use the example in the [USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251](#), and change the VLAN ID. This example uses VLAN 500.

3 Create a downlink single-tagged bridge and a CPE connection in RG-bridged mode.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeConnection</Value>
    </Attribute>
    <!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
    PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
    17, WLAN / DOT11 = 18, TR69 = 20 -->
    <Attribute>
    <Attribute>
      <Name>tpType</Name>
      <Value>20</Value>
    </Attribute>
    <Attribute>
      <Name>phySlotId</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>phyPortId</Name>
      <Value>1</Value>
    </Attribute>
    <!-- If ActiveE port internal ME is not set, it will be
    set to the following -->
    <!-- If already set, following set will be ignored -->
    <Attribute>
      <Name>internalMe</Name>
      <Value>zhone-2648a</Value>
    </Attribute>
    <Attribute>
      <Name>bridgeTemplateName</Name>
      <Value>DownlinkSingleTagged</Value>
```

```

</Attribute>
<!--vlanXlateId here matches the vlanId specified in
uplink bridge--!>
<Attribute>
    <Name>vlanXlateId</Name>
    <Value>500</Value>
</Attribute>
<!-- Brouted = 2, Bridged PPPOE = 4 and bridged = 5 -->
<Attribute>
    <Name>rgMode</Name>
    <Value>5</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Create cpe-system-common profile using OSS XML GW.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeSystemCommon</Value>
</Attribute>
<Attribute>
    <Name>commonIndex</Name>
    <Value>12</Value>
</Attribute>
<Attribute>
    <Name>profileName</Name>
    <Value>sc12</Value>
</Attribute>
<Attribute>
    <Name>acsUserName</Name>
    <Value>test11</Value>
</Attribute>

```



```

<Attribute>
  <Name>acsPassword</Name>
  <Value>zhone</Value>
</Attribute>
<Attribute>
  <Name>acsUrl</Name>
  <Value>www.zhone.com</Value>
</Attribute>
<Attribute>
  <Name>staticProfileIndex</Name>
  <Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```

5 Create a cpe-system profile on a specific ActiveE port with setting the cpe-system-common index created earlier.

```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.
zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>EtherPhysical</Value>
    </Attribute>
    <!-- Handle_Id of parent ActiveE port -->
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeSystem</Value>
    </Attribute>
    <Attribute>
      <Name>commonIndex</Name>
      <Value>12</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

USP on AE: Residential Gateway enabled CPE for Wireless Data service

The following example creates wireless Data service on RG enabled CPE with rg-brouted mode.

Creating an RG enabled Wireless Data service using RG-brouted mode

Creation of RG enabled wireless data service can be done using the following steps.

1 Specifying internal ME for ActiveE port

Internal ME for an ActiveE port can be automatically set using OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ActiveE port, CPE connection add request will not override the internal ME on ActiveE port. For overriding the internal ME, a separate modify operation can be performed on EtherPhysical object to modify meProfile value.

2 Creating an uplink MXK bridge.

Use the example in the [USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251](#), and change the VLAN ID. This example uses VLAN 600.

3 Create a downlink single-tagged bridge and a CPE connection in RG-brouted mode.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmcicpeConnection</Value>
```

```

</Attribute>
<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
    <Name>tpType</Name>
    <Value>18</Value>
</Attribute>
<Attribute>
    <Name>phySlotId</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>phyPortId</Name>
    <Value>1</Value>
</Attribute>
<!-- If ActiveE port internal ME is not set, it will be
set to the following -->
<!-- If already set, following set will be ignored -->
<Attribute>
    <Name>internalMe</Name>
    <Value>zhone-2648a</Value>
</Attribute>
<Attribute>
    <Name>bridgeTemplateName</Name>
    <Value>DownlinkSingleTagged</Value>
</Attribute>
<!--vlanXlateId here matches the vlanId specified in
uplink bridge.--!>
<Attribute>
    <Name>vlanXlateId</Name>
    <Value>600</Value>
</Attribute>
<!-- Brouted = 2, Bridged PPPOE = 4 and bridged = 5 -->
<Attribute>
    <Name>rgMode</Name>
    <Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Create CPE WLAN subscriber profile and associate it with a CPE WLAN common profile or CPE WLAN common advance profile (Optional).

By default, the admin-state of the CPE Wireless LAN (WLAN) UNI port is up after creation of CPE connection on that CPE WLAN UNI port. Because of that, the Data and Video traffic can run on this WLAN UNI port without further configuration.

If you want to change the default WLAN physical configurations, you can create a CPE WLAN subscriber profile manually. The WLAN UNI port ID specified in this command must match the one assigned when you creating downlink bridge and CPE connection.

a Create WLAN common profile

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhong.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeWlan</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>profileName</Name>
      <Value>wlan1</Value>
    </Attribute>
    <Attribute>
      <Name>wlanIndex</Name>
      <Value>4</Value>
    </Attribute>
    <Attribute>
      <Name>maxClients</Name>
      <Value>4</Value>
    </Attribute>
    <Attribute>
      <Name>radiusServerIp</Name>
      <Value>192.168.1.5</Value>
    </Attribute>
    <Attribute>
      <Name>radiusPort</Name>
      <Value>1025</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

b Create WLAN advanced profile

```

<?xml version="1.0" encoding="UTF-8" ?>

```

```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeWlanAdv</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>profileName</Name>
      <Value>wlan1</Value>
    </Attribute>
    <!--wlanIndex here represents WLAN Adv profile
    index.--!>
    <Attribute>
      <Name>wlanIndex</Name>
      <Value>5</Value>
    </Attribute>
    <Attribute>
      <Name>channel</Name>
      <Value>2</Value>
    </Attribute>
    <Attribute>
      <Name>globalMaxClients</Name>
      <Value>10</Value>
    </Attribute>
    <!-- default = 1, all = 2, rateIn2 = 3, stdrates = 4
    -->
    <Attribute>
      <Name>basicRate</Name>
      <Value>2</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

- c Create WLAN subscriber profile and associate it with ActiveE port. Assign the default CPE WLAN common profile and CPE WLAN common advanced profile to WLAN port 1.



Note: The CPE WLAN common advance profile could be used only for WLAN port 1 (i.e. subscriberIndex 1).

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://
www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeWlanSubscriber</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>EtherPhysical</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>14678</Value>
    </Attribute>
    <Attribute>
      <Name>subscriberIndex</Name>
      <Value>1</Value>
    </Attribute>
    <Attribute>
      <Name>ssid</Name>
      <Value>zhone-1</Value>
    </Attribute>
    <Attribute>
      <Name>encryptKey</Name>
      <Value>234567890</Value>
    </Attribute>
    <Attribute>
      <Name>radiusKey</Name>
      <Value>1025</Value>
    </Attribute>
    <Attribute>
      <Name>wlanProfileIndex</Name>
      <Value>4</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

```

        <Name>wlanAdvProfileIndex</Name>
        <Value>5</Value>
    </Attribute>
    <Attribute>
        <Name>internalMe</Name>
        <Value>zhone-2648a</Value>
    </Attribute>
</RequestElement>
</OSS>

```

USP on AE: Static Configuration on the WAN side interfaces (without DHCP)

This section describes how to assign a static IP address to the WAN side interfaces on a zNID.

Assigning a Static IP address to the WAN side interface on the zNID

This example is the continuance of the triple-play services that created in the previous sections.

To assign a static IP address to the WAN side interface, use the following procedure:

- 1 Create services on one ONT (using system defaults).

Refer to [USP on GPON RG: Residential Gateway enabled CPE provisioning for Voice on page 259](#).

- 2 Prepare system wide static IP configuration.

If you want to assign static IP address to the zNID, you need to create a static IP common profile on WAN interface and assign the static IP address to the WAN interface.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>add</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.4</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>Device</Value>
        </Attribute>
    </RequestElement>

```

```

        <Name>Parent_Id</Name>
        <Value>34</Value>
    </Attribute>
    <Attribute>
        <Name>Handle_Type</Name>
        <Value>DynOmciCpeIpServer</Value>
    </Attribute>
    <Attribute>
        <Name>serverIndex</Name>
        <Value>6</Value>
    </Attribute>
    <Attribute>
        <Name>gateway</Name>
        <Value>192.16.45.254</Value>
    </Attribute>
    <Attribute>
        <Name>profileName</Name>
        <Value>static-ip-config-1</Value>
    </Attribute>
    <Attribute>
        <Name>netMask</Name>
        <Value>255.255.255.0</Value>
    </Attribute>
    <Attribute>
        <Name>primaryDns</Name>
        <Value>192.16.45.1</Value>
    </Attribute>
    <!-- 1: dynamic, 2: static -->
    <Attribute>
        <Name>hostIpOption</Name>
        <Value>2</Value>
    </Attribute>
</RequestElement>
</OSS>

```

3 Perform a list operation on EtherPhysical to retrieve the ifIndex of ActiveE port physical.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>list</RequestType>
        <RequestMode>offline</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.4</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Filter_Type</Name>
            <Value>EtherPhysical</Value>

```



```

</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Device_Id=34</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Card_Id_1=1</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>ifTypePortId=1</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Physical_Id_2=1</Value>
</Attribute>
<Attribute>
  <Name>Filter_Retrieve</Name>
  <Value>ifIndex</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 4** Perform a list operation on DynOmciCpeIfVlan to retrieve the WAN side cpe-if-vlan profile given ActiveE port ifIndex and bridge vlan.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.zhone.com/OSSXML
  ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Filter_Type</Name>
      <Value>DynOmciCpeIfVlan</Value>
    </Attribute>
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>Device_Id=34</Value>
    </Attribute>
    <!--ActiveE port ifIndex -->
    <Attribute>
      <Name>Filter_Condition</Name>
      <Value>ifIndex=50</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

```

<!-- vlanXlateId here matches the vlanId specified in
the CPE connection for UniVlan-->
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>vlanXlateId=300</Value>
</Attribute>
<!-- Retrieves WAN side cpe-if-vlan -->
<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>tpType=17</Value>
</Attribute>
</RequestElement>
</OSS>

```

5 Perform the modify operation on DynOmciCpeIfVlan to set the static IP address and IP server on WAN interface.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeIfVlan</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>4229</Value>
</Attribute>
<Attribute>
    <Name>vlanIP</Name>
    <Value>192.168.10.200</Value>
</Attribute>
<Attribute>
    <Name>ipServerProfile</Name>
    <Value>6</Value>
</Attribute>
</RequestElement>
</OSS>

```

USP on AE: Static configuration on the LAN side interfaces with a new DHCP server

This example is the continuance of the triple services that created in the previous sections.

You can set LAN side IP and DHCP server if it is in rg-brouted or rg-bpppoe.

This section configures the following settings on the Brouted LAN interfaces those has same VLAN:

- Assigns a static IP address as the local IP address of the LAN interfaces
- Changes the IP address range for the DHCP server
- Changes the firewall access settings on the LAN interfaces

Assigning a Static IP address to the LAN side interface on the ActiveE port

Assign a Static IP address, a DHCP server, and an IP server to the LAN side interface on the zNID.

- 1 Create services on one ONT (using system defaults).

Refer to [USP on GPON RG: Residential Gateway enabled CPE provisioning for Data on page 251](#).

- 2 Create CPE DHCP server.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeDhcpServer</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

```

<Attribute>
    <Name>serverIndex</Name>
    <Value>2</Value>
</Attribute>
<Attribute>
    <Name>startAddress</Name>
    <Value>192.168.10.50</Value>
</Attribute>
<Attribute>
    <Name>profileName</Name>
    <Value>dhcp1</Value>
</Attribute>
<Attribute>
    <Name>endAddress</Name>
    <Value>192.168.10.254</Value>
</Attribute>
</RequestElement>
</OSS>

```

3 Create IP common profile for Data service.

If you want to assign static IP address to the zNID, you need to create a static IP common profile on LAN interface and assign the static IP address to the LAN interface.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeIpServer</Value>
</Attribute>
<Attribute>
    <Name>serverIndex</Name>
    <Value>5</Value>
</Attribute>

```

```

<Attribute>
  <Name>profileName</Name>
  <Value>HSI-LAN</Value>
</Attribute>
<Attribute>
  <Name>netMask</Name>
  <Value>255.255.255.0</Value>
</Attribute>
<!-- 1:dynamic, 2: static -->
<Attribute>
  <Name>host-ip-option</Name>
  <Value>2</Value>
</Attribute>
<!--Bits values for firewallAccess: 0:http, 1:ping,
2:snmp, 3:snmptrap, 4:ssh, 5:telnet. 11111 means
allow all six protocols on this interface--!>
<Attribute>
  <Name>firewallAccess</Name>
  <Value>11111</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Perform a list operation on EtherPhysical to retrieve the ifIndex of ActiveE port physical.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
  <Name>Filter_Type</Name>
  <Value>EtherPhysical</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Device_Id=34</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>Card_Id_1=1</Value>
</Attribute>
<Attribute>
  <Name>Filter_Condition</Name>
  <Value>ifTypePortId=1</Value>

```

```

</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Physical_Id_2=1</Value>
</Attribute>
<Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>ifIndex</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 5** Perform a list operation on DynOmciCpeIfVlan to retrieve the HandleId of the LAN side cpe-if-vlan profile with the given ActiveE port ifIndex and bridge vlan.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>DynOmciCpeIfVlan</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=34</Value>
</Attribute>
<!--ActiveE port ifIndex -->
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>ifIndex=50</Value>
</Attribute>
<!-- Bridge VLAN -->
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>vlanXlateId=200</Value>
</Attribute>
<!-- Retrieves LAN side cpe-if-vlan -->
<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
<Attribute>
    <Name>Filter_Condition</Name>

```

```

        <Value>tpType=1</Value>
    </Attribute>
</RequestElement>
</OSS>

```

- 6** Perform the modify operation on DynOmciCpeIfVlan to set the DHCP server, static IP address and IP server on the LAN interface.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>modify</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.4</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>DynOmciCpeIfVlan</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Id</Name>
            <Value>4229</Value>
        </Attribute>
        <Attribute>
            <Name>dhcpServerProfile</Name>
            <Value>2</Value>
        </Attribute>
        <Attribute>
            <Name>vlanIP</Name>
            <Value>192.168.10.200</Value>
        </Attribute>
        <Attribute>
            <Name>ipServerProfile</Name>
            <Value>5</Value>
        </Attribute>
    </RequestElement>
</OSS>

```

USP on AE: Configuration of Static Routes

To add static routes to the zNID, user need to create a static route list, and then associate this static route with a CPE system common profile, at last apply this system common profile to the zNID.

Adding a Static Route to the ActiveE port

Adding a static route to the ActiveE port can be done using the following steps.

1 Adding a static route list.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <!-- Handle_Id of parent ActiveE port -->
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeStaticRouteList</Value>
    </Attribute>
    <Attribute>
      <Name>listIndex</Name>
      <Value>3</Value>
    </Attribute>
    <Attribute>
      <Name>profileName</Name>
      <Value>test3</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

2 Adding a static route.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
```



```

<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<!-- Handle_Id of parent device -->
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeStaticRoute</Value>
</Attribute>
<Attribute>
    <Name>listIndex</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>entryIndex</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>destinationIp</Name>
    <Value>10.2.1.0</Value>
</Attribute>
<Attribute>
    <Name>metric</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>netmask</Name>
    <Value>255.255.255.0</Value>
</Attribute>
<Attribute>
    <Name>gateway</Name>
    <Value>10.1.1.253</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 3** Associate this static route with the CPE system common profile. You can either modify an existing CPE system common profile or create a new CPE system common profile. This example creates a new one.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>

```

```

<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeSystemCommon</Value>
</Attribute>
<Attribute>
    <Name>commonIndex</Name>
    <Value>13</Value>
</Attribute>
<Attribute>
    <Name>profileName</Name>
    <Value>mmr</Value>
</Attribute>
</Attribute>
<!--Index of shared static route list profile index -->
<Attribute>
    <Name>staticProfileIndex</Name>
    <Value>2</Value>
</Attribute>
<!--the vlanRouting is disabled by default. This
example enables the crossing VLAN routing on the
static route. If "enabled" is selected for cross VLAN
routing, routing between VLANs is allowed. Route table
lookups ignore the VLAN ID of the ingress and egress
ports. If there is a match, the packet is routed out
the interface specified in the Route table, regardless
of which VLAN it is a member of.enabled=1,
disabled=2--!>
<Attribute>
    <Name>vlanRouting</Name>
    <Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

4 Apply static route to an ActiveE port.

If this ActiveE port does not have system common profile assigned to it, you can create a cpe-system profile on this ActiveE port with the cpe-system-common index created earlier.

```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.
zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>EtherPhysical</Value>
    </Attribute>
    <!-- Handle_Id of parent ActiveE port -->
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeSystem</Value>
    </Attribute>
    <Attribute>
      <Name>commonIndex</Name>
      <Value>13</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

USP on AE: Configuration of Firewall

User can enable or disable firewall on the CPE. Enabling firewall can protect the CPE from unwanted instruction. When firewall is enabled, incoming connections can still be selectively allowed through firewall access and port forwarding settings. The firewall is enabled by default.

Retrieving firewall value

Use the CPE system common profile to retrieve firewall settings. By default, the firewall is enabled.

List example for CPE system common for retrieving firewall value:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>list</RequestType>

```

```

<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Filter_Type</Name>
    <Value>DynOmciCpeSystemCommon</Value>
</Attribute>
<Attribute>
    <Name>Filter_Condition</Name>
    <Value>Device_Id=34</Value>
</Attribute>
<Attribute>
    <Name>Filter_Retrieve</Name>
    <Value>firewall</Value>
</Attribute>
</RequestElement>
</OSS>

```

Configuring firewall access

Firewall access control manages the protocols allowed on the CPE WAN or LAN interfaces. It requires firewall feature to be enabled.

The protocols are listed below:

- HTTP: Web Browser Traffic.
- PING: ICMP Echoes used to test for connectivity.
- SNMP: Simple Network Management Protocol.
- SNMPTRAP: Alarms for Simple Network Management Protocol.
- SSH: Secure Shell.
- TELNET: Remote Terminal support.

By default, those protocol are all enabled on the CPE LAN interfaces. Only PING is enabled on the CPE WAN interfaces.



Note: Modifying a CPE IP Common profile will trigger a partial reconfiguration on each of the CPEs that depend on this profile and may cause service interruptions on these CPEs.

To modify the default firewall access settings on WAN or LAN interfaces, use the following procedure:

- 1 Make sure the CPE has the firewall setting enabled.
- 2 Create an IP common profile with the new firewall access settings:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

```

```

xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
</Attribute>
<Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
</Attribute>
<Attribute>
      <Name>Handle_Type</Name>
      <Value>DynOmciCpeIpServer</Value>
</Attribute>
<Attribute>
      <Name>serverIndex</Name>
      <Value>4</Value>
</Attribute>
<Attribute>
      <Name>gateway</Name>
      <Value>1.2.3.254</Value>
</Attribute>
<Attribute>
      <Name>profileName</Name>
      <Value>test1</Value>
</Attribute>
<Attribute>
      <Name>netMask</Name>
      <Value>255.255.255.0</Value>
</Attribute>
<!--http(0), ping(1), snmp (2), snmptrap (3), ssh(4),
telnet(5)-->
<Attribute>
      <Name>firewallAccess</Name>
      <Value>110000</Value>
</Attribute>
</RequestElement>
</OSS>

```

- 3** Apply the new firewall access settings to the WAN interface:
 - a** Perform a list operation on EtherPhysical to retrieve the ifIndex of ActiveE port physical.

- b** Perform a list operation on DynOmciCpeIfVlan to retrieve the Handle_Id of the WAN side cpe-if-vlan profile by given ActiveE port ifIndex and bridge VLAN.
- c** Perform the modify operation on DynOmciCpeIfVlan to modify the IP common profile index to 4.

Similar examples are provided in [Assigning a Static IP address to the WAN side interface on the zNID on page 271](#).

- 4** Apply the new firewall access settings to the LAN interface:
 - a** Perform a list operation on EtherPhysical to retrieve the ifIndex of ActiveE port physical.
 - b** Perform a list operation on DynOmciCpeIfVlan to retrieve the Handle_Id of the LAN side cpe-if-vlan profile by given ActiveE port ifIndex and bridge VLAN.
 - c** Perform the modify operation on DynOmciCpeIfVlan to modify the IP common profile index to 4.

Similar examples are provided in [Assigning a Static IP address to the LAN side interface on the ONU on page 275](#).

Configuring port forwarding

The CPE port forwarding list reflects the existing port forwarding rules. The port forwarding list only can be created on a WAN interface that has NAT or NAPT enabled. By default, there is no port-forwarding list associated to the WAN interface.

Port forwarding list requires firewall feature to be enabled.

To configure port forwarding on the WAN interfaces, use the following procedure:

- 1** Make sure the CPE has the firewall setting enabled.
- 2** Create a port forwarding list and add one entry into it:

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>add</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
```

```

</Attribute>
<!-- Handle_Id of parent ActiveE port -->
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpePortForward</Value>
</Attribute>
<Attribute>
    <Name>listIndex</Name>
    <Value>1</Value>
</Attribute>
<Attribute>
    <Name>entryIndex</Name>
    <Value>1</Value>
</Attribute>
<!-- dmz (1), portrange (2), portremap (3) -->
<Attribute>
    <Name>fwdType</Name>
    <Value>2</Value>
</Attribute>
<Attribute>
    <Name>portStart</Name>
    <Value>5500</Value>
</Attribute>
<Attribute>
    <Name>portEnd</Name>
    <Value>5500</Value>
</Attribute>
<!-- none (1), tcp (2), udp (3), tcpudp (4), icmp (5),
icmpv4 (6) -->
<Attribute>
    <Name>protocol</Name>
    <Value>3</Value>
</Attribute>
<Attribute>
    <Name>privateIp</Name>
    <Value>192.168.10.2</Value>
</Attribute>
</RequestElement>
</OSS>

```

3 Apply the port forwarding to the WAN interface:

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>

```

```

<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeIfVlan</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>23</Value>
</Attribute>
<!-- Represents the index of shared cpe-port-fwd
profile -->
<Attribute>
    <Name>portFwdListProfile</Name>
    <Value>1</Value>
</Attribute>
</RequestElement>
</OSS>

```

USP on AE: Configuration of Guided VLAN

Guided VLAN is only applicable for RG type CPE connections. There is no guided VLAN support for dynamic OMCI type (i.e. RG Mode = Not Applicable) CPE connections.

A CPE connection is considered to be using guided VLANs if it satisfies the following criteria:

- CPE connection is RG enabled (i.e. RG mode is set to Routed, Bridged PPPoE, or Bridged).
- VLAN ID and SLAN ID of LAN side DynOmciCpeIfVlan (cpe-if-vlan) is 0.
- VLAN ID and Translate VLAN ID of parent CPE connection are different.



Note: Guided SLAN is not supported at this point.

Creating a Data Service Specific CPE connection with setting guided VLAN

Creation of RG enabled CPE connections with enabling guided VLAN is very similar to uni-vlan based RG CPE connections.

- 1 Specifying internal ME for ActiveE port

Internal ME for an ActiveE port can be automatically set using OSS XML GW by providing the valid internal ME value for internalMe in OSS XML requests. If the internal ME is already set for an ActiveE port, CPE connection add request will not override the internal ME on ActiveE port. For overriding the internal ME, a separate modify operation can be performed on EtherPhysical object to modify meProfile value.

2 Creating an uplink MXK bridge.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>addUsingTemplate</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.4</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>Device</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>34</Value>
    </Attribute>
    <Attribute>
      <Name>Template_Type</Name>
      <Value>BridgeLogicalTemplate</Value>
    </Attribute>
    <!-- Tempalte_Id of bridge templates can be retrieved
    by running list operation on BridgeLogicalTemplate
    object -->
    <Attribute>
      <Name>Template_Id</Name>
      <Value>198</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>BridgeLogical</Value>
    </Attribute>
    <Attribute>
      <Name>vlanId</Name>
      <Value>200</Value>
    </Attribute>
    <!--lineGroupId of parent uplink ethernet physical -->
    <Attribute>
      <Name>lineGroupId</Name>
      <Value>19</Value>
    </Attribute>
```

```

<!--lineGroupId of parent uplink ethernet physical -->
<Attribute>
    <Name>lowerIfIndex</Name>
    <Value>19</Value>
</Attribute>
</RequestElement>
</OSS>

```

3 Create a downlink single-tagged bridge and a CPE connection in RG-bridged mode.



Note: For creating CPE connection in RG brouted and bridge PPPoE modes, set rgMode to 2 and 4 respectively in the previous example.



Note: Procedure for creating CPE connection for voice and video services with guided VLAN is analogous to creating services with uni-VLAN and uni-SLAN with the exception that guided VLAN will be set instead of uni-VLAN and uni-SLAN.

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.4</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Parent_Type</Name>
    <Value>Device</Value>
</Attribute>
<Attribute>
    <Name>Parent_Id</Name>
    <Value>34</Value>
</Attribute>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>DynOmciCpeConnection</Value>
</Attribute>
<!-- Ethernet = 1, IPTV=12, H248 = 13, SIP = 14, SIP
PLAR = 15, PWE = 16, virtualEthernetInterfacePoint=
17, WLAN / DOT11 = 18, TR69 = 20 -->
<Attribute>
    <Name>tpType</Name>
    <Value>1</Value>
</Attribute>

```

```

<Attribute>
  <Name>tpIndex</Name>
  <Value>1</Value>
</Attribute>
<Attribute>
  <Name>guidedVlan</Name>
  <Value>203</Value>
</Attribute>
<Attribute>
  <Name>vlanXlateId</Name>
  <Value>200</Value>
</Attribute>
<Attribute>
  <Name>phySlotId</Name>
  <Value>3</Value>
</Attribute>
<Attribute>
  <Name>phyPortId</Name>
  <Value>2</Value>
</Attribute>
<Attribute>
  <Name>bridgeTemplateName</Name>
  <Value>DownlinkSingleTagged</Value>
</Attribute>
<!-- Brouted is 2, Bridged Pppoe is 4 and bridged is 5
-->
<!-- If ActiveE port internal ME is not set, it will be
set to the following -->
<!-- If already set, following set will be ignored -->
<Attribute>
  <Name>internalMe</Name>
  <Value>zhone-2648a</Value>
</Attribute>
<Attribute>
  <Name>rgMode</Name>
  <Value>5</Value>
</Attribute>
</RequestElement>
</OSS>

```

Performance requests

Performance Request: support RSSI, DDM, DBA, and CAC

OSS XML GW can be used to retrieve the information of Connection Admission Control (CAC), Dynamic Bandwidth Allocation (DBA) and Digital Diagnostic Monitoring (DDM) of OLT. It also supports the retrieving of Received Signal Strength Indication (RSSI) information of ONU.

A **getStatus** operation shall be performed on **GponOltPhysical** for retrieving the information of CAC, DBA, and DDM.

A **getStatus** operation shall be performed on **GponOnuPhysical** for retrieving the information of RSSI.

And each **getStatus** operation requires the Handle_Id of corresponding OLT/ONU.

- [Retrieving Handle_Id of OLT on page 332](#)
- [Retrieving Handle_Id of ONU on page 333](#)
- [Retrieving CAC information of OLT on page 333](#)
- [Retrieving DBA information of OLT on page 334](#)
- [Retrieving DDM information of OLT on page 336](#)
- [Retrieving power information of ONU on page 337](#)

Retrieving Handle_Id of OLT

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>GponOltPhysical</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=497</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Card_Id_1=13</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>ifTypePortId=1</Value>
</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>Handle_Id</Value>
</Attribute>
</RequestElement>
</OSS>
```

Retrieving Handle_Id of ONU

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>list</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Filter_Type</Name>
<Value>GponOnuPhysical</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Device_Id=497</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Card_Id_1=13</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>ifTypePortId=1</Value>
</Attribute>
<Attribute>
<Name>Filter_Condition</Name>
<Value>Physical_Id_2=1</Value>
</Attribute>
<Attribute>
<Name>Filter_Retrieve</Name>
<Value>Handle_Id</Value>
</Attribute>
</RequestElement>
</OSS>
```

Retrieving CAC information of OLT

Before running the following query, a list operation shall be performed on GponOltPhysical to know the Handle_Id of the corresponding OLT physical.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>getStatus</RequestType>
```

```

<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
    <Name>Handle_Type</Name>
    <Value>GponOltPhysical</Value>
</Attribute>
<Attribute>
    <Name>Handle_Id</Name>
    <Value>82196</Value>
</Attribute>
<Attribute>
    <Name>Retrieve_Attr</Name>
    <Value>maxAllocIds</Value>
</Attribute>
<Attribute>
    <Name>Retrieve_Attr</Name>
    <Value>availAllocIds</Value>
</Attribute>
<Attribute>
    <Name>Retrieve_Attr</Name>
    <Value>usedAllocIds</Value>
</Attribute>
<Attribute>
    <Name>Retrieve_Attr</Name>
    <Value>maxDbalAllocIds</Value>
</Attribute>
<Attribute>
    <Name>Retrieve_Attr</Name>
    <Value>availDbalAllocIds</Value>
</Attribute>
<Attribute>
    <Name>Retrieve_Attr</Name>
    <Value>usedDbalAllocIds</Value>
</Attribute>
<Attribute>
    <Name>Retrieve_Attr</Name>
    <Value>totalGemPorts</Value>
</Attribute>
</RequestElement>
</OSS>

```

Retrieving DBA information of OLT

Before running the following query, a list operation shall be performed on GponOltPhysical to know the Handle_Id of the corresponding OLT physical.

```
<?xml version="1.0" encoding="UTF-8" ?>
```

```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>getStatus</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value> GponOltPhysical</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>82196</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>totalAvailableBw</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>totalAvailableCompensatedCbrBw</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>allocatedUbrBw</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>allocatedCbrBw</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>allocatedCompensatedCbrBw</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>allocatedAssuredBw</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>allocatedNonAssuredBw</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>allocatedBestEffortBw</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Retrieving DDM information of OLT

Before running the following query, a list operation shall be performed on GponOltPhysical to know the Handle_Id of the corresponding OLT physical.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>getStatus</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.16.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Handle_Type</Name>
<Value>GponOltPhysical</Value>
</Attribute>
<Attribute>
<Name>Handle_Id</Name>
<Value>82196</Value>
</Attribute>
<Attribute>
<Name>Retrieve_Attr</Name>
<Value>opticTemperature</Value>
</Attribute>
<Attribute>
<Name>Retrieve_Attr</Name>
<Value>opticVoltage</Value>
</Attribute>
<Attribute>
<Name>Retrieve_Attr</Name>
<Value>opticTxBiasCurrent</Value>
</Attribute>
<Attribute>
<Name>Retrieve_Attr</Name>
<Value>opticTxPower</Value>
</Attribute>
<Attribute>
<Name>Retrieve_Attr</Name>
<Value>opticStatus</Value>
</Attribute>
</RequestElement>
</OSS>
```


Retrieving power information of ONU

Before running the following query, a list operation shall be performed on GponOnuPhysical to know the Handle_Id of the corresponding ONU physical.

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>getStatus</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>GponOnuPhysical</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>82263</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>rxPower</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>opticRssi</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

Performance Request: PSPPhysicalStats filter getStatus request for EtherPhysical

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>getStatus</RequestType>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.1.2</Version>
    <RequestMode>offline</RequestMode>
  </Request>
</RequestElement>
```

```

        <Attribute>
            <Name>Retrieve_Attr</Name>
            <Value>mauOperStatus</Value>
        </Attribute>
        <Attribute>
            <Name>Retrieve_Attr</Name>
            <Value>statsIndex</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>Ether_Physical</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Id</Name>
            <Value>25</Value>
        </Attribute>
    </RequestElement>
</OSS>

```

Performance Request: operStatus request for EtherPhysical

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhong.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance"
xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd">
    <Request>
        <RequestType>getStatus</RequestType>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.1.2</Version>
        <RequestMode>offline</RequestMode>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Retrieve_Attr</Name>
            <Value>ifOperStatus</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>Ether_Physical</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Id</Name>
            <Value>25</Value>
        </Attribute>
    </RequestElement>
</OSS>

```

Performance Request: Retrieving VDSL Physical status object information

```

<?xml version="1.0" encoding="UTF-8" ?>

```

```

<OSS xmlns="http://www.zhone.com/OSSXML" xmlns:xsi="http://www.w3.org/2001/
XMLSchema-instance" xsi:schemaLocation="http://www.zhone.
com/OSSXML ossxml.xsd">
  <Request>
    <RequestType>getStatus</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.1</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>VdslPhysicalStatus</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>VdslPhysical</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>7450</Value>
    </Attribute>
    <Attribute>
      <Name>zhoneVdslPhysSide</Name>
      <Value>2</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>currAttainableRate</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>currLineRate</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>serialNumber</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>vendorID</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>versionNumber</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>currSnrMargin</Value>
    </Attribute>
    <Attribute>
      <Name>Retrieve_Attr</Name>
      <Value>currAtn</Value>

```

```

</Attribute>
<Attribute>
  <Name>Retrieve_Attr</Name>
  <Value>currStatus</Value>
</Attribute>
<Attribute>
  <Name>Retrieve_Attr</Name>
  <Value>currOutputPwr</Value>
</Attribute>
<Attribute>
  <Name>Retrieve_Attr</Name>
  <Value>chanInterleaveDelay</Value>
</Attribute>
<Attribute>
  <Name>Retrieve_Attr</Name>
  <Value>chanCrcBlockLength</Value>
</Attribute>
<Attribute>
  <Name>Retrieve_Attr</Name>
  <Value>chanCurrTxRate</Value>
</Attribute>
<Attribute>
  <Name>Retrieve_Attr</Name>
  <Value>chanCurrTxSlowBurstProtect</Value>
</Attribute>
<Attribute>
  <Name>Retrieve_Attr</Name>
  <Value>chanCurrTxFastFec</Value>
</Attribute>
</RequestElement>
</OSS>

```

SELT and DELT object retrieval examples

Retrieving SELT noise results

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>getStatus</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>

```

```

        <Value>AdslPhysical</Value>
    </Attribute>
    <Attribute>
        <Name>Handle_Id</Name>
        <Value>5643</Value>
    </Attribute>
    <Attribute>
        <Name>Retrieve_Attr</Name>
        <Value>selTlnfInbandNoise</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Retrieving SELT loop results

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>getStatus</RequestType>
        <RequestMode>offline</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.16.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>AdslSelt</Value>
        </Attribute>
        <!--Handle Id of AdslSelt is same as Device_Id -->
        <!-- upstream is 1 and downstream is 2 -->
        <Attribute>
            <Name>Handle_Id</Name>
            <Value>100</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>AdslPhysical</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>1223</Value>
        </Attribute>
        <Attribute>
            <Name>AdslSelt_Id</Name>
            <Value>1</Value>
        </Attribute>
        <Attribute>
            <Name>Retrieve_Attr</Name>
            <Value>segmentLength</Value>

```

```

</Attribute>
</RequestElement>
</OSS>

```

Retrieving DELT noise results

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>getStatus</RequestType>
    <RequestMode>offline</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.16.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>AdslDelt</Value>
    </Attribute>
    <!--Handle Id of AdslDelt is same as Device_Id '
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>100</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Type</Name>
      <Value>AdslPhysical</Value>
    </Attribute>
    <Attribute>
      <Name>Parent_Id</Name>
      <Value>1223</Value>
    </Attribute>
    <!-- AdslDelt_Id is a mandatory attribute -->
    <!-- upstream is 1 and downstream is 2 -->
    <Attribute>
      <Name>AdslDelt_Id</Name>
      <Value>1</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

N2N bond group examples



Note: : Provisioning EFM bond group is identical to N2N bond groups except that it will take ifTypeExtension as 22. GBond group is also identical to N2N and EFM bond groups except that it will take the ifTypeExtension as 31.

Add request: N2N bond group

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
<Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>2.3.2</Version>
</Request>
<RequestElement>
<Attribute>
<Name>Parent_Type</Name>
<Value>Card</Value>
</Attribute>
<Attribute>
<Name>Parent_Id</Name>
<Value>12</Value>
</Attribute>
<Attribute>
<Name>Handle_Type</Name>
<Value>N2NBond_Physical</Value>
</Attribute>
<Attribute>
<Name>ifAdminStatus</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>ifType</Name>
<Value>1</Value>
</Attribute>
<Attribute>
<Name>ifTypeExtension</Name>
<Value>20</Value>
</Attribute>
<Attribute>
<Name>isActive</Name>
<Value>2</Value>
</Attribute>
<Attribute>
```

```

        <Name>ifTypePortId</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>name</Name>
        <Value>oss_n2n_bg</Value>
    </Attribute>
</RequestElement>
</OSS>

```

List request: N2N bond group

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>list</RequestType>
        <RequestMode>offline</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>2.3.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Parent_Type</Name>
            <Value>Card</Value>
        </Attribute>
        <Attribute>
            <Name>Filter_Condition</Name>
            <Value>Card_Id_1=13</Value>
        </Attribute>
        <Attribute>
            <Name>Parent_Id</Name>
            <Value>20</Value>
        </Attribute>
        <Attribute>
            <Name>Filter_Type</Name>
            <Value>N2NBond_Physical</Value>
        </Attribute>
        <Attribute>
            <Name>Filter_Retrieve</Name>
            <Value>name</Value>
        </Attribute>
    </RequestElement>
</OSS>

```

Modify request: N2N bond group

```

<?xml version="1.0" encoding="UTF-8" ?>

```



```

<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>modify</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.3.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>Physical</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>974</Value>
    </Attribute>
    <Attribute>
      <Name>name</Name>
      <Value>oss_n2n_bg6</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Delete request: N2N bond group

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>delete</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.3.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>N2NBondPhysical</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>971</Value>
    </Attribute>
  </RequestElement>
</OSS>

```

Assign physicals to N2N bond group

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>modify</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.3.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>Physical</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>388</Value>
    </Attribute>
    <!-- Following is the ifIndex of bond group -->
    <Attribute>
      <Name>bondGroupIdx</Name>
      <Value>101</Value>
    </Attribute>
    <!-- Following is the lineGroupId of bond group -->
    <Attribute>
      <Name>bondGroupLgIdx</Name>
      <Value>102</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

Deassign physicals from N2N bond group

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>modify</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>2.3.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
```

```

        <Name>Handle_Type</Name>
        <Value>Physical</Value>
    </Attribute>
    <Attribute>
        <Name>Handle_Id</Name>
        <Value>871</Value>
    </Attribute>
    <!-- Following is the ifIndex of bond group -->
    <Attribute>
        <Name>bondGroupIdx</Name>
        <Value>0</Value>
    </Attribute>
    <Attribute>
        <Name>isActive</Name>
        <Value>1</Value>
    </Attribute>
    <!-- Following is the lineGroupId of bond group -->
    <Attribute>
        <Name>bondGroupLgIdx</Name>
        <Value>0</Value>
    </Attribute>
</RequestElement>
</OSS>

```

Clear statistics requests

ClearStats request: Clearing statistics on a bridge

```

<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
    <Request>
        <RequestType>clearStats</RequestType>
        <RequestMode>online</RequestMode>
        <SessionID>sessionID</SessionID>
        <operName>operName</operName>
        <Version>1.1.2</Version>
    </Request>
    <RequestElement>
        <Attribute>
            <Name>Handle_Type</Name>
            <Value>BridgeLogical</Value>
        </Attribute>
        <Attribute>
            <Name>Handle_Id</Name>
            <Value>4662</Value> <!--HandleId of bridge -->
        </Attribute>
    </RequestElement>
</OSS>

```

ClearStats request: Clearing statistics on a port

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>clearStats</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.1.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>Physical</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>22543</Value> <!-- handleId of physical -->
    </Attribute>
  </RequestElement>
</OSS>
```

Port deletion example

Delete requests: Deleting a port

```
<?xml version="1.0" encoding="UTF-8" ?>
<OSS xmlns="http://www.zhone.com/OSSXML"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.zhone.com/OSSXML
ossxml.xsd">
  <Request>
    <RequestType>delete</RequestType>
    <RequestMode>online</RequestMode>
    <SessionID>sessionID</SessionID>
    <operName>operName</operName>
    <Version>1.1.2</Version>
  </Request>
  <RequestElement>
    <Attribute>
      <Name>Handle_Type</Name>
      <Value>DeletePort</Value>
    </Attribute>
    <Attribute>
      <Name>Handle_Id</Name>
      <Value>2663</Value>
    </Attribute>
  </RequestElement>
</OSS>
```

```
</RequestElement>
</OSS>
```

OSS XML Support on V-Series Devices

List requests

List request: Get Handle_Id of a device using device IP

```
<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhong.com/OSSXML">
-<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Filter_Type</Name>
<Value>Device</Value>
</Attribute>
<!-- Optional filters can be added -->
<!-- <Attribute> <Name>Filter_Condition</Name> <Value>deviceIP=172.16.75.236</Value>
</Attribute> -->
-<Attribute>
<Name>Filter_Condition</Name>
<Value>name=V5816-test</Value>
</Attribute>
</RequestElement>
</OSS>
```

List Request: Device Alarms

```
<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhong.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhong.com/OSSXML">
-<Request>
<RequestType>listAlarm</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Filter_Type</Name>
```

```

<Value>AlarmEvent</Value>
</Attribute>
<!-- To retrieve all the alarms in the network, omit the following filter condition
-->
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DeviceIP=172.16.75.236</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: GPON OLT Ports

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Filter_Type</Name>
<Value>DasanGponOltPort</Value>
</Attribute>
-<Attribute>
<Name>Device_Ip</Name>
<Value>172.16.75.236</Value>
</Attribute>
<!-- To only retrieve a single OLT add the following filter condition -->
<!-- <Attribute> <Name>Filter_Condition</Name> <Value>DasanGponOltPort_Id=1</Value>
</Attribute> -->
</RequestElement>
</OSS>

```

List Request: GPON ONTs

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>

```

```

-<Attribute>
<Name>Filter_Type</Name>
<Value>DasanGponOnu</Value>
</Attribute>
-<Attribute>
<Name>Device_Ip</Name>
<Value>172.16.75.236</Value>
</Attribute>
<!-- To retrieve all the ONUs, omit the following filter conditions -->
<!-- To retrieve all ONUs on a single OLT, only include the DasanGponOltPort_Id filter
-->
<!-- To retrieve a single ONU, include both of the following filters -->
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOltPort_Id=1</Value>
</Attribute>
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOnu_Id=5</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: GPON ONT UNIs

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Filter_Type</Name>
<Value>DasanGponOntUniPort</Value>
</Attribute>
-<Attribute>
<Name>Handle_Type</Name>
<Value>Device</Value>
</Attribute>
-<Attribute>
<Name>Handle_Id</Name>
<Value>172.16.75.236</Value>
</Attribute>
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOltPort_Id=1</Value>
</Attribute>

```

```

-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOnu_Id=6</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: GPON ONT UNI OLTs

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Filter_Type</Name>
<Value>DasanGponOntUniPort</Value>
</Attribute>
-<Attribute>
<Name>Handle_Type</Name>
<Value>Device</Value>
</Attribute>
-<Attribute>
<Name>Handle_Id</Name>
<Value>172.16.75.236</Value>
</Attribute>
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOltPort_Id=1</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: GPON ONT UNI Ports

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>

```



```

-<Attribute>
<Name>Filter_Type</Name>
<Value>DasanGponOntUniPort</Value>
</Attribute>
-<Attribute>
<Name>Handle_Type</Name>
<Value>Device</Value>
</Attribute>
-<Attribute>
<Name>Handle_Id</Name>
<Value>172.16.75.236</Value>
</Attribute>
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOltPort_Id=1</Value>
</Attribute>
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOnu_Id=6</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: GPON ONT ANI

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Filter_Type</Name>
<Value>DasanGponOnuAni</Value>
</Attribute>
-<Attribute>
<Name>Device_Ip</Name>
<Value>172.16.75.236</Value>
</Attribute>
<!-- Without the filters below, records for all active ONUs will be returned. Either
one or both of the filters below can be specified -->
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOltPort_Id=1</Value>
</Attribute>
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOnu_Id=5</Value>

```

```

</Attribute>
</RequestElement>
</OSS>

```

List Request: GPON VoIP Profiles

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Filter_Type</Name>
<Value>DasanGponProfileVoip</Value>
</Attribute>
-<Attribute>
<Name>Handle_Type</Name>
<Value>Device</Value>
</Attribute>
-<Attribute>
<Name>Handle_Id</Name>
<Value>172.16.75.236</Value>
</Attribute>
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOltPort_Id=1</Value>
</Attribute>
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOnu_Id=3</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: GPON VoIP OLTs

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>

```

```

-<RequestElement>
-<Attribute>
<Name>Filter_Type</Name>
<Value>DasanGponProfileVoip</Value>
</Attribute>
-<Attribute>
<Name>Handle_Type</Name>
<Value>Device</Value>
</Attribute>
-<Attribute>
<Name>Handle_Id</Name>
<Value>172.16.75.236</Value>
</Attribute>
-<Attribute>
<Name>Filter_Condition</Name>
<Value>DasanGponOltPort_Id=1</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: GPON Profile DBA

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Filter_Type</Name>
<Value>DasanGponProfileDbA</Value>
</Attribute>
-<Attribute>
<Name>Handle_Type</Name>
<Value>Device</Value>
</Attribute>
-<Attribute>
<Name>Handle_Id</Name>
<Value>172.16.75.236</Value>
</Attribute>
</RequestElement>
</OSS>

```

List Request: GPON Profile ONU

```

<?xml version="1.0" encoding="UTF-8"?>

```

```

-OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-Request>
<RequestType>list</RequestType>
<RequestMode>offline</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-RequestElement>
-Attribute>
<Name>Filter_Type</Name>
<Value>DasanGponProfileOnu</Value>
</Attribute>
-Attribute>
<Name>Handle_Type</Name>
<Value>Device</Value>
</Attribute>
-Attribute>
<Name>Handle_Id</Name>
<Value>172.16.75.236</Value>
</Attribute>
</RequestElement>
</OSS>

```

Add requests

Add request: GPON Profile DBA

```

<?xml version="1.0" encoding="UTF-8"?>
-OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-Request>
<RequestType>add</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-RequestElement>
-Attribute>
<Name>Handle_Type</Name>
<Value>DasanGponProfileDbas</Value>
</Attribute>
-Attribute>
<Name>Device_Ip</Name>
<Value>172.16.75.236</Value>
</Attribute>
-Attribute>
<Name>DasanGponProfileDbas_Id</Name>
<Value>4</Value>
</Attribute>
-Attribute>

```

```

<Name>sleGponProfileDbaNma</Name>
<Value>test2</Value>
</Attribute>
-<Attribute>
<Name>sleGponProfileDbamode</Name>
<Value>2</Value>
</Attribute>
-<Attribute>

```

Modify requests

Modify request: GPON OLT Port

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Handle_Type</Name>
<Value>DasanGponOltPort</Value>
</Attribute>
-<Attribute>
<Name>Device_Id</Name>
<Value>18</Value>
</Attribute>
-<Attribute>
<Name>DasanGponOltPort_Id</Name>
<Value>9</Value>
</Attribute>
-<Attribute>
<Name>sleGponOltDescription</Name>
<Value>my description</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modify request: Assigned Serial Number

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>

```

```

<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Handle_Type</Name>
<Value>DasanGponOnu</Value>
</Attribute>
-<Attribute>
<Name>Device_Ip</Name>
<Value>172.16.75.236</Value>
</Attribute>
-<Attribute>
<Name>DasanGponOltPort_Id</Name>
<Value>2</Value>
</Attribute>
-<Attribute>
<Name>DasanGponOnu_Id</Name>
<Value>6</Value>
</Attribute>
-<Attribute>
<Name>sleGponOnuSerial</Name>
<Value>ZNTS7777777</Value>
</Attribute>
-<Attribute>
<Name>sleGponOnuProfile</Name>
<Value>50mb</Value>
</Attribute>
<!-- the description is optional -->
-<Attribute>
<Name>sleGponOnuDescription</Name>
<Value>customer account number</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modify request: GPON ONUs

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Handle_Type</Name>
<Value>DasanGponOnu</Value>
</Attribute>

```

```

-

```

Modify request: GPON VoIP Profile

```

<?xml version="1.0" encoding="UTF-8"?>
-

```

```

<Name>sleGponProfileVoIPomciAuthName</Name>
<Value>Adam</Value>
</Attribute>
-<Attribute>
<Name>sleGponProfileVoIPomciAuthPasswd</Name>
<Value>password</Value>
</Attribute>
-<Attribute>
<Name>sleGponProfileVoIPomciPhoneNumber</Name>
<Value>777777777</Value>
</Attribute>
-<Attribute>
<Name>sleGponProfileVoIPomciDisplay</Name>
<Value>Adam Benz</Value>
</Attribute>
</RequestElement>
</OSS>

```

Modify request: GPON Profile DBA

```

<?xml version="1.0" encoding="UTF-8"?>
-<OSS xsi:schemaLocation="http://www.zhone.com/OSSXML ossxml.xsd" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns="http://www.zhone.com/OSSXML">
-<Request>
<RequestType>modify</RequestType>
<RequestMode>online</RequestMode>
<SessionID>sessionID</SessionID>
<operName>operName</operName>
<Version>1.1.2</Version>
</Request>
-<RequestElement>
-<Attribute>
<Name>Handle_Type</Name>
<Value>DasanGponProfileDbas</Value>
</Attribute>
-<Attribute>
<Name>Device_Ip</Name>
<Value>172.16.75.236</Value>
</Attribute>
-<Attribute>
<Name>sleGponProfileDbasControlName</Name>
<Value>test</Value>
</Attribute>
-<Attribute>
<Name>sleGponProfileDbasMode</Name>
<Value>2</Value>
</Attribute>
</RequestElement>
</OSS>

```


6

ERROR ATTRIBUTES

This section describes OSSXML Gateway errors. It includes the following sections:

- [Overview, page 361](#)
- [Initialization errors, page 362](#)
- [Operation errors, page 363](#)
- [DataService error definitions, page 397](#)

Overview

When an error occurs while an operation is being executed, an error code is generated. The response includes a name-value pair, where the error status is the name and the value is the error condition.

Attribute values can contain the following types of errors:

Invalid data—The operation failed because of a validation error. An example is when an object cannot be located. The response element contains attributes with invalid data errors:

Name: <variable> Value: Invalid Data

NULL data—The operation failed because it could not be performed. An example is when a service is unreachable. The response element contains attributes with null data errors:

Name: <variable> Value: Null

Missing attributes—The operation failed because the request or request element omitted one or more required attributes. The response element contains attributes with missing attributes errors:

Name: <attribute_name> Value: Missing

Error conditions are:

- Other
- Null
- Missing
- Invalid
- Invalid Attribute

- Invalid Data
- Invalid Request
- Invalid Request Element

Error status messages include:

- SUCCESS—Operation successful
- FAILURE—Operation failed.
- UNKNOWN_STATUS—The status of the operation is unknown.

Initialization errors

CS_DS_INIT_ERROR—Error establishing connection with DataService

CS_FS_INIT_ERROR—Error establishing connection with FaultService of DZMS. Please examine the log files. Please examine the log files. Make sure that the DZMS server initialization has not encountered any error conditions.

CS_METADATA_ERROR—Error retrieving proper metadata

CS_NWS_INIT_ERROR—Error establishing connection with NetworkService

CS_ADS_RESP_ERROR—No response from AdminService

CS_NO_VALID_REQS—No valid requests to process

CS_NWS_PROCESS_ERROR—Error occurred during NWS preprocessing

CS_WRONG_TYPE_ERROR—The data was sent to the wrong business logic handler

CS_MISSING_METADATA—This object type is missing its metadata: \$1

CS_ATTR_MISSING_METADATA—This attribute is missing its metadata:

\$1CS_INVALID_RANGE—\$1 (\$2) must be within range \$3

DR_NWS_INIT_ERROR—Unable to initialize network service

DR_PS_INIT_ERROR—Unable to initialize performance service.

DR_DS_INIT_ERROR—Unable to initialize data service.

DR_CS_INIT_ERROR—Unable to initialize config service.

DR_DS_TRANS_ERROR—Unable to open or close transaction with data service.

DR_DS_LIST_DATA_FAILED—Failed to get data from data service. Request failed.

DR_PS_GET_STATUS_FAILED—Failed to get status info using performance service.

Operation errors

CS_ADD_DISALLOWED—Creation of a new instance of this data is disallowed

CS_ADD_ERROR—Error during add operation

CS_MANAGE_ERROR—Error during manage operation

CS_UNMANAGE_ERROR—Error during unmanage operation

CS_MUST_DELETE_CHILD—Child data must be deleted before this data may be deleted

CS_DELETE_DISALLOWED—Deletion of this data is disallowed

CS_DELETE_ERROR—Error during delete operation

CS_MODIFY_DISALLOWED—Modification of this data is disallowed

CS_GET_ERROR—Error during get operation

CS_LIST_ERROR—Error during list operation

CS_MODIFY_ERROR—Error during modify operation

CS_NO_BL_OPER—Invalid Configuration Data object

CS_NO_CREATE_BASEOPER—Could not create the Base Operation object

CS_NO_CREATE_BLOPER—Could not identify a business object to process the request for the card/port. Probable cause could be the database upgrade failed or DZMS and device versions could be incompatible.

CS_OPER_ERROR—Error during operation

CS_MODIFY_FAULT_DETECTION_TYPE_DISALLOWED—Fault Detection Type can not be modified when the admin status is not DOWN

CS_MOD_ATTR_DISALLOWED—Modification of this attribute is not allowed. A delete and re-create must be performed to change its value.

CS_USE_READONLY_COMMUNITY_NAME_WARNING—Warning: you are using a read-only community name which will disallow any further provisioning.

CS_MUST_DELETE_ATM_XCONN—Associated ATM cross connect(s) must be deleted before this card

CS_ANY_SPEED_DISALLOWED—Only speeds of 9600, 19200, 38400, and 57600 bps are allowed.

CS_FAST_MIN_TX_GT_THAN_MAX_TX_ERR—The fast MAX transmission value must always be greater than the fast MIN transmission value.

CS_INTERLVED_MIN_TX_GT_TH_MAX_TX_ERR—The interleaved MAX transmission value must always be greater than interleaved MIN transmission value.

CS_INVALID_USE—This data object should not be used for this kind of operation.

DR_NULL_DEVICE_GROUP_ID—The group ID in the handle of the device is null.

DR_DS_REMOTE_EXCEPTION—Unable to talk to data service (DS). DS generated a remote exception.

DR_UPDATE_IN_PROGRESS—Unable to obtain device lock. Some update is in progress. Device is busy.

DR_SERVICE_REQUEST_FAILURE—Request failed by another service.

DR_DEVICE_LOCK_NOT_FOUND—Unable to find device lock in the database.

DR_OPER_ERROR—Backup (dump) and restore service operation error.

DR_NO_CREATE_BLOPER—Unable to create BL oper.

DR_DS_REMOTE_EXCEPTION—Unable to talk to data service (DS). DS generated a remote exception.

DR_UPDATE_IN_PROGRESS—Unable to obtain device lock. Some update is in progress. Device is busy.

DR_SERVICE_REQUEST_FAILURE—Request failed by another service.

DR_DEVICE_LOCK_NOT_FOUND—Unable to find device lock in the database.

DR_OPER_ERROR—Backup (dump) and restore service operation error.

DR_NO_CREATE_BLOPER—Unable to create BL oper.

Successful completion

CS_ADD_SUCCESS—Add successful

CS_DELETE_SUCCESS—Delete successful

CS_GET_SUCCESS—Get successful

CS_LIST_SUCCESS—List successful

CS_MODIFY_SUCCESS—Modify successful

Dump errors

DR_DUMP_SUCCESS—Backup (dump) operation completed successfully.

DR_DUMP_FAILURE—Backup (dump) operation failed.

DR_DEVICE_POLL_FAILED—Polling to the device failed.

DR_CREATE_BASE_DIR_FAILED—Failed to create dump and restore base directory under tftpd home directory.

DR_CREATE_DUMP_FILE_FAILED—Failed to create dump file.

DR_NWS_DUMP_REQ_FAILED—Dump request to the device failed.

DR_DUMP_RESTORE_FAILURE—Backup (dump) restore operation failed.

DR_DUMP_DATA_MISSING—Backup (dump) data is missing in the request.

DR_DUMP_FILENAME_MISSING—Filename not specified in the backup request.

DR_INVALID_DUMP_REQUEST—Backup (dump) request is invalid.

DR_DELETE_PARTIAL_DUMP_FILE_FAILED—Unable to delete partial or incomplete dump file.

Restore errors

DR_REBOOT_DEVICE_ERROR—Unable to reboot device after restoring backup file.

DR_DELETE_RESTORE_FILE_ERROR—Unable to delete restore file on the tftpd (DZMS) server.

DR_COPY_RESTORE_FILE_ERROR—Unable to copy restore file into the home directory of the tftpd (DZMS) server.

DR_RESTORE_FAILURE—Restore operation failed.

DR_RESTORE_SUCCESS—Restore operation completed successfully.

DR_INVALID_RESTORE_FILE—Specified invalid restore filename.

DR_COPY_FILE_ERROR—Error occurred during file copying.

DR_NWS_RESTORE_REQ_FAILED—Restore request to the device failed.

DR_NWS_REBOOT_REQ_FAILED—Reboot device request failed.

All Service success

CS_ADD_COMPLETE_SUCCESS—DS&NwS add successful

CS_DELETE_COMPLETE_SUCCESS—DS&NwS delete successful

CS_GET_COMPLETE_SUCCESS—DS&NwS get Successful

CS_LIST_COMPLETE_SUCCESS—DS&NwS list successful

CS_MODIFY_COMPLETE_SUCCESS—DS&NwS modify successful

List errors

DR_LIST_FAILURE—Failed to list backup (dump) files.

DR_LIST_SUCCESS—List operation completed successfully.

DR_INVALID_FILTER_TYPE—Invalid filter type specified in the list request.

DR_INVALID_LIST_REQUEST—List request has incorrect data.

DR_LIST_FILTER_MISSING—The base filter is missing in the request.

Delete errors

DR_DELETE_FAILURE—Delete operation failed.

DR_DELETE_SUCCESS—Delete operation completed successfully.

DR_INVALID_FILE_KEY—File's unique key contains invalid value.

DR_DELETE_FILE_FAILED—Unable to delete the backup (dump) file.

DR_DELETE_RESTORE_COPY_FAILED—Unable to delete the restore copy of the delete file.

DR_INVALID_DELETE_NAME—Name attribute is invalid in the delete request.

Add errors

DR_ADD_SUCCESS—Add operation successful.

DR_ADD_FAILURE—Add operation failed.

DR_ADD_DATA_MISSING—Add request does not contain data.

DR_ADD_SAVE_FILENAME_MISSING—Filename to save the modified file is missing.

DR_INVALID_ADD_REQUEST—Invalid add request received.

DR_ADD_UNIQUE_KEY_MISSING—Unique key is missing in the add request.

DR_ADD_FILENAME_MISSING—Filename is missing in the add request.

DR_ADD_INVALID_IPADDRESS—The specified IP address is invalid.

Modify errors

DR_INVALID_MODIFY_REQUEST—Invalid modify request received.

DR_MODIFY_SUCCESS—Modify operation successful.

DR_MODIFY_FAILURE—Modify operation failed.

DS errors

DS_GET_FAILURE—DS get failure

DS_LIST_FAILURE—DS list failure

DS_ADD_FAILURE—DS add failure

DS_DELETE_FAILURE—DS delete failure

DS_MODIFY_FAILURE—DS modify failure

DS_REQUEST_FAILURE—Failed to execute request

DS_OBJECT_UNKNOWN—The object requested is unknown to the system

DS_DEVICE_LOCKED_ERROR—The database is currently being updated with information from the device.

DS success

DS_GET_SUCCESS—DS get success

DS_LIST_SUCCESS—DS list success

DS_ADD_SUCCESS—DS add success

DS_DELETE_SUCCESS—DS delete success

DS_MODIFY_SUCCESS—DS modify success

Nws errors

CS_NWS_ADD_FAILURE—NWS add failure

CS_NWS_DELETE_FAILURE—NWS delete failure

CS_NWS_GET_ERROR_NAME—getErrorName() returned null

CS_NWS_GET_FAILURE—NWS get failure

CS_NWS_MODIFY_FAILURE—NWS modify failure

CS_NWS_NULL_RESPONSE—Network service is not responding. Response from network service is null.

CS_NWS_REQUEST_EMPTY—Empty Nws Request

Nws success

CS_NWS_ADD_SUCCESS—NetworkService add successful.

CS_NWS_CONFIG_SUCCESS—NetworkService config successful.

CS_NWS_DELETE_SUCCESS—NetworkService delete successful.

CS_NWS_GET_SUCCESS—NetworkService get successful.

CS_NWS_LIST_SUCCESS—NetworkService list successful.

CS_NWS_MODIFY_SUCCESS—NetworkService modify successful.

Service notification

DS_NO_DATA_AVAILABLE—No Data Available

DS_STATUS_NULL—Status was null

DS_FAILURE—DS failure

DS_TIMESTAMP_FAILURE—Modified record is out-of-date

DS_INSTANCE_KEY_FAILURE—No instanceKey attribute was provided

DS_CONNECTION_NOT_AVAILABLE—No database connection was available

Specific error conditions

CS_AAL5_ENCAPSTYPE_ERROR—The value chosen for Aal5EncapsType is not allowed

CS_AAL2_APPID_MODIFY_DISALLOWED—Modification of the AAL2 AppId is disallowed.

CS_INVALID_AAL2_AUDIO_PROFILE_ID—Invalid audio profile identifier specified for the given AppID.

CS_ADMIN_DOWN—The AdminStatus must be down before a delete can be performed

CS_ALR_TOO_LARGE_ERROR—The max allowed line rate must be less than or equal to the max capable line rate

CS_ATM_LOGICAL_ERROR—Invalid ATM Logical index

CS_CARD_DATA_INVALID_SLOT_INFO_ERROR—Invalid slot information in the card data. Both slotsRequired and slotIndex must be set.

CS_CARD_TYPE_UNKNOWN—Unknown card type

CS_CARD_STATUS_DOWN_ERROR—Card in slot \$1 does not appear to be in an operational state

CS_DATA_LIST_NULL—Data List was null

CS_DEV_DATA_NULL—Device Data was null

CS_DEV_RESP_ELEM_NULL—Device Response Elem null

CS_DUPLICATE_IP_ERROR—Duplicate IP Address

CS_DUPLICATE_DESTINATION_ERROR—Duplicate destination found for the same device

CS_DUPLICATE_HOSTNAME—Another host by that name already exists in this routing domain

CS_ERROR_ADD_CARD—Could not add the card to the Database

CS_FETCH_CARD_TYPE_ERROR—Could not obtain a valid card type

CS_FETCH_IFINDEX_FAILED—Could not retrieve the IfIndex

CS_FETCH_IFTYPE_FAILED—Could not retrieve the type of interface

CS_FETCH_IFTYPE_EXTENSION_FAILED—Could not retrieve the ifTypeExtension of the interface

CS_FETCH_IFINDEX_LINEGROUP_FAILED—Could not obtain either the IfIndex or LineGroup

CS_FETCH_IF_INFO_FAILED—Could not obtain interface information such as ifType and/or ifTypeExtension.

CS_FETCH_LGID_FAILURE—Could not obtain the line group ID from the card

CS_JMS_SET_PROPERTIES_ERROR—Could not set JMS properties

CS_IMA_GROUP_ERROR—Could not get IMA group information

CS_IMA_SLOT_ERROR—IMA cards may only be added in slots 1 or 2.

CS_UPLINK_SLOT_ERROR—Uplink cards may only be placed in slots 1 or 2.

CS_PHYSICAL_LOOPBACK_CONFIG_ERROR—The loopback config attribute can only be changed when the admin status is in “testing” mode.

CS_PHYSICAL_LINE_TYPE_ERROR—LineType attribute of the physical contains invalid value.

CS_INVALID_APPID—The T1 Sechtor 100 only supports POTS (appId = casPotsOnlyNoElcp = 10)

CS_INVALID_ATTR_VALUE—The value specified for this attribute is invalid

CS_INVALID_BCAST—The broadcast address is invalid

CS_INVALID_BCAST_ENABLE—The broadcast-enabled setting is invalid

CS_DELETE_V52LINK_DISALLOWED—Deleting a V5.2 link is disallowed when igAdminStatus is inService

CS_DELETE_VOICE_ASSOC_ERROR—Error deleting associated V5.2 voice endpoint

CS_INVALID_ROUTE_DEST—The Static Route Destination is invalid

CS_INVALID_MASK—The Static Route Mask is invalid

CS_INVALID_ROUTE_GW—The Static Route Gateway is invalid

CS_INVALID_ROUTE_DEST—The Static Route Destination is invalid

CS_INVALID_MASK—The Static Route Mask is invalid

CS_INVALID_ROUTE_GW—The Static Route Gateway is invalid

CS_INVALID_DATA—The request contains invalid data or attributes

CS_INVALID_DEST_ADDR—The destination address is invalid

CS_INVALID_IFINDEX—The IfIndex returned from the device is invalid

CS_INVALID_LEASE—For lease times: min <= current <= max

CS_INVALID_NETMASK—The netmask is invalid

CS_WRONG_SUBNET—The IP interface must be on a different subnet than the IP interface on the first Infoservices Card

CS_INVALID_RESPONSE_ELEMENT—The response element contains illegal type of CS Data Object

CS_INVALID_REQUEST_ELEMENT—The request element contains illegal type of CS Data Object

CS_INVALID_REQUEST—The request is not valid for this object

CS_INVALID_RANGE_ERROR—The subnet range is invalid

CS_INVALID_TD_ERROR—The transmit traffic descriptor index of each side of the cross connect must match the receive traffic descriptor index of the other side

CS_INVALID_DIVIDER—The CAC Divider must be greater than 0

CS_INVALID_UPC—The Usage Parameter Control cannot be false when using CBR

CS_INVALID_TRUNKTYPE—A POTS VCL cannot support an UNKNOWN trunk type

CS_SUBNET_RANGE_OVERLAP—Overlapping subnet ranges are not allowed

CS_IP_ADDRESS_ERROR—IP Address is not valid

CS_LIST_ETHERPHYSICAL_FAILED—Could not obtain a list of Ethernet ports from the database

CS_LIST_PHYSICAL_FAILED—Could not obtain a list of Ports from the database. Data service and/or connection to the database may not be in working order.

CS_MANDATORY_MISSING—This mandatory attribute is missing from the request

CS_MISSING_ATTR—At least one required attribute is missing from the request

CS_MISS_COMM_STR—The device is missing a community string. No contact with the device is possible

CS_MISSING_PVCS—The cross connect could not be added since the ATM end points to be connected are missing

CS_MISSING_SUBNET—The network and netmask must be specified

CS_MODIFY_MAU_FAILED—Could not modify MAU ports in the database

CS_MODIFY_TRAFFIC_DESC_DISALLOWED—The admin status must be set to down before the traffic descriptor parameters can be modified

CS_NEXT_INDEX_ERROR—Invalid nextIndex returned from device

CS_NEXT_INDEX_FAILURE—Failed to retrieve nextIndex from device

CS_INVALID_IF_INDEX—Invalid ifIndex. The value is less than or equal to zero.

CS_NO_ELEMENT_IN_GROUPID—No group ID could be obtained

CS_NO_IP_ADDRESS—Device IP Address could not be obtained

CS_NO_OFFLINE_ANYCARD_ADD—Cannot add AnyCard in offline mode

CS_NO_P2P_DHCP—DHCP cannot be enabled on point-to-point interfaces

CS_NOT_SUPPORTED—This operation is not supported in this release

CS_NOTIFICATION_ERROR—Could not publish a response.

CS_OPTIONAL_MISSING—One of these optional attributes is necessary

CS_POST_OPER_ERROR—Error occurs during post-processing of the request

CS_PVC_NOT0—VCI and VPI cannot both be 0

CS_PVC_RXTX_ERROR—The receive and transmit traffic descriptors must be the same

CS_PVC_VCLCASTTYPE_ERROR—The VclCastType can only be point-to-point

CS_RETRIEVE_ERROR—Error retrieving data from request

CS_ROLLBACK_NOT_ALLOWED—The rollback operation is not allowed for this type of request

CS_STATUS_NULL—Status was null

CS_T1_MODIFY_FAILED—Could not modify the T1 Port

CS_UNSUPPORTED_CARD_TYPE—Unsupported card type

CS_UNSUPPORTED_APPID—The specified Application ID is not supported

CS_T1_UPDATE_FAILED—Failed to update the T1

CS_T1_LIST_FAILED—Could not obtain a list of T1s from the database for a given card

CS_T1_ADD_DISALLOWED—T1 creation is disallowed. T1 already exists.

CS_T1_DELETE_DISALLOWED—T1 cannot be deleted. Delete parent card.

CS_T3_ADD_DISALLOWED—T3 creation is disallowed. T3 already exists on the card.

CS_T3_DELETE_DISALLOWED—T3 cannot be deleted. Delete parent card.

CS_SYS_OBJECT_ID_MISSING—The sysObjectID is missing or not set.

CS_SYS_OBJECT_ID_MIS_MATCH—The sysObjectID from the network device did not match with the sysObjectID of the device in the request.

CS_GET_DHCP_DISALLOWED—The information requested may only be retrieved when the IP interface's DHCP setting is something other than 'noDhcp'.

CS_DHCP_SERVER_GROUP_REQ_MATCH_STR— The vendor match string and the client match string cannot both be empty.

CS_IP_DHCP_ERROR—IP has DHCP server enabled, IP address mode must be static

CS_CLEAR_DHCP_CLIENTRES_ERROR—The DHCP client resource record on this card could not be cleared. The device and DZMS are now out of sync. Please synchronize with the device to resolve the issue.

CS_AAL2_PVC_CAS_INCORRECT_ENABLE—The Channel Associated Signalling cannot be set to “Enabled” for the given AppId.

CS_INVALID_DSL_LINE_TYPE—DSL line type is not valid for this device version

CS_ATM_PVC_SCR_RATE_OVERSUBSCRIBED_ERROR— Either the SCR Rate in the associated traffic descriptor(s) caused an over subscription of total available bandwidth (verify if an atmVclBandwidthUnavailable trap is in the trap log), or the device did not accept one or more of the values sent.

CS_ZEDGE_AAL2PVC_ADD_DISALLOWED—Z-Edge 64 only supports one Aal2Vcl. One Aal2Vcl already exists on the device.

CS_ATM_PVC_EXCEEDS_BANDWIDTH—The cell rate specified for this ATM end point exceeds the available bandwidth. Please verify the traffic descriptor settings.

CS_LINE_GROUP_ERROR—Line group is not specified for IP Logical

CS_SYNC_INTERFACES_FAILED—Failed to sync up interfaces: physicals, logicals, etc.

CS_SYNC_PHYSICALS_FAILED—Failed to sync up physicals from the device.

CS_SYNC_LINE_GROUPS_FAILED—Failed to sync up LineGroups from the device.

CS_SYNC_ATM_LOGICALS_FAILED—Failed to sync up ATM Logicals from the device.

CS_UNSUPPORTED_DEVICE_TYPE—Failed to recognize the device on which the operation is being performed. This generally happens when the request type has not been set correctly in the incoming request from the client.

Device errors

CS_GET_COMMUNITY_NAME_FAILED—Unable to get community name data from the device.

CS_ADD_COMMUNITY_NAME_FAILED—Unable to add community name data in the device.

CS_ADD_READ_ONLY_COMMUNITY_PROFILE_DENIED—A community profile with access privilege equal to read-write/admin has to be created as the first profile.

CS_SETUP_COMMUNITY_NAME_FAILED—Failed to setup default community name in the device.

CS_DELETE_PREF_COMMUNITY_NAME_DENIED—Please change the device's preferred community name before deleting CommunityName. Preferred community name cannot be deleted.

CS_GET_SYS_OBJECT_ID_FAILURE—Failed to get sysObjectID from the device.

CS_ADD_DEVICE_FULL_UPDATE_FAILED—Device add operation succeeded but a full update request to the device failed. Please check the log files for more information on the problem, and resolve any issues before continuing device provisioning.

CS_VERIFY_DEVICE_IDENTITY_FAILED—Failed to verify device identity.

1. Check whether the community name is valid for the device.
2. Check whether the device is reachable using the IP Address specified.
3. Make sure that device is SNMP Manageable.

CS_ADD_TRAP_DESTINATION_FAILED—Unable to add trap destination in the device.

CS_DELETE_TRAP_DESTINATION_FAILED—Unable to delete trap destination in the device.

CS_CREATE_DEVICE_DATA_FAILED—Failed to create device data.

CS_ZEDGE_TOO_MANY_CARDS—More than one card appears to belong to this Z-Edge 64, but there should only be one.

CS_DEVICE_ALREADY_EXISTS—A device with specified IP Address already exists in DZMS.

CS_COMM_NAME_ERROR—Community Name is invalid.

CS_UNSUPPORTED_DEVICE_TYPE—The IP address specified is not of a DZMS-manageable device. Please enter the IP address of a BAN, MALC, Raptor, Sechtor 100, or Z-Edge 64.

CS_MISSING_DEFAULT_DEVICE_TMPL— Could not locate the default device template in the database.

CS_DEVICE_NOT_FOUND—The device with IP address \$1 could not be located in the database.

Subscriber voice connection errors

CS_MANDATORY_ENDPOINTS_MISSING—Mandatory voice connection end points are missing in the request

CS_VOICE_CONN_EMPTY—Unable to retrieve voice connection(s) for the given subscriber

CS_END_POINTS_EMPTY—Unable to retrieve end point(s) for the given voice connections

CS_VOICE_CONN_INDEX_FAILURE—Unable to retrieve voice connection index from the device

CS_END_POINTS_INDEX_FAILURE—Unable to retrieve voice connection end point index from the device

CS_INVALID_END_POINTS—Invalid end points specified for the given voice connection.

CS_VOICE_INVALID_MODIFY_CONN_ENDPTS—Subscriber voice connection end point type cannot be modified.

CS_END_POINT_PARENT_MISSING—The parent voice connection is not available.

CS_GET_SUB_FAILURE—Failure to retrieve subscriber information from the database

CS_CUSTOMER_HAS_SUBSCRIBERS—Please delete all subscribers before delete Customer

CS_CARD_HAS_SUBSCRIBERS—Please delete all SubscriberPorts provisioned on this card before deleting this card from the device

CS_MISSING_DEFAULT_SUBSCRIBER_PORT—Could not locate the default SubscriberPort template in the database

CS_MISSING_DEFAULT_SUBVOICE_CONN—Could not locate the default SubVoiceConn template in the database

CS_MISSING_DEFAULT_SUBSCRIBER_ENDPT—Could not locate the default SubscriberPort End Point template in the database

CS_MISSING_DEFAULT_GR303CRV—Could not locate the default GR-303 CRV template in the database

CS_MISSING_DEFAULT_V52USERPORT—Could not locate the default V5.2 USERPORT template in the database

CS_MISSING_SUBSCRIBER_PORT—Could not locate the subscriber port associated with the voice connection.

CS_BULK_POTS_V52_PRIMARY_LINK_MISSING—Could not locate the primary link of the V5.2 Interface Group.

CS_BULK_ISDN_V52_PRIMARY_LINK_MISSING—Could not locate the primary link of the V5.2 Interface Group for ISDN<->V5.2 voice provisioning.

CS_BULK_POTS_GR303_PRIMARY_LINEMAP_MISSING—Could not locate the primary line map of the GR-303 Interface Group.

CS_GR303_AAL2_VC_FAILURE—Provisioning a SubVoiceConnection between the GR-303 and AAL2 VCL has failed.

CS_GR303_AAL2_VC_FAILURE_INFO—Provisioning a SubVoiceConnection between GR-303 (IG name: \$1, igcrv: \$2) and AAL2 (slot: \$3, port: \$4, VPI: \$5, VCI: \$6, CID: \$7) has failed.

CS_V52_AAL2_VC_FAILURE—Provisioning a SubVoiceConnection between the V5.2 and AAL2 VCL has failed.

CS_V52_AAL2_VC_FAILURE_INFO—Provisioning a SubVoiceConnection between V5.2 (IG n\$1, igcrv: \$2) and AAL2(slot: \$3, port: \$4, VPI: \$5, VCI: \$6, CID: \$7) has failed.

CS_V52_ELCPAAL2_VC_FAILURE—Provisioning a SubVoiceConnection between the V5.2 and ELCPAAL2 VCL has failed.

CS_V52_ELCPAAL2_VC_FAILURE_INFO—Provisioning a SubVoiceConnection between V5.2 (IG name: \$1, user port: \$2) and AAL2ELCP (slot: \$3, port: \$4, VPI: \$5, VCI: \$6, portid: \$7) has failed.

CS_POTS_AAL2_VC_FAILURE—Provisioning a SubVoiceConnection between the POTS and AAL2 VCL has failed.

CS_POTS_AAL2_VC_FAILURE_INFO—Provisioning a SubVoiceConnection between POTS (slot: \$1, port: \$2) and AAL2 (slot: \$3, port: \$4, VPI: \$5, VCI: \$6, CID: \$7) has failed.

CS_ISDN_AAL2_VC_FAILURE—Provisioning a SubVoiceConnection between the ISDN and AAL2 VCL has failed.

CS_ISDN_AAL2_VC_FAILURE_INFO—Provisioning a SubVoiceConnection between ISDN (slot: \$1, port: \$2) and AAL2 (slot: \$3, port: \$4, VPI: \$5, VCI: \$6, CID: \$7) has failed.

CS_POTS_GR303_VC_FAILURE—Provisioning a SubVoiceConnection between the POTS and AAL2 VCL has failed.

CS_POTS_GR303_VC_FAILURE_INFO—Provisioning a SubVoiceConnection between POTS (slot: \$1, port: \$2) and GR-303 (IG name: \$3, igcrv: \$4) has failed.

CS_POTS_V52_VC_FAILURE—Provisioning a SubVoiceConnection between the POTS and V5.2 VCL has failed.

CS_POTS_V52_VC_FAILURE_INFO—Provisioning a SubVoiceConnection between POTS (slot: \$1, port: \$2) and V5.2 (IG name: \$3, user port: \$4) has failed.

CS_MISSING_GR303IG—Could not locate GR-303 IG\$1 under device with IP \$2

CS_ISDN_V52_VC_FAILURE—Provisioning a SubVoiceConnection between the ISDN and V5.2 VCL has failed.

CS_ISDN_V52_VC_FAILURE_INFO—Provisioning a SubVoiceConnection between ISDN (slot: \$1, port: \$2) and V5.2 (IG name: \$3, user port: \$4) has failed.

CS_ISDN_V52_END_PT_MAP_FETCH_FAILURE—Failed to retrieve V5.2 Endpoint map information.

CS_ADMIN_VOICE_DELETE_ERROR—Please admin down voice connection(s) prior to delete operation.

Bulk summary errors

CS_BULK_VCXC_SUMMARY—During this bulk provisioning:

\$1 VcCrossConns were successfully provisioned

\$2 VcCrossConns failed provisioning

CS_BULK_SUB_VOICE_CONN_SUMMARY—During this bulk provisioning: \$1 SubVoiceConnections were successfully provisioned \$2 SubVoiceConnections failed provisioning

CS_BULK_VCL_SUMMARY—During this bulk provisioning: \$1 VCLs were successfully provisioned \$2 VCLs failed provisioning

CS_ROLLBACK_SUCCESS—Rollback successful.

CS_ROLLBACK_FAILURE—Rollback failure.

CS_DS_ROLLBACK_FAILURE—Failed to restore the system to the previous good state. Rollback attempt failed. Please perform “Synchronize with device...” operation.

CS_ENTITY_ROLLBACK_FAILED—Failed to rollback entity \$1. Please perform sync operation with the device.

Card profile errors

CS_CARD_SW_FILENAME_MISSING— The card specific software filename is not specified, nor is the default card software filename. Please provide at least one.

CS_GET_DEF_CARD_SW_FILENAME_FAILED—Unable to get the default card software filename from the device.

CS_SLOT_NOT_EXIST—Trying to add card on a non-existent slot.

CS_CARD_PORTS_FAILURE—The physical ports of the card are not fully provisioned. Not all ifIndices/lgids are valid in the card’s physicals. Please perform sync operation.

CS_CARD_LOGICAL_FAILURE—The logical ports of the card are not fully provisioned. Some of the logical ports have zero IfIndices. Please perform sync operation.

CS_GET_GEN_CARD_ATTRS_FAILED—Unable to get the card specific attributes values from the device.

CS_CARD_MISSING_IN_DB—Card profile object is missing in the database. Configsync may be required to get the card state in sync.

CS_CARD_UPDATE_FAILED_IN_DB—Error while updating card data object in database.

CS_CHANGE_CARD_ADMIN_STATUS_TO_MAINT—Please change the card admin status to disabled mode.

CS_GET_CARD_ADMIN_STATUS_FAILED—Unable to fetch card admin status from the device.

CS_GET_INFO_SERV_CARD_ENTRIES_FAILED—Failed to get all the InfoServices card resource entries from the device.

CS_GET_INFO_SERV_CARD_PROFILE_FAILED—Unable to get a particular InfoServices card profile entry from the device.

CS_DEFAULT_IS_CARD_SW_FILENAME_MISSING—The default InfoServices config card profile name is not specified in the CS resource file.

CS_ADD_INFO_SERV_CARD_PROFILE_FAILED—Unable to add InfoServices card profile in the device.

CS_GET_MALC_UPLINK_CARD_ENTRIES_FAILED—Failed to get all the MALC uplink card resource entries from the device.

CS_GET_MALC_UPLINK_CARD_PROFILE_FAILED—Unable to get a particular MALC uplink card profile entry from the device.

CS_DEFAULT_MALC_UPLINK_CARD_SW_FILENAME_MISSING—The default MALC uplink card software filename is not specified in the CS resource file.

CS_ADD_DS1_IMA_8PORT_CARD_PROFILE_FAILED—Unable to add Ds1Ima8Port card profile in the device.

CS_GET_CARD_SW_RUNNING_VERSION_FAILED—Unable to get the card software running version from the device.

CS_INVALID_CARD_LINE_TYPE—Received invalid card line type value from the device.

CS_INVALID_CLOCK_SOURCE_TYPE—Primary Clock Source is either down or it must be set to Loop or Local Timing.

CS_INVALID_PHYSICAL_TYPE_FOR_PRIMARY_CLOCK_SOURCE—Can not use specified physical port for primary clock source.

CS_UNABLE_TO_OBTAIN_CLOCK_SOURCE_ON_PHYSICAL—Unable to obtain clock source on physical port.

CS_MODIFY_ATM_CONFIGURATION_DISALLOWED—Cannot modify atmConfiguration attribute. It may only be set during an add operation.

CS_INVALID_ATM_CONFIGURATION—The AtmConfiguration attribute contains an invalid value. Please refer to DZMS documentation.

CS_CARD_GROUP_FULL—The card group to which this card would belong is already full to capacity.

CS_LIST_IMA_GROUP_PHYSICAL_FAILED—Failed to list IMA physical group objects from the database.

CS_IMA_GROUP_INVALID_DATA—IMA physical group has invalid ifIndex/LgId.

CS_DELETE_PLACE HOLDER_CARD_FAILED—The delete operation of the place holder card failed.

CS_CARD_SW_ENABLE_SET_FAILED—Failed to configure card software enable flag.

CS_MANUAL_CARD_GROUPING_ERROR—The redundant card has to be in an adjacent slot.

CS_MUST_DELETE_VOICE_CONNECTION—Please delete all Subscriber Voice Connections involving all POTS Physicals.

CS_DELETE_LAST_UPLINK_ERROR—Deleting the last uplink card profile is not allowed.

CS_INVALID_APS_CONFIG—The MALC device with ID: \$1 has inconsistent APS Group Configurations.

CS_UNSUPPORTED_CARD_TYPE—Unsupported card type

CS_MUST_DELETE_ATM_XCONN—Associated ATM cross connect(s) must be deleted before this card

CS_CARD_REBOOT_AFTER_ADD_ERROR—Failed to reboot card after adding the card profile.

CS_CARD_STATUS_FETCH_ERROR—Failed to fetch card status from the device.

CS_CARD_GET_ATM_CC_ERROR—Failed to get ATM Cross Connect from the database for ATM Logical verification. Data service is not responding.

CS_CARD_GET_NUM_SLOTS_ERROR—Could not determine number of slots for current card that is being added.

CS_CARD_LIST_CLOCK_PROFILE_ERROR—Failed to list system card profiles on the card during delete operation.

CS_CARD_ATM_XCONN_VERIFY_ERROR—Error occurred while verifying ATM cross connects on the card that is being deleted.

CS_CARD_GET_SW_FILENAME_ERROR—Unexpected error occurred while fetching default software filename.

CS_CARD_NOT_FOUND_DURING_LG_PROCESSING—Inserted card not found in the DZMS database. Failed to process line group information for the inserted card.

ATM cross connect and PVC errors

CS_DELETE_PVC_DISALLOWED—The ATM end point cannot be deleted because it is part of a cross-connect. Please delete the associated ATM cross-connect first.

CS_DELETE_CCPVC_FAILED_VALS—The cross connect was deleted successfully, but the ATM end points associated to the cross connect could not be deleted index \$1, VPI \$2, VCI \$3; index \$4, VPI \$5, VCI \$6

CS_DELETE_CCVPL_FAILED_PVALS—The cross connect was deleted successfully, but the ATM end points associated to the cross connect could not be deleted slot \$1, port \$2, VPI \$3; slot \$4, port \$5, VPI \$6

CS_DELETE_CCPVC_FAILED_PVALS—The cross connect was deleted successfully, but the ATM end points associated to the cross connect could not be deleted slot \$1, port \$2, VPI \$3, VCI \$4; slot \$5, port \$6, VPI \$7, VCI \$8

CS_DELETE_CCPVC_FAILED—The cross connect was deleted successfully, but the ATM end points associated to the cross connect could not be deleted.

CS_MODIFY_PVC_CROSS_CONN_ID_FAILED—Unable to modify the cross-connect IDs in the ATM end points of the cross-connect.

CS_PVCS_INUSE—At least one of the ATM end points specified for creation are already in use.

CS_MODIFY_PVC_CROSS_CONN_DISALLOWED—Modification of an ATM end point is not allowed when it is part of a cross-connect.

CS_XCVCL_ADMIN_UP—The cross connect could not be added because one of the ATM end points was admin down. Please admin up the following end point: index \$1, VPI \$2, VCI \$3

CS_XCVCL_ADMIN_BOTH—The cross connect could not be added because both of the ATM end points were admin down. Please admin up the end points: index \$1, VPI \$2, VCI \$3; index \$1, VPI \$2, VCI \$3

CS_XCVPL_ADMIN_UP—The cross connect could not be added because one of the ATM end points was admin down. Please admin up the following end point: index \$1, VPI \$2

CS_XCVPL_ADMIN_BOTH—The cross connect could not be added because both of the ATM end points were admin down. Please admin up the end points: index \$1, VPI \$2; index \$1, VPI \$2

CS_MAX_PORT_VPI_IN_USE—The maximum number of VPIs on this port has already been reached.
CS_MAX_PORT_VPI_IN_USE_INFO—The maximum number of VPIs on slot/port \$1/\$2 has already been reached. Please select an existing VPI, and try the add again. Please select an existing VPI, and try the add again.

CS_MAX_PORT_VPI_IN_USE_INFO—The maximum number of VPIs on slot/port \$1/\$2 has already been reached. Please select an existing VPI, and try the add again.

CS_VPL_MISSING_ATM_VPI—The AtmVpi for this VPL is missing. Please create one, then attempt adding this VPL.

CS_MAX_PORT_VC_VPI_IN_USE—The maximum number of VC switched VPIs on this port has already been reached.

CS_VPI_RANGE_ERROR—The VPI value (\$1) must be within range \$2 - \$3.

CS_VPI_RANGE_ERROR_INFO—The VPI value (\$1) must be within range \$2 - \$3 on slot \$4, port \$5.

CS_VCI_RANGE_ERROR—The VCI value (\$1) must be within range \$2 - \$3.

CS_VCI_RANGE_ERROR_INFO—The VCI value (\$1) must be within range \$2 - \$3 on slot \$4, port \$5.

CS_ATM_PVC_SCR_RATE_OVERSUBSCRIBED_ERROR—Either the SCR Rate in the associated traffic descriptor(s) caused an over subscription of total available bandwidth (verify if an atmVclBandwidthUnavailable trap is in the trap log), or the device did not accept one or more of the values sent.

CS_ZEDGE_AAL2PVC_ADD_DISALLOWED—Z-Edge 64 only supports one AAL2 VCL. One AAL2 VCL already exists on the device.

CS_ATM_PVC_EXCEEDS_BANDWIDTH—The cell rate specified for this ATM end point exceeds the available bandwidth. Please verify the traffic descriptor settings.

CS_UNSUPPORTED_APPID—The specified Application ID is not supported

CS_AAL2_PVC_CAS_INCORRECT_ENABLE—The Channel Associated Signalling cannot be set CS_AAL5_ENCAPSTYPE_ERROR—The value chosen for Aal5EncapsType is not allowed to “Enabled” for the given AppId.

CS_AAL2_APPID_MODIFY_DISALLOWED—Modification of the AAL2 AppId is disallowed.

CS_INVALID_AAL2_AUDIO_PROFILE_ID—Invalid audio profile identifier specified for the given AppId.

CS_INVALID_AAL2_APID_VAL—The audio profile identifier for this application ID should be \$1

CS_INVALID_AAL2_TD_TYPE—Traffic descriptors with the service category UBR cannot be used with AAL2 VCLs.

CS_PVC_NOT0—VCI and VPI cannot both be 0

CS_PVC_NOTZERO_INFO—VCI and VPI cannot both be 0 (slot/port/VPI/VCI) \$1/\$2/\$3/\$4

CS_PVC_RXTX_ERROR—The receive and transmit traffic descriptors must be the same

CS_PVC_VCLCASTTYPE_ERROR—The VclCastType can only be point-to-point

CS_PVC_VCLCONNKIND_ERROR—The VclConnKind can only be PVC

CS_OAM_F5_PING_ERROR—OAM F5 ping failed on the device. Request was successfully sent to the device but device failed to execute the ping request. Probable Cause: Internal device error.

CS_OAM_F5_PING_UNSUPPORTED—OAM F5 ping is not supported by the parent card of the VCL.

ATM traffic descriptor errors

CS_DELETE_ATM_TRAFF_DISALLOWED—ATMTrafficDesc cannot be deleted. Valid ATM end points are referencing this traffic descriptor, and they must be deleted first.

CS_INVALID_ATM_TRAFFIC_VCL_RATE—The ATM Traffic Descriptor's VCL rate is invalid.

CS_ATM_TRAFFIC_VCL_RATE_OVERSUBSCRIBED_ERROR—The specified VCL rate probably caused an oversubscription of total bandwidth. Please check the trap log for a atmVclBandwidthUnavailable trap to verify.

CS_ATM_TRAFF_PARAM_ERROR—ATMTrafficDesc parameter value is not valid

CS_ATM_TRAFF_UNSUPP_SC_FOR_DEVICE—ATMTrafficDesc service category not supported on this device

CS_ATM_TRAFF_PARAM_MBS_ERROR—ATMTrafficDesc MBS parameter can not be less than 2

CS_ATM_TRAFF_PARAM_CDVT_ERROR—ATMTrafficDesc CDVT parameter can not be less than 20

CS_ATM_TRAFF_PARAM1_LESS_THAN_PARAM2_ERROR—ATMTrafficDesc parameter1 (PCR CLP—0+1) can not be less than parameter2 (SCR / PCR CLP—0)

CS_ATM_TRAFF_PARAM1_LE_TO_PARAM2_ERROR—ATMTrafficDec parameter1 (PCR CLP=0+1) must be greater than parameter2 (SCR)

CS_ATM_TRAFF_PARAM_ERR_MBS_PCR_LO—Given your MBS and SCR settings, the PCR value cannot be less than \$1 or, given your PCR and SCR settings, the MBS value cannot be less than \$2

CS_ATM_TRAFF_PARAM_ERR_MBS_PCR_HI—Given your MBS and SCR settings, the PCR value cannot be greater than \$1 or given your PCR and SCR settings, the MBS value cannot be greater than \$2

CS_MODIFY_TRAFFIC_DESC_DISALLOWED—The admin status must be set to down before the traffic descriptor parameters can be modified

CS_ATM_TRAFF_PARAM_ERR_CDVT_LO—Given your traffic descriptor settings, the CDVT value cannot be less than \$1

CS_ATM_TRAFF_PARAM_ERR_CDVT_HI—Given your traffic descriptor settings, the CDVT value cannot be greater than \$1

CS_ATM_TRAFFIC_MOD_ERROR—Traffic Descriptors may not be modified while associated VCLs are Admin up

CS_INVALID_TD_ERROR—The transmit traffic descriptor index of each side of the cross connect must match the receive traffic descriptor index of the other side

CS_INVALID_DIVIDER—The CAC Divider must be greater than 0

CS_INVALID_UPC—The Usage Parameter Control cannot be false when using CBR

CID errors

CS_INVALID_CIDS—The min or max CID value is outside the allowed range

CS_INVALID_CID—The CID value is invalid.

CS_INVALID_CID_SSCS_TYPE—Invalid SSCS type specified for the given CID.

CS_INVALID_CID_INDEX—The CID index (\$1) must be within \$2 - \$3

CS_CID_OUTSIDE_AAL2_RANGE—The min or max CID value is outside the allowed AAL2 Application ID range (\$1 to \$2)

CS_CID_PARENT_AAL2_NOT_FOUND—CID's parent AAL2 VCL not found.

CS_INVALID_AAL2_SSCS_TYPE—Invalid combination of default SSCS type and AAL2 AppId values.

CS_BCHAN_CREATE_ERROR—B1/B2 channel CID autocreation failed

CS_BCHAN_DEL_DISALLOWED—B1/B2 channel CID manual deletion is not allowed.

CS_CID_BCHAN_DEL_ERROR—Failed Auto B1/B2 channel CID deletion

CS_BCHAN_GET_ERROR—Could not fetch B Channel's CID from device

CS_JET_255_ERROR—A CID index of 255 is not allowed for this AAL2 Application ID.

CS_MISSING_LINE_GROUP—Could not find the associated LineGroup in the database

CS_ASSOC_VOICE_CONN_ERROR—The voice connection currently associated with this CID must be deleted before this CID can be removed.

CS_JET_255_DELETE_DISALLOWED—Delete of CID 255 is not allowed. Only CIDs in the range 1 to 254 for the given AppId may be deleted.

CS_PCM_ERROR—The PCM Encoding must be set to the same value as the parent AAL2 VCL's PCM Encoding. Changing the AAL2 VCL's encoding will also update the PCM encoding of all related CIDs.

CS_ELCP_PARENT_AAL2_NOT_FOUND—Elcp's parent AAL2 VCL not found.

CS_AAL2ELCP_CREATE_DISALLOWED—Create Aal2Elcp is disallowed when the port type does not match the AAL2's AppId.

CS_INVALID_SUBNET—The DHCP subnet has not been provisioned properly.

CS_NO_SUBNET—Could not find a proper DHCP subnet.

ATM VPI errors

CS_INVALID_SWITCH_TYPE—The ATM VP or VC switch type must be selected

CS_INVALID_ATMVPI_DEVICE—The ATM VPI can not be created on that device type

CS_MAXVCI_TOO_LOW— The chosen maxVci is too low, and would cause current VCLs to be deleted upon card reboot

CS_EXISTING_VCL_ERROR—This VPI can not have its switch type changed since there are existing AtmEndPoints using it

CS_VPL_MISSING_VPI—An ATM VPI must be created on this port before an ATM VPL may be created

CS_MISSING_ATM_VPI—An ATM VPI must be created on this port before an AtmEndPoint may be created

CS_MISSING_ATM_VPI_INFO—An ATM VPI must be created on this port before an AtmEndPoint may be created (slot/port/VPI) \$1/\$2/\$3

CS_WRONG_ATM_EP_TYPE—This ATM End Point does not match the ATM VPI switch type

CS_WRONG_ATM_EP_TYPE_INFO—ATM End Point does not match the ATM VPI switch type (slot/port/VPI): \$1/\$2/\$3

CS_VCI_OUT_OF_RANGE—This VCI is greater than the maximum specified by the ATM VPI: \$1

CS_VCI_OUT_OF_RANGE_INFO—This VCI (slot/port/VPI/VCI) \$1/\$2/\$3/\$4 is greater than the maximum specified by the ATM VPI: \$5

CS_INVALID_VPI—The Specified VPI is invalid. Please choose a value between 0 and 255.

CS_MAX_ATMENDPOINTS—The limit of AtmEndPoint connections has been reached for this card, so no more may be added

CS_REBOOT_REQUIRED—This card must be rebooted before AtmEndPoint provisioning can continue

CS_VPI_MUST_DEL_EPS—AtmVpis may not be deleted until the VPL or VCLs using this are removed.

CS_BULK_POTS_GR303_CRV_RANGE_ERROR—Cannot accommodate CRVs for all the POTS ports from the stating CRV. Some CRVs not available.

CS_BULK_POTS_V52_USERPORT_RANGE_ERROR—Cannot accommodate UserPorts for all the POTS ports from the stating UserPort. Some UserPorts are not available in the given range.

CS_BULK_ISDN_V52_USERPORT_RANGE_ERROR—Cannot accommodate UserPorts for all the ISDN ports from the stating UserPort. Some UserPorts are not available in the given range.

Bulk processing errors

CS_MISSING_TRUNK_PORT—The mandatory trunk port was missing from the request. Please select a valid trunk port and resend the request.

CS_MISSING_LINE_PORTS—At least one line card or port must be selected in order to cross connect between a trunk and line side.

CS_MISSING_TRNK_VCL—The request did not contain the proper upstream VCL. Please select a valid VCL and resubmit the request.

CS_MISSING_LINE_VCL—The request did not contain the proper downstream VCL. Please select a valid VCL and resend the request.

CS_CARD_ATM_FULL—Based on this card's current ATM configuration, no more ATM EndPoints may be added to this card. Please add more AtmVpis or select another card.

CS_VCI_RANGE_FULL—All valid VCI values for the specified VPI are already in use. Please select another VPI.

CS_START_VCI_IN_USE—The startVci value is already being used on this port. Please select a different startVci.

CS_MISSING_SUBSCRIBER_PORT— Could not locate the subscriber port associated with the voice connection.

CS_BULK_POTS_V52_PRIMARY_LINK_MISSING— Could not locate the primary link of the V52 Interface Group.

CS_BULK_POTS_GR303_PRIMARY_LINEMAP_MISSING—Could not locate the primary line map of the GR-303 Interface Group.

Bulk statistics errors

CS_BULKSTATS_ADD_DISALLOWED—Currently, there can only be one Bulk Stats Config added on the device

Bulk data cross connect processing errors

CS_INVALID_AUTO_CREATION_VALUE—The autoCreation value (\$1) must be either \$2 or \$3.

CS_MISSING_LINE_VCI—The line-side VCI value must be specified.

CS_MISSING_LINE_VPI—The line-side VPI value must be specified.

CS_MISSING_TD_TYPE—The traffic descriptor type must be specified.

CS_MISSING_TD_SERCAT—The traffic descriptor service category must be specified.

CS_MISSING_TRUNK_TX_TD—The trunk side transmit traffic descriptor index must be specified.

CS_MISSING_TRUNK_RX_TD—The trunk side receive traffic descriptor index must be specified.

CS_MISSING_TRUNK_PORT—The mandatory trunk port was missing from the request. Please select a valid trunk port and resend the request.

CS_MISSING_LINE_PORTS—At least one line card or port must be selected in order to cross connect between a trunk and line side.

CS_MISSING_PORT—Could not locate the port in the database via the specified line group ID. This generally occurs if the device configuration has changed prior to the DZMS server beginning this provisioning (for example, a card could have been removed). Please verify that DZMS and the device are in sync, and attempt to provision again.

CS_MISSING_TRNK_VCL—The request did not contain the proper upstream VCL. Please select a valid VCL and resubmit the request.

CS_MISSING_LINE_VCL—The request did not contain the proper downstream VCL. Please select a valid VCL and resend the request.

CS_MISSING_DB_TRUNK_VCL—The trunk-side VCL could not be located in the database.

CS_MISSING_REAL_TD—When existing TD indices are used, both must be valid and present in the database. Please select valid existing TD indices.

CS_TD_NEXT_INDEX_ERROR—The next index for the traffic descriptor could not be retrieved from the device.

CS_VCXC_NEXT_INDEX_ERROR—The next index for the cross connect could not be retrieved from the device.

CS_MISSING_DEVIP—The device IP could not be retrieved from the database. Please verify that the device has been configured with an IP address.

CS_CARD_ATM_FULL—Based on this card's current ATM configuration, no more ATM EndPoints may be added to this card. Please add more AtmVpis or select another card.

CS_VCI_RANGE_FULL—All valid VCI values for the specified VPI are already in use. Please select another VPI.

CS_START_VCI_IN_USE—The startVci value is already being used on this port. Please select a different startVci.

CS_MODIFY_VCLCCID_ERROR—The database could not update the cross connect ID in each VCL. They are in use by the cross connect, but the UI may not show it at this point.

CS_LOOKUP_CARD_FROM_PHYSICAL_ERROR—The card in slot \$1 could not be located via port \$2 in the database.

CS_LOOKUP_LOGICAL_FROM_PHYSICAL_ERROR—The ATMLogical associated with slot \$1, port \$2 could not be retrieved from the database.

CS_FOUND_EXISTING_VCL—A VCL may potentially have been on the device but not in DZMS. This VCL (slot/port/VPI/VCI) \$1/\$2/\$3/\$4 has been retrieved from the device and added to DZMS. Please verify that ZMS and the device are in sync before attempting further provisioning.

Physical connections

CS_PHYSICAL_TYPE_MISMATCH—The physical port type did not match the expected type.

CS_LINE_ALREADY_USED—The line is already used.

CS_MISSING_TD—Both Traffic descriptor indices must be specified

CS_T1_MODIFY_FAILED—Could not modify the T1 Port

CS_T1_UPDATE_FAILED—Failed to update the T1

CS_T1_LIST_FAILED—Could not obtain a list of T1s from the database for a given card

CS_T1_ADD_DISALLOWED—T1 creation is disallowed. T1 already exists.

CS_T1_DELETE_DISALLOWED—T1 cannot be deleted. Delete parent card.

CS_T3_ADD_DISALLOWED—T3 creation is disallowed. T3 already exists on the card.

CS_T3_DELETE_DISALLOWED—T3 cannot be deleted. Delete parent card.

AccessLists

CS_DELETE_ACCESSLIST_DISALLOWED—Delete AccessList is not allowed when it is part of a provisioned community name.

Miscellaneous errors

CS_PHYSICAL_MULTIPLE—There were multiple physicals returned, should be only one.

CS_NOT_ENOUGH_INFO—There is not enough info in the trap to process it properly.

CS_QUERY_FAILED—The DS query failed, no data retrieved.

CS_PARSE_ERROR—There are parsing errors.

CS_TRAP_REC_PARSE_ERROR—Error parsing the varBinds list from CSTrapRecords.properties.

CS_INSTANCE_PARSE_ERROR—Error parsing the instance attributes defined in metadata.

CS_TRAP_INSTANCE_PARSE_ERROR—Error parsing instance information in the trap.

CS_GET_METADATA—Error in retrieving metadata.

CS_NULL_OBJECT_TYPE—Null Object.

CS_METADATA_INSTANCE_ID_MISMATCH—The instance attributes in metadata does not match the trap.

CS_GET_MD_INSTANCE_ATTR—Cannot fetch the instance attribute from metadata.

CS_REMOTE_EXCEPTION—Remote exception occurred.

CS_INSTANTIATE_FAILED—Instantiation of the object failed.

CS_NOT_SUPPORTED_ON_DEVICE_VERSION— Value of attribute not supported on the version of this device.

CS_ACTION_IN_TRANSITION— An action has been initiated on the device. Waiting for trap acknowledgement.

CS_USED_AS_CLOCK_SOURCE—Since this physical port is being used as the primary clock source, the value may not change.

CS_TRANSMIT_CLOCK_SOURCE_IS_THROUGH—Since transmit clock source of this Port or Channel is configured as 'Through Timing', the port or channel is not allowed as System Clock Source.

CS_IMA_PROVISIONING_WARNING— If you do not set 'Admin Status' of IMA physical group to 'Down' before modifying parameters, traffic may be impacted. Do you want to continue?

CS_IMA_MIN_RX_LINKS_RAPTOR100_ERROR—Minimum Receive Links should be between 1 and 4 on Raptor100

CS_IMA_MIN_TX_LINKS_RAPTOR100_ERROR—Minimum Transmit Links should be between 1 and 4 on Raptor100

CS_NAME_TOO_LONG—Name is too long. The number of characters has exceeded the limit.

CS_NAME_TOO_LONG_VAL—The number of characters in the name has exceeded the \$1 character limit.

CSS_UNKNOWN—Unknown error

CSS_DEVICE_NOT_FOUND—Updated device not found

CSS_TRANSLATION—Could not translate record to DZMS object

CSS_VALIDATION—Validation error

CSS_NO_RECORDS—No updated records were found

CSS_UNKNOWN_ACTION—Unknown update action in update

CSS_UNKNOWN_UPDATED_DEVICE_TYPE—Unknown updated device type

CSS_NOT_IMPLEMENTED—

CSS_CREATE_CLASS_OBJECT—Could not create class

CSS_NO_TRANS_SUPPORT—

CSS_INVALID_IF_TYPE—Invalid ifType in update

CSS_UNKNOWN_REQUEST—Unknown request

CSS_UNKNOWN_RECOR—Unknown record in update

CSS_UNKNOWN_CARD_TYPE—Unknown card type in record

CSS_UNKNOWN_INTERFACE_TYPE—Unknown card interface type in card status trap

CSS_UNKNOWN_CPE_DEVICE—Unknown CPE device

CSS_GENERATE_ALARM—Generation of FS alarm failed

CSS_UPDATE_IN_PROGRESS—A Config Sync Update is already in progress. Please try later.

CSS_ADD_DNS_RESOLVER_NOT_ALLOWED—Add not allowed for dns resolver

CSS_FILE_SYNTAX_ERROR—Update file contains syntax error

CSS_IF_TYPE_ATTR_NOT_FOUND—The ifType attribute was not found in the update file

CSS_IF_TYPE_EXT_ATTR_NOT_FOUND—The ifTypeExtension attribute was not found in the update file

CSS_IF_STACK_RECORDS_NOT_FOUND—If Stack Record entries are missing from the update file

CSS_IF_TRANSLATE_RECORDS_NOT_FOUND—If Translate Record entries are missing from the update file

CSS_IF_TYPE_NOT_SUPPORTED—IfType and IfTypeExtension not supported in CSS for this device

CSS_FETCH_IF_INDEX_FROM_IF_TRANSLATE—ZhonelfIndex attribute not found in If Translate Record in update file

CSS_LOCATE_IF_TRANSLATE_RECORD_FAILURE—Could not locate If Translate Record in update file

CSS_LOCATE_IF_STACK_RECORD_FAILURE—Could not locate If_Stack_Record in update file

CSS_MAP_OBJECT_TO_RECORD_FAILURE—Could not map DZMS object to device record

CSS_IF_INDEX_ATTR_NOT_FOUND—Could not find the ifIndex attribute in the device record

CSS_ATTRIBUTE_NOT_FOUND—Attribute not found in update file record

CSS_UNSUPPORTED_DS1_LINE_TYPE—DS1 line type specified in update file is not valid

CSS_OUTSTANDING_UPDATE_NOT_FOUND—Could not locate outstanding update

CSS_IP_LOGICAL_VPI_VCI_NOT_FOUND—VPI/VCI values for IpLogical were not found

CSS_IP_LOGICAL_LOWER_LAYER_NOT_FOUND—Lower layer interface of IpLogical was not found

CSS_DUPLICATE_OBJECTS_IN_CACHE—Duplicate objects were found in cache

CSS_UNKNOWN_TYPE_EXTENSION—An unknown ifTypeExtension value was found in the update file

Version errors

CS_VERSION_ERROR—The current software version does not support the request made

CS_VERSION_INFO_ERROR—The current software version (\$1) must be at least \$2 to support the request made.

CS_IP_LOGICAL_LOWER_IF_INDEX—Can not add IP Logical when lowerIfIndex equals 0

CS_MISSING_CARD_VERSION—Card version is missing on the device.

Logical errors

CS_BRIDGE_LOGICAL_LOWER_IF_INDEX—Can not add Bridge Logical when lowerIfIndex equals 0

CS_BRIDGE_LOGICAL_ENCAPS_TYPE_ERROR— AAL5 VCL encapsulation must be 'Other', to add a Bridge Logical

CS_MISSING_CARD_VERSION—Card version is missing on the device.

Quick cross connect errors

CS_MISSING_PORT—Could not locate the port on the device via the given port number

CS_MISSING_LGID—Could not locate the port on the device via the specified port lgid

CS_INVALID_PORT_NUM—Both port numbers must exist in the database, and be greater than 0

CS_MISSING_ATMLOGICAL—Could not locate the ATM Logical interface via the port number given

CS_MISSING_DEFAULT_PVC—Could not locate the default ATM end point template in the database

CS_MISSING_DEFAULT_CID—Could not locate the default CID template in the database

CS_MISSING_DEFAULT_TD—Could not locate the default TD template in the database

CS_MISSING_DEFAULT_ELCP—Could not locate the default ELCP template in the database

CS_MISSING_DEFAULT_SUBSCRIBER_PORT—Could not locate the default SubscriberPort template in the database

CS_MISSING_DEFAULT_SUBVOICE_CONN—Could not locate the default SubVoiceConn template in the database

CS_MISSING_DEFAULT_SUBSCRIBER_ENDPT—Could not locate the default SubscriberPort End Point template in the database

CS_MISSING_DEFAULT_GR303CRV—Could not locate the default GR-303 CRV template in the database

CS_MISSING_DEFAULT_V52USERPORT—Could not locate the default V5.2 USERPORT template in the database

CS_MISSING_VPIVCI—All VPI and VCI parameters must be set

CS_MISSING_VPI—All VPI parameters must be set

CS_MISSING_CC_PVCS—Please either select 2 existing ATM end points, or create them both. A combination of new and existing is not allowed.

CS_MISSING_VPI_VALUE—The VPI parameter must be set

CS_MISSING_AAL5_PROXY—Could not locate the Aal5 proxy port

CS_VCLRATES_INUSE—VCL rates currently in use by traffic descriptors can not be modified

CS_VCLRATE_INUSE—This VCL rate appears to be used by at least one traffic descriptor, so it can not be modified

MTAC card issues

CS_E1MTAC—The uplink card appears to contain a non-E1-compatible interface

CS_T1MTAC—The uplink card appears to contain a non-T1-compatible interface

CS_MTACFC_SLOT_ERROR—This card may only be added to slot 10.

CS_VCLRATES_INUSE—VCL rates currently in use by traffic descriptors can not be modified

CS_VCLRATE_INUSE—This VCL rate appears to be used by at least one traffic descriptor, so it can not be modified

Data/handle related errors

CSS_HANDLE_NOT_FOUND—Handle not found

CSS_AUTO_ADDED_OBJ_HANDLE_NOT_FOUND—Automatically added object not found in DZMS database

CSS_DATA_NOT_FOUND—Data object not found

CSS_PARENT_HANDLE_NOT_FOUND—Parent not found

CSS_EMPTY_BASE_DATA—Empty base data

CSS_GET_HANDLE_CONSTRUCTOR—Could not get handle constructor

CSS_CREATE_PHYSICAL_FAILURE—Could not create physical

CSS_CREATE_PVC_FAILURE—Could not create AtmPvc

CSS_CREATE_LOGICAL_FAILURE—Could not create logical: failed to set ifIndex

CSS_INVALID_CPE_TRAP_DATA—Invalid data in CPE trap

CSS_ADD_ACCESS_LIST—Could not add access list

CSS_FILTER_NOT_UNIQUE—Filter did not uniquely identify data object

CSS_DEVICE_LOCK_NOT_FOUND—Could not locate device lock in database

CSS_PHYSICAL_NOT_FOUND—Could not locate physical in database

CSS_INTERFACE_NOT_FOUND—Could not locate interface in database

CSS_DSL_PROFILE_NOT_FOUND—Could not locate DSL profile in database

CSS_ADSL_PROFILE_NOT_FOUND—Could not locate ADSL profile in database

CSS_ISDN_ALARM_PROFILE_NOT_FOUND—Could not locate ISDN alarm profile in database

CSS_IF_TYPE_NOT_FOUND—ifType was not found in the update file

CSS_POTS_CONFIG_DATA_NOT_FOUND—Could not locate autocreated POTS configuration data in database

CSS_IMA_PHYSICAL_GROUP_NOT_FOUND—Could not locate autocreated IMA Physical Group in database

CSS_SKYZHONE_LINK_NOT_FOUND—Could not locate SkyZhone link in DZMS database

CSS_MTAC_NOT_FOUND—MTAC object not found in DZMS database

CSS_ATM_LOGICAL_NOT_FOUND—Could not locate AtmLogical in DZMS database

Update file related errors

CSS_PARSE_INSTANCE_ID—Could not parse instance ID

CSS_PARSE_SSPTS—Could not parse shelf/slot/port/subport

CSS_UPDATE_FILE_NOT_FOUND—Update file not found

CSS_INVALID_UPDATED_ATTR_DATA—Invalid updated attribute data

CSS_READ_FILE_DATA—Could not read update file data

CSS_READ_DELIMITERS—Error reading update file delimiters

CSS_READ_RECORD_DATA—Could not read record data

CSS_CLOSE_UPDATE_FILE—Could not close update file

CSS_POLL_UPDATE_STATUS—Error polling device for update file transfer status

CSS_SYNC_ERROR—Device error transferring update file to DZMS

CSS_SYNC_STATUS_TIMEOUT—Transfer of update file from device to DZMS timed out

CSS_FILE_TRANSFER_TIMEOUT—Transfer of update file from device to DZMS timed out

CSS_LINE_TYPE_ATTR_NOT_FOUND—DS3 line type attribute not found in update file

CSS_UNKNOWN_DS3_INTERFACE_TYPE—DS3 profile in update file does not contain enough information to determine interface type

CSS_DS3_LINE_TYPE_NOT_FOUND—DS3 profile in update file does not contain line type

CSS_INVALID_ZMS_CARD_INTERFACE_TYPE—The Card interface type in the ZMS is invalid

CSS_CARD_TYPE_NOT_IN_TRAP—Card type was not present in the card status trap

CSS_POTS_INTERFACE_TYPE_NOT_FOUND—The POTS interface type was not found in update file

Tokenizing errors

CSS_TOKENIZE_RECORD_DATA—Could not tokenize record data
 CSS_TOKENIZE_ATTRIBUTE_DATA—Could not tokenize attribute data
 CSS_TOKENIZE_RECORD_INFO—Could not tokenize record information

Service related failures

CSS_INIT_CS_SERVICE—Could not initialize CS
 CSS_INIT_CSS—Could not initialize CSS
 CSS_INIT_ADS—Could not initialize ADS
 CSS_INIT_DS—Could not initialize DS
 CSS_INIT_FS—Could not initialize FS
 CSS_INIT_NWS—Could not initialize NWS
 CSS_AUTHENTICATE_ERROR—Could not authenticate user
 CSS_CREATE_REQUEST—Could not create request
 CSS_REMOTE_EXCEPTION—Remote exception
 CSS_SUBMIT_REQUEST—Error submitting request to service
 CSS_GET_ACTIVE_UPDATES—Could not retrieve active updates
 CSS_CS_MODIFY—Configuration Service could not process modify
 CSS_CS_DELETE—Configuration Service could not process delete
 CSS_CS_ADD—Configuration Service could not process add
 CSS_PROCESS_UPDATE—Could not process update
 CSS_UPDATE_REQUEST—Could not submit update request
 CSS_DS_QUERY—Error submitting DS query
 CSS_UNIQUE_DEVICE_NOT_FOUND—Unique updated device not found in database
 CSS_EMPTY_SERVICE_RESPONSE—Server response was empty
 CSS_INVALID_REQUEST—Invalid request
 CSS_NULL_MD_INSTANCE_ATTRS—Instance attribute in metadata is null
 CSS_TRANSACTION_ROLLBACK_ERROR—Could not rollback transaction
 CSS_BEGIN_TRANSACTION—Could not get database connection
 CSS_DS_ADD—DS could not add object
 CSS_DS_DELETE—DS could not delete object

CSS_WRONG_ZMS_IP_ADDR—Device's DZMS IP address does not match actual DZMS IP Address

CSS_ERROR_ON_ZMSIP_GET—Error occurred trying to get DZMS IP address from device

Metadata errors

CSS_GET_METADATA—Could not get metadata

CSS_METADATA_INSTANCE_ID_MISMATCH—Record instance ID mismatch with metadata instance ID

CSS_MD_ATTRNAME_NOT_FOUND—Attribute name not found in metadata

CSS_GET_MD_INSTANCE_ATTR—Error retrieving instance ID from metadata

Trap errors

CSS_REGISTER_FOR_FS_TRAPS—Error registering for traps

CSS_TRAP_NAME_NOT_FOUND—Trap name not found

CSS_CARD_STATUS_TRAP_CONTENTS—Invalid card status trap contents

CSS_TRAP_DATA_NOT_FOUND—Trap data not found

CSS_GET_TRAP_OID_FAILURE—Failure retrieving trap OID

Persist errors

CSS_PERSIST_ACTIVE_UPDATE—Could not persist active update

CSS_PERSIST_COMPLETED_UPDATE—Could not persist completed update

Null pointer errors

CSS_INVALID_SESSION—Invalid session

CSS_NULL_DEVICE_DATA—Null device data

CSS_NULL_SYS_OID—Null sysObjectID

CSS_NULL_BASE_DATA—Null base data

CSS_NULL_DEVICE_HANDLE—Null device handle

CSS_NULL_GROUP_ID—Null group ID

CSS_NULL_RECORD_NAME—Null record name

CSS_NULL_OBJECT_TYPE—Null object type

CSS_NULL_SERVICE_RESPONSE—Null service response

CSS_NULL_SERVICE_RESPONSE_STATUS—Null service response status

CSS_NULL_FILE_DATA—Null file data

CSS_NULL_ATTR_NAME—Null attribute name

CSS_NULL_PARENT_HANDLE—Null parent handle

CSS_NULL_RECORD_DATA—Null record data

CSS_NULL_ATTR_DATA—Null attribute data

CSS_NULL_LIST_FILTER—Null list filter

Database debug definitions

Database errors

DB_INVALID_REQUEST—Invalid database request

DB_RESULT_SUCCESS—Database result is valid

DB_UNIQUE_CONSTRAINT_VIOLATION—Object is not uniquely identified

DB_TIMESTAMP_FAILURE—Modified record is out of date

DB_DEVICE_SYNC_IN_PROGRESS—Synchronization with the device is in progress.

DB_SUBSCRIBER_EXISTS—Cannot delete device because subscribers exist

DB_DEADLOCK—Database locked, please try again

DB_OPER_ERROR—Database operation error

Error strings defined for diagnostic service

DGS_ONE_ELEMENT—Only one request element is allowed for diagnostic request.

DGS_NO_VALIDATOR—Unable to obtain a BL object, object type unknown.

DGS_NO_VALID_INDEX—Unable to obtain a valid index to start diag request.

DGS_TEST_STARTED—Cannot start test that is already in progress.

DGS_TEST_NOT_STARTED—Cannot start test that is already in progress.

DGS_ACTION_REQUIRED—Diagnostic action attribute required to perform update.

DGS_NO_TEST_STARTED—No test has been initialized, there is no data to retrieve.

DGS_INVALID_TEST—Test type specified is invalid for object type.

DGS_ACTION_REQUIRED—Action is required to process update request.

DGS_OPER_INVALID—Must set AdminStatus to maintenance before starting a diag test.

DGS_OPER_NO_RESULT—Unable to retrieve operational status, can not start test.

ADS errors

Authentication

ADS_ERROR—ADS error

ADS_INVALID_REQ—Invalid request type

Authorization

ADS_LOGIN_ERROR—User name and/or password is incorrect.

ADS_BADLOGIN_LIMIT—Too many attempts by user to login, account is deactivated, contact your administrator.

ADS_FAILURE—Authorization failure

ADS_INVALID_SESSION—Operation not authorized since request session ID is invalid (unknown).

Property get and set errors

ADS_MISSING_FILENAME—The name of the properties file must be specified to retrieve its property list

ADS_INVALID_PROP_DATA—In order to retrieve or update properties, an ADSPropertyData must be used

User

ADS_OPERATOR_INACTIVE—Specified user is not active, contact your administrator.

ADS_RESERVED_USERNAME—User name 'zhone' is reserved.

ADS_ADMIN_RESERVED—User name 'admin' is reserved.

ADS_ADMIN_CAN_NOT_DELETE—Can not delete administrator user 'admin'.

ADS_ADMIN_CAN_NOT_MODIFY—Can only modify the password for user 'admin'.

ADS_USER_MODIFY_INVALID—Users can only modify their password.

ADS_OPEN_SESSION—This user is currently logged in.

ADS_INVALID_PASSWORD—The password specified is invalid. Please choose another.

ADS_USER_INVALID—The user specified by the request session ID does not exist, or has been removed from the system.

ADS_INVALID_ADD_SELF—A user cannot add themselves to the system.

ADS_INVALID_DEL_SELF—A user cannot delete themselves from the system.

Group

ADS_GROUP_HAS_OPERATORS—There are users in this group.

ADS_GROUP_HAS_REGIONS—There are regions in this group.

ADS_GROUP_HAS_CUSTOMERS—There are customers in this group.

ADS_GROUP_HAS_SUBSCRIBERS—There are subscribers in this group.

ADS_DEFAULT_GROUP_DELETE—Can not delete a default group.

ADS_GROUP_INVALID—The user's group specified is invalid. Please choose another.

ADS_GROUP_PERMISSIONS_ERROR—This user does not have permission to modify attributes of users in other groups.

Access errors

ADS_ACCESS_DENIED—Insufficient privileges to perform the action.

ADS_SUPER_USER_ADMIN_ACCESS—Administrator can only perform operations on users and groups.

ADS_GROUP_OBJECT_ACCESS—Only administrators can perform operations on group objects.

ADS_OPERATOR_OBJECT_ACCESS—Only global administrators and group Administrators can perform operations on users.

Filename list errors

ADS_FILENAME_LIST_ERROR—Error reading list of filenames

DataService error definitions

Initialization errors

DS_INIT_ERROR—Error establishing connection with Database.

DataService errors

DS_GET_FAILURE—DS get failure
DS_LIST_FAILURE—DS list failure
DS_ADD_FAILURE—DS add failure
DS_DELETE_FAILURE—DS delete failure
DS_MODIFY_FAILURE—DS modify failure
DS_REQUEST_FAILURE—Failed to execute request
DS_OBJECT_UNKNOWN—The object requested is unknown to the system

DataService success

DS_GET_SUCCESS—DS get success
DS_LIST_SUCCESS—DS list success
DS_ADD_SUCCESS—DS add success
DS_DELETE_SUCCESS—DS delete success
DS_MODIFY_SUCCESS—DS modify success

Service notification

DS_NO_DATA_AVAILABLE—No Data Available
DS_STATUS_NULL—Status was null
DS_FAILURE—DS failure
DS_TIMESTAMP_FAILURE—Modified record is out of date
DS_INSTANCE_KEY_FAILURE—No instanceKey attribute provided
DS_CONNECTION_NOT_AVAILABLE—No database connection available

Class problems

DS_NOT_FOUND—Error: dynamic class not found
DS_NOT_ACCESSABLE—Error: dynamic class not accessible
DS_NOT_CREATED—Error: dynamic class not instantiated
DS_GET_MAXVCI_FAILURE—The card's maxVci value could not be read from the database. Please ensure that the card has been properly provisioned by synchronizing with the device.
DS_GET_KEY_ERROR—Failed to get a unique key from the database sequence.

DS_INSERT_FAILED_ERROR—The object could not be added to the database.

DS_UPDATE_FAILED_ERROR—The object could not be modified in the database.

DS_DELETE_GET_ERROR—The row retrieval from the database prior to the delete failed. This object has either already been deleted, or has not yet been added to the database.

DS_DELETE_FAILED_ERROR—The object could not be deleted from the database.

DS_GET_FAILED_ERROR—The object could not be retrieved from the database.

DS_CUSTOM_QUERY_FAILED_ERROR—The custom request failed.

DS_UNKNOWN_QUERY_ERROR—The query could not be processed since its request type was unknown.

DS_MANAGE_FAILED_ERROR—An error was encountered while attempting to manage the device.

DS_UNMANAGE_FAILED_ERROR—An error was encountered while attempting to unmanage the device.

DS_NO_AVAILABLE_DS1_LOGICAL_ID—No DS1 logical IDs are currently available.

DS_UNSUPPORTED_DB_OPERATION_ERROR—This operation is not supported.

DS_LIST_FAILED_ERROR—The list request failed.

DS_CONN_INTERNAL_USE_ONLY—This connection is available only during adapter initialization

Unique constraint violation errors (per db table)

DS_UCV_ACCESS_LIST—This name is already used, please select another

DS_UCV_ACCESS_LIST_ENTRY—This name or IP Address is already used, please select another

DS_UCV_ACCESS_LIST_ENTRY_TMPL—This name or IP Address is already used, please select another

DS_UCV_ACCESS_LIST_TMPL—This name is already used, please select another

DS_UCV_ACTION—This name is already used, please select another

DS_UCV_ACTIVE_UPDATE—This index is already used, please select another

DS_UCV_ADSL_ALARM_PROFILE—This name is already used, please select another

DS_UCV_ADSL_ALARM_PROFILE_TMPL—This name is already used, please select another

DS_UCV_ADSL_CONFIG_PROFILE—This name is already used, please select another

DS_UCV_ADSL_CONFIG_PROFILE_TMPL—This name is already used, please select another

DS_UCV_ADSL_FAST_PHYSICAL—This name is already used, please select another

DS_UCV_ADSL_FAST_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_ADSL_INTERLEAVED_PHYSICAL—This name is already used, please select another

DS_UCV_ADSL_INTERLEAVED_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_ADSL_PHYSICAL—One of the following is already being used: name, alarm profile name, config profile name.

DS_UCV_ADSL_PHYSICAL_TMPL—One of the following is already being used: name, alarm profile name, config profile name.

DS_UCV_ADS_GROUP—This name is already used, please select another

DS_UCV_ADS_ROLE—This name is already used, please select another

DS_UCV_ALARM_FORWARD_HOST—This forward IP address is already used, please select another

DS_UCV_ALARM_LOG—This name is already used, please select another

DS_UCV_ALARM_PROPAGATION—This name is already used, please select another

DS_UCV_APS_CHANNEL_CONFIG—The group name or channel number is already used, please select another

DS_UCV_APS_CHANNEL_CONFIG_TMPL—The group name or channel number is already used, please select another

DS_UCV_APS_GROUP_CONFIG—This name is already used, please select another

DS_UCV_APS_GROUP_CONFIG_TMPL—This name is already used, please select another

DS_UCV_ATM_CID—Either this name is already used, or another CID already exists with that index.

DS_UCV_ATM_CID_TMPL—Either this name is already used, or another CID already exists with that index.

DS_UCV_AAL2_ELCP—Either this name is already used, or another ELCP already exists with that index.

DS_UCV_AAL2_ELCP_TMPL—Either this name is already used, or another Elcp already exists with that index.

DS_UCV_ATM_END_POINT—This index, VPI, VCI combination is already in use on this port, please select another.

DS_UCV_ATM_END_POINT_TMPL—This index, VPI, VCI combination is already in use on this port, please select another.

DS_UCV_ATM_LOGICAL—This name is already used, please select another

DS_UCV_ATM_LOGICAL_TMPL—This name is already used, please select another

DS_UCV_ATM_TRAFFIC_DESC—This name or index is already used, please select another

DS_UCV_ATM_TRAFFIC_DESC_TMPL—This name is already used, please select another

DS_UCV_ATM_VC_CROSS_CONN—This name or cross connect ID is already used, please select another

DS_UCV_ATM_VC_CROSS_CONN_TMPL—This name is already used, please select another

DS_UCV_ATM_VPI—This index, VPI combination is already in use on this port, please select another

DS_UCV_ATM_VPI_TMPL—This name is already used, please select another

DS_UCV_ATM_VP_CROSS_CONN—This name or cross connect ID is already used, please select another

DS_UCV_ATM_VP_CROSS_CONN_TMPL—This name is already used, please select another

DS_UCV_BRIDGE_LOGICAL—This name is already used, please select another

DS_UCV_BRIDGE_LOGICAL_TMPL—This name is already used, please select another

DS_UCV_BULK_VC_CROSS_CONN—This name is already used, please select another

DS_UCV_BULK_VC_CROSS_CONN_TMPL—This name is already used, please select another

DS_UCV_CARD—This name or slot number is already in use, please select another

DS_UCV_CARD_ALARM_CLEAR—This ID is already used, please select another

DS_UCV_CARD_ALARM_EVENT—This ID is already used, please select another

DS_UCV_CARD_DEFAULT—This ID is already used, please select another

DS_UCV_CARD_DEFAULT_TMPL—This name is already used, please select another

DS_UCV_CARD_EXTERNAL_RELAY—This name is already used, please select another

DS_UCV_CARD_EXTERNAL_RELAY_TMPL—This name is already used, please select another

DS_UCV_CARD_GROUP—This name or ID is already used, please select another

DS_UCV_CARD_GROUP_ENTRY—This name or ID is already used, please select another

DS_UCV_CARD_GROUP_ENTRY_TMPL—This name is already used, please select another

DS_UCV_CARD_GROUP_TMPL—This name is already used, please select another

DS_UCV_CARD_TMPL—This name is already used, please select another

DS_UCV_COMMUNITY_NAME—This name is already used, please select another

DS_UCV_COMMUNITY_NAME_TMPL—This name is already used, please select another

DS_UCV_COMPLETED_UPDATE—This request ID is already being used, please select another

DS_UCV_COMPOSITE—This name is already used, please select another

DS_UCV_COMPOSITE_TMPL—This name is already used, please select another

DS_UCV_COMPOSITE_TMPL_ASSOC—This name is already used, please select another

DS_UCV_CUSTOMER—This ID is already used, please select another

DS_UCV_CUSTOMER_TMPL—This name is already used, please select another

DS_UCV_DEVICE—This IP address is already in use, please select another

DS_UCV_DEVICE_ALARM_CLEAR—This ID is already used, please select another

DS_UCV_DEVICE_ALARM_EVENT—This ID is already used, please select another

DS_UCV_DEVICE_LOCK—This name is already used, please select another

DS_UCV_DEVICE_TMPL—This name is already used, please select another

DS_UCV_DHCP_GROUP—This name is already used, please select another

DS_UCV_DHCP_GROUP_TMPL—This name is already used, please select another

DS_UCV_DHCP_HOST—This ID is already used, please select another

DS_UCV_DHCP_HOST_TMPL—This name is already used, please select another

DS_UCV_DHCP_LEASE—This name is already used, please select another

DS_UCV_DHCP_LEASE_TMPL—This name is already used, please select another

DS_UCV_DHCP_SERV_DEFAULT—This ID is already used, please select another

DS_UCV_DHCP_SERV_DEFAULT_TMPL—This name is already used, please select another

DS_UCV_DHCP_SUBNET—This name is already used, please select another

DS_UCV_DHCP_SUBNET_TMPL—This name is already used, please select another

DS_UCV_DS1_ATM_PHYSICAL—This ID is already used, please select another

DS_UCV_DS1_ATM_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_DS1_CHANNEL—This name is already used, please select another

DS_UCV_DS1_CHANNEL_TMPL—This name is already used, please select another

DS_UCV_DS1_PHYSICAL—This name is already used, please select another

DS_UCV_DS1_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_DS3_UNCHANNELIZED_PHYSICAL—This name is already used, please select another

DS_UCV_DS3_UNCHANNELIZED_PHYSICA_TMPL—This name is already used, please select another

DS_UCV_DSL_ALARM_PROFILE—This name is already used, please select another

DS_UCV_DSL_ALARM_PROFILE_TMPL—This name is already used, please select another

DS_UCV_DSL_PHYSICAL—This name is already used, please select another

DS_UCV_DSL_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_E1_ATM_PHYSICAL—This name is already used, please select another

DS_UCV_E1_ATM_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_E1_PHYSICAL—This name is already used, please select another

DS_UCV_E1_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_E3_PHYSICAL—This name is already used, please select another

DS_UCV_E3_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_E3_UNCHANNELIZED_PHYSICAL—This name is already used, please select another

DS_UCV_E3_UNCHANNELIZED_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_END_PT—This ID is already used, please select another

DS_UCV_END_PT_TMPL—This name is already used, please select another

DS_UCV_ETHER_PHYSICAL—This name is already used, please select another

DS_UCV_ETHER_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_EVENT_ALARM_ASSOC—This ID is already used, please select another

DS_UCV_FILTER—This name is already used, please select another

DS_UCV_FILTER_GROUP—This name is already used, please select another

DS_UCV_GR303CRV—This name or CRV index is already used, please select another

DS_UCV_GR303CRV_TMPL—This name or CRV index is already used, please select another

DS_UCV_GR303IG—This name is already used, please select another

DS_UCV_GR303IG_TMPL—This name is already used, please select another

DS_UCV_GR303_LINE_MAP—This name, or ig name / physical ID combination is already used, please select another

DS_UCV_GR303_LINE_MAP_TMPL—This name is already used, please select another

DS_UCV_GROUP_ROLE—This name is already used, please select another

DS_UCV_HDSL2_CONFIG_PROFILE—This name is already used, please select another

DS_UCV_HDSL2_CONFIG_PROFILE_TMPL—This name is already used, please select another

DS_UCV_IMA_GROUP—This name is already used, please select another

DS_UCV_IMA_GROUP_TMPL—This name is already used, please select another

DS_UCV_IMA_LINK—This name is already used, please select another

DS_UCV_IMA_LINK_TMPL—This name is already used, please select another

DS_UCV_IMA_PHYSICAL_GROUP—This name is already used, please select another

DS_UCV_IMA_PHYSICAL_GROUP_TMPL—This name is already used, please select another

DS_UCV_IP_ALIAS_ADDR—This name is already used, please select another

DS_UCV_IP_ALIAS_ADDR_TMPL—This name is already used, please select another

DS_UCV_IP_CONFIG—This name is already used, please select another

DS_UCV_IP_CONFIG_TMPL—This name is already used, please select another

DS_UCV_IP_DHCP_CLIENT_IF—This name is already used, please select another

DS_UCV_IP_DHCP_CLIENT_IF_TMPL—This name is already used, please select another

DS_UCV_IP_DHCP_CLIENT_RESOURCE—This ID is already used, please select another

DS_UCV_IP_DHCP_CLIENT_RESOURCE_TMPL—This name is already used, please select another

DS_UCV_IP_DNS_RESOLVER_CONFIG—This name is already used, please select another

DS_UCV_IP_DNS_RESOLVER_CONFIG_TMPL—This name is already used, please select another

DS_UCV_IP_DNS_RESOLVER_HOST—This name or host name is already used, please select another

DS_UCV_IP_DNS_RESOLVER_HOST_TMPL—This name or host name is already used, please select another

DS_UCV_IP_FILTER_SPEC—This name is already used, please select another

DS_UCV_IP_FILTER_SPEC_TMPL—This name is already used, please select another

DS_UCV_IP_FILTER_STMT—This name is already used, please select another

DS_UCV_IP_FILTER_STMT_TMPL—This name is already used, please select another

DS_UCV_IP_LOGICAL—This name is already used, please select another

DS_UCV_IP_LOGICAL_TMPL—This name is already used, please select another

DS_UCV_IP_NET_TO_MEDIA—This name is already used, please select another

DS_UCV_IP_NET_TO_MEDIA_TMPL—This name is already used, please select another

DS_UCV_IP_RIP2_CONFIG—This name is already used, please select another

DS_UCV_IP_RIP2_CONFIG_TMPL—This name is already used, please select another

DS_UCV_IP_RIP2_GLOBAL—This name is already used, please select another

DS_UCV_IP_RIP2_GLOBAL_TMPL—This name is already used, please select another

DS_UCV_IP_ROUTING_DOMAIN—This name is already used, please select another

DS_UCV_IP_ROUTING_DOMAIN_TMPL—This name is already used, please select another

DS_UCV_IP_STATIC_ROUTE—Either this name is already used, or the combination of dest, NetMask, next hop, interface number, metric, region, device, routing domain ID is already used.

DS_UCV_IP_STATIC_ROUTE_TMPL—This name is already used, please select another

DS_UCV_ISDN_ALARM_PROFILE—This ID is already used, please select another

DS_UCV_ISDN_ALARM_PROFILE_TMPL—This name is already used, please select another

DS_UCV_ISDN_BRI_PHYSICAL—This index is already used, please select another

DS_UCV_ISDN_BRI_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_ISDN_PHYSICAL—This index is already used, please select another

DS_UCV_ISDN_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_LEA_CONTROLLER—This IP Address is already used, please select another

DS_UCV_LEA_CONTROLLER_TMPL—This name is already used, please select another

DS_UCV_LEA_RELAY_PORT—This controller ID and relay index combination is already used, please select another

DS_UCV_LEA_RELAY_PORT_TMPL—This name is already used, please select another

DS_UCV_LINE_GROUP—This name is already used, please select another

DS_UCV_LINE_GROUP_ENTRY—This name is already used, please select another

DS_UCV_LINE_GROUP_ENTRY_TMPL—This name is already used, please select another

DS_UCV_LINE_GROUP_TMPL—This name is already used, please select another

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DS_UCV_LOGICAL_ALARM_CLEAR—This ID is already used, please select another

DS_UCV_LOGICAL_ALARM_EVENT—This ID is already used, please select another

DS_UCV_LOGICAL_TMPL—This name is already used, please select another

DS_UCV_MAU_PHYSICAL—This name is already used, please select another

DS_UCV_MAU_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_MTAC—This name is already used, please select another

DS_UCV_MTAC_TMPL—This name is already used, please select another

DS_UCV_NAT_CONFIG—This ID is already used, please select another

DS_UCV_NAT_CONFIG_TMPL—This name is already used, please select another

DS_UCV_NTP_CONFIG—This name is already used, please select another

DS_UCV_NTP_CONFIG_TMPL—This name is already used, please select another

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DS_UCV_OPERATOR_GROUP_ROLE—This name is already used, please select another

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DS_UCV_POTS_EM_CONFIG_TMPL—This name is already used, please select another

DS_UCV_POTS_FXO_CONFIG—This ID is already used, please select another

DS_UCV_POTS_FXO_CONFIG_TMPL—This name is already used, please select another

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DS_UCV_POTS_FXS_CONFIG_TMPL—This name is already used, please select another

DS_UCV_POTS_PHYSICAL—This ID is already used, please select another

DS_UCV_POTS_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_PPP_AUTH—This ID is already used, please select another

DS_UCV_PPP_AUTH_TMPL—This name is already used, please select another

DS_UCV_PPP_LCP—This ID is already used, please select another

DS_UCV_PPP_LCP_TMPL—This name is already used, please select another

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DS_UCV_PPP_LOGICAL_TMPL—This name is already used, please select another

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DS_UCV_PVC_ALARM_CLEAR—This ID is already used, please select another

DS_UCV_PVC_ALARM_EVENT—This ID is already used, please select another

DS_UCV_REGION—This name is already used, please select another

DS_UCV_REGION_ALARM_CLEAR—This ID is already used, please select another

DS_UCV_REGION_ALARM_EVENT—This ID is already used, please select another

DS_UCV_REGION_TMPL—This name is already used, please select another

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DS_UCV_RS232_ASYNC_CONFIG_TMPL—This name is already used, please select another

DS_UCV_RS232_PHYSICAL—This name is already used, please select another

DS_UCV_RS232_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_SDSL_CONFIG_PROFILE—This name is already used, please select another

DS_UCV_SDSL_CONFIG_PROFILE_TMPL—This name is already used, please select another

DS_UCV_SHDSL_CONFIG_PROFILE—This name is already used, please select another

DS_UCV_SHDSL_CONFIG_PROFILE_TMPL—This name is already used, please select another

DS_UCV_SHELF—This name is already used, please select another

DS_UCV_SHELF_ALARM_CLEAR—This ID is already used, please select another

DS_UCV_SHELF_ALARM_EVENT—This ID is already used, please select another

DS_UCV_SHELF_TMPL—This name is already used, please select another

DS_UCV_SOFTWARE_DOWNLOAD—This name is already used, please select another

DS_UCV_SOFTWARE_DOWNLOAD_TMPL—This name is already used, please select another

DS_UCV_SONET_OC3_PHYSICAL—This name is already used, please select another

DS_UCV_SONET_OC3_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_SONET_PHYSICAL—This name is already used, please select another

DS_UCV_SONET_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_SUBSCRIBER—This name is already used, please select another

DS_UCV_SUBSCRIBER_PORT—This ID is already used, please select another

DS_UCV_SUBSCRIBER_PORT_TMPL—This name is already used, please select another

DS_UCV_SUBSCRIBER_TMPL—This name is already used, please select another

DS_UCV_SUB_DATA_CONN—This ID is already used, please select another

DS_UCV_SUB_DATA_CONN_TMPL—This name is already used, please select another

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DS_UCV_SUB_VOICE_AAL2_END_PT_TMPL—This name is already used, please select another

DS_UCV_SUB_VOICE_CONN—This ID is already used, please select another

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DS_UCV_SUB_VOICE_GR303_END_PT—This ID is already used, please select another

DS_UCV_SUB_VOICE_GR303_END_PT_TMPL—This name is already used, please select another

DS_UCV_SUB_VOICE_ISDN_END_PT—This ID is already used, please select another

DS_UCV_SUB_VOICE_ISDN_END_PT_TMPL—This name is already used, please select another

DS_UCV_SUB_VOICE_POTS_END_PT—This ID is already used, please select another

DS_UCV_SUB_VOICE_POTS_END_PT_TMPL—This name is already used, please select another

DS_UCV_SUB_VOICE_V52_END_PT—This ID is already used, please select another

DS_UCV_SUB_VOICE_V52_END_PT_TMPL—This name is already used, please select another

DS_UCV_SYSTEM_CLOCK—This name is already used, please select another

DS_UCV_SYSTEM_CLOCK_TMPL—This name is already used, please select another

DS_UCV_T1E1_PHYSICAL—This name is already used, please select another

DS_UCV_T1E1_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_T1_ATM_PHYSICAL—This name is already used, please select another

DS_UCV_T1_ATM_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_T1_CHANNEL—This name is already used, please select another

DS_UCV_T1_CHANNEL_TMPL—This name is already used, please select another

DS_UCV_T1_PHYSICAL—This name is already used, please select another

DS_UCV_T1_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_T3_PHYSICAL—This name is already used, please select another

DS_UCV_T3_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_T3_UNCHANNELIZED_PHYSICAL—This name is already used, please select another

DS_UCV_T3_UNCHANNELIZED_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_TEMPLATE—This name is already used, please select another

DS_UCV_TRAP_CONFIG—This destination is already used, please select another

DS_UCV_TRAP_CONFIG_TMPL—This name is already used, please select another

DS_UCV_TRAP_EVENT—This ID is already used, please select another

DS_UCV_TRAP_FORWARD_HOST—This IP Address is already used, please select another

DS_UCV_UNNUMBERED_IP_ALIAS—This ID is already used, please select another

DS_UCV_UNNUMBERED_IP_ALIAS_TMPL—This name is already used, please select another

DS_UCV_UNT_DS3_PHYSICAL—This name is already used, please select another

DS_UCV_UNT_DS3_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_UNT_PHYSICAL—This name is already used, please select another

DS_UCV_UNT_PHYSICAL_TMPL—This name is already used, please select another

DS_UCV_V52_CCHANNEL—The ig name, channel number, and time slot index combination is already used, please select another.

DS_UCV_V52_CCHANNEL_TMPL—This name is already used, please select another

DS_UCV_V52_CPATH—This name is already used, or the combination of ig name, path type, and channel ID is already used, please select another.

DS_UCV_V52_CPATH_TMPL—This name is already used, please select another

DS_UCV_V52_IG—This name is already used, please select another

DS_UCV_V52_IG_TMPL—This name is already used, please select another

DS_UCV_V52_LINK—This name is already used, please select another

DS_UCV_V52_LINK_TMPL—This name is already used, please select another

DS_UCV_V52_PROTECTION_CPATH—This name is already used, please select another

DS_UCV_V52_PROTECTION_CPATH_TMPL—This name is already used, please select another

DS_UCV_V52_USER_PORT—This name, user port type, and address combination is already used, please select another.

DS_UCV_V52_USER_PORT_TMPL—This name is already used, please select another

DS_UCV_VASP_DEFAULT_CONFIG—This name is already used, please select another

DS_UCV_VASP_DEFAULT_CONFIG_TMPL—This name is already used, please select another

DS_UCV_VCL_RATE—This ID is already used, please select another

DS_UCV_VCL_RATE_TMPL—This name is already used, please select another

DS_UCV_XDSL_PHYSICAL—This name is already used, please select another

DS_UCV_XDSL_PHYSICAL_TMPL—This name is already used, please select another

Error strings defined for FaultService

FS_ADD_SUCCESS—FaultService add request successful.

FS_DELETE_SUCCESS—FaultService delete request successful.

FS_LIST_SUCCESS—FaultService list request successful.

FS_GET_SUCCESS—FaultService get request successful.

FS_MODIFY_SUCCESS—FaultService modify request successful.

FS_ADS_AUTH_ERROR—FaultService Authorization error from ADS.

Error strings defined for network protocol layer

NWS_TIME_OUT_ERROR—SNMP ERROR: The request to the device timed out. Please verify that the device is reachable before attempting further provisioning.

NWS_AUTHORIZATION_ERROR—SNMP ERROR: The community name used was not authorized to perform the request on the device.

NWS_INVALID_VALUE_ERROR—SNMP ERROR: At least one of the values sent in the request to the device is invalid.

NWS_COMMIT_ERROR—SNMP ERROR: The device indicates that it could not store the change requested.

NWS_GENERAL_ERROR—SNMP ERROR: The device had difficulty processing the request. Please verify that the parameter values are within acceptable ranges.

NWS_INCONSISTENT_NAME_ERROR—SNMP ERROR: The device indicates that at least one of the parameter names was invalid. Please verify that the device version supports all the parameters in the request.

NWS_INCONSISTENT_VALUE_ERROR—SNMP ERROR: The device indicates that either the object being added already exists, or at least one of the parameter values was not within an acceptable range.

NWS_NO_ACCESS_ERROR—SNMP ERROR: The community name used was not authorized to perform the request on the device.

NWS_CREATION_ERROR—SNMP ERROR: The device could not create the profile as requested.

NWS_ATTR_EXIST_ERROR—SNMP ERROR: An attribute specified in the request is not available in this version of device software. Please verify that the device software version is correctly updated to support this attribute before attempting the request again.

NWS_OBJ_EXIST_ERROR—SNMP ERROR: The device does not recognize the object being queried.

NWS_NO_ERROR—The request to the device was successful.

NWS_NOT_WRITABLE_ERROR—SNMP ERROR: The device indicates that at least one of the parameters specified in the request was not writable (modifiable).

NWS_READ_ONLY_ERROR—SNMP ERROR: The device indicates that at least one of the parameters specified in the request is read-only and should not be modified.

NWS_RESOURCE_ERROR—SNMP ERROR: The device indicates that no further resources could be allocated to complete the request.

NWS_TOO_BIG_ERROR—SNMP ERROR: The response from the device was too big to be sent via SNMP. Please decrease the number of requested attributes, and send the request again.

NWS_UNDO_ERROR—SNMP ERROR: The device software encountered an error while processing the request, but it was unable to undo some of its provisioning. Please synchronize with the device to obtain the partially created device profiles before proceeding with further provisioning.

NWS_WRONG_ENCODING_ERROR—SNMP ERROR: The device indicates that one or more attributes in the request were not encoding properly.

NWS_WRONG_LENGTH_ERROR—SNMP ERROR: The device indicates that the data for one or more of the attributes was too long for its type.

NWS_WRONG_TYPE_ERROR—SNMP ERROR: The device indicates that at least one of the attributes in the request had an incorrect data type.

NWS_WRONG_VALUE_ERROR—SNMP ERROR: The device indicates that at least one of the parameters was not within an acceptable range.

NWS_SNMP_SEND_ERROR—DZMS was unable to send the SNMP PDU to the device. Please verify that network congestion levels are at tolerable levels.

NWS_SNMP_OPEN_ERROR—DZMS was unable to Open an SNMP Session. Please verify that the DZMS server has sufficient resources to open SNMP sockets.

NWS_NO_DATA_ERROR—No data was specified in the request to the device.

NWS_NO_OID_ERROR—The required MIB variable is not loaded. Please verify that DZMS was able to properly complete its startup initialization.

NWS_NO_MIBS_LOADED—There are no MIBs loaded, so the request PDU could not be created. Please verify that DZMS was able to properly complete its startup initialization.

NWS_END_OF_MIB_VIEW—Reached the end of the viewable MIB.

NWS_NO_METADATA_HT—Unable to find METADATA Hashtable for EntityData Type, could not find DeviceRep.

NWS_NO_METADATA—Unable to find METADATA for Attribute Name, could not find DeviceRep.

NWS_NOT_CLONABLE—Unable to create new data object for getNext request.

NWS_NO_INSTANCE_ATTR—Need to specify attributes in order to create the Instance.

NWS_NULL_OID—The OID in the metadata is null.

NWS_NO_OID_SYNTAX—Unable to get syntax for OID.

NWS_NO_SNMP_TYPE_SPECIFIED—The SNMP PDU request type was not specified, so the PDU could not be sent.

NWS_PARSE_ERROR—An invalid value from the device caused a parsing error on this attribute. The network or the device may currently be too busy to handle SNMP requests.

NWS_OPER_ERROR—An error occurs during SNMP operation

Error strings defined for PerformanceService

PS_AUTHORIZATION_ERROR—Not authorized to perform request.

PS_NO_ERROR—Request to performance service was successful.

PS_NO_DATA—There was no data specified in request to performance service.

PS_NO_LEASE—Unable to obtain lease for request.

PS_ERROR_RENEWING_LEASE—Unable to renew lease.

PS_NO_IPADDR—Unable to obtain IP address for device.

PS_NO_TEMPLATE_FOUND—No Template found for name specified.

PS_NO_DS_REFERENCE—Unable to obtain reference to DataService

PS_ATTR_NOT_APPLICABLE—Attribute is not applicable on Object specified.

PS_ATTR_NOT_POLLABLE—Attribute is not pollable.

PS_NO_VALID_INTERVAL—Unable to obtain valid intervals from device.

PS_NOT_VALID_INTERVAL—Interval specified is not valid on device.

PS_NO_FILTER—Filter needs to be specified in a PS_LIST request.

PS_NO_ATTRIBUTES—At least one attribute must be specified in the filter of a list request.

Error strings defined for Common

COMMON_NO_ERROR—No error.

COMMON_HANDLE_CREATE—Unable to create handle.

COMMON_INVALID_HANDLE—Handle hierarchy is invalid.

COMMON_NO_FILTER—Filter needs to be specified in a LIST request.

COMMON_NO_IPADDR—Unable to obtain IP address for device.

COMMON_NO_DATA—There was no data specified in request.

COMMON_ATTR_NOT_APPLICABLE—Attribute is not applicable on object specified.

COMMON_CREATE_DATA_OBJ—There is no metadata found to create BaseData specified.

COMMON_REQUEST_NULL—The request provided is null.

COMMON_NO_IP_ADDR—Unable to obtain device IP address.

COMMON_UNKNOWN—Unknown error occurred.

OperationStatus defaults

COMMON_INITIAL—Initial state.

COMMON_SUCCESS—The operation completed successfully.

COMMON_PARTIAL—The operation was only partially successful.

COMMON_FAILURE—The operation failed.

COMMON_WARNING—Warning: A minor error occurred during processing.

Event manager

EVENTMANAGER_REGISTRATION_ERROR—Failed to register subject with Event Manager.

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